

Large Language Model Enhanced Recommender Systems

Qidong Liu^{1,2}, Xiangyu Zhao², Yuhao Wang², Yejing Wang², Zijian Zhang², Yuqi Sun¹,
Xiang Li, Maolin Wang², Pengyue Jia², Chong Chen, Wei Huang, Feng Tian¹

¹ Xi'an Jiaotong University

² City University of Hong Kong

Aug 3, 2025

Agenda

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang

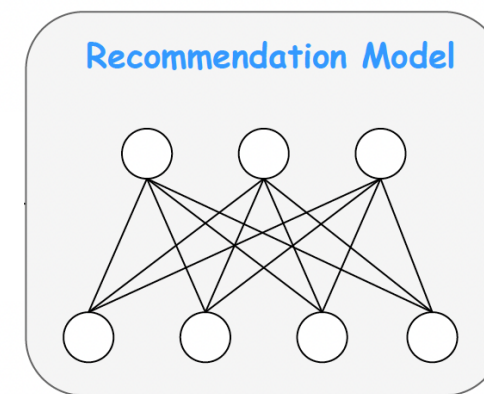
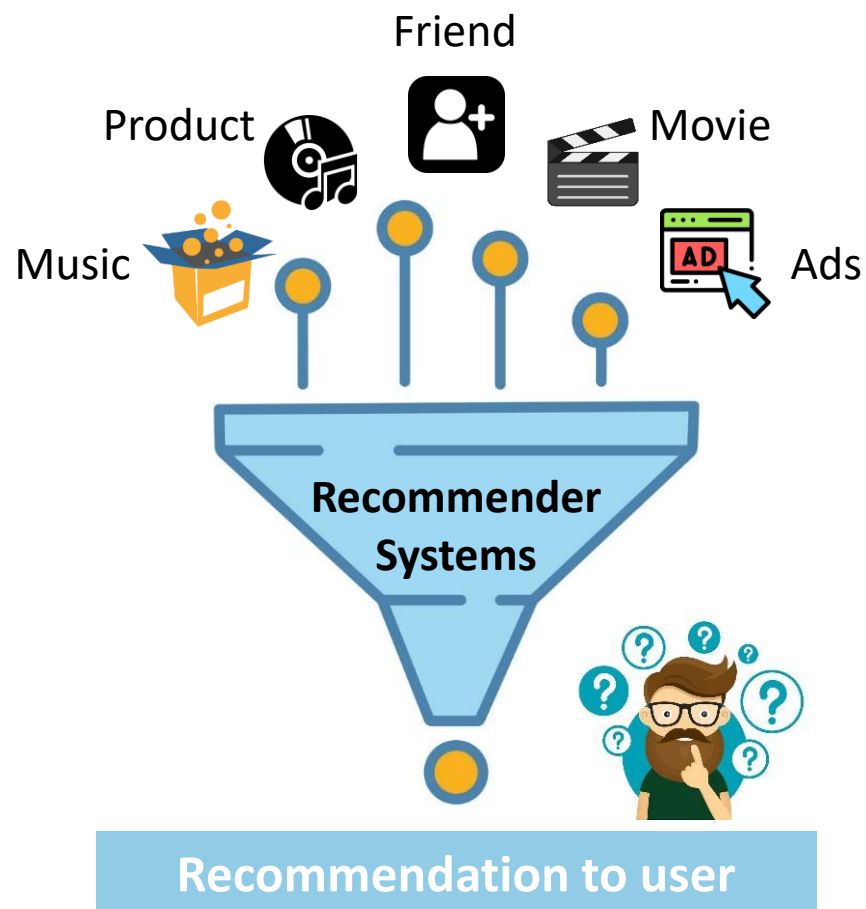


5 Conclusion

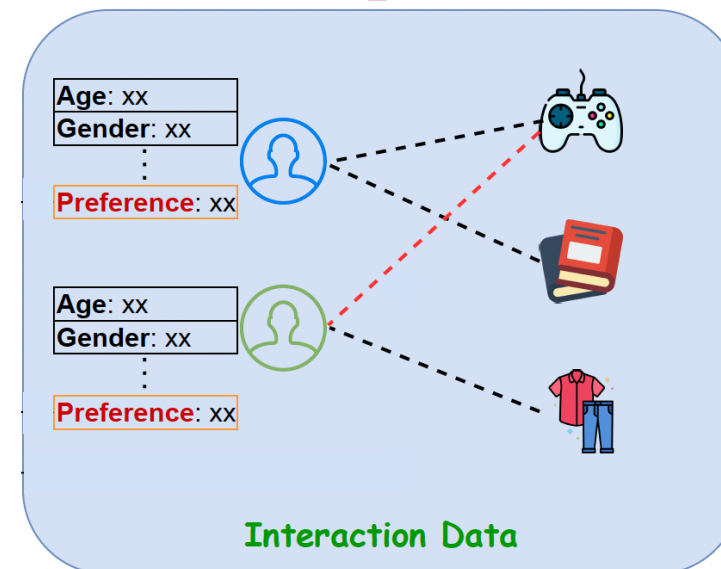


Yejing Wang

Information overload

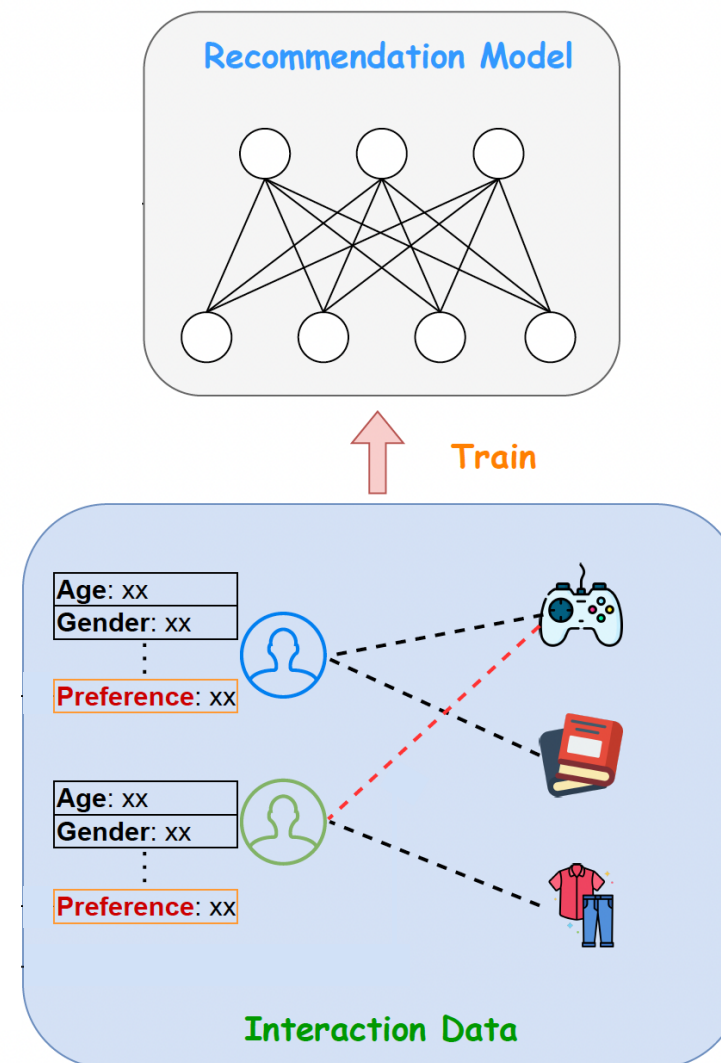


↑ Train



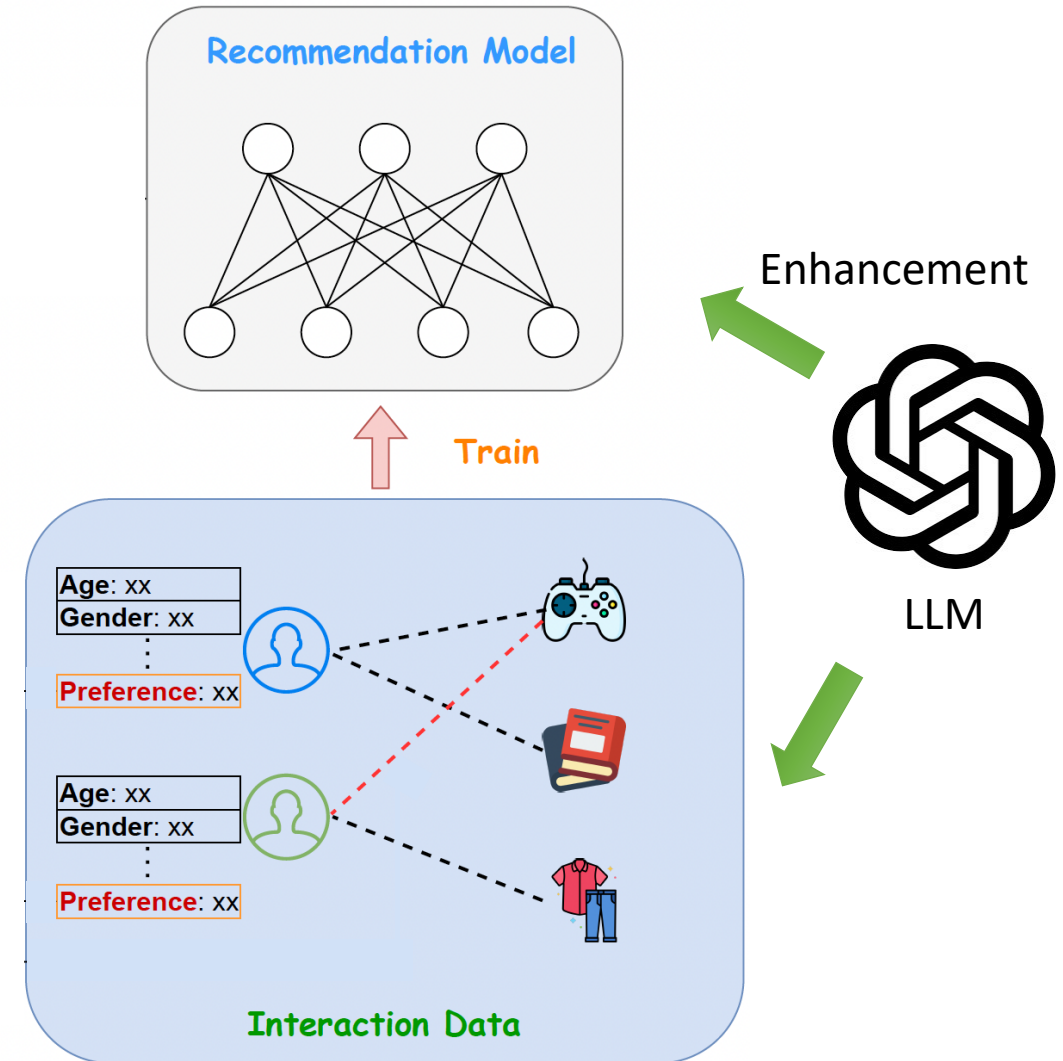
Limitations of RS

- Feature level: Lack of reasoning and knowledge in feature representations
- Interaction level: Data sparsity in user-item interactions
- Model level: Overlooking semantic understanding



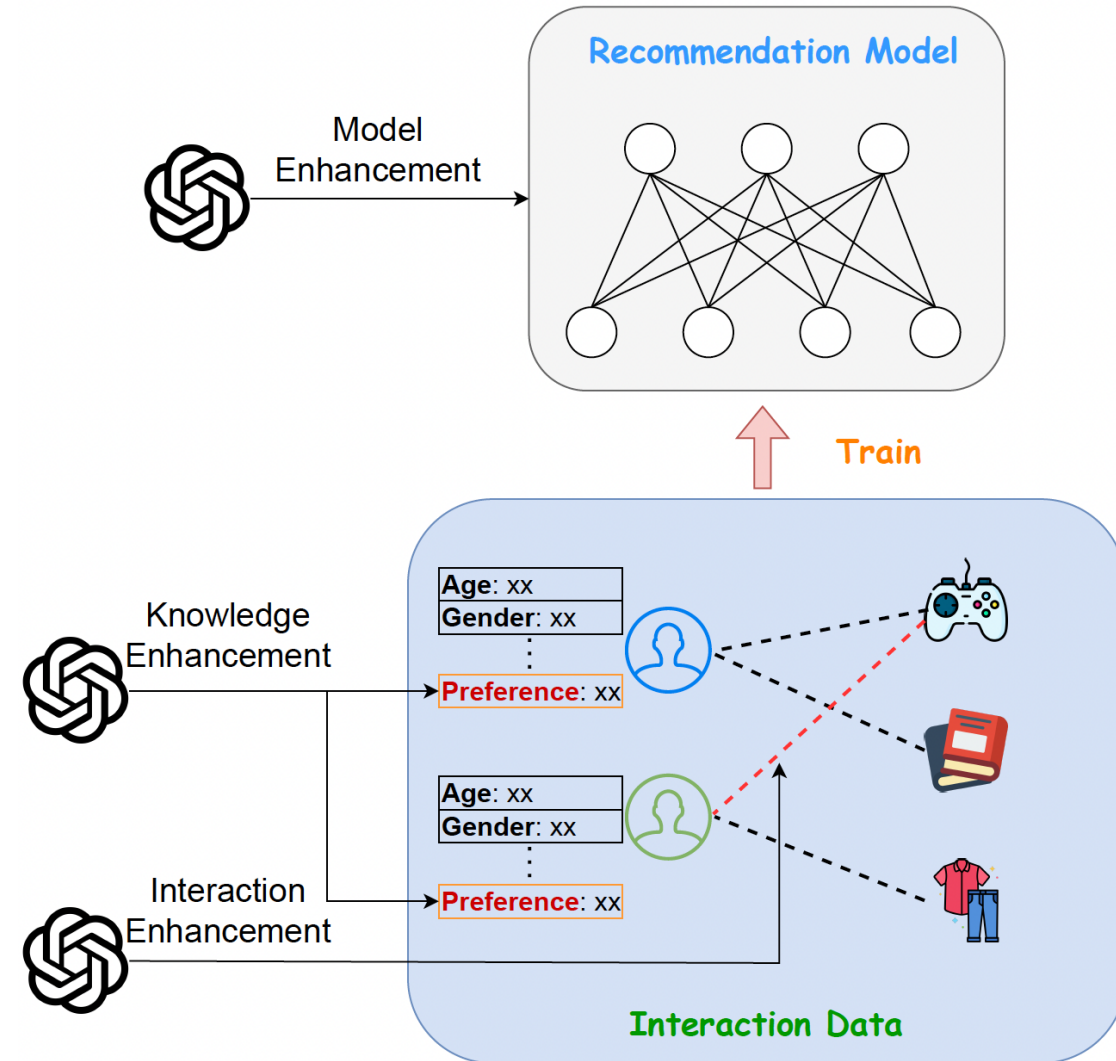
Advantages of LLM

- Feature level: Reasoning abilities and world knowledge to derive textual descriptions
- Interaction level: Approximate users and derive new user-item interactions
- Model level: Analyze interactions from a semantic view



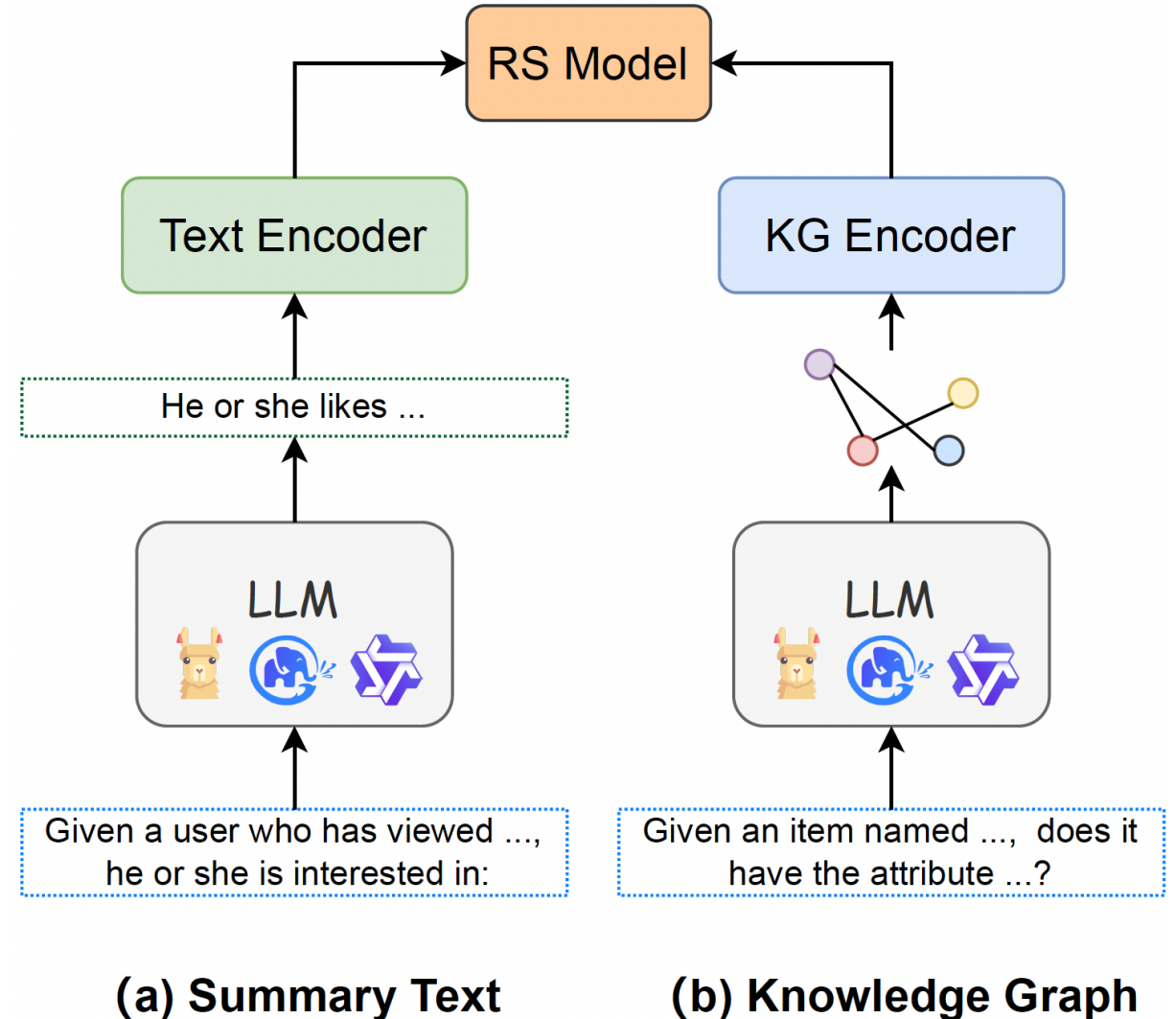
LLM-enhanced RS (LLMERS)

- Knowledge enhancement
- Interaction enhancement
- Model enhancement



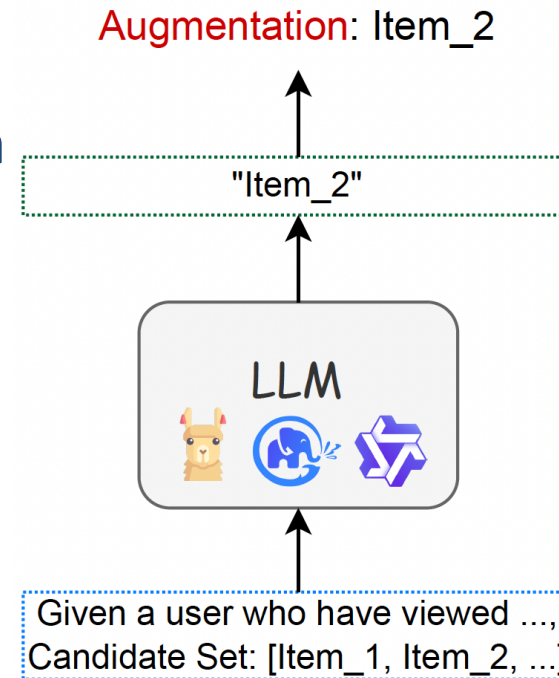
Knowledge enhancement

- Summary Text: Utilizes LLM to summarize characteristics of items or reason for user preferences.
- Knowledge Graph: Applies LLM to generate or augment structured Knowledge Graphs

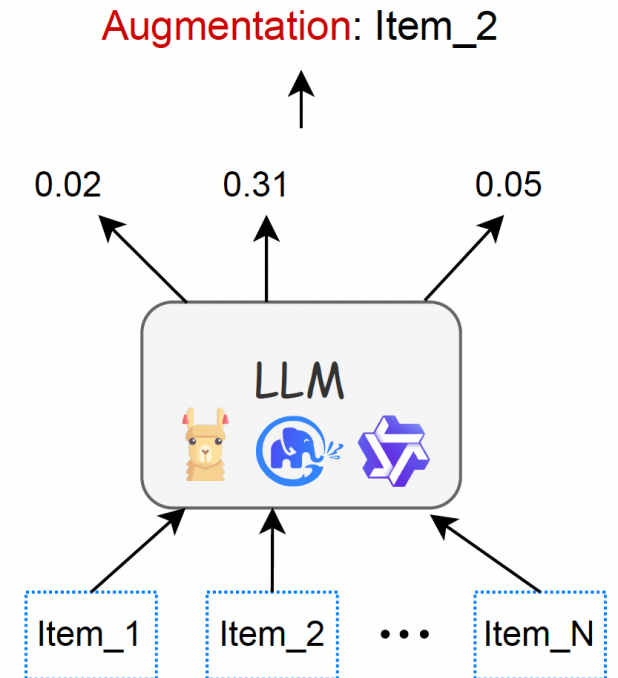


Interaction enhancement

- Text-based: LLM outputs names of pseudo-interacted items as augmentation
- Score-based: LLM derives probabilities of possible interactions for augmentation



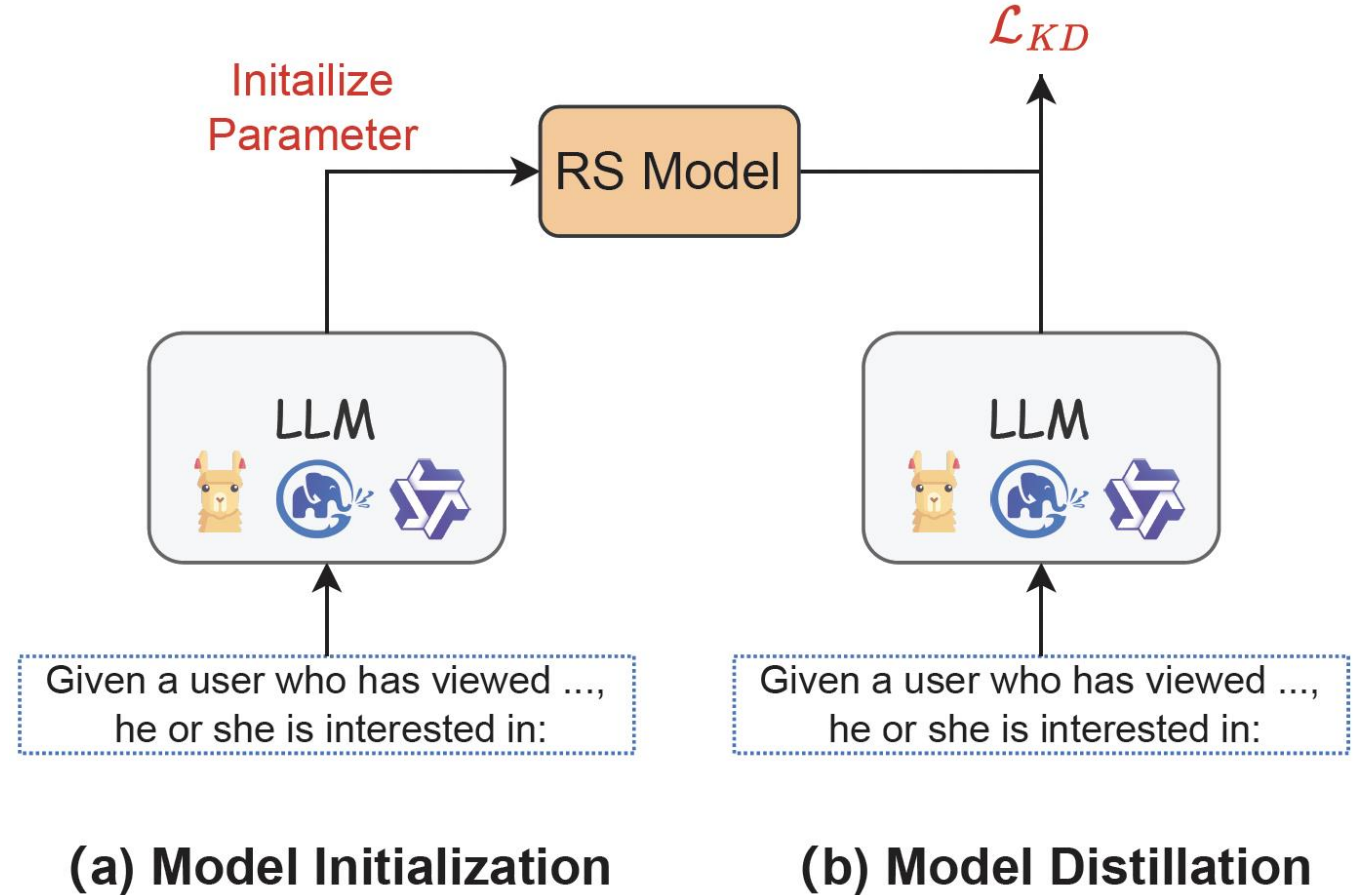
(a) Text-based



(b) Score-based

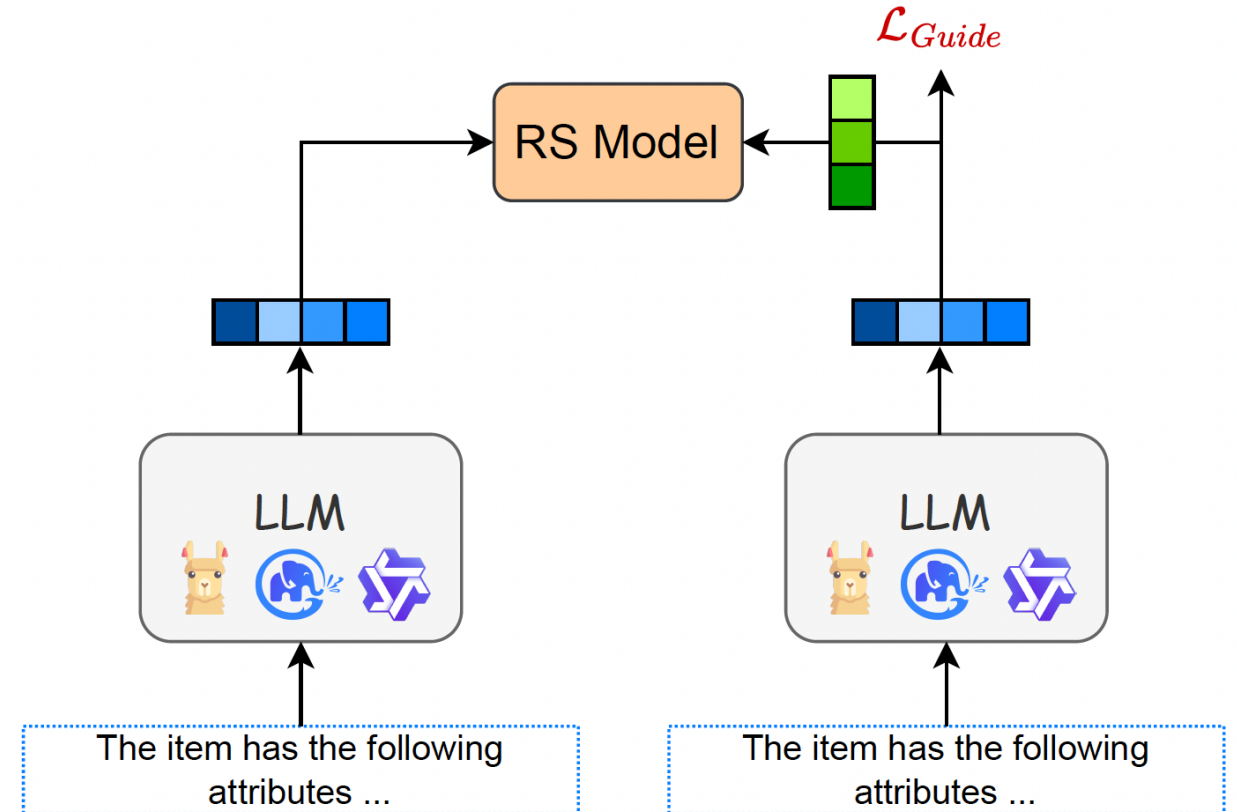
Model enhancement

- Model Initialization: Pretrain or initialize RS model weights with LLM's semantics before training
- Model Distillation: Transfer powerful abilities of LLM to smaller RS models



Model enhancement

- **Embedding Utilization:** Directly use LLM-derived embeddings as a semantic supplement for RS
- **Embedding Guidance:** Use LLM embeddings as guidance for training or parameter synthesis of RS models



(c) Embedding Utilization

(d) Embedding Guidance

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang



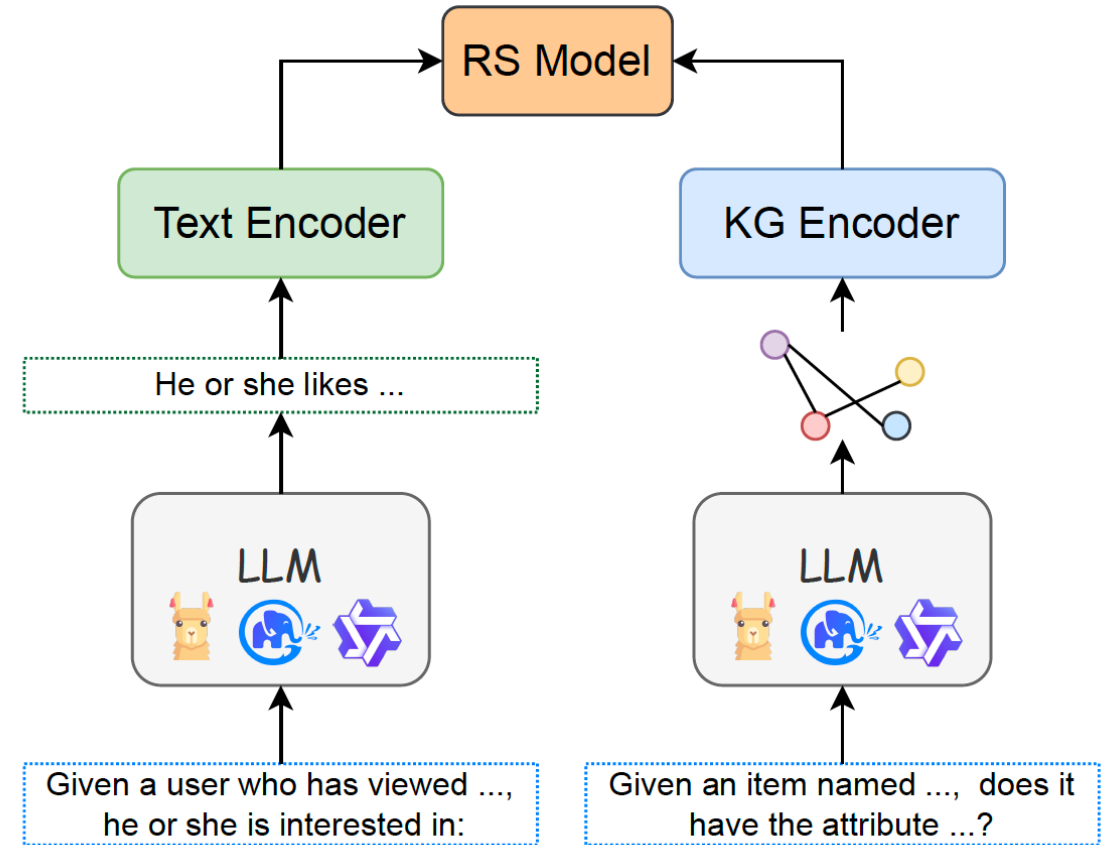
5 Conclusion



Yejing Wang

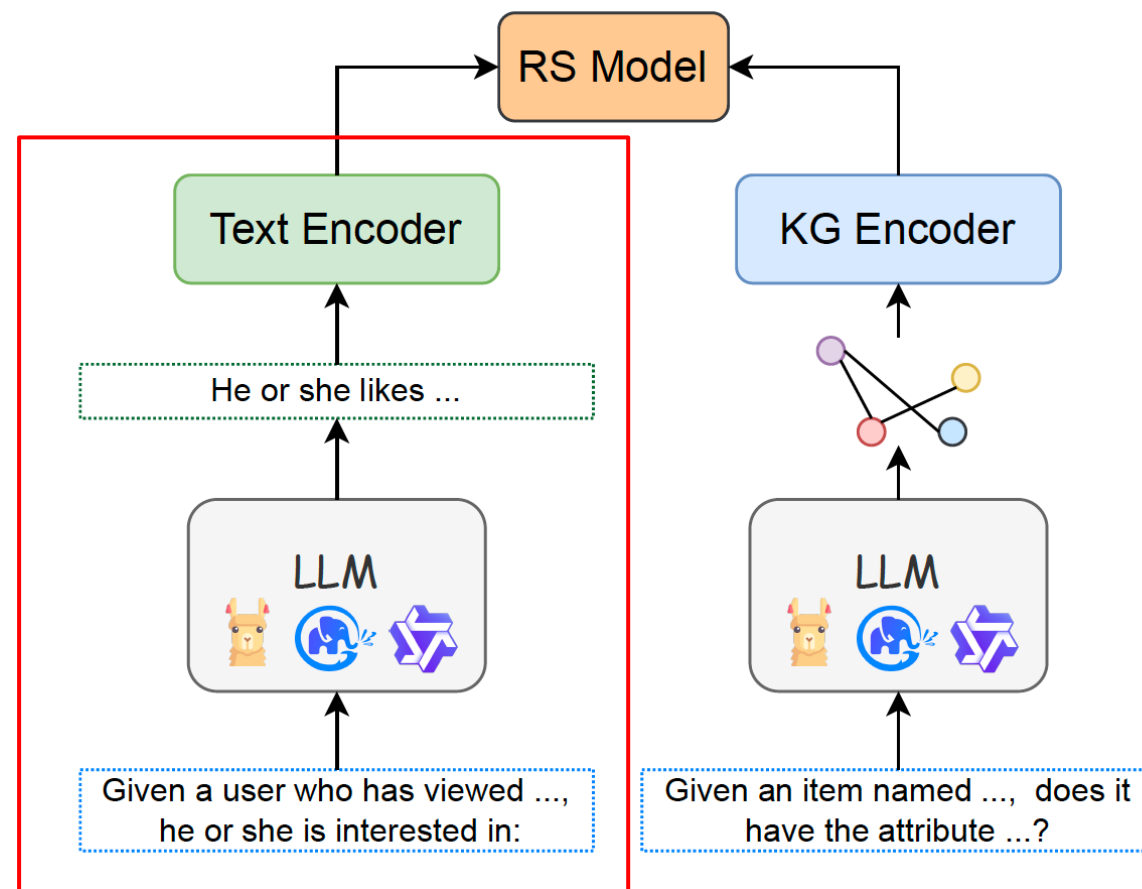
Knowledge Enhancement

- LLM owns extensive world knowledge and powerful reasoning abilities, which can supplement the RS with external knowledge.
- Categories
 - Summary Text
 - Knowledge Graph
 - Combination



Summary Text

- This subcategory refers to utilizing LLM to summarize the characteristics of items or to reason for the preference of users.
- For example, the prompt for summarization can be “Given a user who has viewed <Browsing History>, please explain what he or she is interested in: ”.
- Categories
 - User Only
 - Item Only
 - User & Item



- Background
 - The explainability of recommendation systems is crucial for enhancing user trust and satisfaction.
- Motivation
 - Fine-tuning LLM models for recommendation tasks incurs high computational costs and alignment issues with existing systems, limiting the application potential of proven proprietary/closed-source LLM models, such as GPT-4.
- Key Components
 - Semantic embedding
 - User multi-preference extraction using zero-shot prompting
 - Semantic alignment

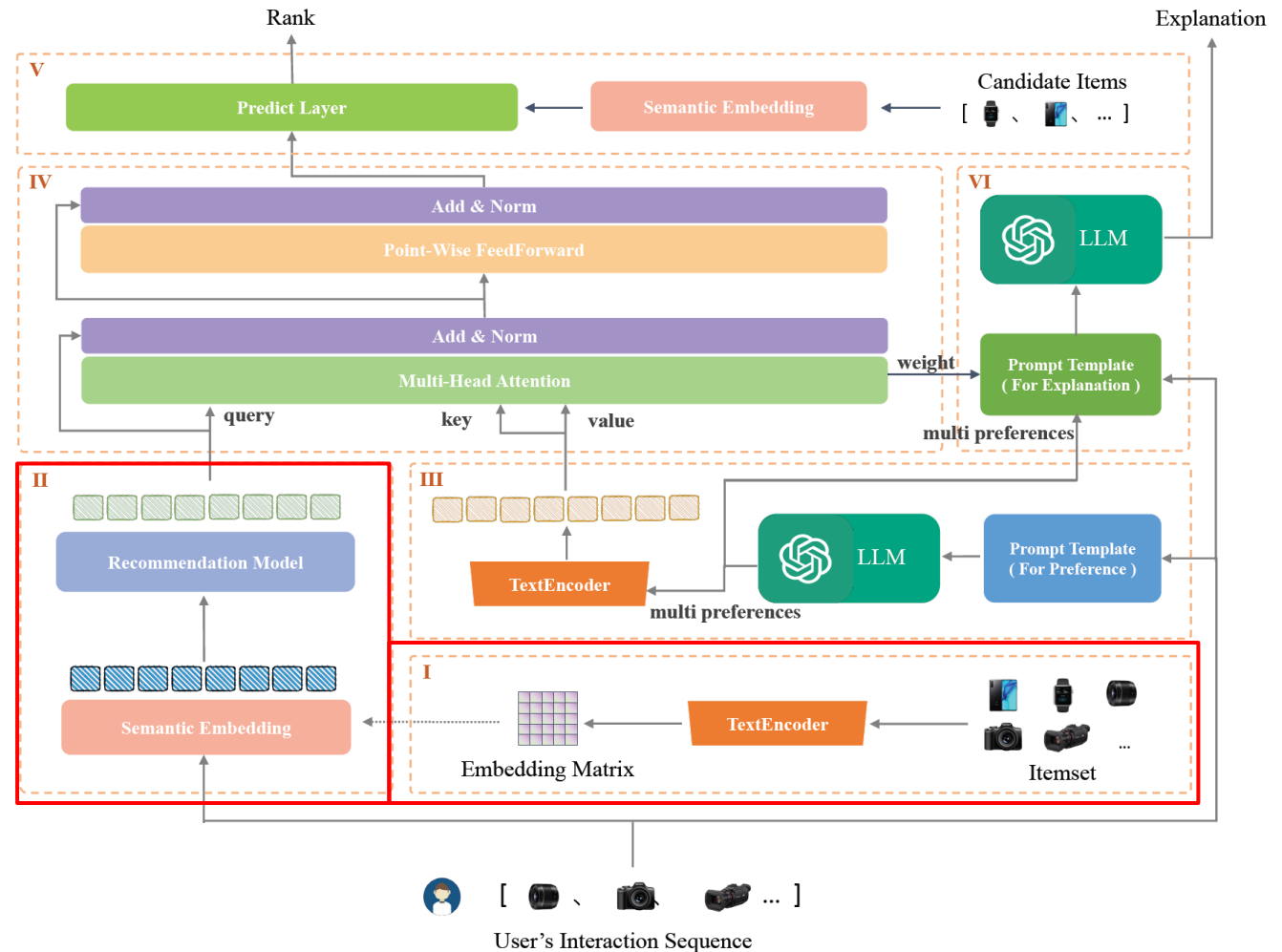
• Semantic Embedding

- Unlike traditional recommender systems that use item IDs, LANE encodes user history as semantic vectors using a text encoder

$$\mathbf{M} = \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \vdots \\ \mathbf{m}_n \end{bmatrix} = \text{TextEncoder} \left(\begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{bmatrix} \right)$$

• Integrated Model Module

- LANE incorporates a standard sequential RS, but adapts its embedding layer to use the semantic vectors generated previously.



- Extracting Multi-Preferences using LLM Zero-Shot Prompting

$$\text{Preference}^u = \text{LLM}(\text{prompt}_p(S^u)),$$

$$\mathbf{P}^u = \text{TextEncoder}(\text{Preference}^u)$$

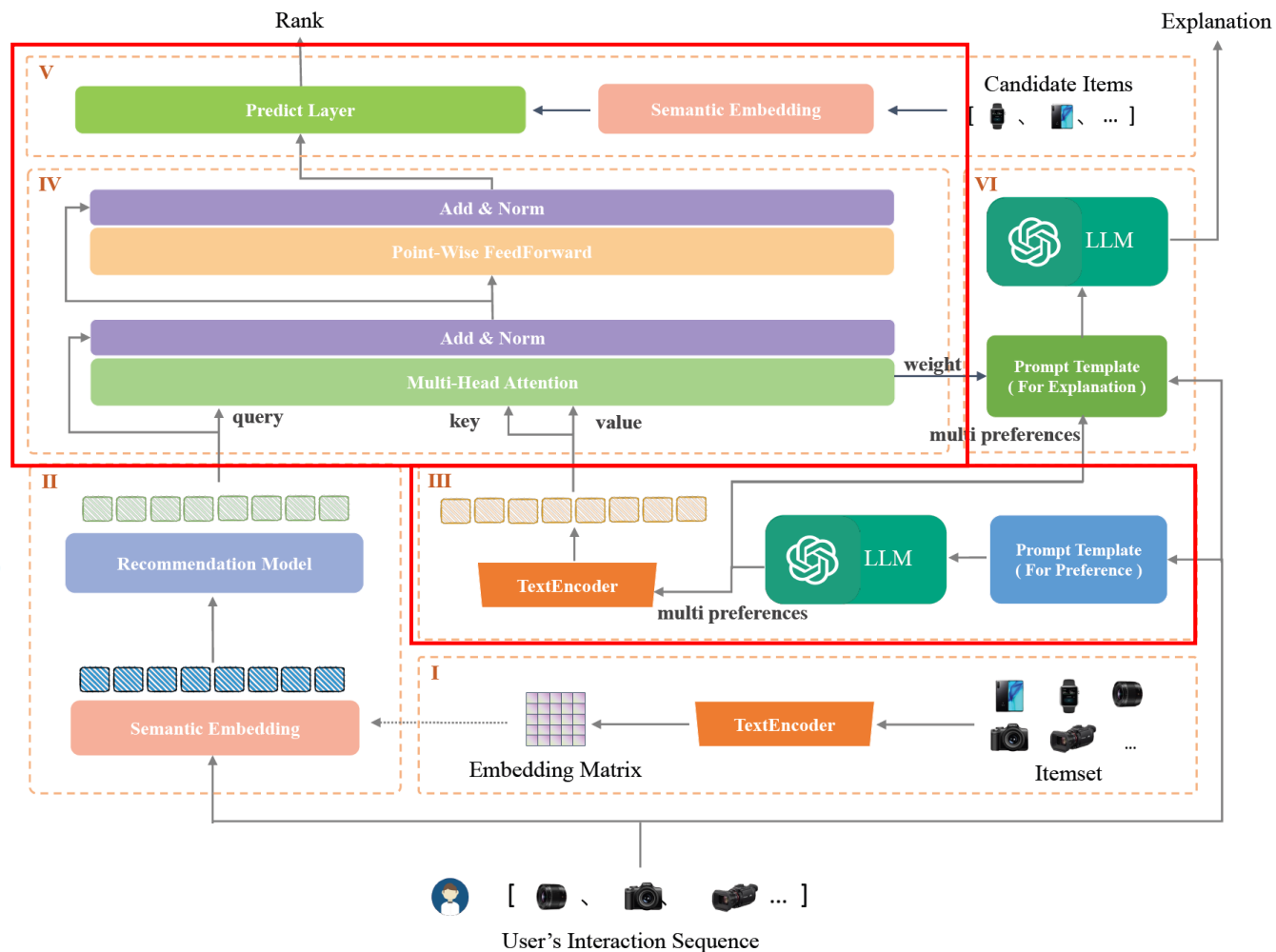
- Semantic Alignment via Multi-Head Attention

$$\text{att}^u = \text{LayerNorm}(\text{Multihead}(\mathbf{Q}^u, \mathbf{P}^u, \mathbf{P}^u)) + \mathbf{Q}^u,$$

$$\mathbf{F}^u = \begin{bmatrix} \mathbf{f}_1^u \\ \mathbf{f}_2^u \\ \vdots \\ \mathbf{f}_n^u \end{bmatrix} = \text{LayerNorm}(\text{FNN}(\text{att}^u)) + \text{att}^u$$

- Prediction

$$\mathbf{r}_{t,i}^u = \mathbf{f}_t^u \cdot \mathbf{m}_i,$$



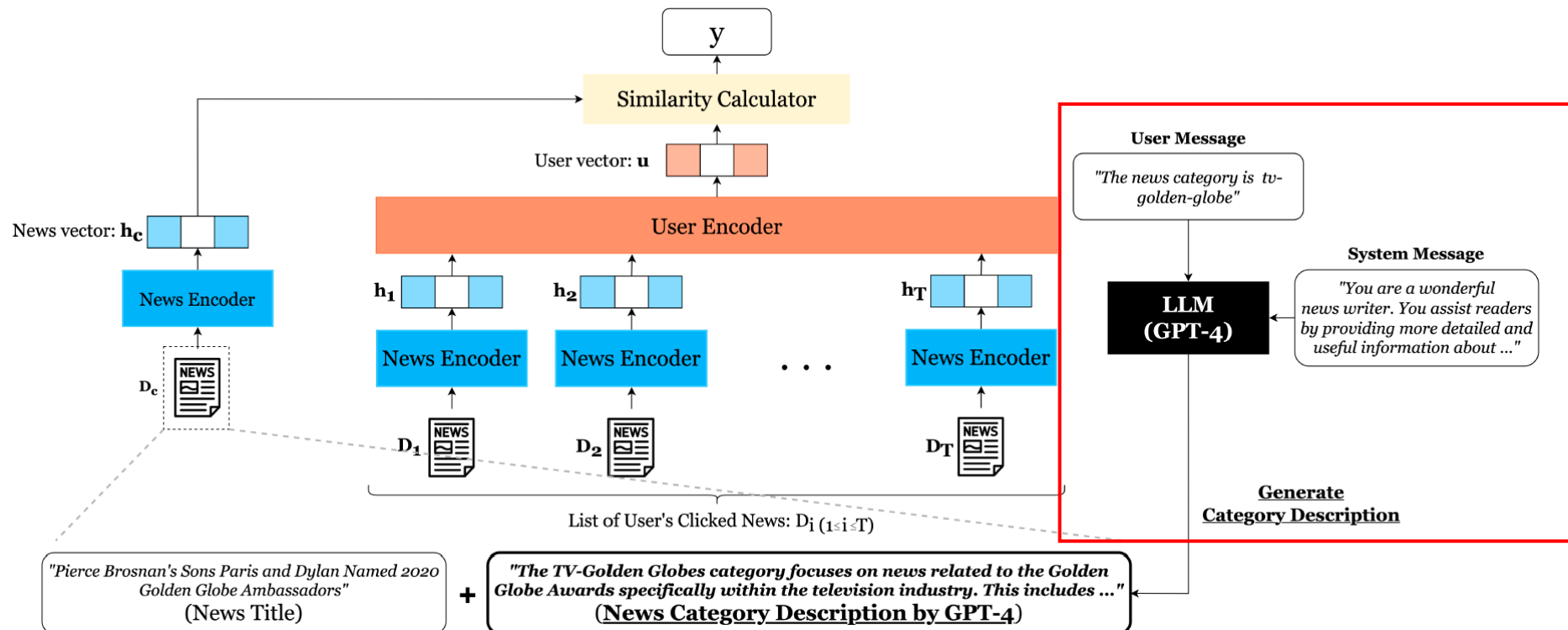
- LANE consistently achieves significant improvements in both NDCG@10 and HR@10 across all datasets and baseline models.

Dataset	Beauty		Steam		ML-1M	
	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10
GRU4Rec	0.2853	0.4422	0.5025	0.7492	0.5277	0.7614
LANE-GRU4Rec	0.3238	0.4825	0.5109	0.7598	0.5667	0.7844
Improv.	13.49%	9.11%	1.67 %	1.41 %	7.39%	3.02%
BERT4Rec	0.224	0.3808	0.477	0.7261	0.4611	0.7151
LANE-ERT4Rec	0.2734	0.4526	0.4982	0.7495	0.5111	0.7646
Improv.	22.05%	18.86%	4.44%	3.22%	10.84%	6.92%
SASRec	0.2831	0.423	0.4789	0.728	0.5701	0.7983
LANE-SASRec	0.3511	0.5172	0.5649	0.803	0.5888	0.8106
Improv.	24.02%	22.27%	17.96%	10.30%	3.28%	1.54%

- **Practical Values:** LANE works without fine-tuning LLMs, making it easy to integrate into existing systems while also providing explainable recommendations.

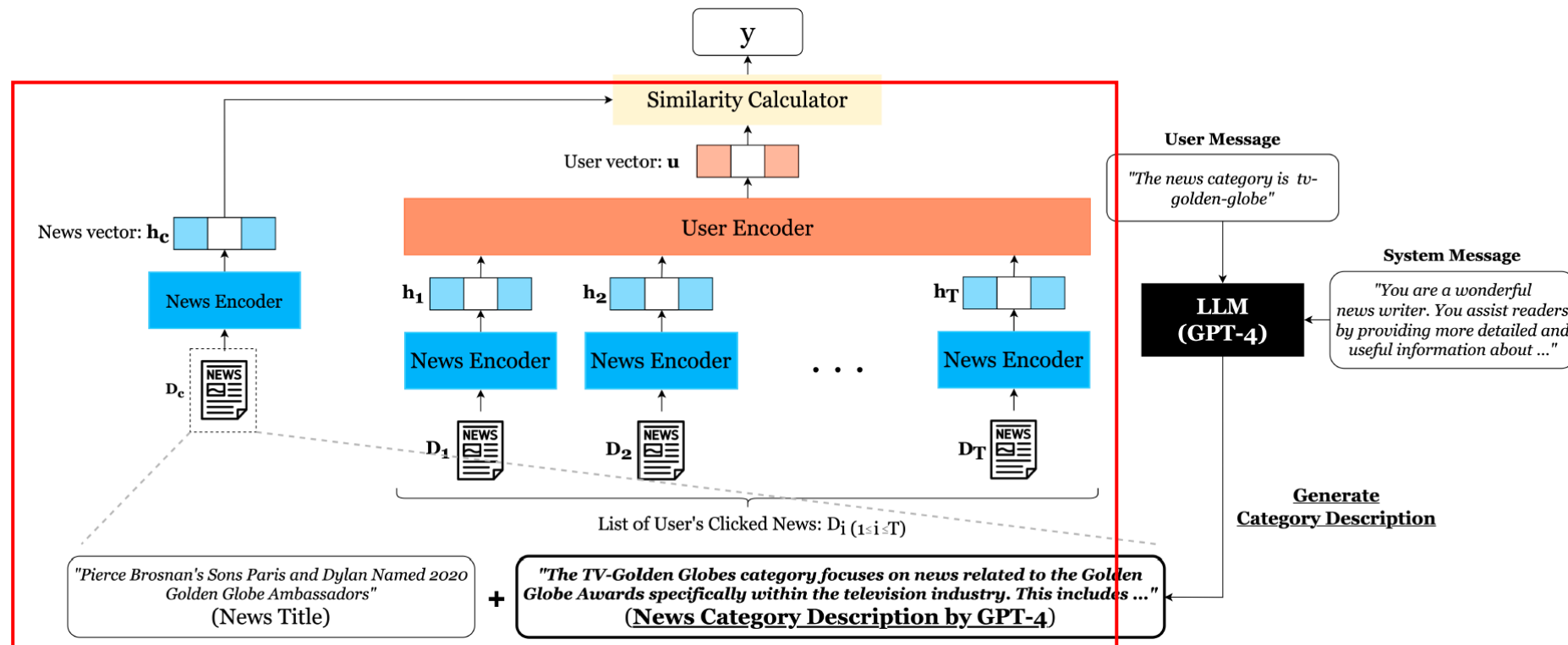
- Background
 - The explainability and informativeness of news metadata—especially category information—play a crucial role in personalized news recommendation.
- Motivation
 - Manually creating detailed descriptions for news categories is labor-intensive.
 - Existing category templates are too generic and provide insufficient semantic guidance.
 - Pretrained language models often fail to interpret short category names effectively without richer context.
- Contribution
 - Generating descriptions with LLMs
 - Incorporating descriptions into recommendation models

- Category Description Generation
 - User Message + System Message -> LLM -> Generated Category Description



- Integration into News Encoder

- $D_{input} = D_{title} [sep] D_{desc}$, encoding with bert-style news encoder



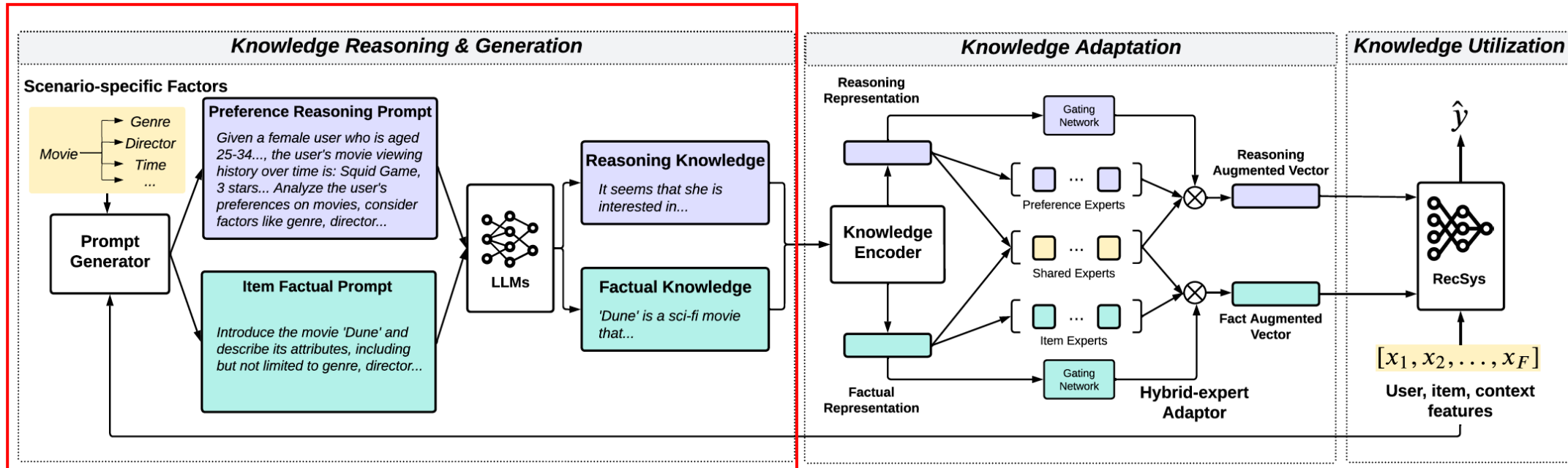
- The proposed method consistently outperforms both the title-only and template-based baselines across all models and settings, achieving up to 5.8% AUC improvement.

Recommendation Model	PLM	Method	AUC	MRR	nDCG@5	nDCG@10
NAML [19]	DistilBERT	<i>title only</i>	0.675	0.292	0.317	0.384
		<i>title + template-based</i>	0.690	0.295	0.327	0.393
		<i>title + generate-description (ours)</i>	0.713	0.326	0.363	0.425
	BERT	<i>title only</i>	0.700	0.318	0.350	0.414
		<i>title + template-based</i>	0.696	0.308	0.340	0.405
		<i>title + generate-description (ours)</i>	0.707	0.322	0.357	0.420
NRMS [21]	DistilBERT	<i>title only</i>	0.674	0.297	0.322	0.387
		<i>title + template-based</i>	0.675	0.311	0.341	0.400
		<i>title + generate-description (ours)</i>	0.707	0.324	0.359	0.422
	BERT	<i>title only</i>	0.689	0.306	0.336	0.400
		<i>title + template-based</i>	0.667	0.301	0.329	0.389
		<i>title + generate-description (ours)</i>	0.706	0.320	0.355	0.418
NPA [20]	DistilBERT	<i>title only</i>	0.700	0.311	0.344	0.408
		<i>title + template-based</i>	0.698	0.309	0.342	0.407
		<i>title + generate-description (ours)</i>	0.707	0.319	0.354	0.417
	BERT	<i>title only</i>	0.689	0.301	0.332	0.398
		<i>title + template-based</i>	0.694	0.314	0.345	0.410
		<i>title + generate-description (ours)</i>	0.710	0.324	0.360	0.422

- Practical Values:** LLM-generated category descriptions offer a simple and generalizable way to enrich news representations, enabling better recommendations without additional training or domain-specific effort.

- Background
 - Classical recommender systems are limited to closed-domain data and lack access to open-world knowledge, constraining their reasoning and generalization abilities.
- Motivation
 - Directly using LLMs as recommenders leads to poor accuracy, high latency, and difficulty handling compositional reasoning.
 - It is challenging to extract useful, aligned, and reliable knowledge from LLMs that is compatible with recommendation models.
- Contribution
 - Propose KAR, a model-agnostic framework that extracts both reasoning knowledge about users and factual knowledge about items from LLMs.
 - Design a hybrid-expert adaptor to transform textual knowledge into dense augmented vectors aligned with recommendation space.

- Knowledge Reasoning and Generation
 - Use LLMs with **factorized prompts** to extract two types of open-world knowledge:
 - Reasoning knowledge $k_i^{(p)}$ for user i
 - Factual knowledge $k_i^{(l)}$ for item i



- Knowledge Adaptation and Utilization

- Use a knowledge encoder (e.g., BERT) to encode LLM outputs
- Apply hybrid-expert adaptor with gating networks and shared/dedicated experts to transform to

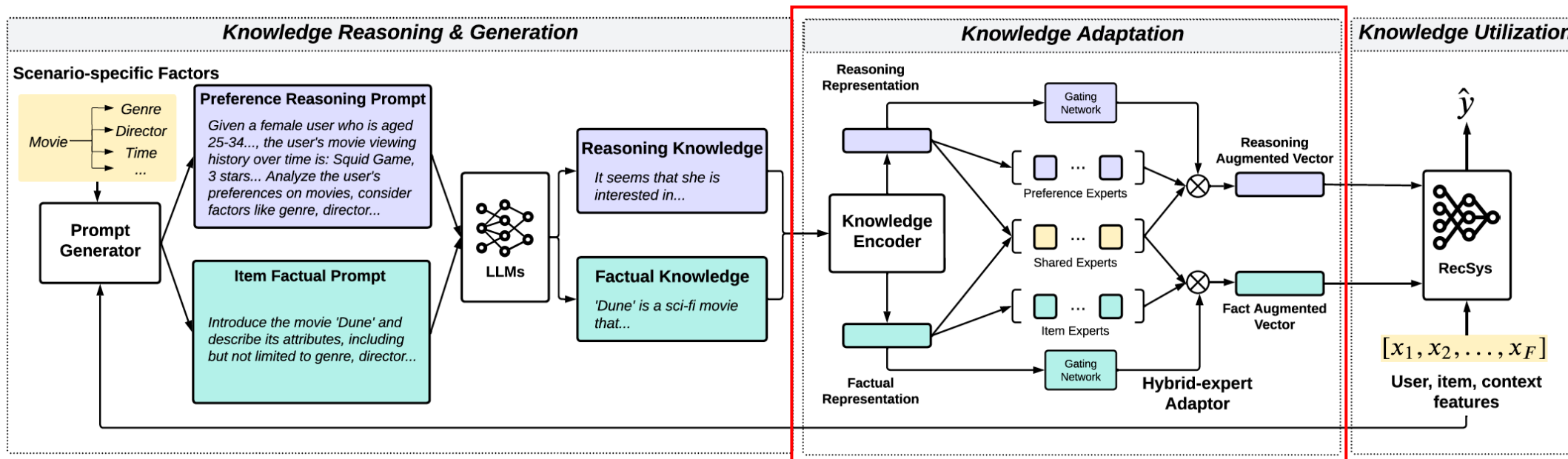
$$\alpha_i^p = \text{Softmax}(g^p(r_i^p)), \quad \alpha_i^l = \text{Softmax}(g^l(r_i^l)),$$

$$r_i^{(p)} = \text{Aggr}(\text{Encoder}(k_i^{(p)}))$$

$$r_i^{(l)} = \text{Aggr}(\text{Encoder}(k_i^{(l)}))$$

$$\hat{r}_i^p = \sum_{e \in S_s} \alpha_{i,e}^p \times e(r_i^p) + \sum_{e \in S_p} \alpha_{i,e}^p \times e(r_i^p),$$

$$\hat{r}_i^l = \sum_{e \in S_s} \alpha_{i,e}^l \times e(r_i^l) + \sum_{e \in S_l} \alpha_{i,e}^l \times e(r_i^l),$$



- KAR consistently improves AUC and logloss across 9 CTR models, with up to **1.6% AUC** and **3.1% logloss gain**

Backbone Model	AUC			Logloss		
	base	KAR	improv.	base	KAR	improv.
DCNv2	0.7924	0.8049*	1.58 %	0.5451	0.5315*	2.50 %
DCN	0.7929	0.8043*	1.46 %	0.5457	0.5319*	2.53 %
DeepFM	0.7928	0.8041*	1.44 %	0.5462	0.5321*	2.57 %
FiBiNet	0.7925	0.8051*	1.59 %	0.5450	0.5310*	2.56 %
AutoInt	0.7934	0.8060*	1.59 %	0.5440	0.5297*	2.65 %
FiGNN	0.7944	0.8054*	1.39 %	0.5424	0.5307*	2.16 %
xDeepFM	0.7942	0.8041*	1.25 %	0.5457	0.5317*	2.57 %
DIEN	0.7960	0.8059*	1.25 %	0.5469	0.5298*	3.13 %
DIN	0.7975	0.8066*	1.15 %	0.5387	0.5304*	1.55%

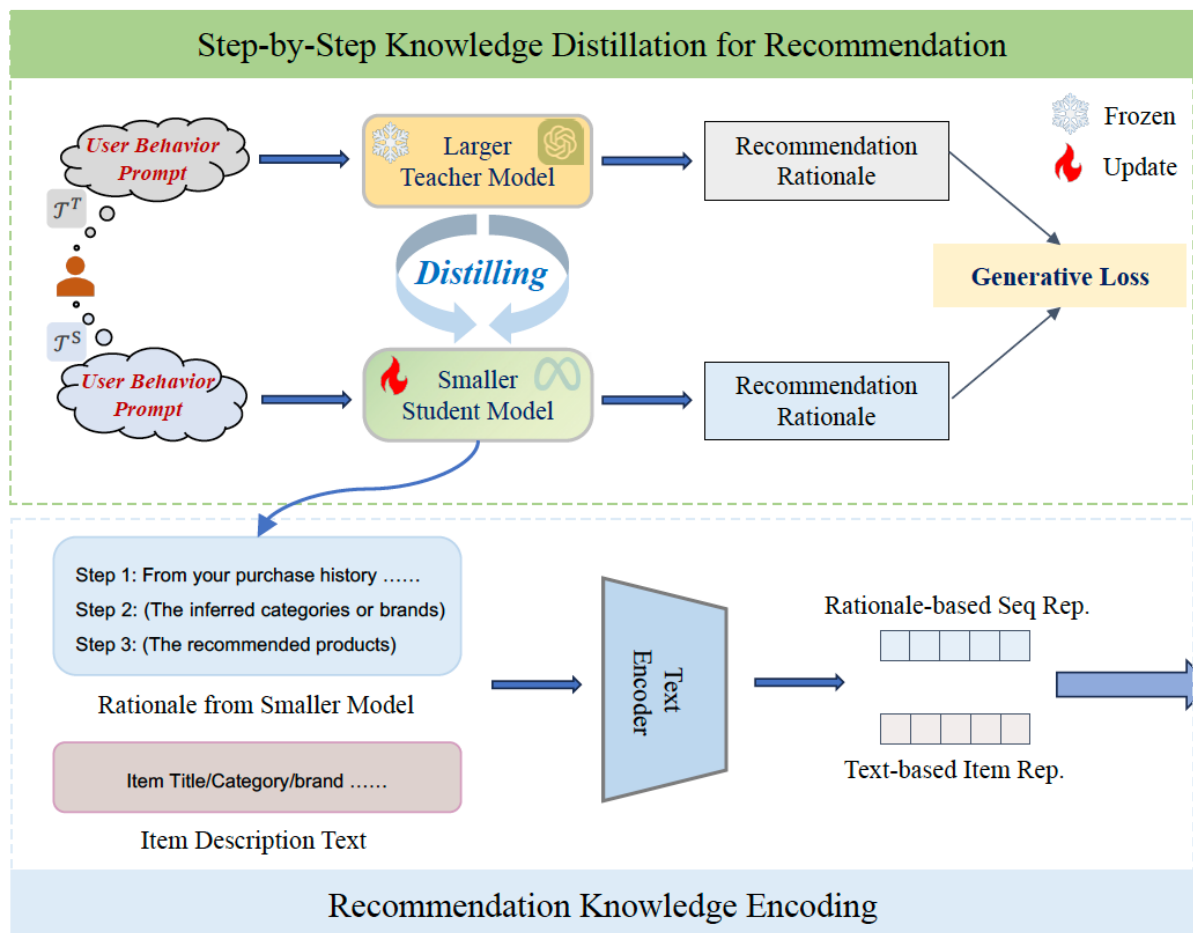
* denotes statistically significant improvement (t-test with p -value < 0.05) over the backbone model.

- Practical Values:** KAR offers a practical, modular, and efficient solution to enhance recommenders using LLM-generated knowledge, without modifying their internal architecture.

- Background
 - LLMs show strong reasoning ability and knowledge but are expensive and hard to deploy in real systems.
- Motivation
 - One-shot generation by LLMs often produces irrelevant or incorrect recommendations.
 - The scale and inference cost of LLMs are prohibitive for production use.
- Contribution
 - Propose SLIM, use CoT prompting with LLMs to generate rationales.
 - Distill these rationales into a small LLM for efficient deployment.
 - Enable both ID-based and ID-agnostic recommendation by encoding rationales and item descriptions via a text encoder.

- Step-by-Step Knowledge Distillation
 - Chain-of-Thought Prompting on Teacher Model with user prompt:
 - Summarize preferences
 - Infer categories or brands
 - Recommend specific products
- Fine-tune Student Model with Rationales:
 - The objective is to minimize generative loss

$$\mathcal{L}_{\text{distill}} = \sum_{u \in U'} \sum_{t=1}^{|r'_u|} \log P_{\theta}(r'_{u,t} | p'_u, r'_{u,<t})$$



- Rationale Encoding and Enhanced Recommendation

- Text Encoding:

- Convert student-generated rationales r_u and item descriptions f_i into embeddings

$$\mathbf{z}_{\text{text},i} = \text{TextEncoder}(f_i), \quad \mathbf{s}_{\text{text},u} = \text{TextEncoder}(r_u)$$

- ID-Based Integration:

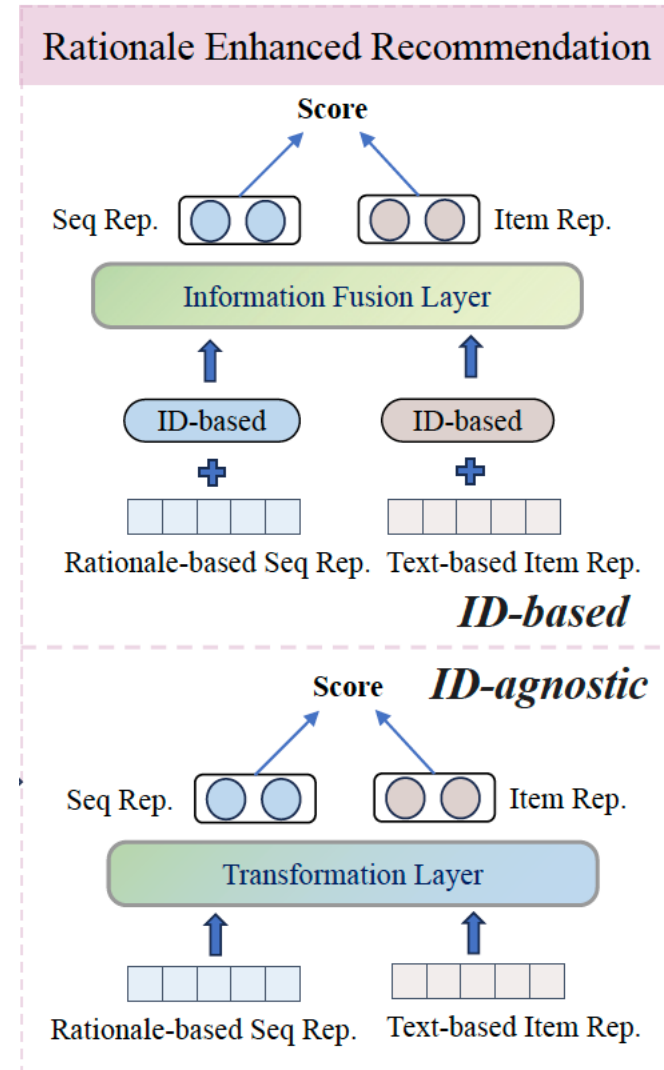
- Use an Information Fusion Layer to combine text and ID embeddings

$$\mathbf{z}_i = g_f([g_l(\mathbf{z}_{\text{text},i}); \mathbf{z}_{\text{id},i}]), \quad \mathbf{s}_u = g_f([g_l(\mathbf{s}_{\text{text},u}); \mathbf{s}_{\text{SeqEnc},u}])$$

- ID-Agnostic Matching:

- Apply a transformation to match rationale and item vectors

$$\mathbf{z}_i = g_t(\mathbf{z}_{\text{text},i}), \quad \mathbf{s}_u = g_t(\mathbf{s}_{\text{text},u})$$

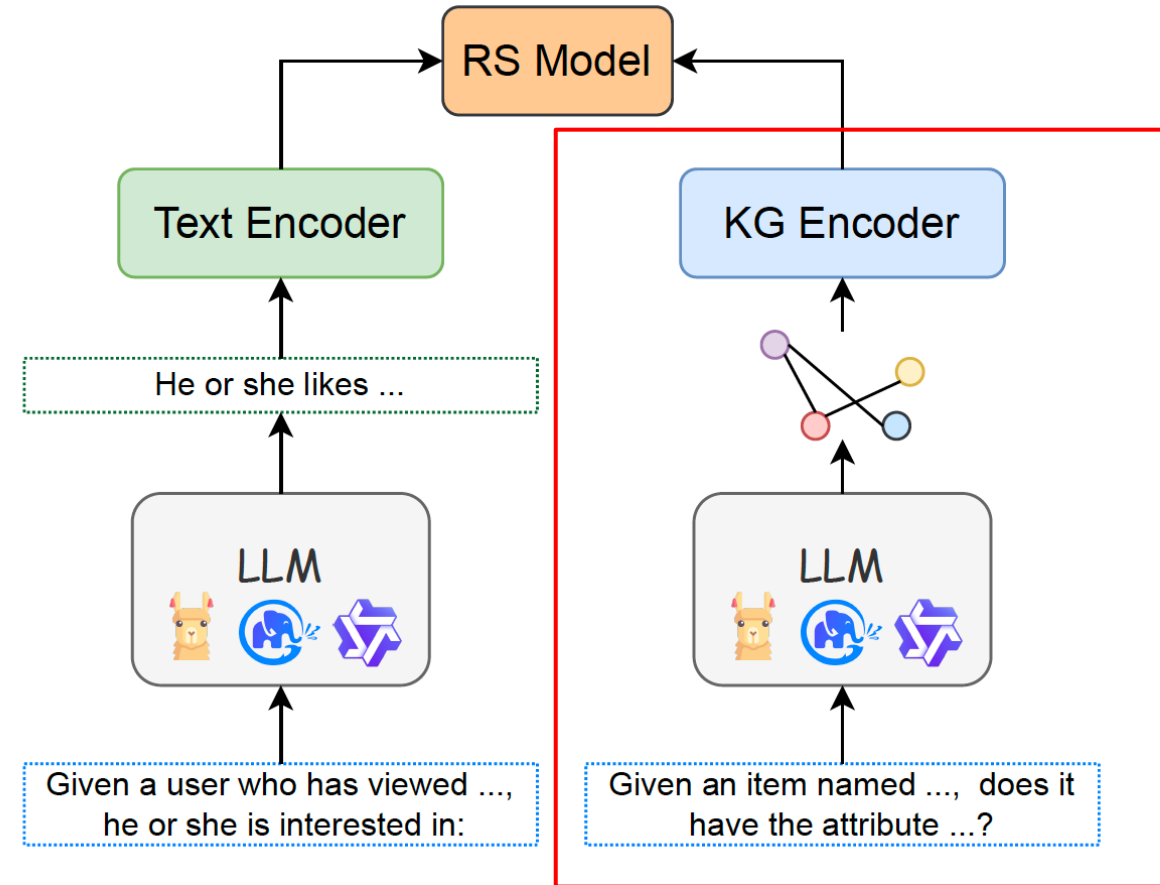


- SLIM consistently improves performance across three datasets and three backbone models (GRU4Rec, SASRec, SRGNN), outperforming both traditional and ChatGPT-augmented baselines.

Methods	Games			Food			Home		
	NDCG@10	Hit @10	Hit @20	NDCG@10	Hit @10	Hit @20	NDCG@10	Hit @10	Hit @20
GRU4Rec	17.61 $\pm$ 0.18	30.87 $\pm$ 0.56	42.39 $\pm$ 0.62	9.10 $\pm$ 0.30	15.27 $\pm$ 0.58	19.51 $\pm$ 0.24	2.19 $\pm$ 0.21	4.17 $\pm$ 0.39	7.53 $\pm$ 0.64
GRU4Rec ⁺	27.33 $\pm$ 0.53	44.06 $\pm$ 0.79	56.53 $\pm$ 1.24	17.75 $\pm$ 0.78	31.10 $\pm$ 1.09	45.01 $\pm$ 1.46	12.19 $\pm$ 1.02	26.76 $\pm$ 2.58	49.10 $\pm$ 3.82
SLIM ⁻	27.70 $\pm$ 0.47	45.13 $\pm$ 0.56	57.70 $\pm$ 0.37	17.97 $\pm$ 0.70	31.78 $\pm$ 1.47	46.88 $\pm$ 2.10	13.59 $\pm$ 1.05	30.30 $\pm$ 2.01	55.85 $\pm$ 3.55
SLIM	28.37 $\pm$ 0.41	45.68 $\pm$ 0.53	58.09 $\pm$ 0.58	18.32 $\pm$ 0.53	32.56 $\pm$ 1.30	46.92 $\pm$ 1.82	15.64 $\pm$ 0.51	34.33 $\pm$ 1.53	62.93 $\pm$ 3.46
Improv.	3.81%	3.68%	2.76%	3.21%	4.69%	4.24%	28.3%	28.29%	28.17%
SASRec	22.73 $\pm$ 0.28	37.77 $\pm$ 0.52	51.53 $\pm$ 0.39	26.78 $\pm$ 0.24	35.78 $\pm$ 0.36	43.32 $\pm$ 0.55	2.66 $\pm$ 0.22	5.56 $\pm$ 0.72	14.93 $\pm$ 1.53
SASRec ⁺	27.46 $\pm$ 0.19	44.88 $\pm$ 0.63	58.90 $\pm$ 0.38	30.95 $\pm$ 0.38	44.98 $\pm$ 0.53	55.61 $\pm$ 1.12	5.58 $\pm$ 0.10	11.09 $\pm$ 0.16	20.69 $\pm$ 0.77
SLIM ⁻	31.58 $\pm$ 0.35	50.83 $\pm$ 0.62	63.45 $\pm$ 0.71	32.65 $\pm$ 0.15	48.01 $\pm$ 0.48	59.25 $\pm$ 0.56	5.95 $\pm$ 0.32	11.83 $\pm$ 0.62	22.43 $\pm$ 0.54
SLIM	31.43 $\pm$ 0.39	51.11 $\pm$ 0.82	64.10 $\pm$ 0.26	32.80 $\pm$ 0.40	48.27 $\pm$ 0.64	59.30 $\pm$ 0.89	6.01 $\pm$ 0.19	12.01 $\pm$ 0.38	22.29 $\pm$ 0.85
Improv.	14.46%	13.88%	8.83%	5.98%	7.31%	6.64%	7.71%	8.3%	7.73%
SRGNN	16.45 $\pm$ 0.22	29.29 $\pm$ 0.14	40.99 $\pm$ 0.52	10.99 $\pm$ 2.07	20.32 $\pm$ 4.30	32.14 $\pm$ 6.55	5.04 $\pm$ 0.83	13.48 $\pm$ 2.23	37.22 $\pm$ 3.85
SRGNN ⁺	21.54 $\pm$ 0.64	36.77 $\pm$ 1.05	49.11 $\pm$ 1.54	11.91 $\pm$ 0.71	21.39 $\pm$ 1.91	33.63 $\pm$ 3.41	11.61 $\pm$ 1.14	25.22 $\pm$ 2.46	43.85 $\pm$ 3.58
SLIM ⁻	22.35 $\pm$ 1.48	37.69 $\pm$ 1.53	51.29 $\pm$ 0.59	12.92 $\pm$ 0.78	23.80 $\pm$ 1.60	37.22 $\pm$ 2.75	11.25 $\pm$ 1.42	24.28 $\pm$ 3.19	44.05 $\pm$ 5.97
SLIM	23.77 $\pm$ 0.20	39.81 $\pm$ 0.52	52.34 $\pm$ 0.63	12.38 $\pm$ 0.51	22.98 $\pm$ 1.30	36.44 $\pm$ 1.72	12.29 $\pm$ 1.39	26.51 $\pm$ 2.71	47.01 $\pm$ 3.61
Improv.	10.35%	8.27%	6.58%	3.95%	7.43%	8.36%	5.86%	5.11%	7.21%

- Practical Values:** SLIM enables small language models to act as effective reasoners in sequential recommendation by distilling step-by-step rationales from LLMs at a fraction of the computational cost.

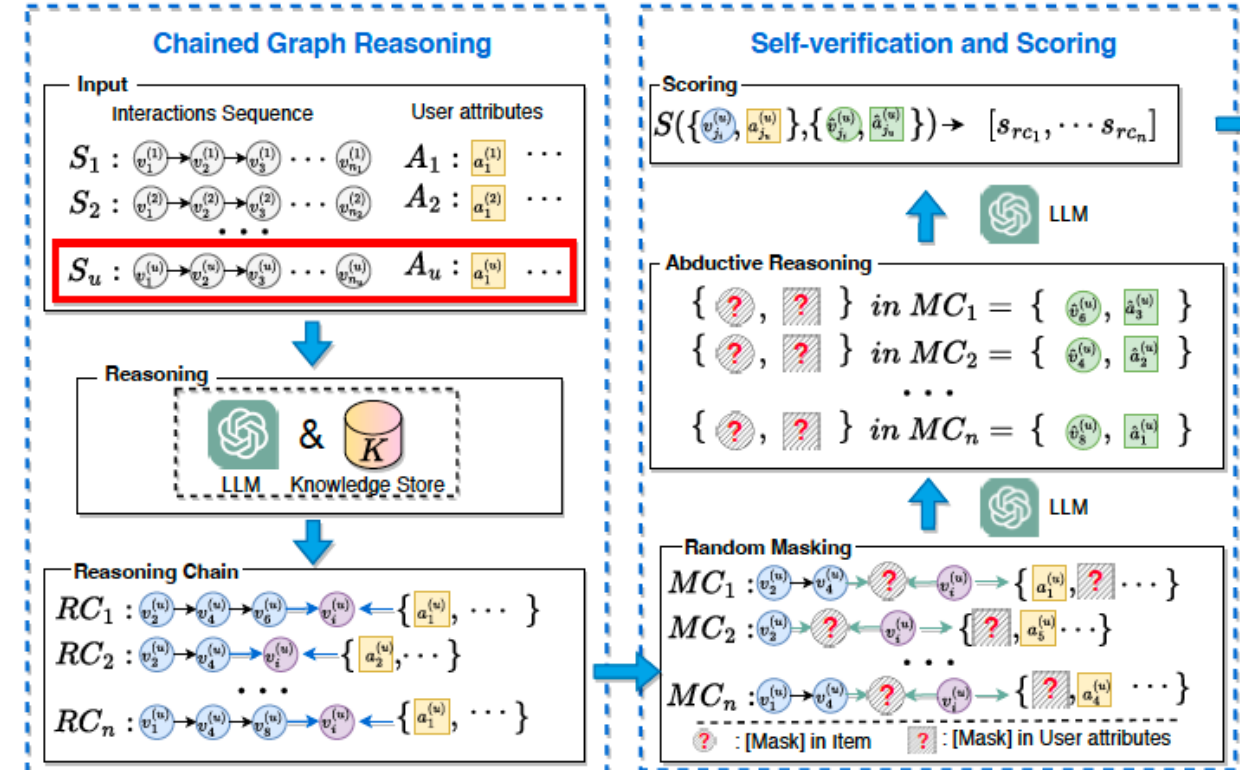
- Summary text is a type of unstructured knowledge, while the **structured Knowledge Graph (KG)** may drive **better integration**.
- Many work explore how to **apply the LLM to generate a KG** or **augment an existing KG** for enhancing RS.
- Categories
 - Generation
 - Completion & Fusion



- Background
 - Recent works use knowledge graphs or behavior graphs to enhance reasoning, but most rely on pre-defined structures and cannot flexibly adapt to personalized needs.
 - LLMs exhibit powerful reasoning and abductive inference ability.
- Motivation
 - Can LLMs generate personalized multi-hop reasoning chains for sequential RS?
 - How can we verify, score, and incorporate these LLM-generated chains into real-world recommenders?
 - Can this be done without relying on explicit external knowledge graphs?
- Contributions
 - Adaptive Reasoning Module

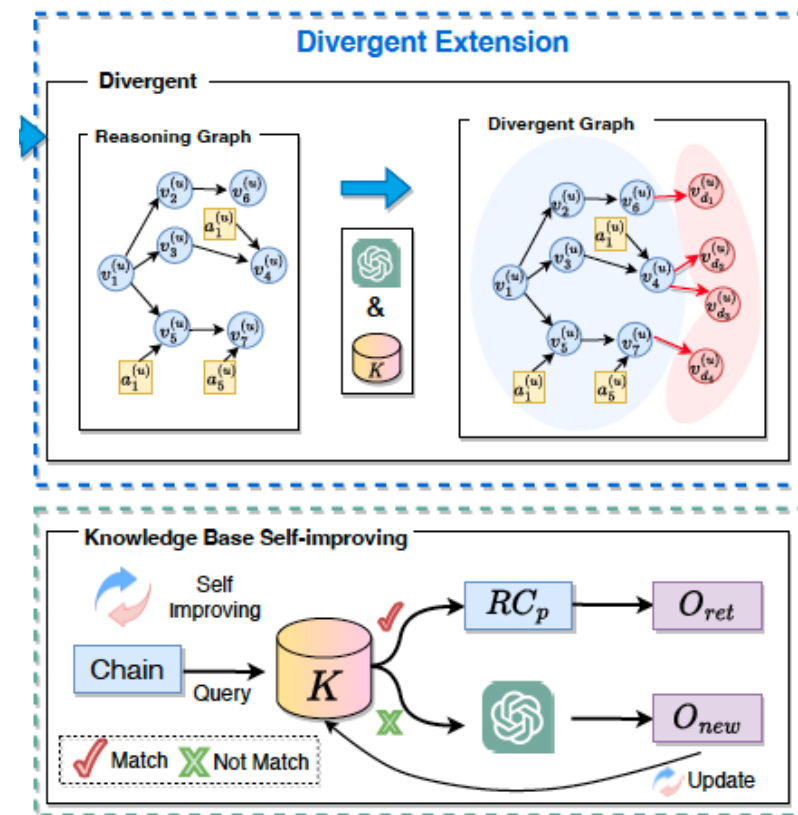
- Chain Construction and Verification
 - Chained Graph Reasoning
 - LLM builds reasoning chains RC from interaction sequence S_u and attribute A_u
 - Random Masking & Abductive Reasoning
 - Each chain is perturbed into masked candidates MC_i .
 - The LLM performs abductive reasoning to infer whether masked parts still lead to the correct reasoning path.
- Scoring

$$S(RC_i, A_u, S_u) \rightarrow [s_{RC_1}, \dots, s_{RC_n}]$$



- Reasoning Enhancement and Model Fusion
 - Divergent Extension:
 - From the reasoning graph, LLM identifies alternative paths by querying unexplored nodes, forming divergent reasoning graphs.
 - Self-Improving Knowledge Base:
 - Mismatches between retrieved reasoning and newly inferred reasoning are used to update the knowledge base.
 - Final Fusion:

$$E_{\text{fusion}} = \text{Fusion}(E_{\text{base}}, E_{\text{ori}}, E_{\text{div}})$$



- LLMRG improves NDCG and HR by over 25–30% on ML-1M, Amazon Beauty, and Amazon Clothing, outperforming all baselines.

Dataset	Metric	FDSA			BERT4Rec			CL4SRec			DuoRec		
		Original	GPT3.5	GPT4	Original	GPT3.5	GPT4	Original	GPT3.5	GPT4	Original	GPT3.5	GPT4
ML-1M	HR@5	0.0909	+ 20.70%	+ 25.79%	0.1124	+ 26.67%	+ 32.56%	0.1141	+ 19.98%	+ 21.02%	0.2011	+ 12.87%	+ 14.76%
	HR@10	0.1631	+ 17.93 %	+ 22.87%	0.1910	+ 13.52 %	+ 16.49 %	0.1866	+ 17.30 %	+ 19.31 %	0.2837	+ 14.10 %	+ 15.53 %
	NDCG@5	0.0599	+ 21.33%	+ 30.27 %	0.0713	+ 25.74 %	+ 32.82 %	0.0721	+ 14.97 %	+ 16.78 %	0.1265	+ 23.55 %	+ 26.01 %
	NDCG@10	0.0878	+ 21.78%	+ 28.25%	0.0980	+ 23.34%	+ 28.06%	0.1013	+ 17.67%	+ 20.42%	0.1663	+ 12.86%	+ 13.77%
Amazon Beauty	HR@5	0.0237	+ 13.89 %	+ 17.53 %	0.0201	+ 19.17 %	+ 23.22 %	0.0398	+ 11.15 %	+ 14.15 %	0.0552	+ 9.31 %	+ 11.93 %
	HR@10	0.0418	+ 15.02 %	+ 17.78 %	0.0413	+ 17.79 %	+ 22.14 %	0.0664	+ 10.22 %	+ 11.32 %	0.0839	+ 5.14 %	+ 6.61 %
	NDCG@5	0.0195	+ 16.20 %	+ 18.64 %	0.0192	+ 14.21 %	+ 17.63 %	0.0221	+ 8.45 %	+ 10.18 %	0.0350	+ 7.42 %	+ 9.24 %
	NDCG@10	0.0275	+ 14.78 %	+ 17.64 %	0.0263	+ 11.53 %	+ 14.76 %	0.0322	+ 8.17 %	+ 9.68 %	0.0447	+ 6.67 %	+ 7.95 %
Amazon Clothing	HR@5	0.0119	+ 20.67 %	+ 23.92 %	0.0128	+ 16.09 %	+ 19.10 %	0.0166	+ 7.90 %	+ 10.92 %	0.0190	+ 9.98 %	+ 11.40 %
	HR@10	0.0197	+ 14.45 %	+ 17.88 %	0.0202	+ 10.52 %	+ 13.72 %	0.0273	+ 11.21 %	+ 14.99 %	0.0311	+ 7.65 %	+ 9.48 %
	NDCG@5	0.0073	+ 8.16 %	+ 10.86 %	0.0081	+ 7.39 %	+ 10.39 %	0.0093	+ 6.02 %	+ 9.09 %	0.0118	+ 6.74 %	+ 9.19 %
	NDCG@10	0.0109	+ 6.01 %	+ 8.13 %	0.0113	+ 5.21 %	+ 5.94 %	0.0125	+ 4.32 %	+ 8.07 %	0.0155	+ 7.89 %	+ 9.29 %

- **Practical Values:** LLMRG enables LLM-based, adaptive, multi-hop reasoning without external knowledge graphs, improving recommendation via personalized and verifiable chains.

- Background
 - Existing KG-based recommenders often convert textual information into IDs, ignore semantic connections, and struggle to capture high-order relations due to GNNs' limitations.
- Motivation
 - KGs are incomplete or too sparse
 - Converting entities into IDs discards rich semantic information, and GNNs fail to effectively propagate information over long distances in the KG.
- Contributions
 - Item-item semantic graph
 - Representation alignment and neighbor augmentation modules

- KG Comprehension via LLM

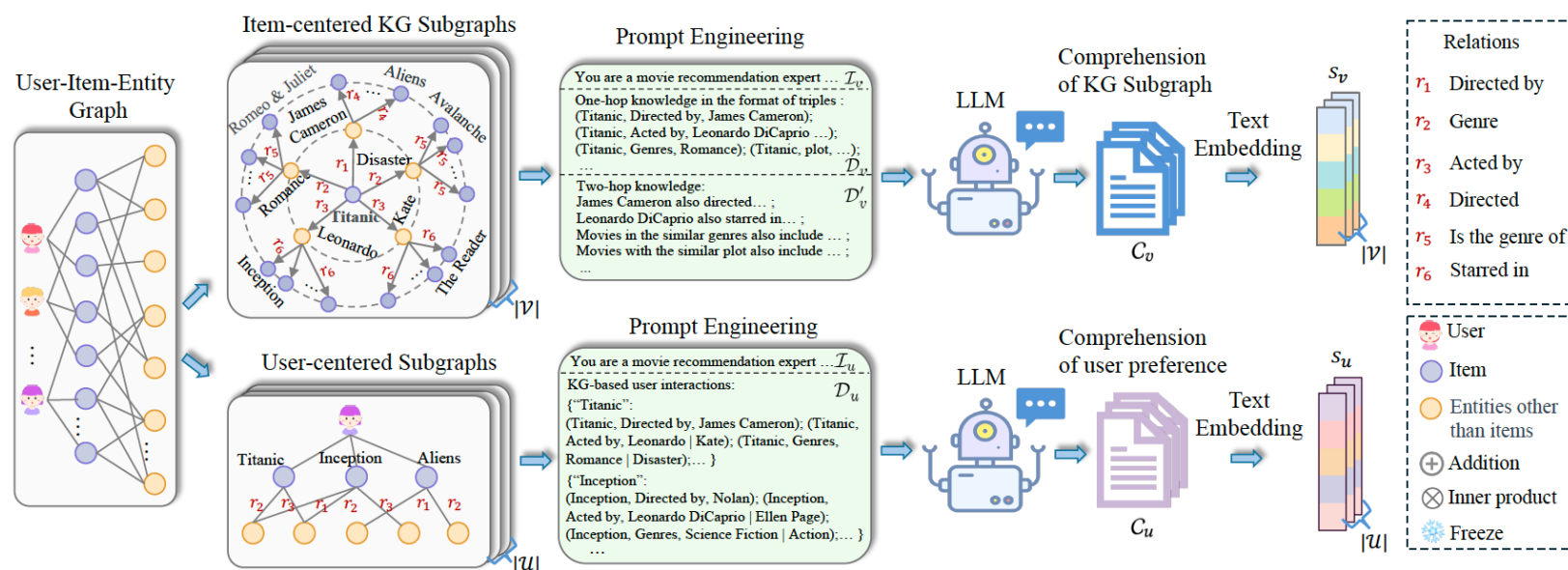
- Item-centered KG Subgraph Construction
- Text Conversion & Prompting
- Embedding Generation

- Use pre-trained model P to obtain semantic embedding

$$T_v = \{(v, r, e)\}$$

$$C_v = \text{LLMs}(I_v, D_v, D'_v)$$

$$s_v = P(C_v)$$



- Semantic Graph Construction & Integration

- Semantic Item-Item Graph

- Use cosine similarity to define edge weights
 - Construct graph

$$r(v_i, v_j) = \text{sim}(s_{v_i}, s_{v_j})$$

$$\mathcal{G}_v = \{(v_i, r(v_i, v_j), v_j)\}$$

- Representation Integration

- Align semantic embeddings to ID space
 - Fuse with ID embeddings

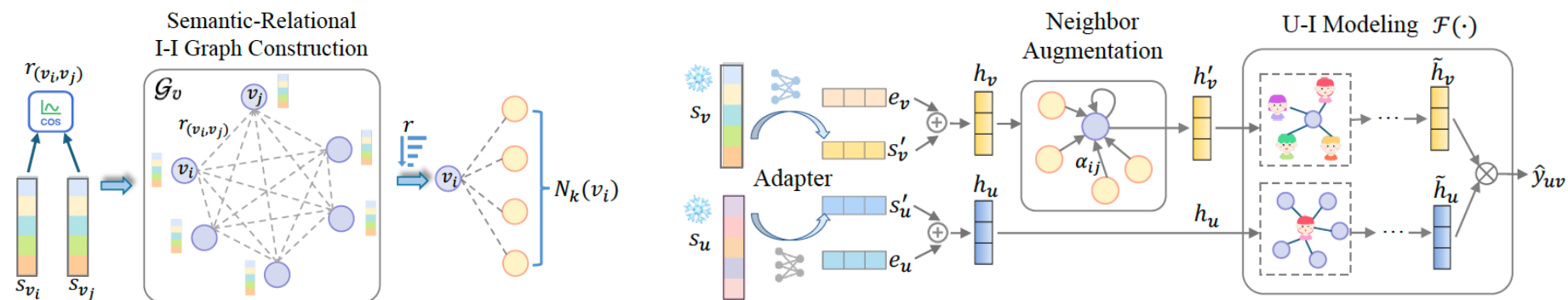
$$s'_v = \sigma(W_1 s_v), \quad s'_u = \sigma(W_2 s_u)$$

$$h_v = \frac{1}{2}(e_v + s'_v), \quad h_u = \frac{1}{2}(e_u + s'_u)$$

- Neighbor Augmentation

$$w_{ij} = a(W s_{v_i} \| W s_{v_j}), \quad \alpha_{ij} = \text{softmax}_j(w_{ij})$$

$$\mathbf{h}'_{v_i} = \sigma \left(\frac{1}{2} \left(\mathbf{h}_{v_i} + \sum_{j \in \mathcal{N}_k(v_i)} \alpha_{ij} \mathbf{h}_{v_j} \right) \right)$$

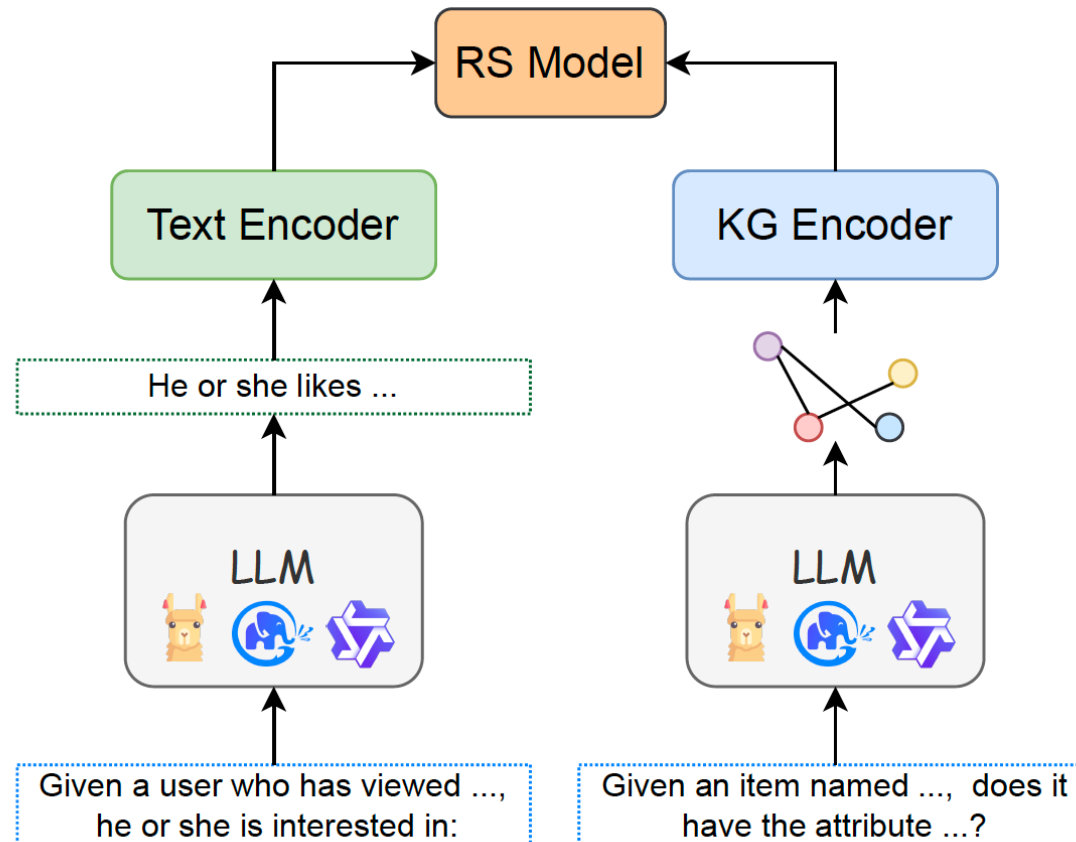


- CoLaKG achieves the best performance across all four datasets, demonstrating strong generalization and semantic reasoning ability.

Model	MovieLens				Last-FM				MIND				Funds			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
BPR-MF	0.1257	0.3100	0.2048	0.3062	0.1307	0.1352	0.1971	0.1685	0.0315	0.0238	0.0537	0.0310	0.4514	0.3402	0.5806	0.3809
NFM	0.1346	0.3558	0.2129	0.3379	0.2246	0.2327	0.3273	0.2830	0.0495	0.0356	0.0802	0.0458	0.4388	0.3187	0.5756	0.3651
LightGCN	0.1598	0.3901	0.2512	0.3769	0.2589	0.2799	0.3642	0.3321	0.0624	0.0492	0.0998	0.0609	0.4992	0.3778	0.6353	0.4204
CKE	0.1524	0.3783	0.2373	0.3609	0.2342	0.2545	0.3266	0.3001	0.0526	0.0417	0.0822	0.0510	0.4926	0.3702	0.6294	0.4130
RippleNet	0.1415	0.3669	0.2201	0.3423	0.2267	0.2341	0.3248	0.2861	0.0472	0.0364	0.0785	0.0451	0.4764	0.3591	0.6124	0.4003
KGAT	0.1536	0.3782	0.2451	0.3661	0.2470	0.2595	0.3433	0.3075	0.0594	0.0456	0.0955	0.0571	0.5037	0.3751	0.6418	0.4182
KGIN	0.1631	0.3959	0.2562	0.3831	0.2562	0.2742	0.3611	0.3215	0.0640	0.0518	0.1022	0.0639	0.5079	0.3857	0.6428	0.4259
KGCL	0.1554	0.3797	0.2465	0.3677	<u>0.2599</u>	0.2763	<u>0.3652</u>	0.3284	<u>0.0671</u>	<u>0.0543</u>	<u>0.1059</u>	<u>0.0670</u>	0.5071	0.3877	0.6355	0.4273
KGRec	<u>0.1640</u>	<u>0.3968</u>	<u>0.2571</u>	<u>0.3842</u>	0.2571	0.2748	0.3617	0.3251	0.0627	0.0506	0.1003	0.0625	<u>0.5104</u>	<u>0.3913</u>	<u>0.6467</u>	<u>0.4304</u>
RLMRec	0.1613	0.3920	0.2524	0.3787	0.2597	<u>0.2812</u>	0.3651	<u>0.3335</u>	0.0619	0.0486	0.0990	0.0602	0.4988	0.3784	0.6351	0.4210
CoLaKG	0.1699	0.4130	0.2642	0.3974	0.2738	0.2948	0.3803	0.3471	0.0698	0.0562	0.1087	0.0684	0.5273	0.4012	0.6524	0.4392

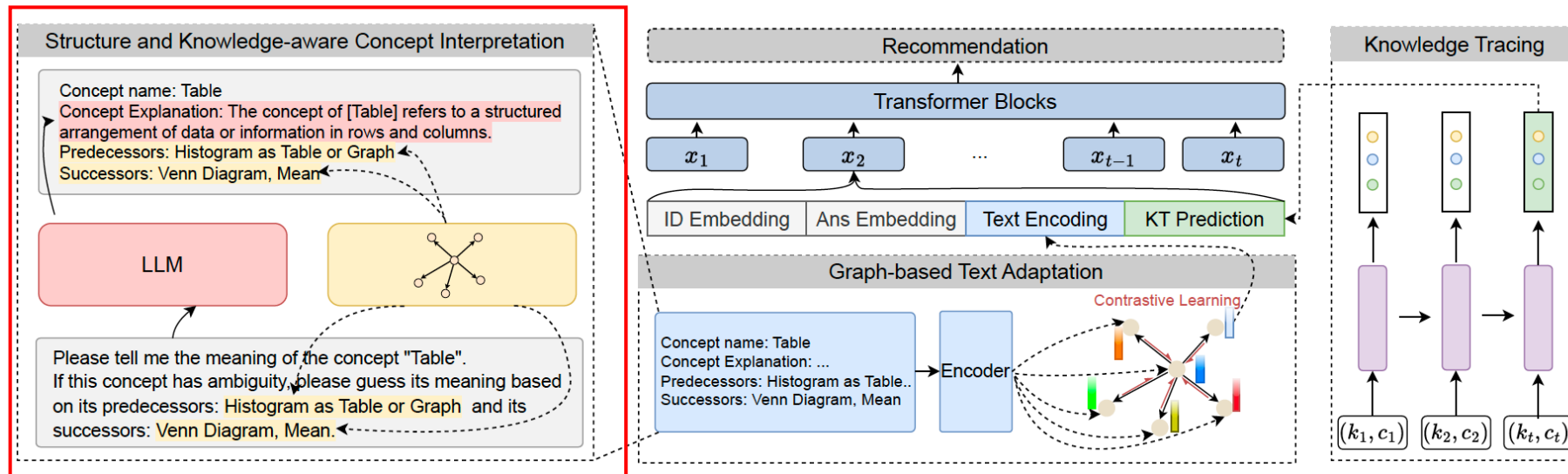
- Practical Values:** By leveraging LLMs to comprehend and augment KG information, CoLaKG bridges the gap between structured knowledge and collaborative filtering, resulting in more robust and semantically aligned recommendation

- In consideration of the effectiveness of Summary text and Knowledge graph, some work also resort to combine both of them.

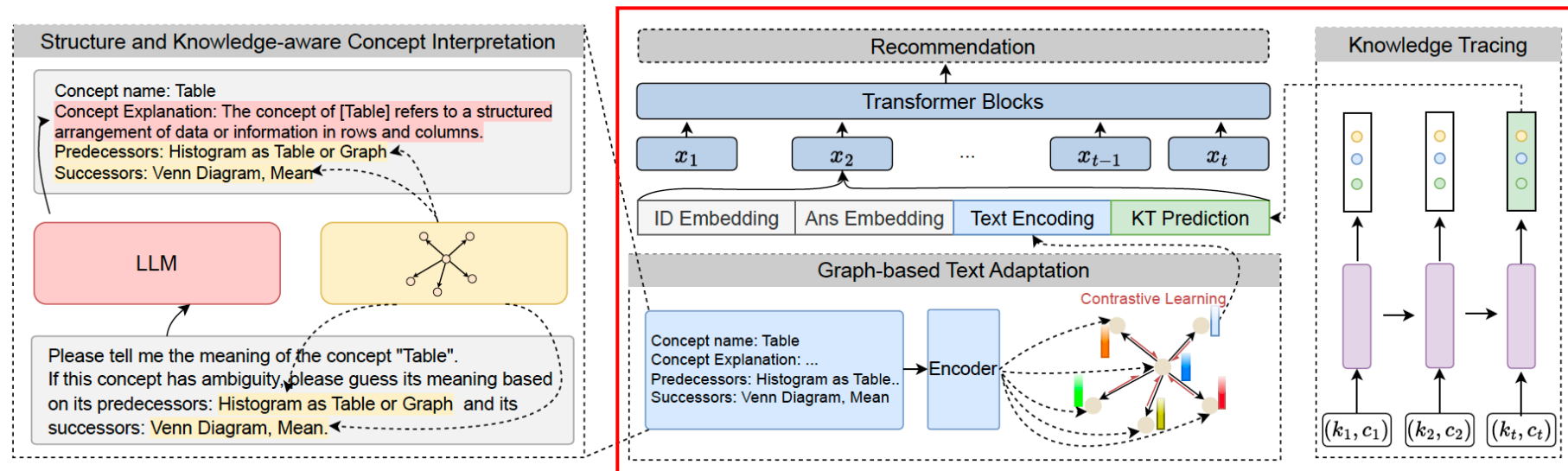


- Background
 - Existing KT and recommendation models often rely on ID-based concept representations, which fail to generalize and lack interpretability.
- Motivation
 - Prior methods overlook the textual and relational information between concepts, which leads to limited generalization, especially when dealing with unseen or ambiguous concepts.
- Contributions
 - Structure and Knowledge-aware Concept Interpretation
 - Graph-based Text Adaptation

- Structure and Knowledge-aware Concept Interpretation
 - For a given concept c , SKarRec constructs prompt with (1) name and explanation (2) its predecessors and successors in the concept graph
 - Prompt is sent to LLM to generate rich interpretation
 - The interpreted concept text is used as semantic anchor for downstream tasks



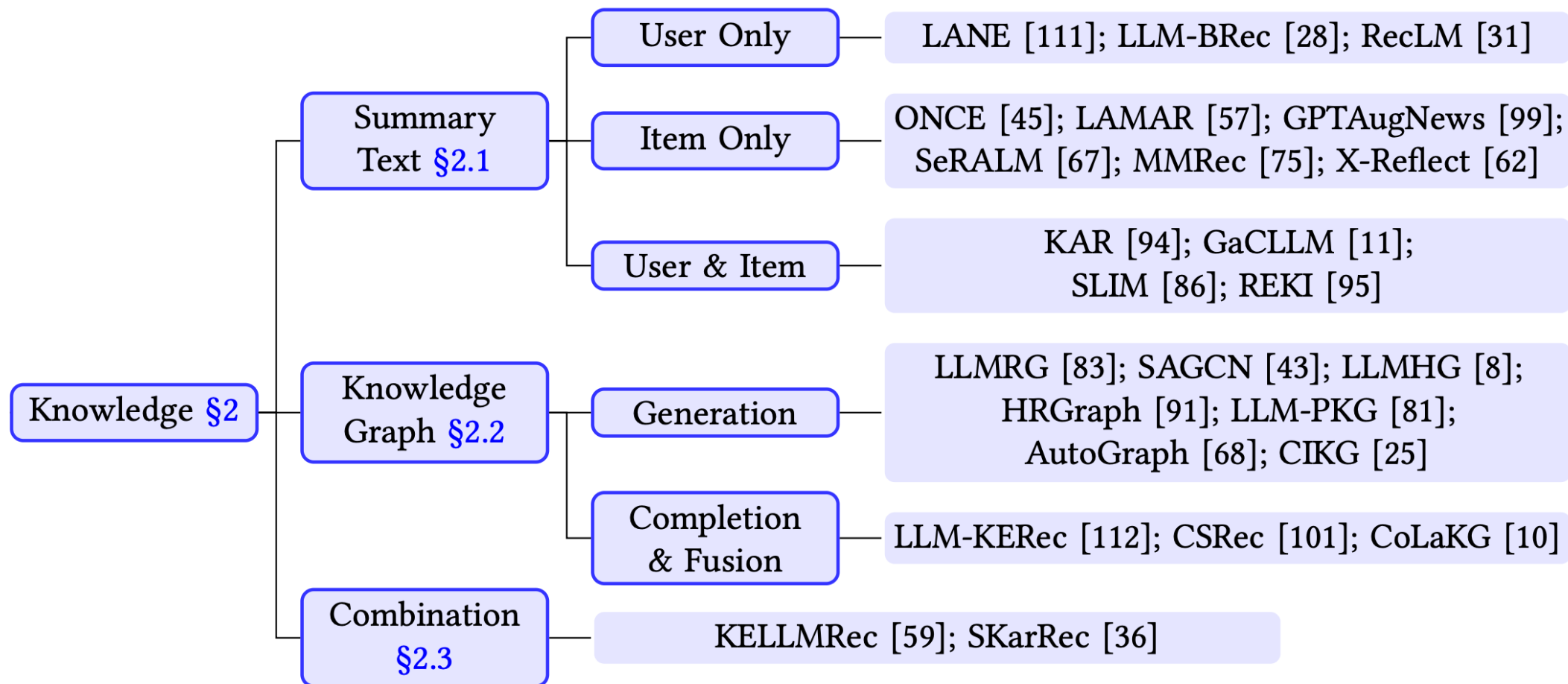
- Graph-based Text Adaptation for Recommendation and Tracing
 - Interpreted concept texts are encoded into embeddings using a transformer encoder.
 - Graph G over concepts is built using their relations
 - Contrastive learning: positive pair (connected nodes), negative pair (unconnected)



- SKarRec achieves the best performance on ASSIST09, ASSIST12, and Junyi across all three metrics, significantly outperforming prior graph-based, text-only, and ID-only methods.

Dataset	Metric	Graph-based Methods		ID-Only Methods		Text-Only Methods		ID-Text Methods			
		ACKRec	GCARec	SASRec	BERT4Rec	ZESRec	RECFORMER	FDSA	S3-Rec	UniSRec	SKarRec
ASSIST09	HR@1	0.2408	0.2513	0.7649	0.1935	0.5809	0.6072	0.7627	<u>0.7661</u>	0.7597	0.7922*
	NDCG@5	0.3787	0.3904	<u>0.8736</u>	0.2203	0.7269	0.6540	0.8722	0.8731	0.8647	0.8838*
	MRR	0.3627	0.3756	0.8503	0.2342	0.6998	0.6711	0.8484	<u>0.8505</u>	0.8439	0.8646*
ASSIST12	HR@1	0.1976	0.2119	0.2838	0.2540	0.2596	0.1990	0.2861	<u>0.2904</u>	0.2888	0.2933*
	NDCG@5	0.3078	0.3169	0.4895	0.4247	0.4649	0.2055	0.4886	0.4957	0.4934	<u>0.4954</u>
	MRR	0.3064	0.3182	0.4574	0.4047	0.4343	0.2536	0.4579	<u>0.4633</u>	0.4617	0.4645*
Junyi	HR@1	0.1447	0.1284	0.8469	0.3550	0.8546	0.8279	0.8492	<u>0.8608</u>	0.8407	0.8730*
	NDCG@5	0.1872	0.1650	0.8907	0.4826	0.8930	0.7625	0.8917	<u>0.9022</u>	0.8890	0.9076*
	MRR	0.1918	0.1740	0.8825	0.4661	0.8860	0.8497	0.8840	<u>0.8942</u>	0.8799	0.9014*

- Practical Values:** By integrating structural graph knowledge and language-model-based semantic reasoning, SKarRec offers a robust framework for educational recommendation and knowledge tracing with superior generalization and interpretability



Agenda

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang



5 Conclusion



Yejing Wang

Agenda

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang



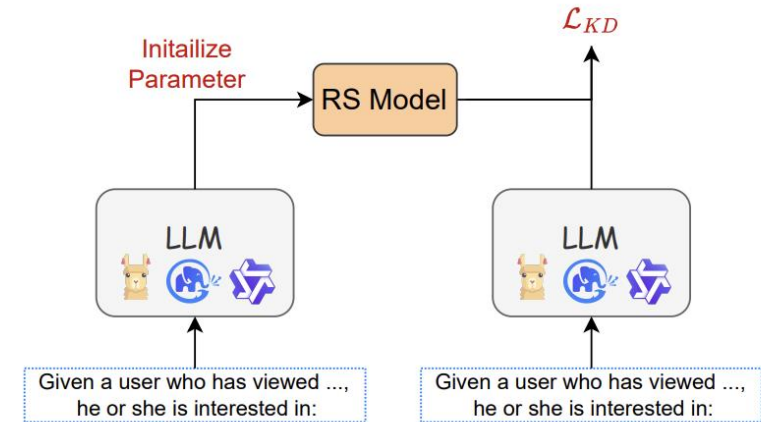
5 Conclusion



Yejing Wang

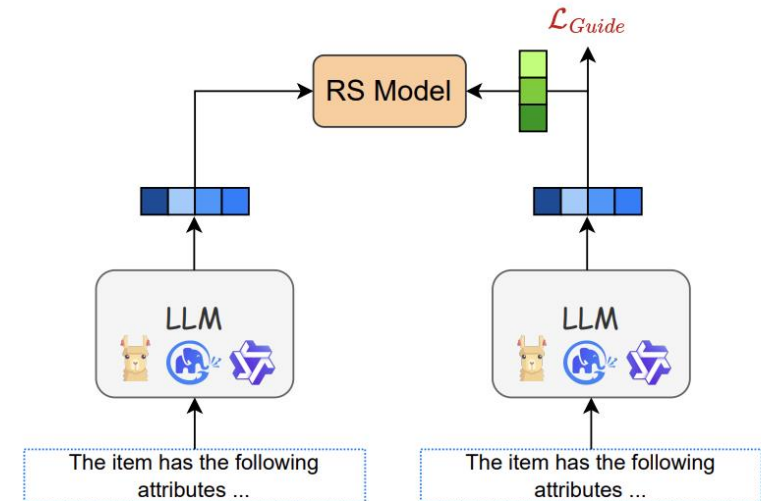
Model Enhancement

- LLMs possess powerful semantic capabilities that can be directly integrated into recommendation systems to enhance models.
- Categories
 - Model Initialization
 - Model Distillation
 - Embedding Utilization
 - Embedding Guidance



(a) Model Initialization

(b) Model Distillation

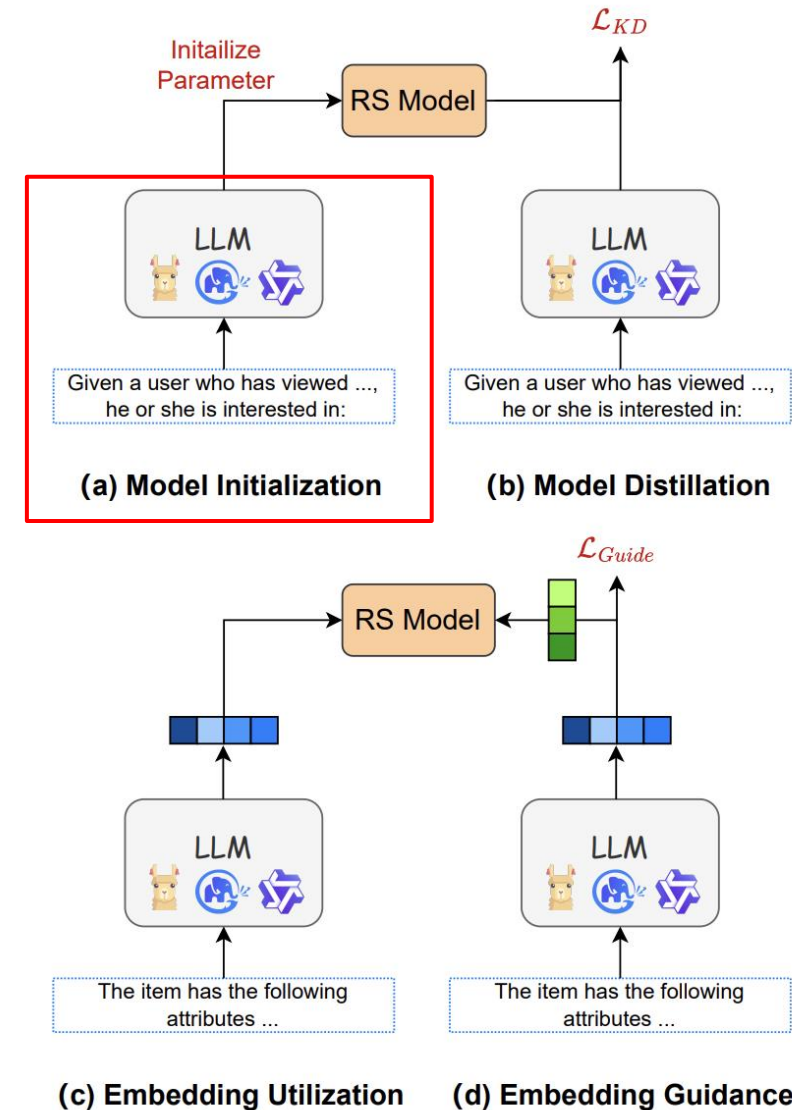


(c) Embedding Utilization

(d) Embedding Guidance

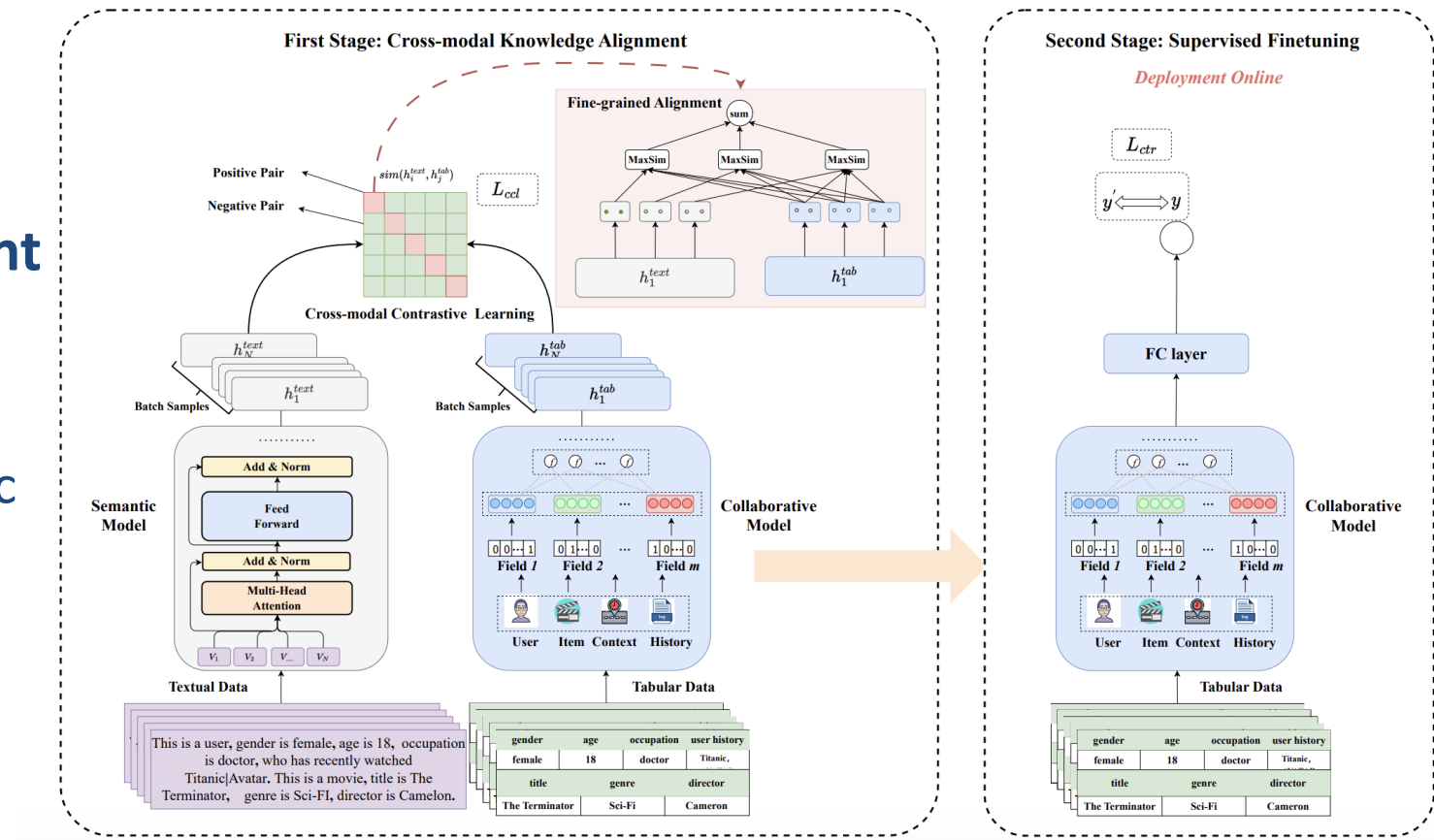
Model Initialization

- This subcategory refers to utilizing LLM-derived semantics to initialize the weights of recommendation models before training begins.
- This approach accelerates model convergence while preserving semantic knowledge from LLM pretraining for downstream recommendation tasks.
- Categories
 - Whole
 - Embedding



- Background
 - CTR prediction is a core task in recommender systems, where traditional collaborative models capture user preferences through feature co-occurrence.
- Motivation
 - One-hot encoding discards semantic relationships, leading to poor performance in cold-start scenarios.
 - Direct use of Pre-trained Language Models is computationally expensive and fails to meet low-latency requirements for online inference.
- Key Stages
 - Cross-modal Knowledge Alignment
 - Supervised Fine-tuning

- CTRL proposes a two-stage framework to effectively integrate semantic knowledge into collaborative.
- Cross-modal Knowledge Alignment**
 - Tabular data feeds collaborative model, text prompts feed PLM. Contrastive learning aligns representations to distill semantic knowledge.
- Supervised Fine-tuning**
 - Enhanced collaborative model fine-tunes on CTR task. Only lightweight collaborative model deploys online, no PLM needed.



- CTRL significantly outperforms SOTA collaborative and semantic models on multiple public datasets and an industrial system, while maintaining high inference efficiency.

Category	Model	MovieLens			Amazon			Alibaba		
		AUC	Logloss	RelaImpr	AUC	Logloss	RelaImpr	AUC	Logloss	RelaImpr
Collaborative Models	Wide&Deep	0.8261	0.4248	3.52%	0.6968	0.4645	5.30%	0.6272	0.1943	5.19%
	DeepFM	0.8268	0.4219	3.30%	0.6969	0.4645	5.33%	0.6280	0.1951	4.53%
	DCN	<u>0.8313</u>	<u>0.4165</u>	1.90%	0.6999	0.4642	3.75%	<u>0.6281</u>	0.1949	4.45%
	PNN	0.8269	0.4220	3.27%	0.6979	0.4657	4.80%	0.6271	0.1956	5.27%
	AutoInt	0.8290	0.4178	2.61%	<u>0.7012</u>	<u>0.4632</u>	3.08%	0.6279	<u>0.1948</u>	4.61%
	FiBiNet	0.8196	0.4188	5.63%	0.7003	0.4704	3.54%	0.6270	0.1951	5.35%
	xDeepFM	0.8296	0.4178	2.43%	0.7009	0.4642	3.23%	0.6272	0.1959	5.19%
Semantic Models	P5	0.7583	0.4912	30.70%	0.6923	0.4608	7.85%	0.6034	0.3592	29.40%
	CTR-BERT	0.7650	0.4944	27.40%	0.6934	0.4629	7.24%	0.6005	0.3620	33.13%
	P-Tab	0.8031	0.4612	11.38%	0.6942	0.4625	6.80%	0.6112	0.3584	20.32%
CTRL		0.8376*	0.4025*	-	0.7074*	0.4577*	-	0.6338*	0.1890*	-

- Practical Values:** CTRL achieves superior performance by fusing collaborative and semantic signals, maintains efficient inference, and is industrial-friendly with flexible model compatibility.

- Background
 - CTR prediction is a core task in recommender systems, where traditional ID-based models use one-hot encoding and PLMs use textual modality.
- Motivation
 - One-hot encoding discards semantic information, leading to poor performance in sparse scenarios.
PLMs struggle with field-wise collaborative signals and have high computational costs for online inference.
- Key Components
 - Modality Transformation
 - Modality Alignment Pretraining
 - Adaptive Finetuning

• Modality Transformation

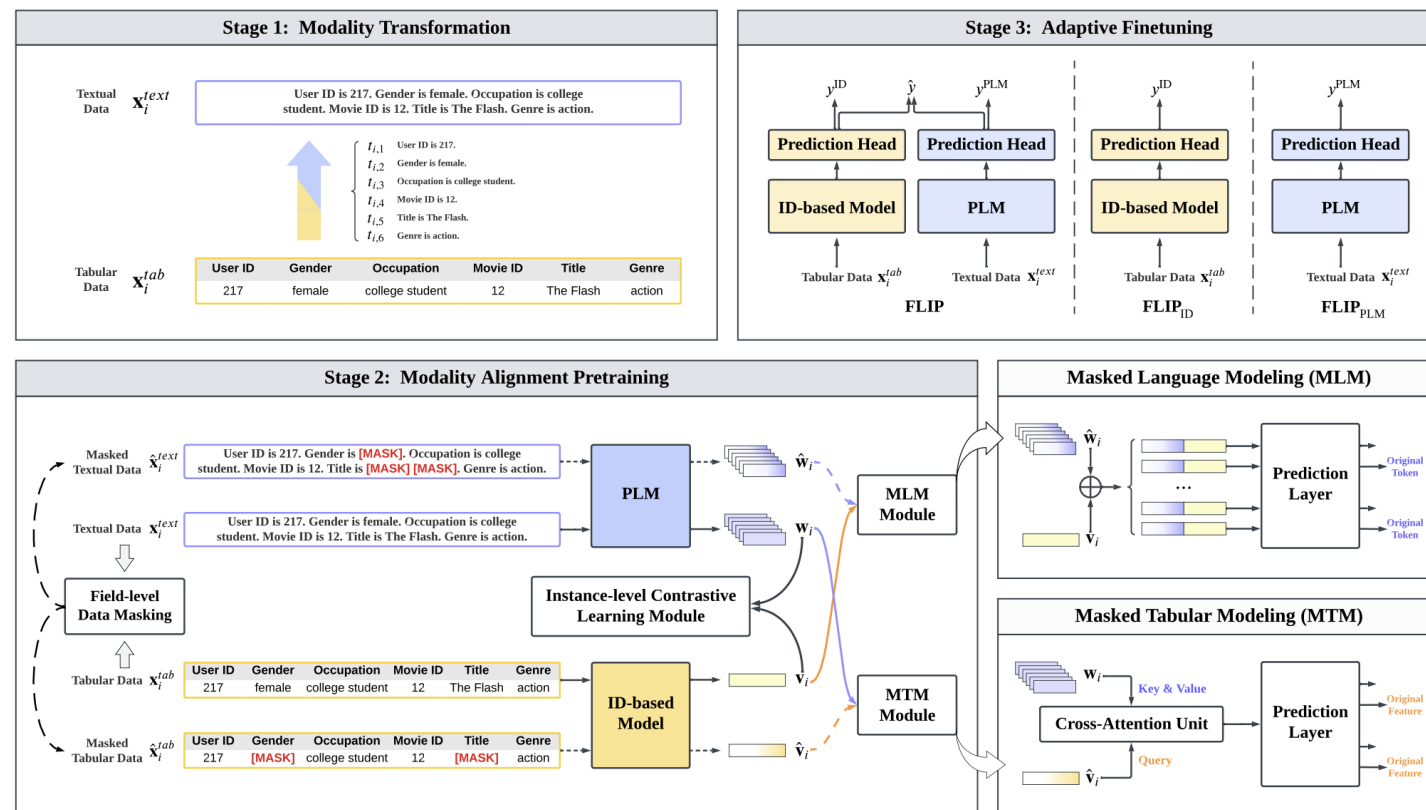
- Convert tabular features to textual sentences using simple templates, preserving semantic information in natural language format.

• Modality Alignment Pretraining

- Field-level masking with MLM, MTM, and ICL for fine-grained cross-modal alignment.

• Adaptive Finetuning

- Joint training with learnable weight α combining both model outputs for CTR prediction.

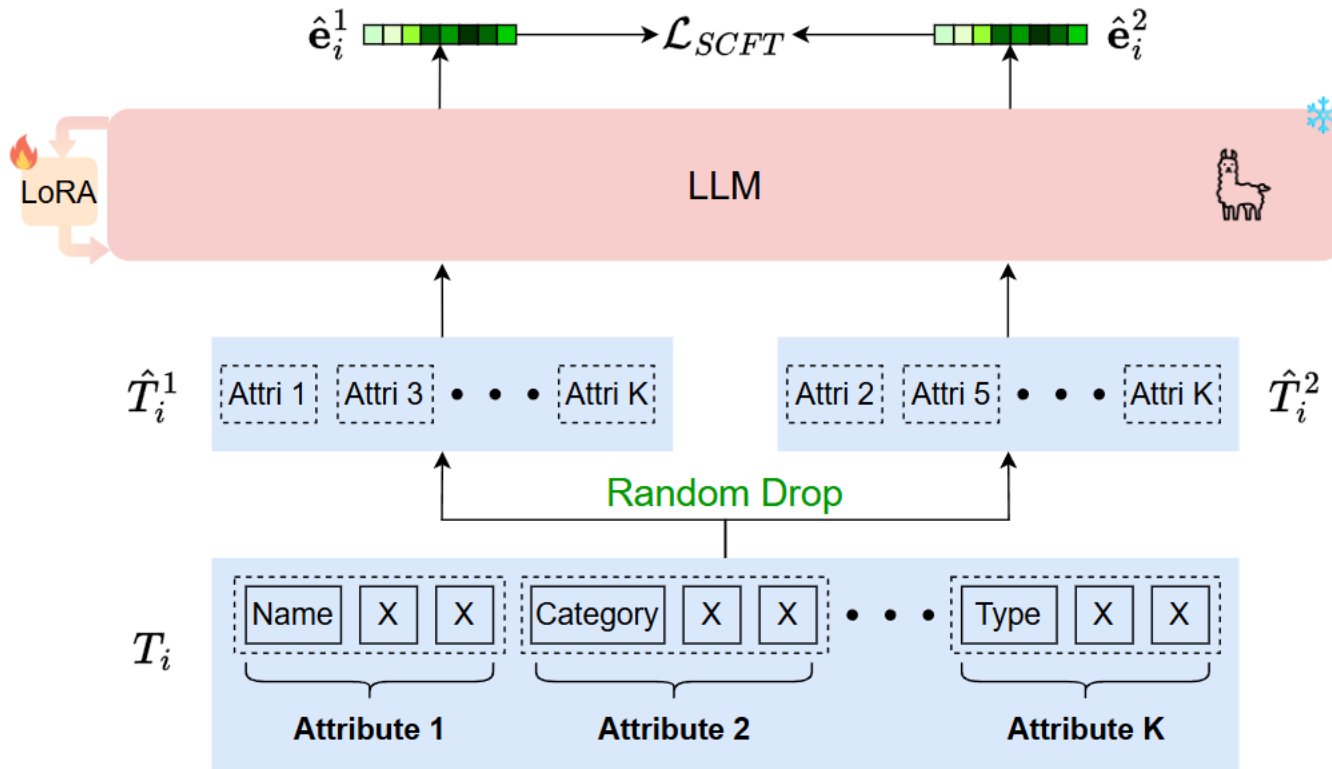


- **FLIP** significantly outperforms SOTA on three public datasets. It also shows strong compatibility with various ID models and PLMs.

Model		MovieLens-1M			BookCrossing			GoodReads		
		AUC	Logloss	Rel.Impr	AUC	Logloss	Rel.Impr	AUC	Logloss	Rel.Impr
ID-based	AFM	0.8449	0.3950	1.79%	0.7946	0.5116	1.06%	0.7630	0.5160	1.96%
	PNN	0.8546	0.3946	0.63%	0.7956	0.5131	0.93%	<u>0.7725</u>	<u>0.5055</u>	0.70%
	Wide&Deep	0.8509	0.3957	1.07%	0.7951	0.5116	1.00%	0.7684	0.5090	1.24%
	DCN	0.8509	0.4056	1.07%	<u>0.7957</u>	0.5108	0.92%	0.7693	0.5086	1.12%
	DeepFM	0.8539	0.3905	0.71%	0.7947	0.5122	1.04%	0.7671	0.5138	1.41%
	xDeepFM	0.8454	0.3934	1.72%	0.7953	0.5108	0.97%	0.7720	0.5079	0.77%
	AFN	0.8525	<u>0.3868</u>	0.88%	0.7932	0.5139	1.24%	0.7654	0.5118	1.64%
	AutoInt	0.8509	0.4013	1.07%	0.7953	0.5118	0.97%	0.7716	0.5071	0.82%
	DCNv2	<u>0.8548</u>	0.3893	0.61%	0.7956	<u>0.5103</u>	0.93%	0.7724	0.5057	0.72%
	FLIP_{ID} (Ours)	0.8600*	0.3802*	-	0.8030*	0.5043*	-	0.7779*	0.5014*	-
PLM-based	CTR-BERT	0.8304	<u>0.4131</u>	1.88%	0.7795	0.5300	1.65%	0.7385	0.5316	1.23%
	P5	0.8304	0.4173	1.88%	0.7801	<u>0.5261</u>	1.58%	0.7365	0.5336	1.51%
	PTab	<u>0.8426</u>	0.4195	0.41%	<u>0.7880</u>	0.5384	0.56%	<u>0.7456</u>	<u>0.5268</u>	0.27%
	FLIP_{PLM} (Ours)	0.8460*	0.4127*	-	0.7924*	0.5304*	-	0.7476*	0.5255*	-
ID+PLM	CTRL	<u>0.8572</u>	<u>0.3838</u>	0.57%	0.7985	0.5101	0.95%	<u>0.7741</u>	<u>0.5045</u>	0.59%
	MoRec	0.8561	0.3896	0.70%	<u>0.7990</u>	<u>0.5087</u>	0.89%	0.7731	0.5085	0.72%
	FLIP (Ours)	0.8621*	0.3788*	-	0.8061*	0.5004*	-	0.7787*	0.5001*	-

- **Practical Values:** FLIP's fine-grained alignment (joint MLM+MTM) outperforms instance-level methods, enables meaningful feature-level interactions, and enhances diverse ID-based models (DeepFM, AutoInt, DCNv2) and PLMs of varying sizes.

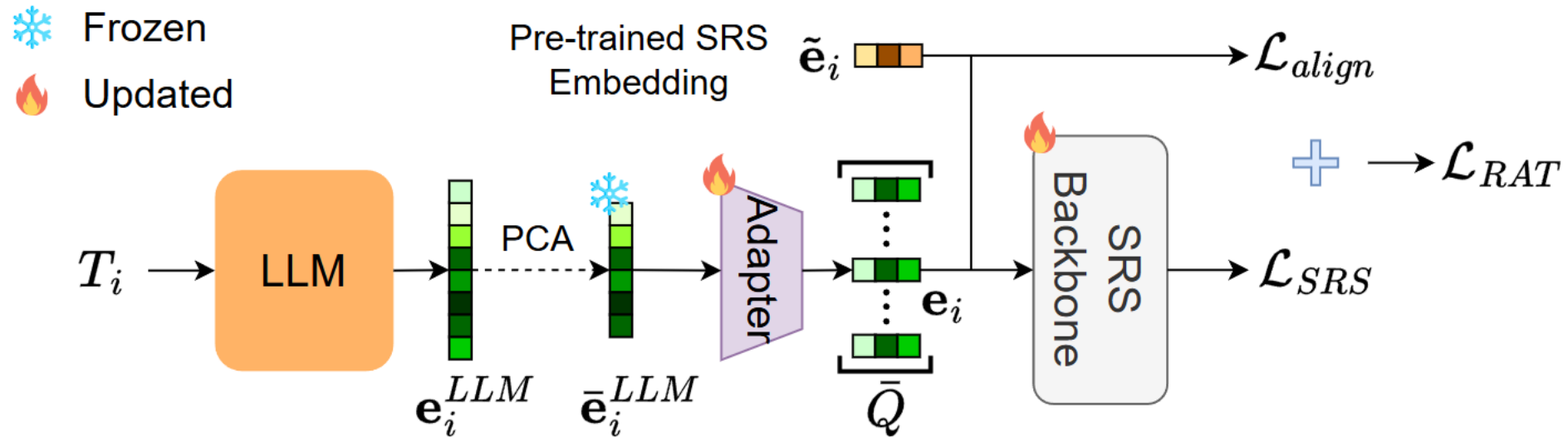
- Background
 - Sequential recommender systems (SRS) predict user preferences through interaction history using learned item embeddings
- Motivation
 - SRS models lack semantic understanding of textual item attributes.
 - LLMs have strong semantics but poor item-level distinction for recommendations.
- Key Stages
 - Supervised Contrastive Fine-Tuning (SCFT)
 - Recommendation Adaptation Training (RAT)



Stage 1: Supervised Contrastive Fine-Tuning (SCFT)

- **Build prompts for each item**
 - Includes instruction + attribute-value pairs.
- **Data Augmentation**
 - Randomly drop attributes to create two prompt views per item (as positive pairs).
- **Contrastive Learning**
 - Use contrastive loss to bring same-item embeddings closer, push different-item embeddings apart.
- **Outcome**
 - LLM learns to encode subtle attribute differences, providing stronger item representations.

Stage 2: Recommendation Adaptation Training (RAT)



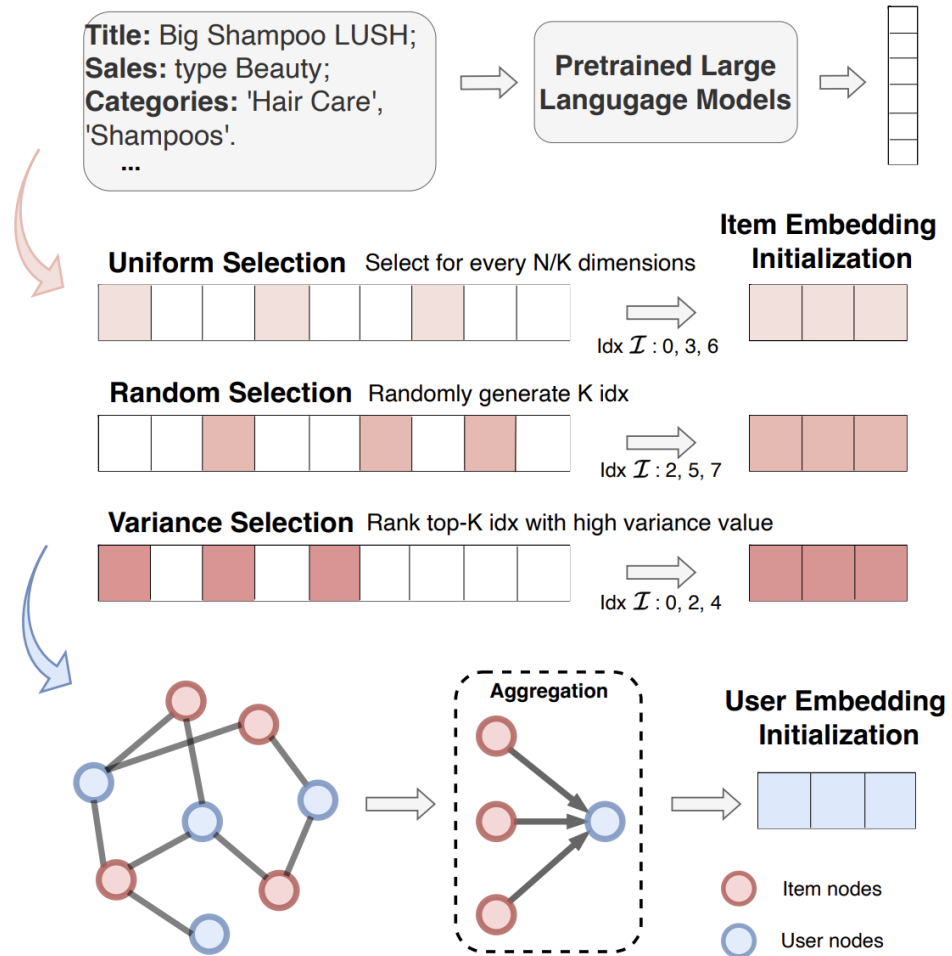
- **Embedding Transformation:** Use PCA to reduce dimensionality, then an adapter (two-layer MLP) to match the SRS size.
- **Adaptation:** Freeze LLM, train the adapter and SRS backbone together using SRS loss.
- **Collaborative Alignment:** Align adapter output with embeddings from a trained SRS, using contrastive loss to prevent overfitting.

- **LLMEmb** outperforms all baselines on Yelp, Amazon Beauty, and Fashion, especially improving recommendations for less popular (long-tail) items.

Backbone Model		Yelp				Fashion				Beauty			
		Overall		Tail		Overall		Tail		Overall		Tail	
		H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10
GRU4Rec	- None	0.4879	0.2751	0.0171	0.0059	0.4798	0.3809	0.0257	0.0101	0.3683	0.2276	0.0796	0.0567
	- MELT	<u>0.4985</u>	<u>0.2825</u>	0.0201	0.0079	0.4884	0.3975	0.0291	0.0112	0.3702	0.2161	0.0009	0.0003
	- LLM2X	0.4872	0.2749	0.0201	0.0072	0.4881	0.4100	0.0264	0.0109	0.4151	<u>0.2713</u>	0.0896	0.0637
	- SAID	0.4891	0.2764	0.0180	0.0062	<u>0.4920</u>	<u>0.4168</u>	<u>0.0347</u>	<u>0.0151</u>	<u>0.4193</u>	0.2621	<u>0.0936</u>	<u>0.0661</u>
	- TSLRec	0.4528	0.2509	<u>0.0255</u>	<u>0.0095</u>	0.4814	0.4042	0.0149	0.0071	0.3119	0.1865	0.0750	0.0474
	- LLMEmb	0.5270*	0.2980*	0.1116*	0.0471*	0.5062*	0.4329*	0.1046*	0.0477*	0.4445*	0.2726	0.3183*	0.1793*
Bert4Rec	- None	0.5307	0.3035	0.0115	0.0044	0.4668	0.3613	0.0142	0.0067	0.3984	0.2367	0.0101	0.0038
	- MELT	<u>0.6206</u>	0.3770	0.0429	0.0149	0.4897	0.3810	0.0059	0.0019	0.4716	0.2965	0.0709	0.0291
	- LLM2X	0.6199	<u>0.3781</u>	0.0874	0.0330	0.5109	<u>0.4159</u>	0.0377	0.0169	0.5029	0.3209	0.0927	0.0451
	- SAID	0.6156	0.3732	<u>0.0973</u>	<u>0.0382</u>	<u>0.5135</u>	0.4124	<u>0.0694</u>	<u>0.0433</u>	<u>0.5127</u>	<u>0.3360</u>	<u>0.1124</u>	<u>0.0664</u>
	- TSLRec	0.6069	0.3680	0.0969	<u>0.0388</u>	0.5078	0.4143	0.0418	0.0182	0.4936	0.3178	0.1013	0.0589
	- LLMEmb	0.6294*	0.3881*	0.1876*	0.1094*	0.5244*	0.4238*	0.1485*	0.0764*	0.5247*	0.3485*	0.2430*	0.1224*
SASRec	- None	0.5940	0.3597	0.1142	0.0495	0.4956	0.4429	0.0454	0.0235	0.4388	0.3030	0.0870	0.0649
	- MELT	0.6257	0.3791	0.1015	0.0371	0.4875	0.4150	0.0368	0.0144	0.4334	0.2775	0.0460	0.0172
	- LLM2X	<u>0.6415</u>	<u>0.3997</u>	<u>0.1760</u>	<u>0.0789</u>	0.5210	0.4486	0.0768	0.0473	0.5043	0.3319	<u>0.1608</u>	<u>0.0940</u>
	- SAID	0.6277	0.3841	0.1548	0.0669	<u>0.5316</u>	<u>0.4619</u>	<u>0.0901</u>	<u>0.0540</u>	<u>0.5097</u>	0.3343	0.1549	0.0906
	- TSLRec	0.6152	0.3795	0.1383	0.0620	0.5125	0.4594	0.0652	0.0382	0.4977	<u>0.3366</u>	0.1211	0.0789
	- LLMEmb	0.6647*	0.4113*	0.2951*	0.1456*	0.5521*	0.4730*	0.1513*	0.0826*	0.5277*	0.3460*	0.4194*	0.2595*

- **Practical Values:** LLMEmb's SCFT+RAT framework bridges semantic understanding with collaborative filtering, enables efficient precomputed embeddings, and enhances sequential recommenders without runtime overhead.

- Background
 - Collaborative filtering models predict user preferences through interaction patterns but face cold-start and embedding collapse challenges.
- Motivation
 - CF models capture interaction patterns but lack semantic understanding.
 - LLMs provide rich semantics but struggle with item-level distinctions.
- Key Stages
 - Selective Sampling
 - Efficient Initialization



- **LLMInit** bridges the LLM-CF embedding gap using selective initialization with sampling strategies to transfer knowledge from large LLM embeddings to lightweight CF models.
- **Selective Sampling**
 - Extract K-dimensional embeddings from LLM representations using variance-based and random selection strategies.
- **Efficient Initialization**
 - Initialize CF models with LLM embeddings to inherit semantic knowledge while maintaining scalability.

- **LLMInit** significantly outperforms baseline CF models across all datasets (Beauty, Toys-Games, Tools-Home, Office-Products).

Method	Beauty		Toys-Games		Tools-Home		Office-Products	
	Recall@10	NDCG@10	Recall@10	NDCG@10	Recall@10	NDCG@10	Recall@10	NDCG@10
LightGCN [7]	0.0910	0.0432	0.0775	0.0360	0.0574	0.0283	0.0745	0.0365
+LLMInit-Rand	0.0960 (+5.5%)	0.0467 (+8.1%)	0.0805 (+3.9%)	0.0387 (+7.50%)	0.0612 (+6.6%)	0.0313 (+10.6%)	0.0773 (+3.8%)	0.0387 (+6.0%)
+LLMInit-Uni	0.1006 (+10.6%)	0.0469 (+8.6%)	0.0806 (+4.0%)	0.0388 (+7.8%)	0.0633 (+10.3%)	0.0319 (+12.7%)	0.0791 (+6.2%)	0.0395 (+8.2%)
+LLMInit-Var	0.1019 (+12.0%)	0.0485 (+12.3%)	0.0808 (+4.3%)	0.0389 (+8.1%)	0.0633 (+10.3%)	0.0317 (+12.0%)	0.0816 (+9.5%)	0.0414 (+13.4%)
SGL [23]	0.1017	0.0474	0.0832	0.0380	0.0580	0.0284	0.0669	0.0297
+LLMInit-Rand	0.1069 (+5.1%)	0.0520 (+9.7%)	0.0885 (+6.4%)	0.0418 (+10.0%)	0.0692 (+19.3%)	0.0337 (+18.7%)	0.0810 (+21.1%)	0.0426 (+43.4%)
+LLMInit-Uni	0.1101 (+8.3%)	0.0513 (+8.2%)	0.0920 (+10.6%)	0.0424 (+11.6%)	0.0676 (+16.6%)	0.0333 (+17.3%)	0.0773 (+15.6%)	0.0350 (+17.9%)
+LLMInit-Var	0.1106 (+8.8%)	0.0530 (+11.8%)	0.0927 (+11.4%)	0.0427 (+12.4%)	0.0686 (+18.3%)	0.0339 (+19.4%)	0.0794 (+18.7%)	0.0421 (+41.8%)
SGCL [32]	0.1027	0.0499	0.0828	0.0382	0.0585	0.0294	0.0647	0.0298
+LLMInit-Rand	0.1094 (+6.5%)	0.0512 (+2.6%)	0.0929 (+12.2%)	0.0418 (+9.4%)	0.0651 (+11.3%)	0.0326 (+10.9%)	0.0770 (+19.0%)	0.0365 (+22.5%)
+LLMInit-Uni	0.1115 (+8.6%)	0.0513 (+2.8%)	0.0923 (+11.5%)	0.0422 (+10.5%)	0.0650 (+11.1%)	0.0327 (+11.2%)	0.0742 (+14.7%)	0.0354 (+18.8%)
+LLMInit-Var	0.1104 (+7.5%)	0.0522 (+4.6%)	0.0941 (+13.6%)	0.0421 (+10.2%)	0.0646 (+10.4%)	0.0327 (+11.2%)	0.0776 (+19.9%)	0.0366 (+22.8%)

- **Practical Values:** LLMInit provides a "free lunch" approach for enhancing CF models with LLM semantic knowledge while maintaining high efficiency and scalability for real-world deployment.

- Background
 - Real-world LLM deployment in recommendation is hindered by the Domain Gap and Catastrophic Forgetting.
- Motivation
 - Domain Gap: Mismatch between LLM's open-world knowledge and recommendations' collaborative knowledge.
 - Catastrophic Forgetting: Fine-tuning LLMs to their full extent causes them to forget pre-trained knowledge.
- Key Components
 - Content-Embedding Generation (CEG)
 - Preference Comprehension (PCH)

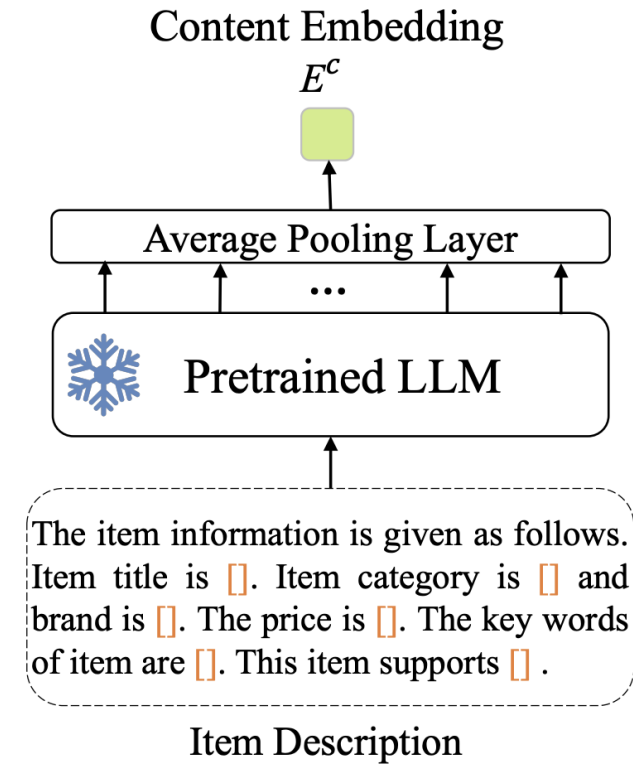
LEARN is a twin-tower framework that adapts LLM knowledge by freezing the LLM and using self-supervised training.

- **Content-Embedding Generation**

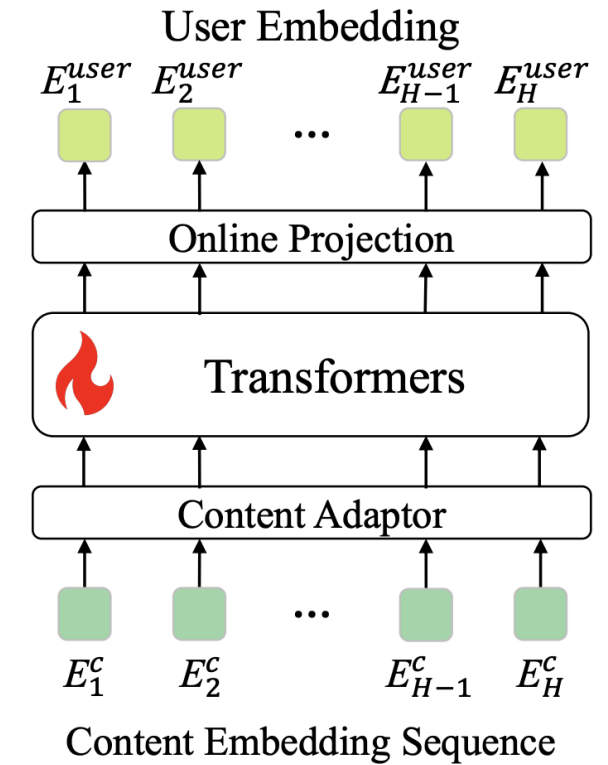
- Uses a frozen LLM as an item encoder to convert text into rich content embeddings. Freezing is key to preventing catastrophic forgetting.

- **Preference Comprehension**

- A lightweight Transformer learns user preferences from item embeddings via contrastive learning.



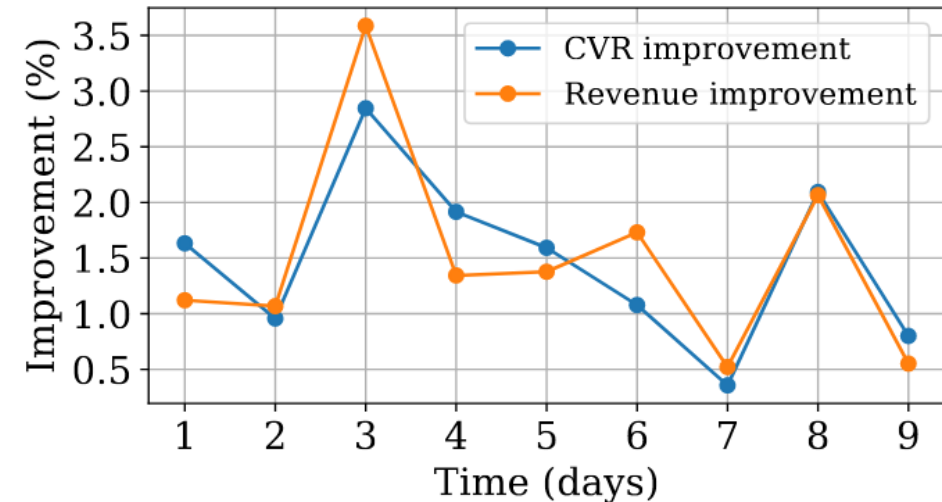
(a) Content Extraction Module



(b) Preference Alignment Module

- **LEARN** achieves SOTA performance on large-scale offline datasets and delivers significant business impact in real-world online A/B tests on Kuaishou.

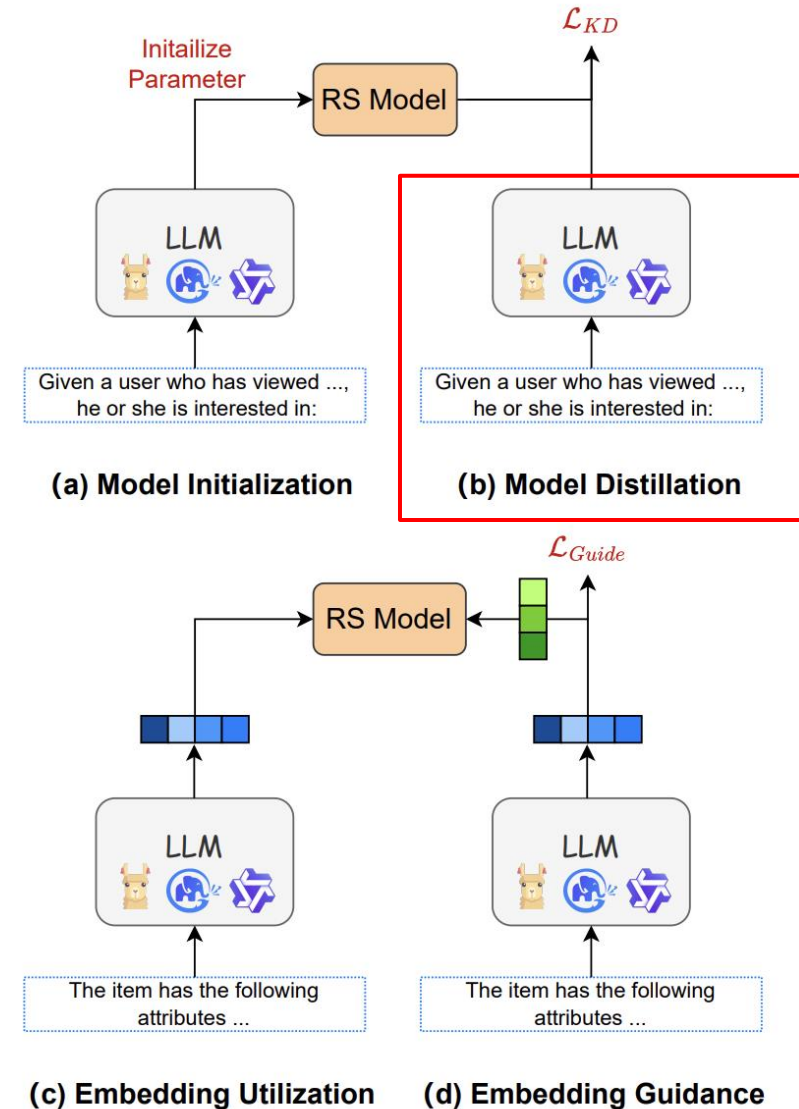
Method	H@10	H@50	H@200
SASRec [20]	0.0306	0.0754	0.1431
HSTU [43]	0.0416 (+35.95%)	0.0957 (+26.92%)	0.1735 (+21.24%)
LEARN (Ours)	0.0407 (+33.01%)	0.0979 (+29.84%)	0.1874 (+30.96%)
–	N@10	N@50	N@200
SASRec [20]	0.0164	0.0260	0.0362
HSTU [43]	0.0227 (+38.41%)	0.0344 (+32.31%)	0.0461 (+27.35%)
LEARN (Ours)	0.0224 (+36.59%)	0.0371 (+42.69%)	0.0483 (+33.43%)



- **Practical Values:** LEARN is the first LLM-enhanced solution successfully deployed and monetized in a large-scale industrial recommender system, setting a new standard for practical application.

Model Initialization

- This subcategory refers to compressing LLM knowledge into smaller RS models through distillation techniques.
- This approach transfers LLM capabilities to lightweight models while maintaining recommendation quality for efficient deployment.
- Categories
 - Feature-based
 - Response-based



- Background
 - Medication recommendation provides prescription suggestions to doctors. LLMs have two advantages: medical semantic understanding and cold-start capabilities.
- Motivation
 - Complex medication names cause LLMs to output recommendations not in drug databases, leading to failures.
 - Resource-intensive LLM inference challenges resource-constrained medical institutions in using LLMs for drug recommendations.
- Key Stages
 - Improve for LLMs
 - Distilling LLMs

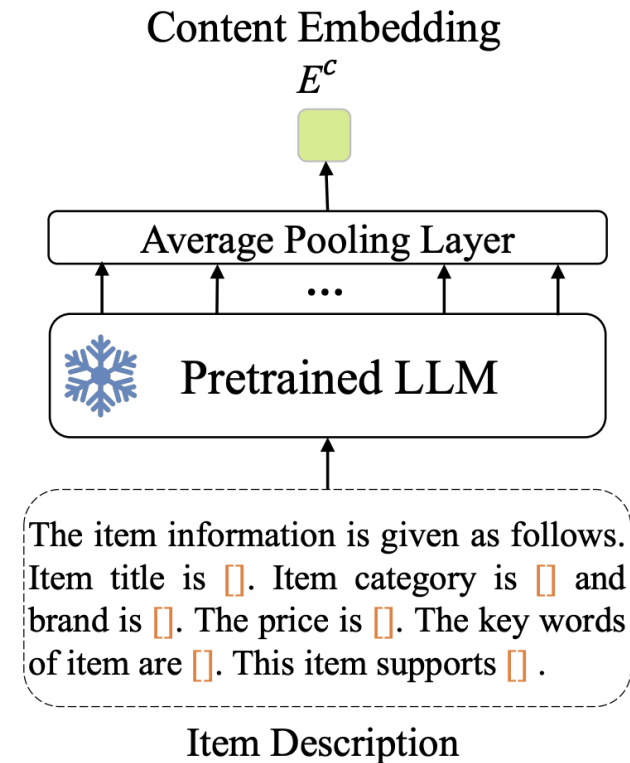
LEADER improved the recommendation performance of LLMs by enhancing the output layer.

- **Improve for LLMs**

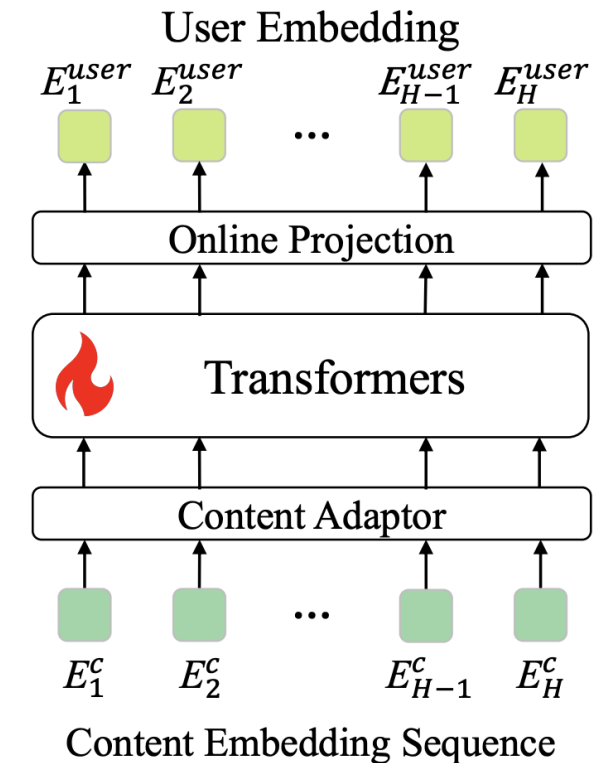
- Replace the original text generation head layer of the large model with a new classification layer, and fine-tune the model using BCE loss.

- **Distilling LLMs**

- Design the distillation process where the hidden states from the final layer of the LLMs are used as guidance to distill semantic understanding into the designed smaller model.



(a) Content Extraction Module



(b) Preference Alignment Module

- **LEADER** significantly outperforms baseline. The distilled smaller model demonstrates excellent performance in both overall and cold-start scenarios.

Model	Overall			Multi-visit			Single-visit		
	PRAUC	Jaccard	F1	PRAUC	Jaccard	F1	PRAUC	Jaccard	F1
RETAIN	0.7513 $\pm$ 0.0025	0.4943 $\pm$ 0.0023	0.6516 $\pm$ 0.0022	0.7580 $\pm$ 0.0020	0.5106 $\pm$ 0.0023	0.6674 $\pm$ 0.0022	0.7337 $\pm$ 0.0067	0.4811 $\pm$ 0.0053	0.6403 $\pm$ 0.0049
G-Bert	-	-	-	0.6904 $\pm$ 0.0017	0.4578 $\pm$ 0.0019	0.6186 $\pm$ 0.0018	-	-	-
GAMENet	0.7605 $\pm$ 0.0011	0.5024 $\pm$ 0.0010	0.6595 $\pm$ 0.0008	0.7638 $\pm$ 0.0023	0.5070 $\pm$ 0.0028	0.6635 $\pm$ 0.0025	0.7451 $\pm$ 0.0053	0.4840 $\pm$ 0.0038	0.6442 $\pm$ 0.0036
SafeDrug	0.7582 $\pm$ 0.0020	0.5054 $\pm$ 0.0024	0.6621 $\pm$ 0.0021	0.7623 $\pm$ 0.0029	0.5095 $\pm$ 0.0027	0.6658 $\pm$ 0.0024	0.7416 $\pm$ 0.0044	0.4900 $\pm$ 0.0043	0.6481 $\pm$ 0.0042
MICRON	-	-	-	0.7651 $\pm$ 0.0027	0.5110 $\pm$ 0.0025	0.6741 $\pm$ 0.0023	-	-	-
COGNet	-	-	-	0.7771 $\pm$ 0.0019	0.5275 $\pm$ 0.0021	0.6805 $\pm$ 0.0019	-	-	-
REFINE	-	-	-	0.7791 $\pm$ 0.0017	0.5235 $\pm$ 0.0018	0.6794 $\pm$ 0.0017	-	-	-
LEADER(T)	0.7816 $\pm$ 0.0015*	0.5391 $\pm$ 0.0015*	0.6921 $\pm$ 0.0014*	0.7854 $\pm$ 0.0015*	0.5450 $\pm$ 0.0021*	0.6971 $\pm$ 0.0018*	0.7590 $\pm$ 0.0046*	0.5090 $\pm$ 0.0044*	0.6668 $\pm$ 0.0041*
LEADER(S)	0.7795 $\pm$ 0.0025*	0.5175 $\pm$ 0.0022*	0.6737 $\pm$ 0.0019*	0.7830 $\pm$ 0.0019	0.5208 $\pm$ 0.0020	0.6768 $\pm$ 0.0017	0.7631 $\pm$ 0.0056*	0.5038 $\pm$ 0.0062*	0.6614 $\pm$ 0.0057*

- **Practical Values:** LEADER can be used for recommending prescriptions for patients who first visit the hospital (cold-start scenarios).

- Background
 - LLM-based recommenders face serious inference inefficiency issues, posing substantial challenges to their practical applications.
- Motivation
 - Teacher Knowledge Reliability: The teacher's knowledge isn't always reliable.
 - Model Capacity Gap: The student struggles to assimilate teacher knowledge.
 - Semantic Space Divergence: Different "languages" make direct alignment counterproductive.
- Key Stages
 - Importance-aware Ranking Distillation
 - Collaborative Embedding Distillation

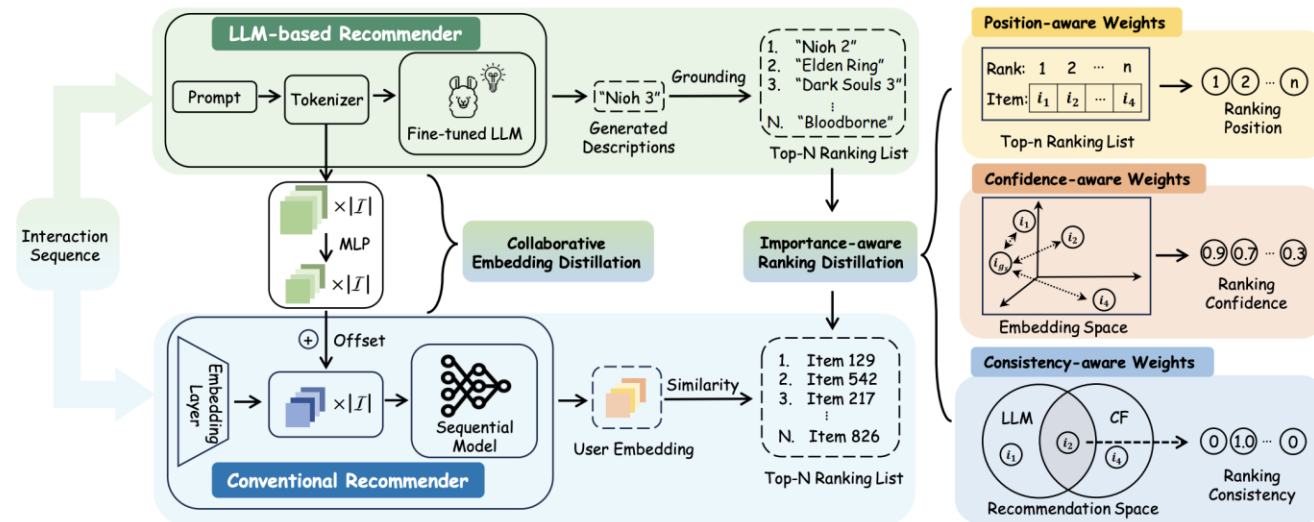
DLLM2Rec is designed to effectively distill knowledge from LLM-based recommenders to conventional recommenders.

- **Importance-aware Ranking Distillation**

- DLLM2Rec introduces importance weights to selectively learn reliable knowledge rather than blindly mimicking teacher rankings.

- **Collaborative Embedding Distillation**

- DLLM2Rec performs collaborative fusion to bridge semantic space divergence between teacher and student.



- **DLLM2Rec's** lightweight student model not only achieves a million-fold speedup in inference but also outperforms the massive LLM teacher on several datasets.

Backbone	Model	Games		MovieLens		Toys	
		HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
Teacher	BIGRec	0.0532	0.0341	0.0541	0.0370	0.0420	0.0207
	+None	0.0305	0.0150	0.0608	0.0236	0.0172	0.0081
	+Hint	0.0284	0.0120	0.0646	0.0240	0.0128	0.0058
	+HTD	0.0299	0.0128	0.0578	0.0229	0.0155	0.0062
	+RD	0.0398	0.0177	0.0667	0.0254	0.0157	0.0076
GRU4Rec	+CD	0.0306	0.0149	0.0699	0.0256	0.0126	0.0052
	+RRD	0.0359	0.0163	0.0657	0.0243	0.0215	0.0097
	+DCD	0.0427	0.0190	0.0666	0.0263	0.0262	0.0114
	+UnKD	0.0370	0.0170	0.0607	0.0226	0.0235	0.0114
	KAR	0.0307	0.0149	0.0603	0.0229	0.0184	0.0079
	LLM-CF	0.0393	0.0174	0.0677	0.0246	0.0132	0.0058
	+DLLM2Rec	0.0446	0.0205	0.0815	0.0308	0.0281	0.0118
Gain.S		+46.17%	+36.94%	+34.05%	+30.43%	+63.88%	+42.18%
Gain.B		+4.56%	+7.64%	+16.60%	+16.80%	+7.40%	+1.27%

Dataset	Model	HR@20	NDCG@20	Inference time
Games	BIGRec	0.0532	0.0341	2.3×10^4 s
	DLLM2Rec	0.0751	0.0331	1.8s
	Gain	+37.41%	-2.99%	$+1.3 \times 10^6$ %
MovieLens	BIGRec	0.0541	0.0370	1.8×10^4 s
	DLLM2Rec	0.1063	0.0437	1.7s
	Gain	+96.49%	+18.18%	$+1.1 \times 10^6$ %
Toys	BIGRec	0.0420	0.0207	1.1×10^4 s
	DLLM2Rec	0.0463	0.0225	1.6s
	Gain	+10.24%	+8.70%	$+6.8 \times 10^5$ %

- **Practical Values:** DLLM2Rec provides the industry with a path to integrate the powerful capabilities of large language models into existing recommendation systems in an economical, efficient, and feasible manner.

Agenda

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang



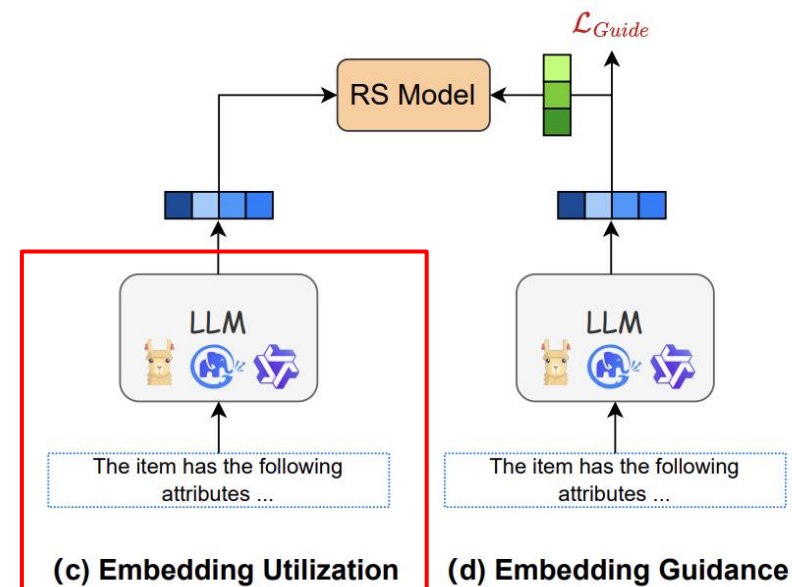
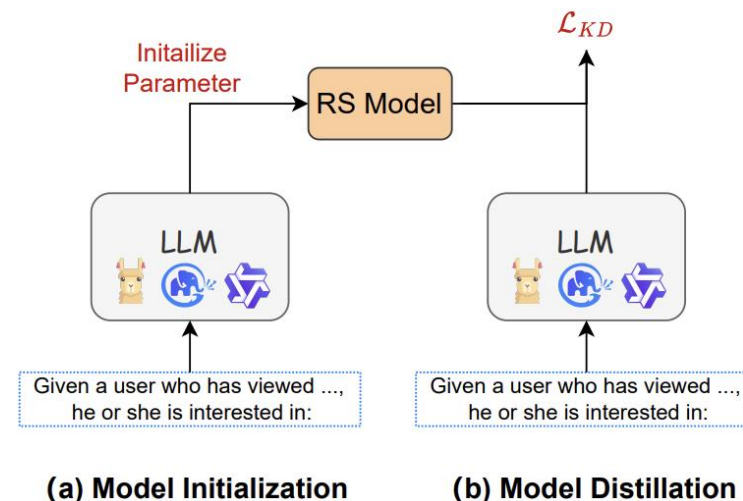
5 Conclusion



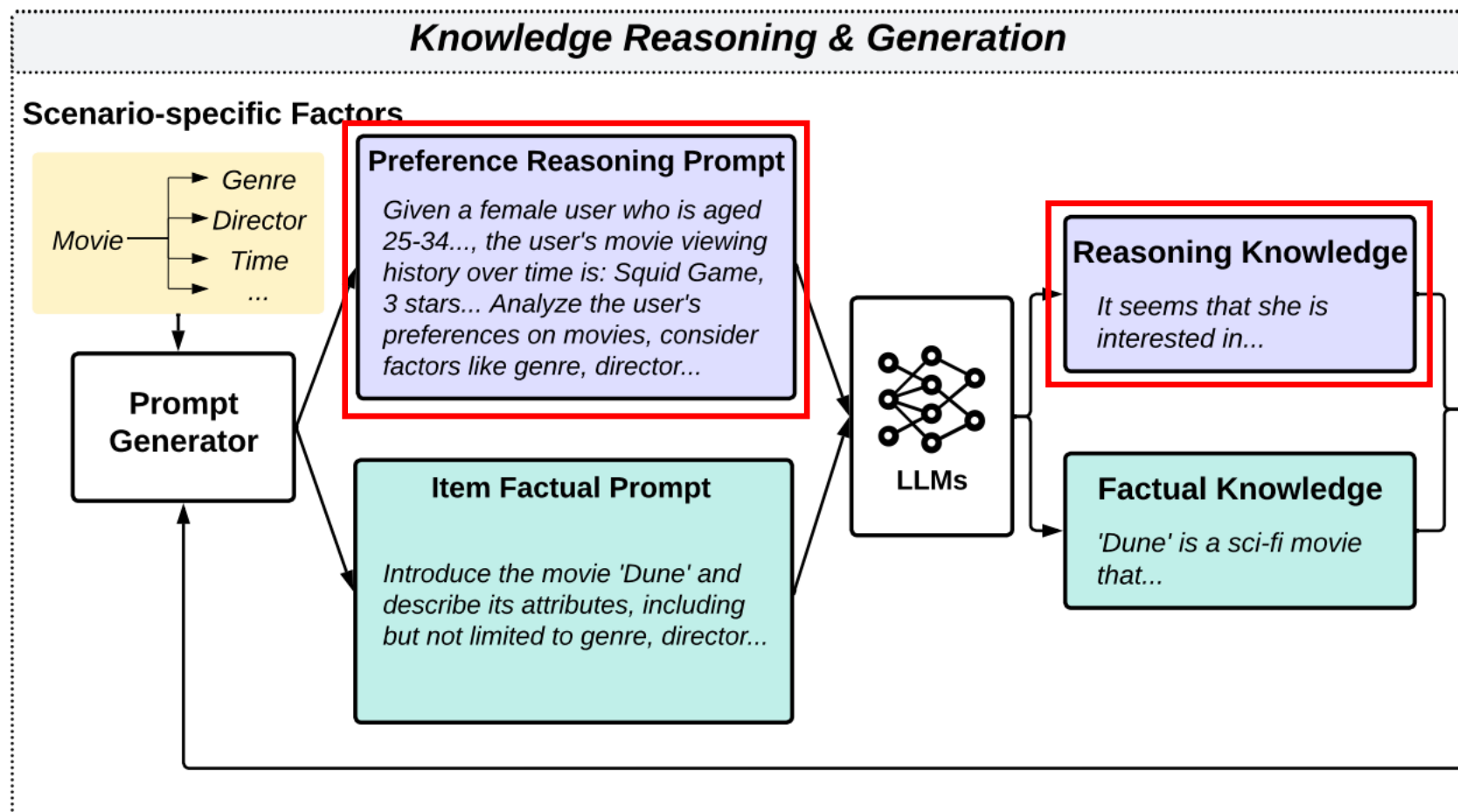
Yejing Wang

Embedding Utilization

- This subcategory refers to utilizing the embeddings derived from LLM to enhance traditional RS as a semantic supplement.
- This approach tackles the deficiency that textual outputs are often difficult and inefficient to be integrated into RS directly.
- Categories
 - User Only
 - Item Only
 - User & Item

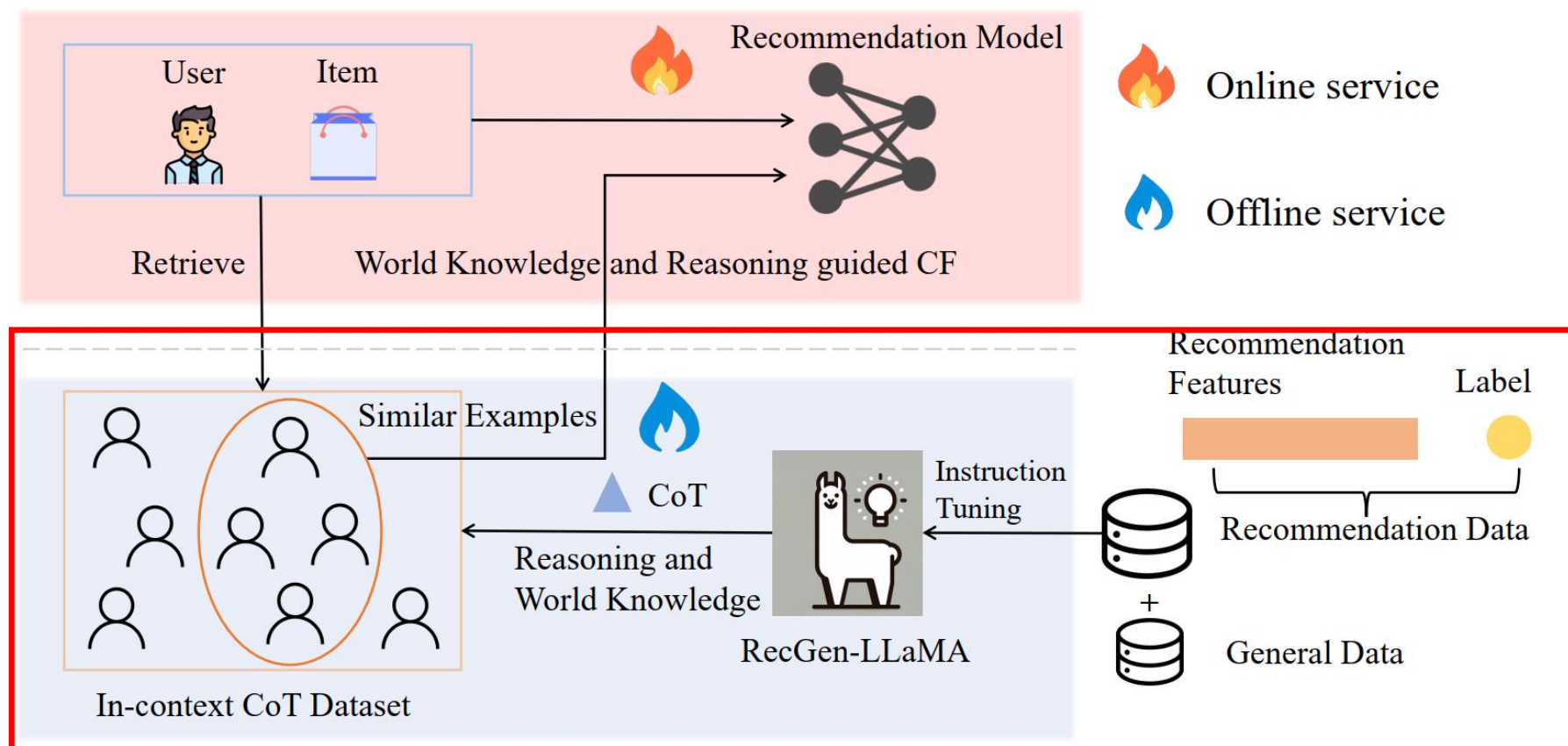


- Reasoning about user preferences given **user profile** and **interactions**



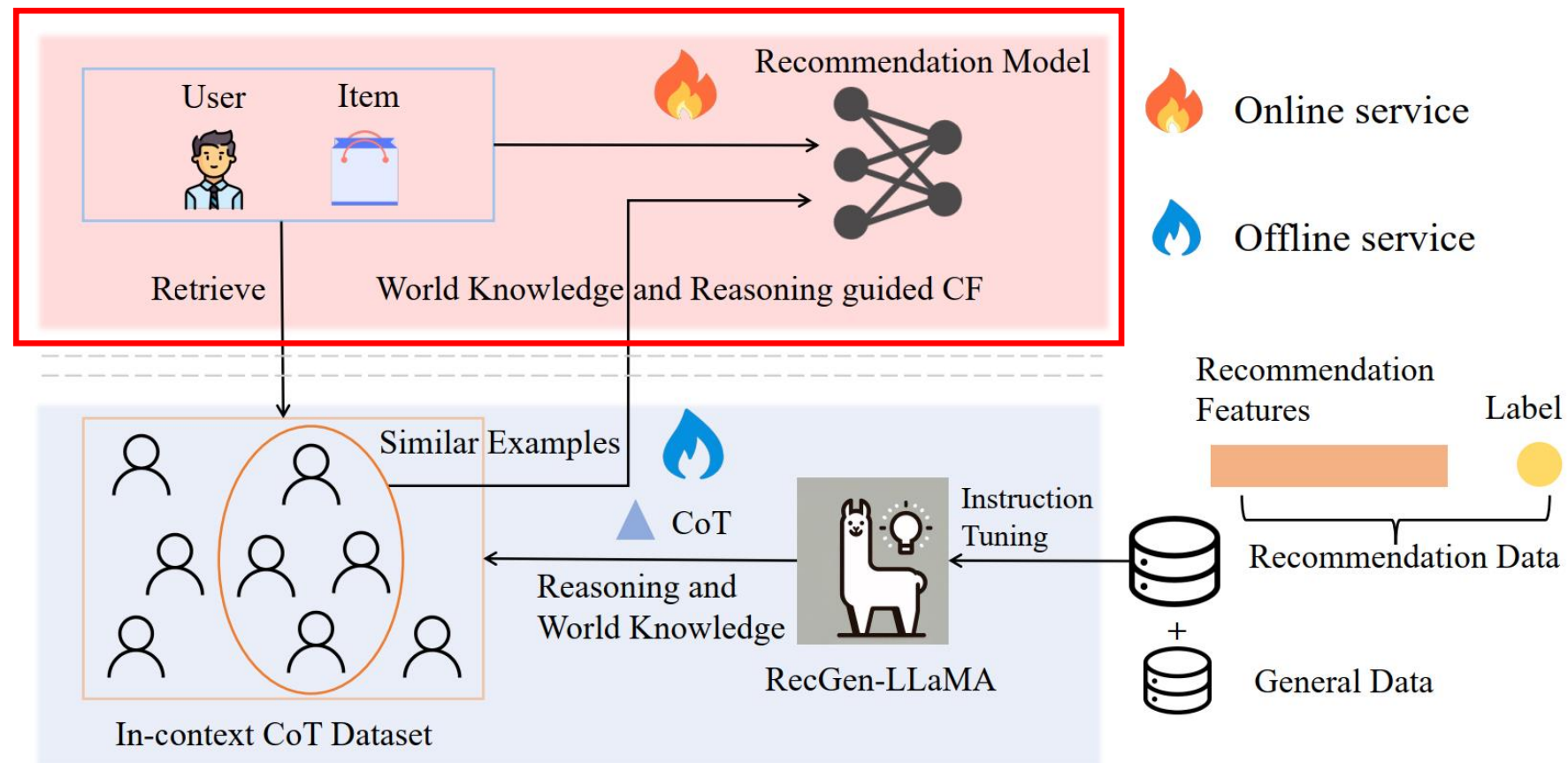
Offline service

- Chain of Thought (CoT) reasoning
- In-context CoT dataset construction



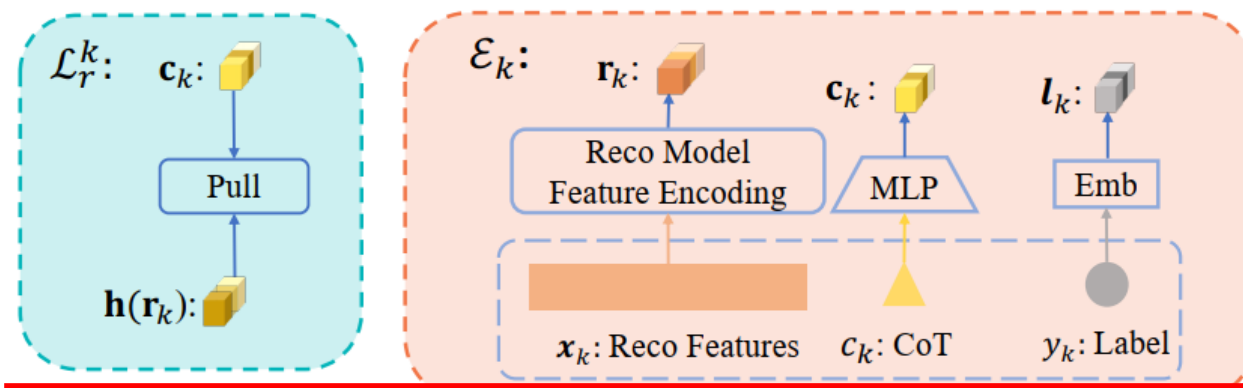
• Online service

- In-context CoT examples Retrieval
- World knowledge & reasoning guided CF
- Feature enhanced RS



• CoT Prompt

- Generating CoT of the user's decision making on the target item



Chain of Thought Prompt

<<SYS>> As an AI model developed for analyzing consumer behavior, your task is to generate a chain of thought that considers the following points:

1. Utilize the user's interaction history and review comments to summarize their profiles.
2. Introduce the target new item and detail its features precisely. In addition, integrate information about items related to the current target new item ...
3. Contemplate the alignment between the user's profile and the features of the target item.
4. Reflect on the user's potential desire for diversity in their purchases.

Your output should be a clear and logical chain of thought... Ensure your analysis is impartial... Focus should be on understanding factors that influence the user's decision-making regarding the target item. </SYS>

Please generate a chain of thought based on the user's ... considering how these might relate to their interest in the target new item.

{Recommendation Features}

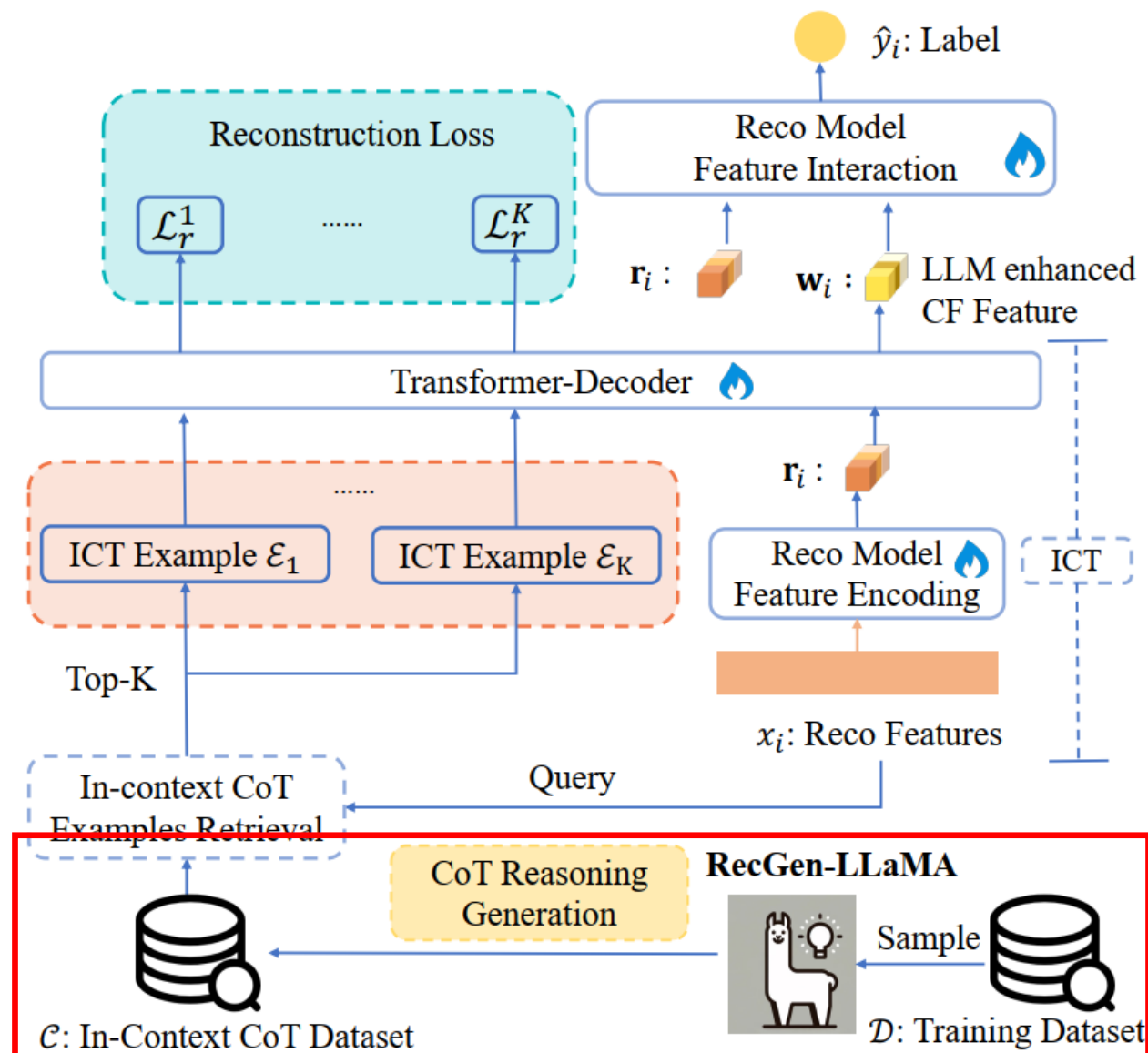
User's decision-making: The user {Ground-Truth Label} the target new item.

Let's think step by step and develop the chain of thought for the above considerations.

Commence with the chain of thought immediately:

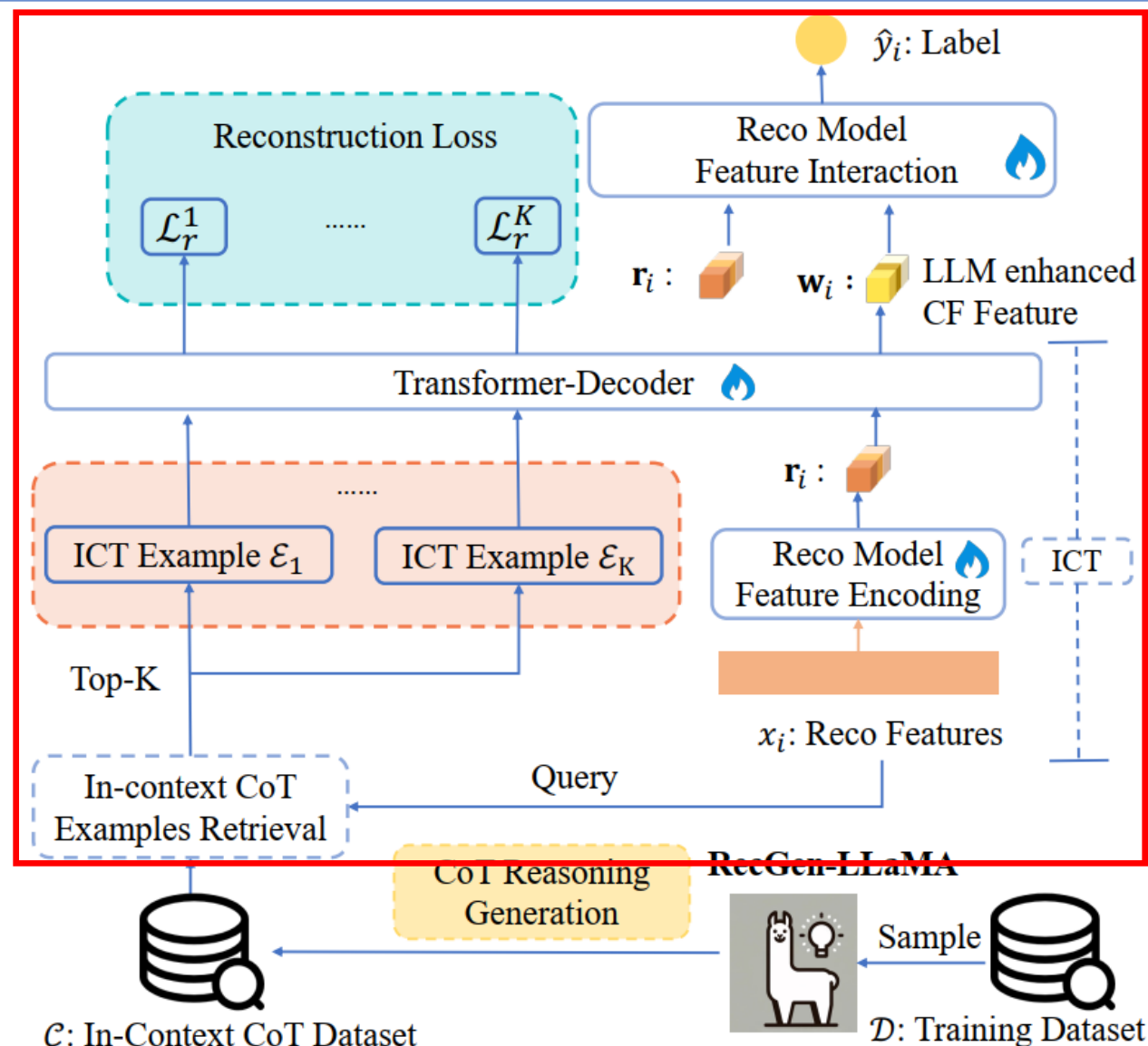
• Offline Service

- Instruction tuning on LLaMA2
- RecGen-LLaMA generating CoT forming ICT dataset



• Online Service

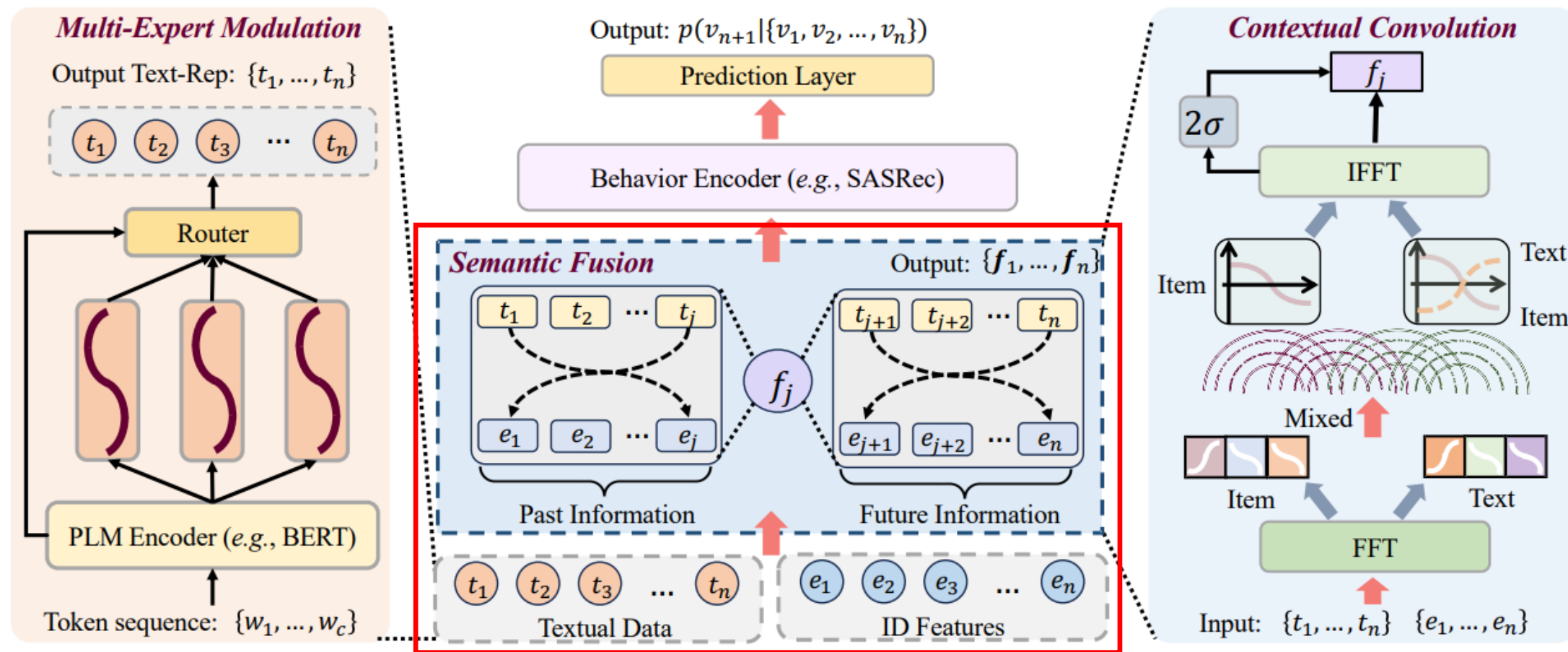
- Embedding-based retrieval forming ICT examples
- Learning world-knowledge and reasoning guided CF feature



Overall performance

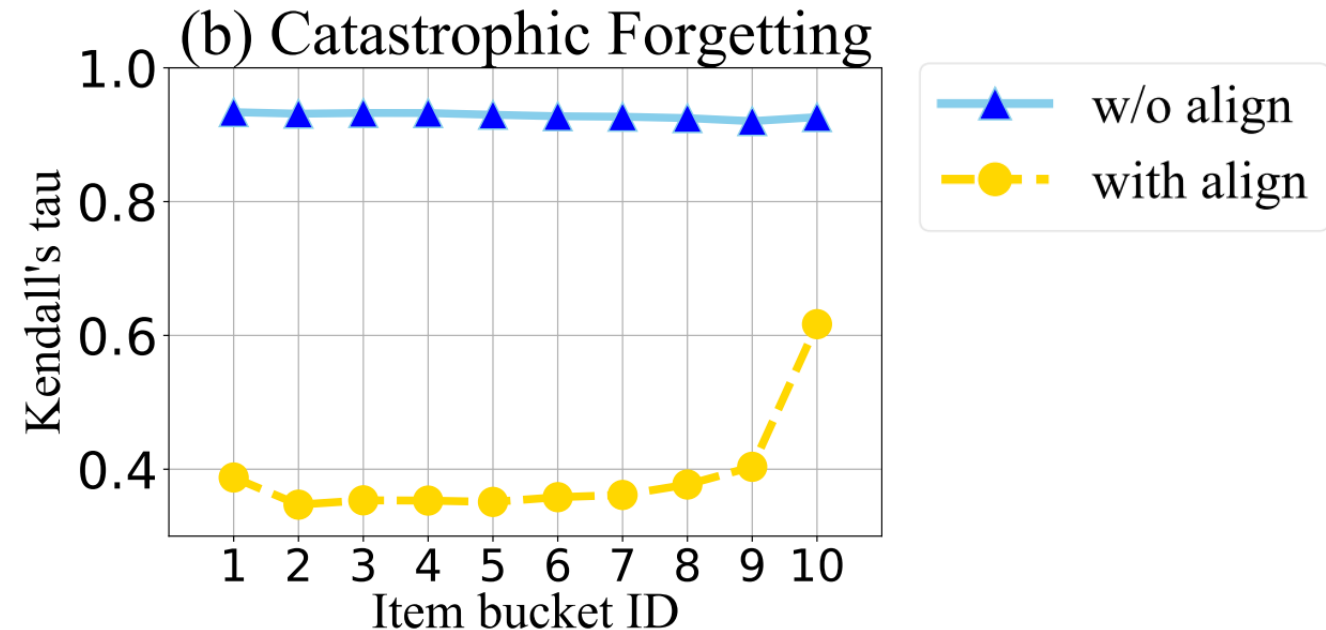
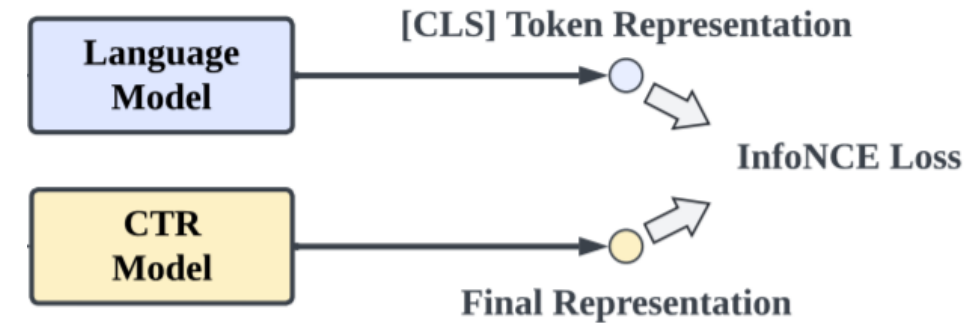
Backbone	Framework	Sports			Beauty			Toys		
		AUC↑	Logloss↓	RelaImpr↑	AUC↑	Logloss↓	RelaImpr↑	AUC↑	Logloss↓	RelaImpr↑
DeepFM	None	0.7990	0.5471	0.000%	0.7853	0.5545	0.000%	0.7681	0.5770	0.000%
	KD	<u>0.8043</u>	<u>0.5404</u>	1.773%	<u>0.7959</u>	<u>0.5442</u>	3.715%	<u>0.7713</u>	<u>0.5716</u>	1.194%
	KAR	0.7991	0.5469	0.033%	0.7870	0.5546	0.596%	0.7698	0.5718	0.634%
	LLM-CF	0.8137*	0.5306*	4.916%	0.8044*	0.5366*	6.695%	0.7881*	0.5581*	7.460%
xDeepFM	None	0.8158	0.5318	0.000%	0.8065	0.5359	0.000%	0.7836	0.5589	0.000%
	KD	<u>0.8169</u>	0.5298	0.348%	<u>0.8104</u>	0.5345	1.272%	0.7865	0.5553	1.023%
	KAR	0.8161	0.5279	0.094%	0.8101	0.5315	1.175%	0.7898	0.5529	2.186%
	LLM-CF	0.8196*	0.5248*	1.203%	0.8113	0.5311	1.566%	0.7947*	0.5473*	3.914%
AutoInt	None	0.8003	0.5444	0.000%	0.7949	0.5469	0.00%	0.7630	0.5770	0.000%
	KD	0.8012	0.5439	0.300%	<u>0.7961</u>	<u>0.5444</u>	0.407%	0.7635	0.5770	0.190%
	KAR	0.8039	0.5390	1.199%	0.7939	0.5476	-0.339%	0.7683	0.5741	2.015%
	LLM-CF	0.8088*	0.5391	2.831%	0.8090*	0.5321*	4.781%	0.7754*	0.5685*	4.714%
DCNv1	None	0.8023	0.5442	0.000%	0.8146	0.5255	0.000%	0.7621	0.5831	0.000%
	KD	<u>0.8040</u>	<u>0.5441</u>	0.562%	0.8147	0.5286	0.031%	<u>0.7652</u>	0.5847	1.183%
	KAR	0.8024	0.5469	0.033%	0.8165	0.5229	0.604%	0.7651	0.5821	1.144%
	LLM-CF	0.8092*	0.5368*	2.282%	0.8182*	0.5216*	1.144%	0.7702*	0.5745*	3.090%
DCNv2	None	0.8110	0.5331	0.000%	0.8028	0.5378	0.00%	0.7774	0.5650	0.000%
	KD	<u>0.8112</u>	<u>0.5320</u>	0.064%	0.8057	0.5343	0.958%	0.7827	0.5609	1.911%
	KAR	0.8087	0.5363	-0.739%	0.8003	0.5404	-0.825%	0.7759	0.5662	-0.541%
	LLM-CF	0.8131*	0.5307*	0.675%	0.8033	0.5372	0.165%	0.7812	0.5619	1.370%
DIN	None	0.7986	0.5519	0.000%	0.7861	0.5613	0.000%	0.7586	0.5885	0.000%
	KD	<u>0.8023</u>	<u>0.5422</u>	1.239%	<u>0.7934</u>	<u>0.5518</u>	2.551%	<u>0.7652</u>	<u>0.5847</u>	2.552%
	KAR	0.7971	0.5525	-0.502%	0.7861	0.5604	0.000%	0.7620	0.5874	1.315%
	LLM-CF	0.8089*	0.5374*	3.449%	0.7967*	0.5492*	3.705%	0.7783*	0.5699*	7.618%

- LLM-CF achieves improvement on different CTR backbones.

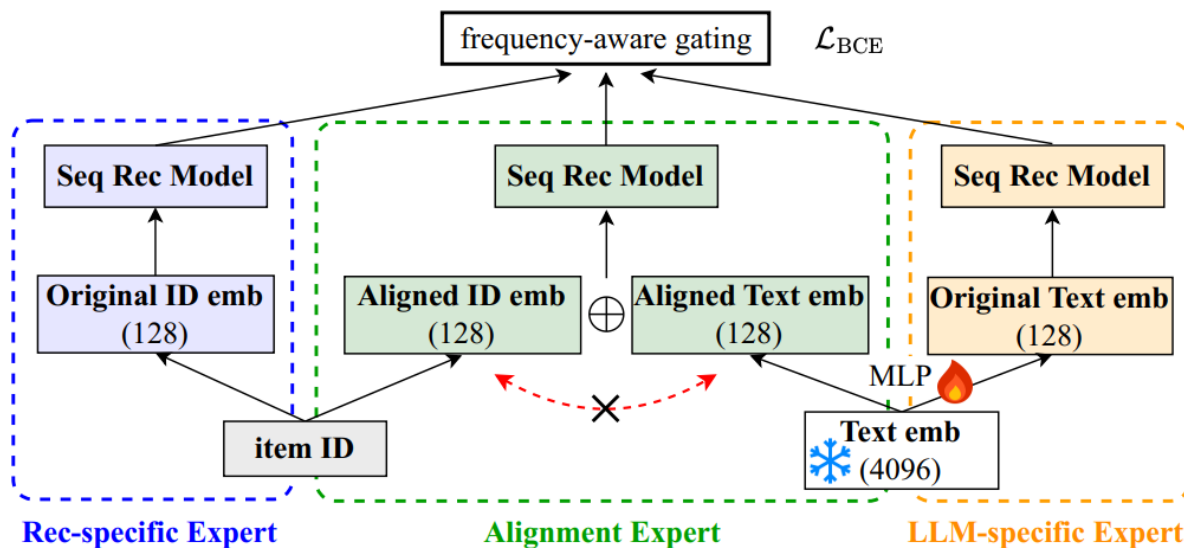
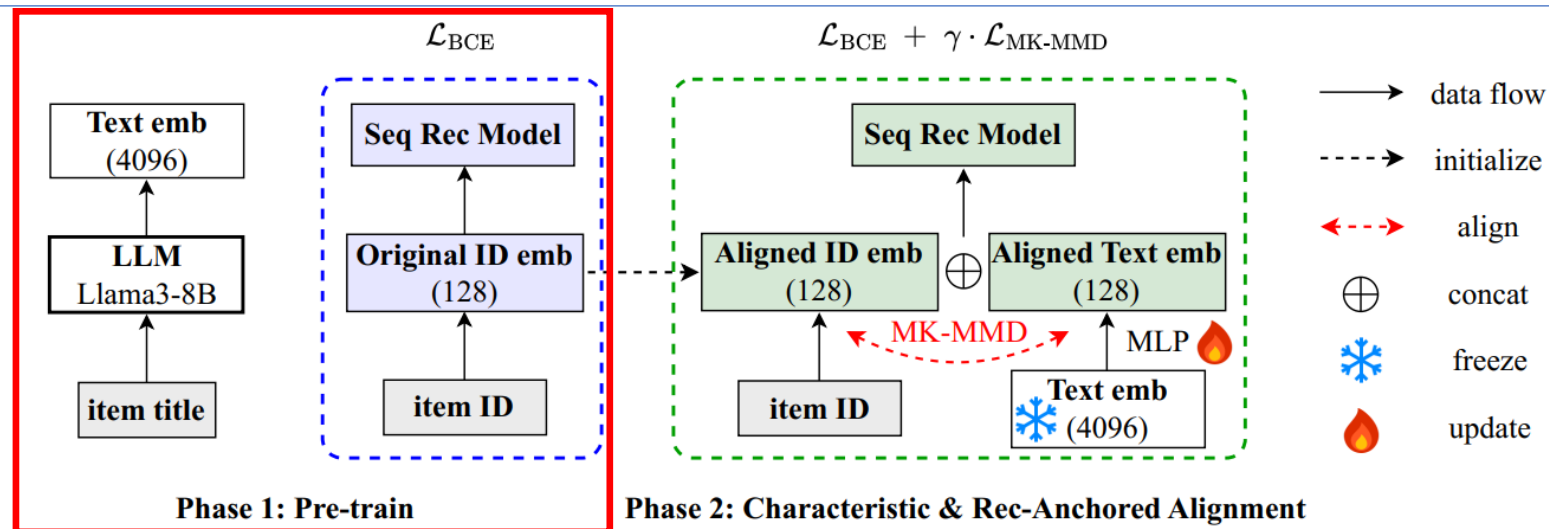


- TedRec applies Fourier Transform to conduct semantic fusion in the frequency domain.

- **Challenges for Alignment + SFT**
 - **Inability to Capture All Statistics of Data Distribution:** Contrastive learning with common cosine kernel is not optimal
 - **Catastrophic Forgetting in Alignment:** With only one set of collaborative embeddings, the alignment leads to catastrophic forgetting on the collaborative embeddings



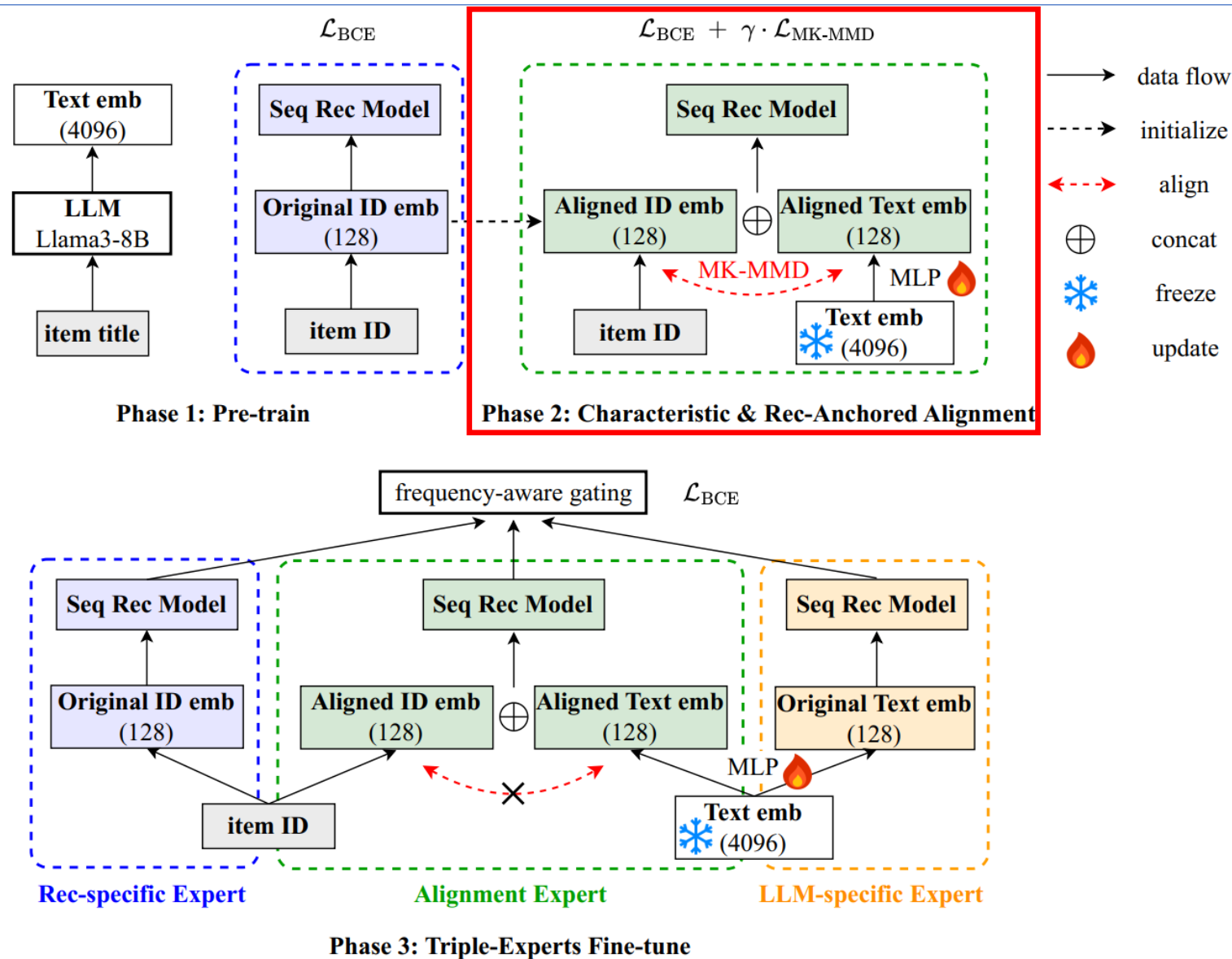
- **Pre-train**
 - Text & Collaborative emb



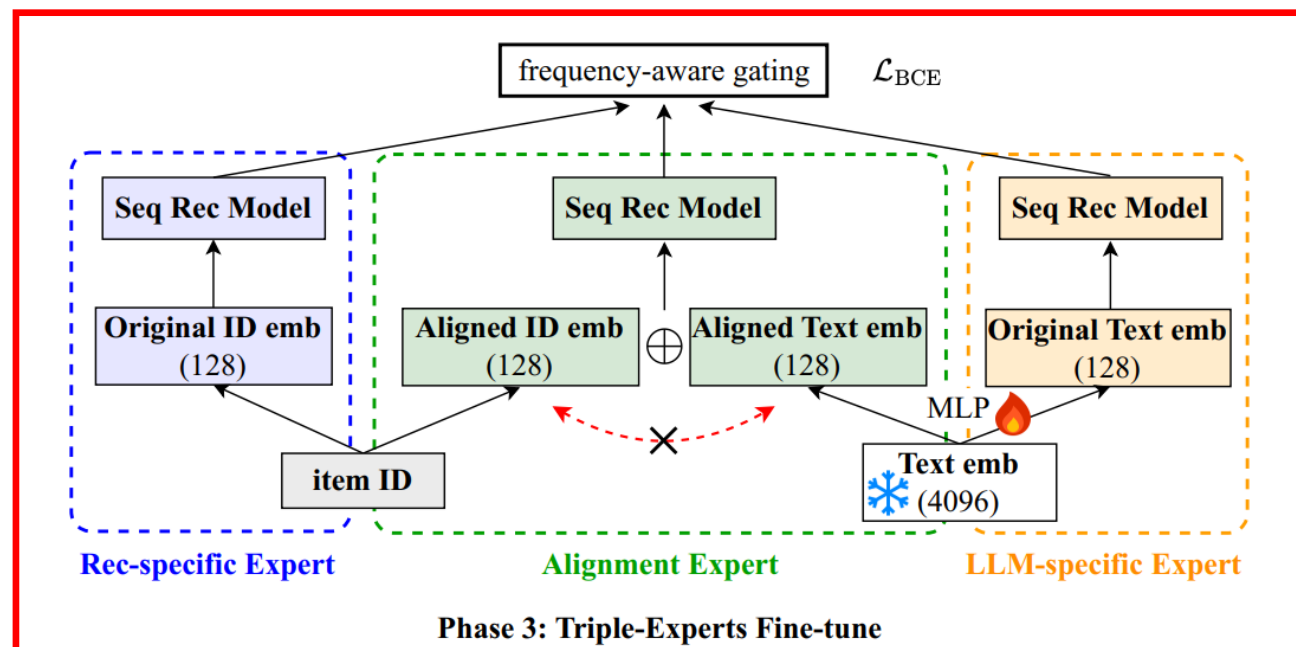
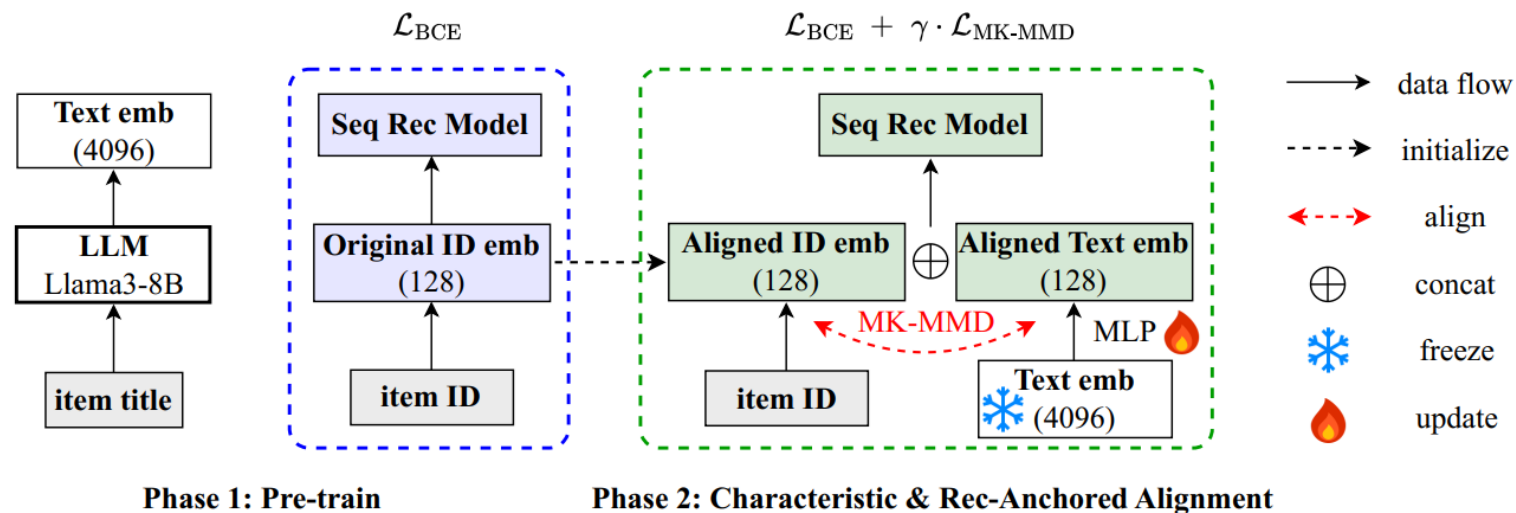
Phase 3: Triple-Experts Fine-tune

Align

- Recommendation as anchor
- Multi-kernel maximum mean discrepancy as alignment loss based on characteristic kernel



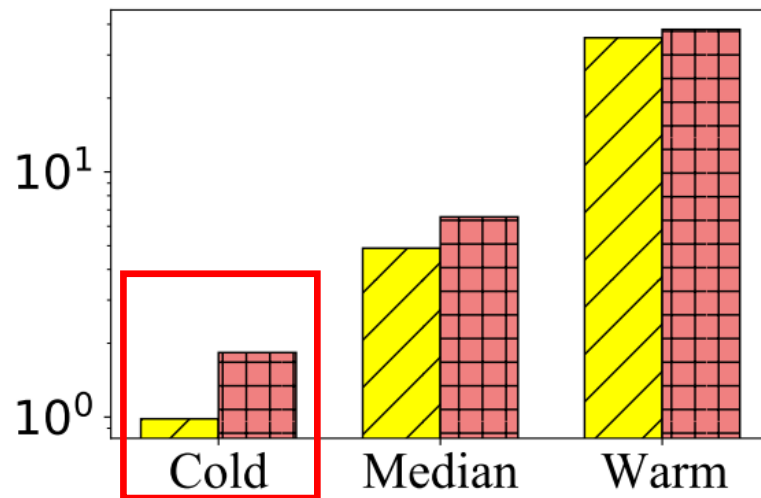
- **Disentangle**
 - Triple-Experts & Multi-Emb



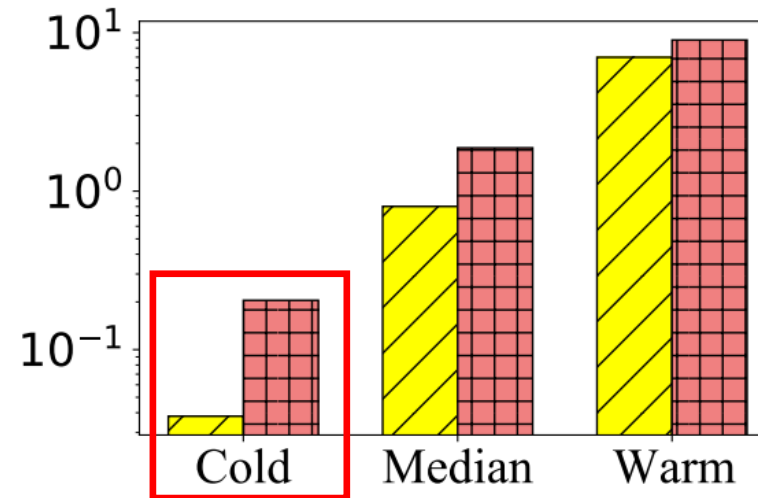
Datasets	Metric	SASRec	Hybrid	MoRec	CTRL	MAKE	DisCo	SMEM	Ours	Impr.
MIND	HR@10	16.8437	17.3385	14.1235	18.1318	16.4317	17.3367	<u>18.4319</u>	18.6703[★]	10.84%
	nDCG@10	9.0520	9.3321	7.7090	9.7996	8.9684	9.4131	<u>10.0003</u>	10.1515[★]	12.15%
Electronics	HR@10	1.6754	1.5673	0.9562	1.9468	0.9495	1.8148	<u>2.3492</u>	2.4804[★]	48.05%
	nDCG@10	0.7938	0.8385	0.4721	0.9370	0.4611	0.9025	<u>1.4287</u>	1.5373[★]	93.67%
Prime Pantry	HR@10	2.6338	3.1000	3.1699	2.8048	3.3408	2.8358	<u>3.4341</u>	3.8303[★]	45.43%
	nDCG@10	1.2926	1.5606	1.6013	1.3700	1.6379	1.3451	<u>1.7440</u>	1.9104[★]	47.80%

- PAD surpasses all baselines and achieves significant improvement.

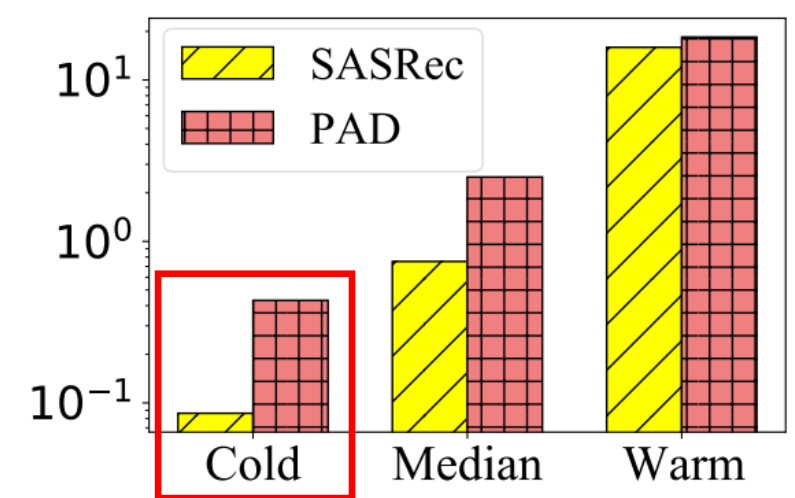
MIND



Electronics



Prime Pantry



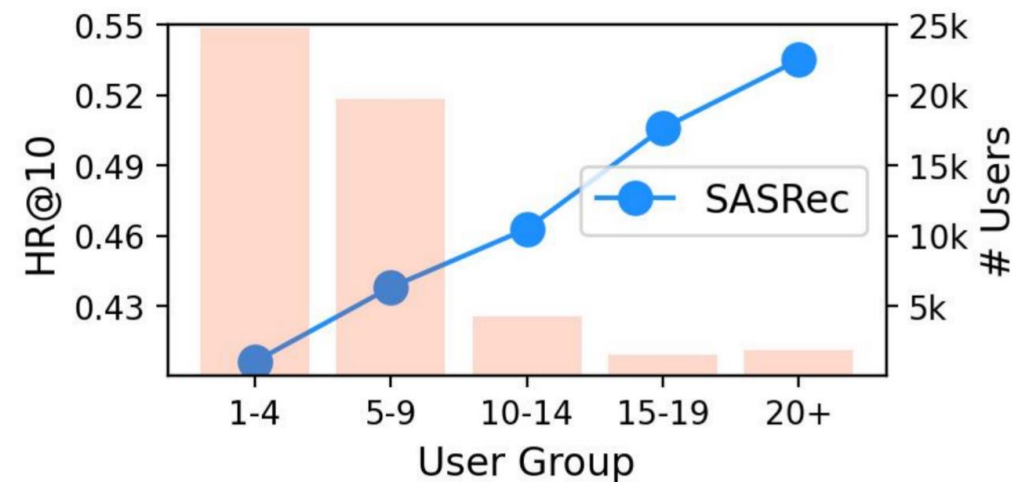
- PAD can mitigate cold-start problem with LLM knowledge.

• Long-tail Challenges for SR

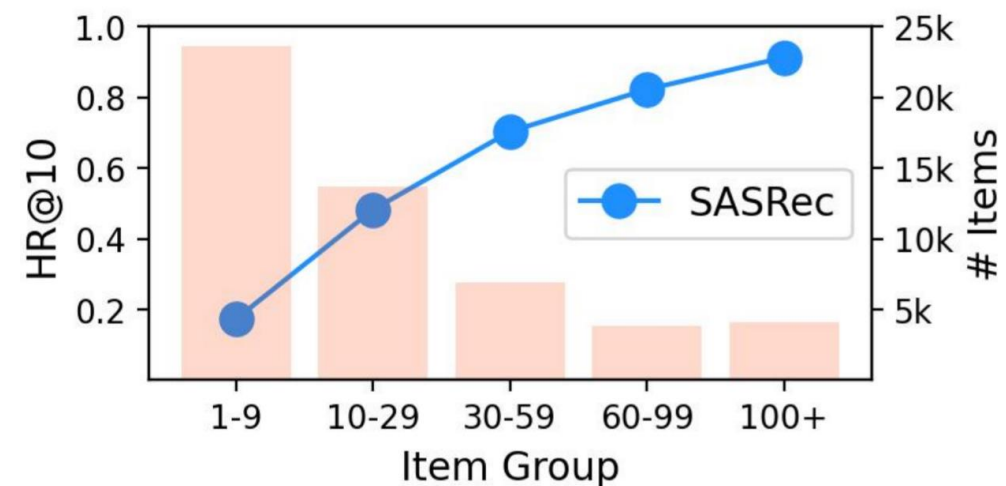
- **Long-tail User Challenge:** The majority of users receive less than optimal recommendation services
→ Poor experience for **new users**

- **Long-tail Item Challenge:** Current SRS models often underperform those unpopular items
→ Less profits for **small sellers**

- **LLMs are promising to address the long-tail challenges from semantic view!**

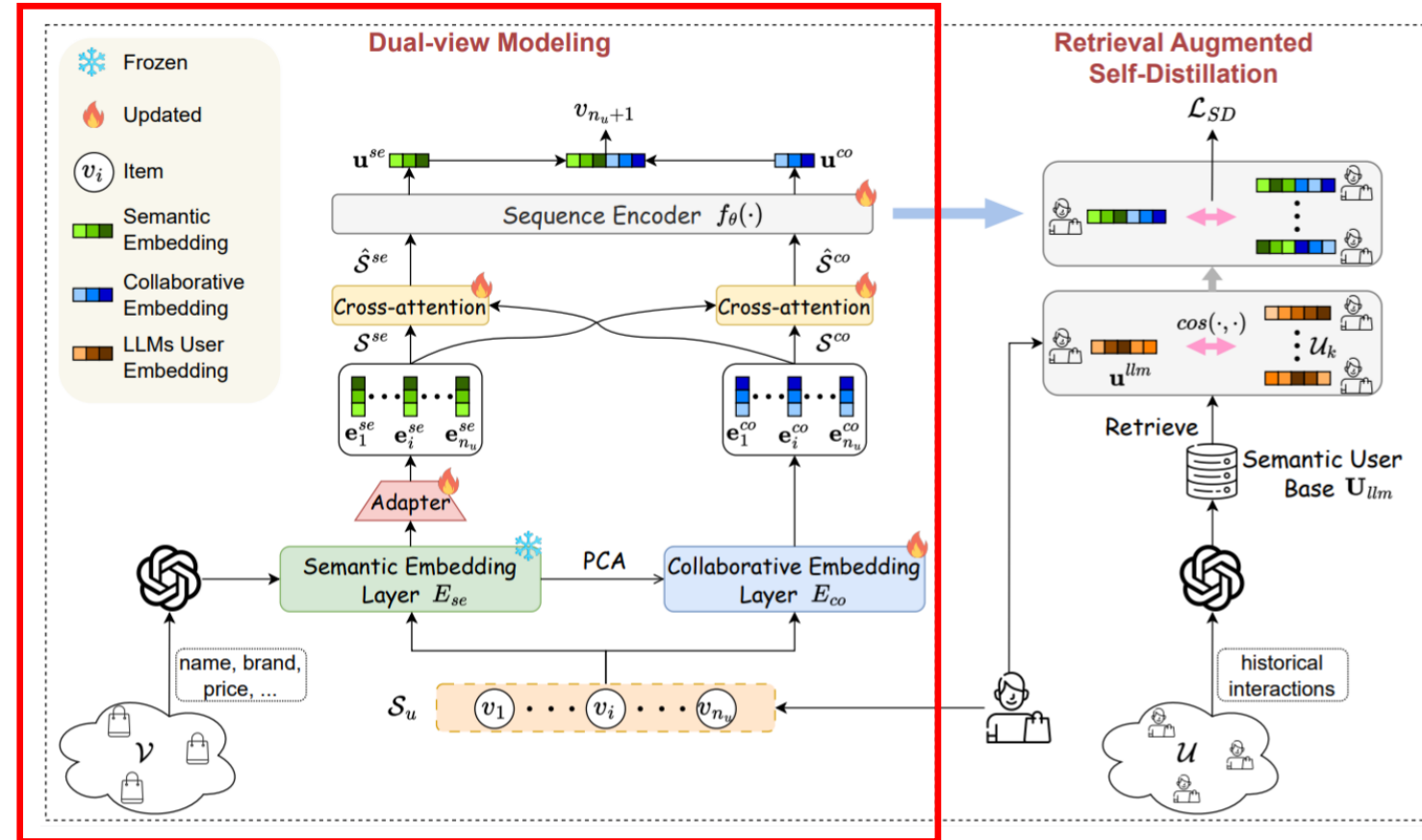


(a) Long-tail User Challenge

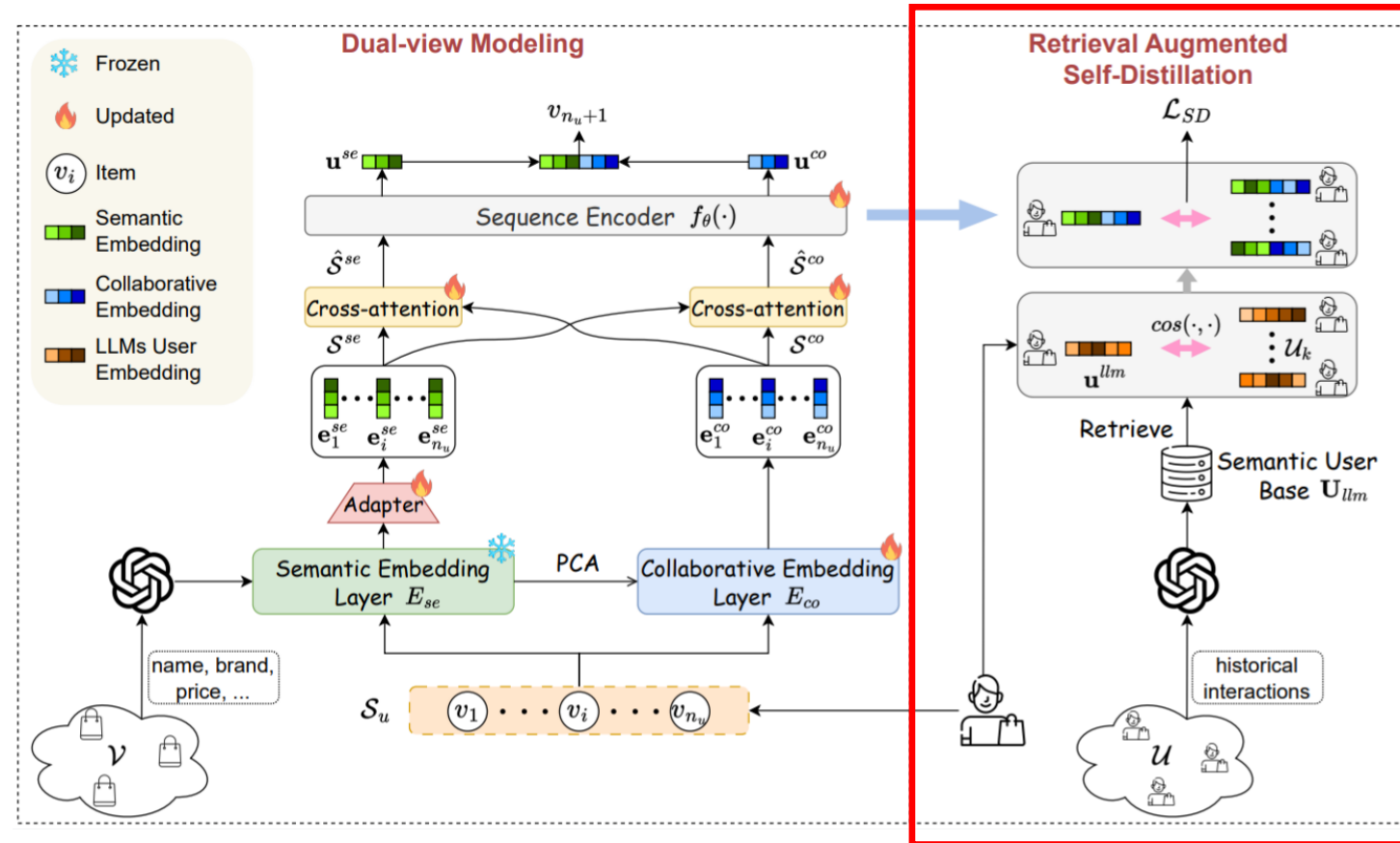


(b) Long-tail Item Challenge

- **Dual-view Modeling:** Consisting of semantic-view modeling and collaborative-view modeling to address the long-tail item issue



- Retrieval Augmented Self-Distillation:** Enhancing the SRS model to address the long-tail user problem

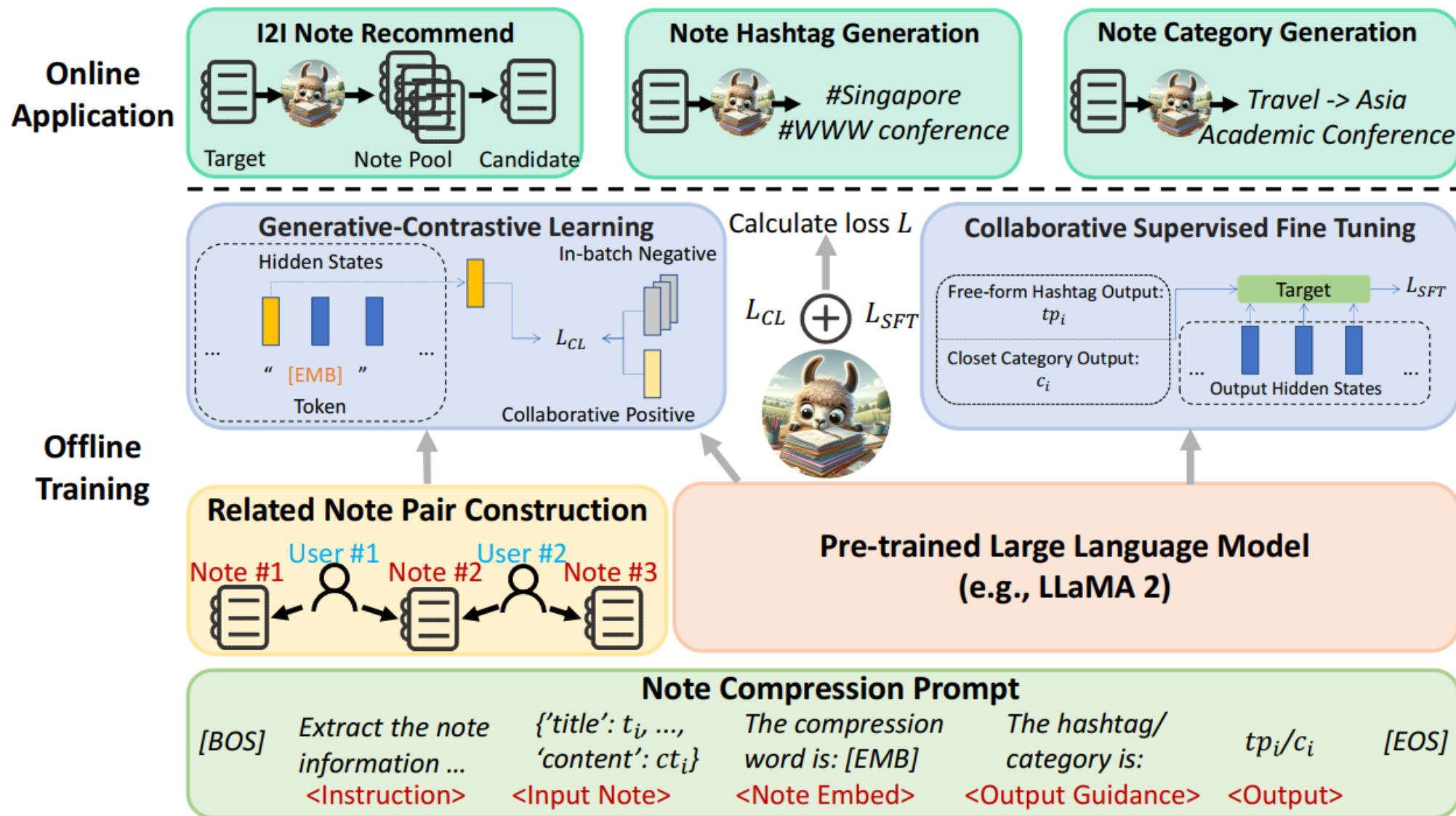


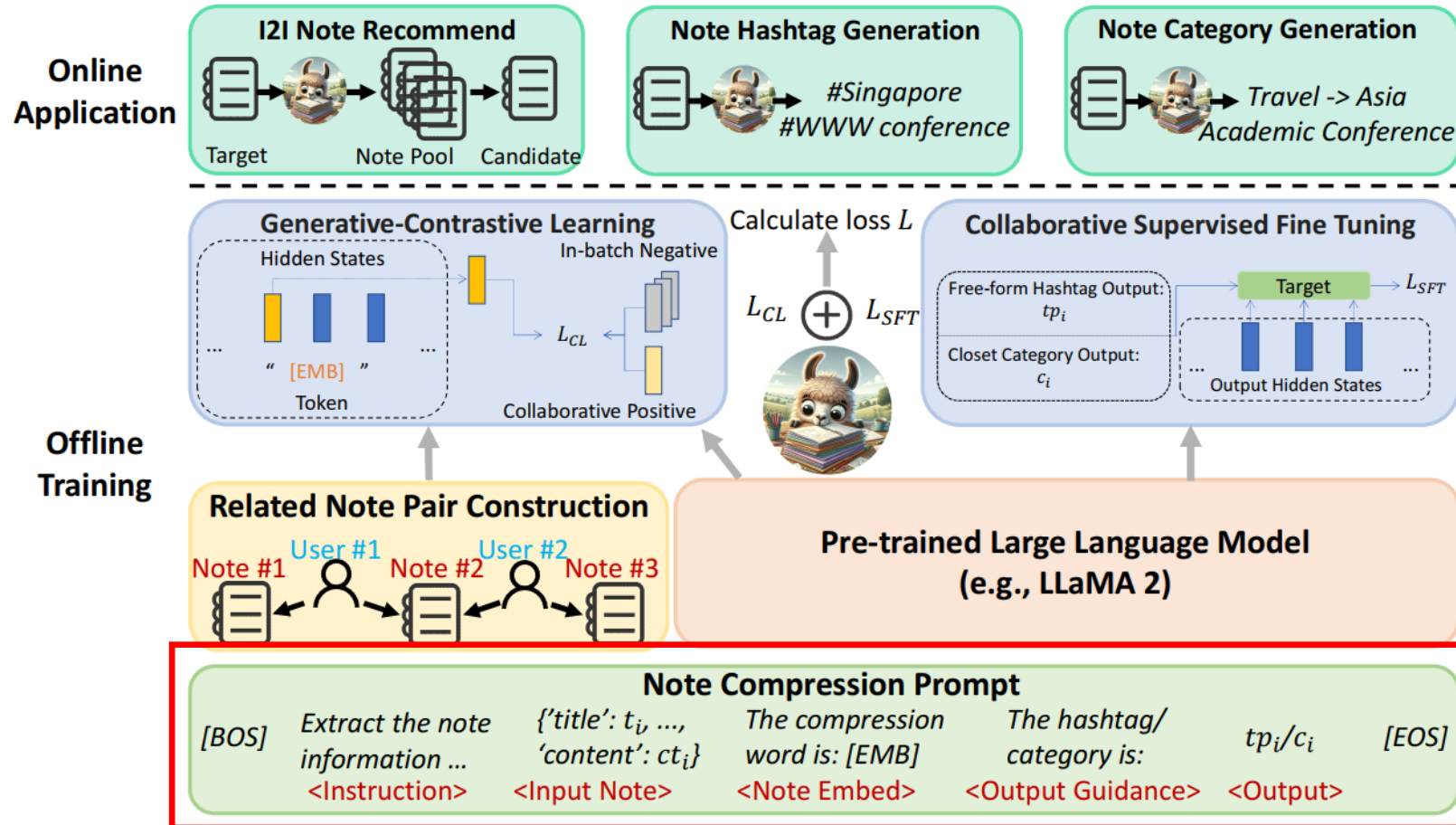
Dataset	Model	Overall		Tail Item		Head Item		Tail User		Head User	
		H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10
Yelp	GRU4Rec	0.4879	0.2751	0.0171	0.0059	0.6265	0.3544	0.4919	0.2777	0.4726	0.2653
	- CITIES	0.4898	0.2749	0.0134	0.0051	0.6301	0.3543	0.4936	0.2783	<u>0.4756</u>	0.2618
	- MELT	<u>0.4985</u>	<u>0.2825</u>	<u>0.0201</u>	<u>0.0079</u>	<u>0.6393</u>	<u>0.3633</u>	<u>0.5046</u>	<u>0.2865</u>	<u>0.4750</u>	<u>0.2671</u>
	- RLMRec	0.4886	0.2777	0.0188	0.0067	0.6269	0.3574	0.4920	0.2804	<u>0.4756</u>	<u>0.2671</u>
	- LLMInit	0.4872	0.2749	<u>0.0201</u>	0.0072	0.6246	0.3537	0.4908	0.2775	<u>0.4732</u>	0.2647
	- LLM-ESR	0.5724*	0.3413*	0.0763*	0.0318*	0.7184*	0.4324*	0.5782*	0.3456*	0.5501*	0.3247*
	Bert4Rec	0.5307	0.3035	0.0115	0.0044	0.6836	0.3916	0.5325	0.3047	0.5241	0.2988
	- CITIES	0.5249	0.3015	0.0041	0.0014	0.6783	0.3899	0.5274	0.3032	0.5155	0.2954
	- MELT	<u>0.6206</u>	0.3770	0.0429	0.0149	<u>0.7907</u>	<u>0.4836</u>	<u>0.6210</u>	0.3780	<u>0.6191</u>	<u>0.3733</u>
	- RLMRec	0.5306	0.3039	0.0104	0.0040	0.6938	0.3922	0.5351	0.3065	0.5137	0.2936
	- LLMInit	0.6199	<u>0.3781</u>	<u>0.0874</u>	<u>0.0330</u>	0.7766	0.4797	0.6204	<u>0.3796</u>	0.6178	0.3723
	- LLM-ESR	0.6623*	0.4222*	0.1227*	0.0500*	0.8212*	0.5318*	0.6637*	0.4247*	0.6571*	0.4127*
	SASRec	0.5940	0.3597	0.1142	0.0495	0.7353	0.4511	0.5893	0.3578	0.6122	0.3672
	- CITIES	0.5828	0.3540	0.1532	0.0700	0.7093	0.4376	0.5785	0.3511	0.5994	0.3649
	- MELT	0.6257	0.3791	0.1015	0.0371	<u>0.7801</u>	0.4799	0.6246	0.3804	0.6299	0.3744
	- RLMRec	0.5990	0.3623	0.0953	0.0412	0.7474	0.4568	0.5966	0.3613	0.6084	0.3658
	- LLMInit	<u>0.6415</u>	<u>0.3997</u>	<u>0.1760</u>	<u>0.0789</u>	0.7785	<u>0.4941</u>	<u>0.6403</u>	<u>0.4010</u>	<u>0.6462</u>	<u>0.3948</u>
	- LLM-ESR	0.6673*	0.4208*	0.1893*	0.0845*	0.8080*	0.5199*	0.6685*	0.4229*	0.6627*	0.4128*

- LLM-ESR leads the overall performance, which indicates better enhancing effects.

Dataset	Model	Overall		Tail Item		Head Item		Tail User		Head User	
		H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10
Yelp	GRU4Rec	0.4879	0.2751	0.0171	0.0059	0.6265	0.3544	0.4919	0.2777	0.4726	0.2653
	- CITIES	0.4898	0.2749	0.0134	0.0051	0.6301	0.3543	0.4936	0.2783	0.4756	0.2618
	- MELT	0.4985	0.2825	0.0201	0.0079	0.6393	0.3633	0.5046	0.2865	0.4750	0.2671
	- RLMRec	0.4886	0.2777	0.0188	0.0067	0.6269	0.3574	0.4920	0.2804	0.4756	0.2671
	- LLMInit	0.4872	0.2749	0.0201	0.0072	0.6246	0.3537	0.4908	0.2775	0.4732	0.2647
	- LLM-ESR	0.5724*	0.3413*	0.0763*	0.0318*	0.7184*	0.4324*	0.5782*	0.3456*	0.5501*	0.3247*
	Bert4Rec	0.5307	0.3035	0.0115	0.0044	0.6836	0.3916	0.5325	0.3047	0.5241	0.2988
	- CITIES	0.5249	0.3015	0.0041	0.0014	0.6783	0.3899	0.5274	0.3032	0.5155	0.2954
	- MELT	0.6206	0.3770	0.0429	0.0149	0.7907	0.4836	0.6210	0.3780	0.6191	0.3733
	- RLMRec	0.5306	0.3039	0.0104	0.0040	0.6938	0.3922	0.5351	0.3065	0.5137	0.2936
	- LLMInit	0.6199	0.3781	0.0874	0.0330	0.7766	0.4797	0.6204	0.3796	0.6178	0.3723
	- LLM-ESR	0.6623*	0.4222*	0.1227*	0.0500*	0.8212*	0.5318*	0.6637*	0.4247*	0.6571*	0.4127*
	SASRec	0.5940	0.3597	0.1142	0.0495	0.7353	0.4511	0.5893	0.3578	0.6122	0.3672
	- CITIES	0.5828	0.3540	0.1532	0.0700	0.7093	0.4376	0.5785	0.3511	0.5994	0.3649
	- MELT	0.6257	0.3791	0.1015	0.0371	0.7801	0.4799	0.6246	0.3804	0.6299	0.3744
	- RLMRec	0.5990	0.3623	0.0953	0.0412	0.7474	0.4568	0.5966	0.3613	0.6084	0.3658
	- LLMInit	0.6415	0.3997	0.1760	0.0789	0.7785	0.4941	0.6403	0.4010	0.6462	0.3948
	- LLM-ESR	0.6673*	0.4208*	0.1893*	0.0845*	0.8080*	0.5199*	0.6685*	0.4229*	0.6627*	0.4128*

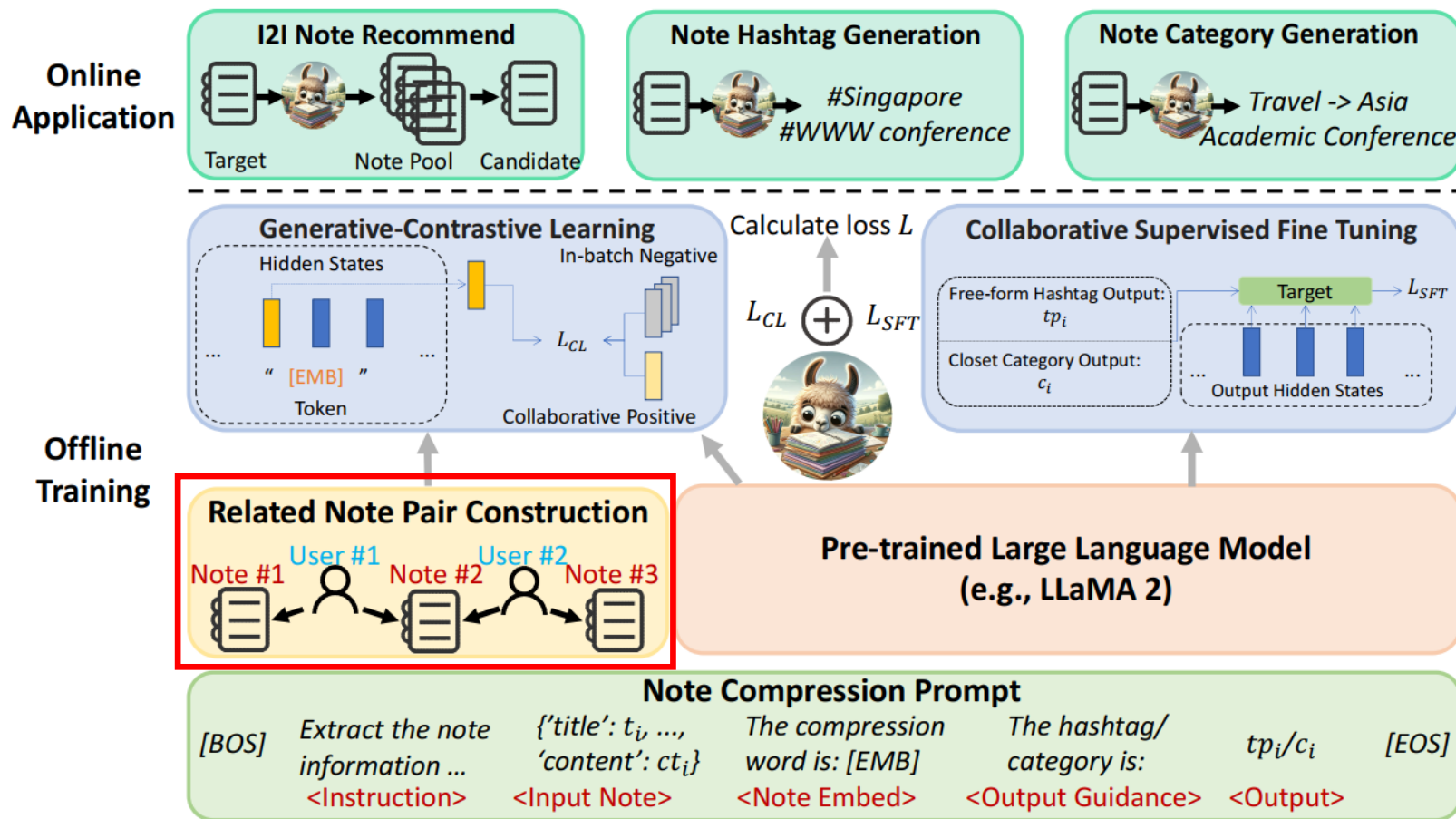
- LLM-ESR achieves the best on tail and popular item group.
- LLM-ESR can augment the tail user group better.





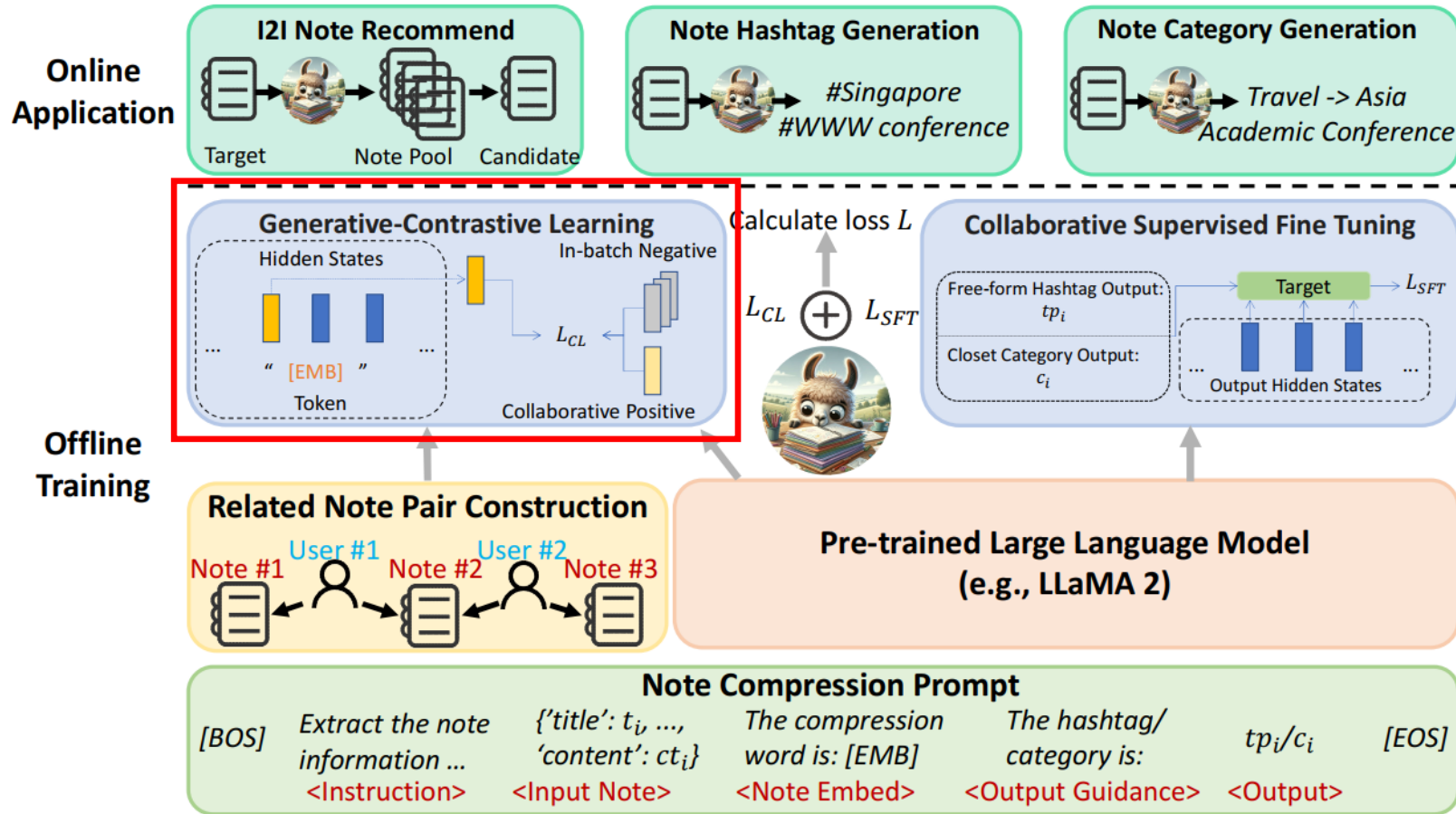
• Note Compression Prompt

- Compressing the note content
- Generating hashtags/categories

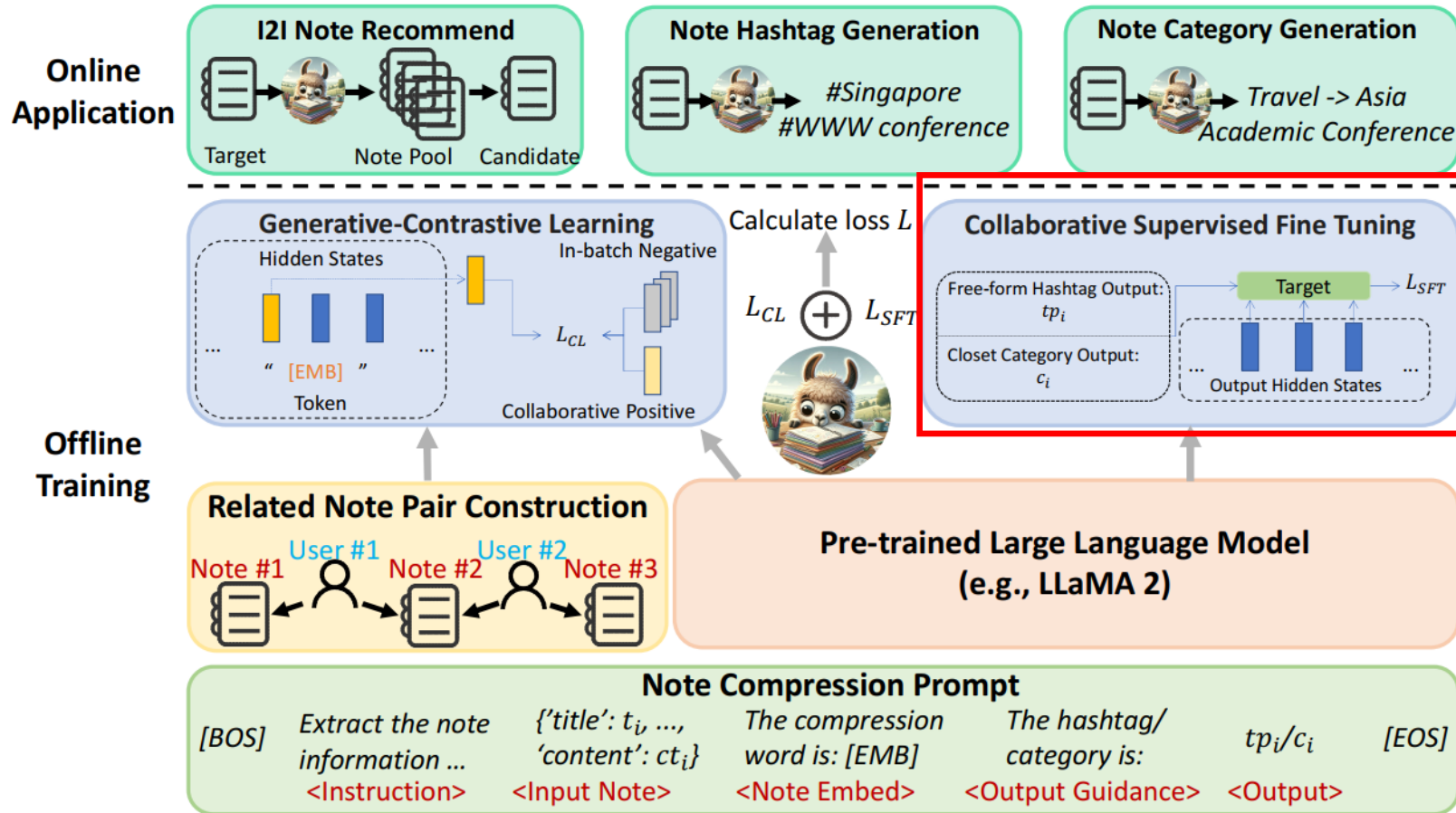


• Constructing related note pairs

- Counting co-occurrence scores
- Selecting notes with highest scores as related notes



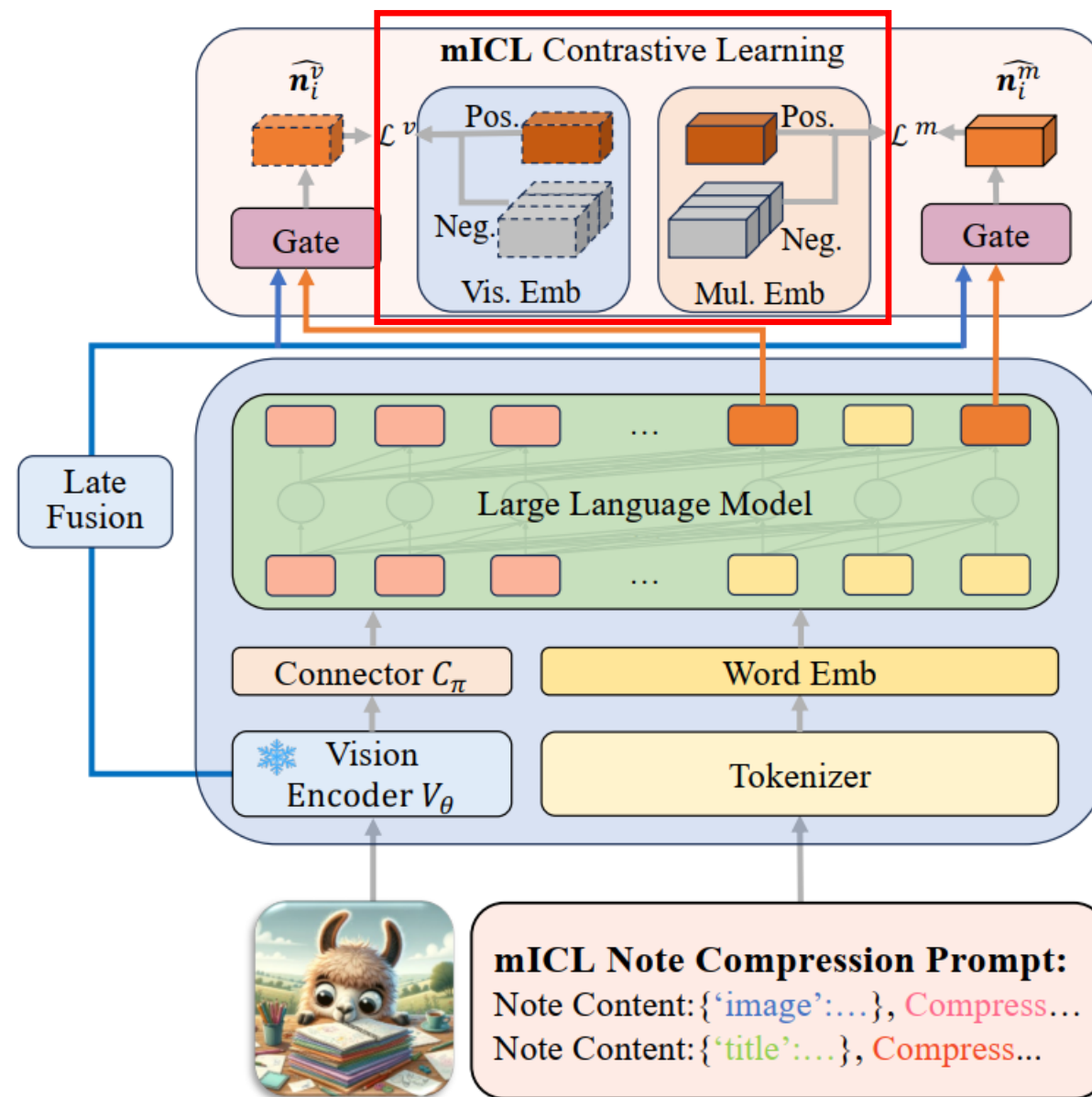
- **Generative-Contrastive Learning**
 - Identifying related notes from in-batch negatives



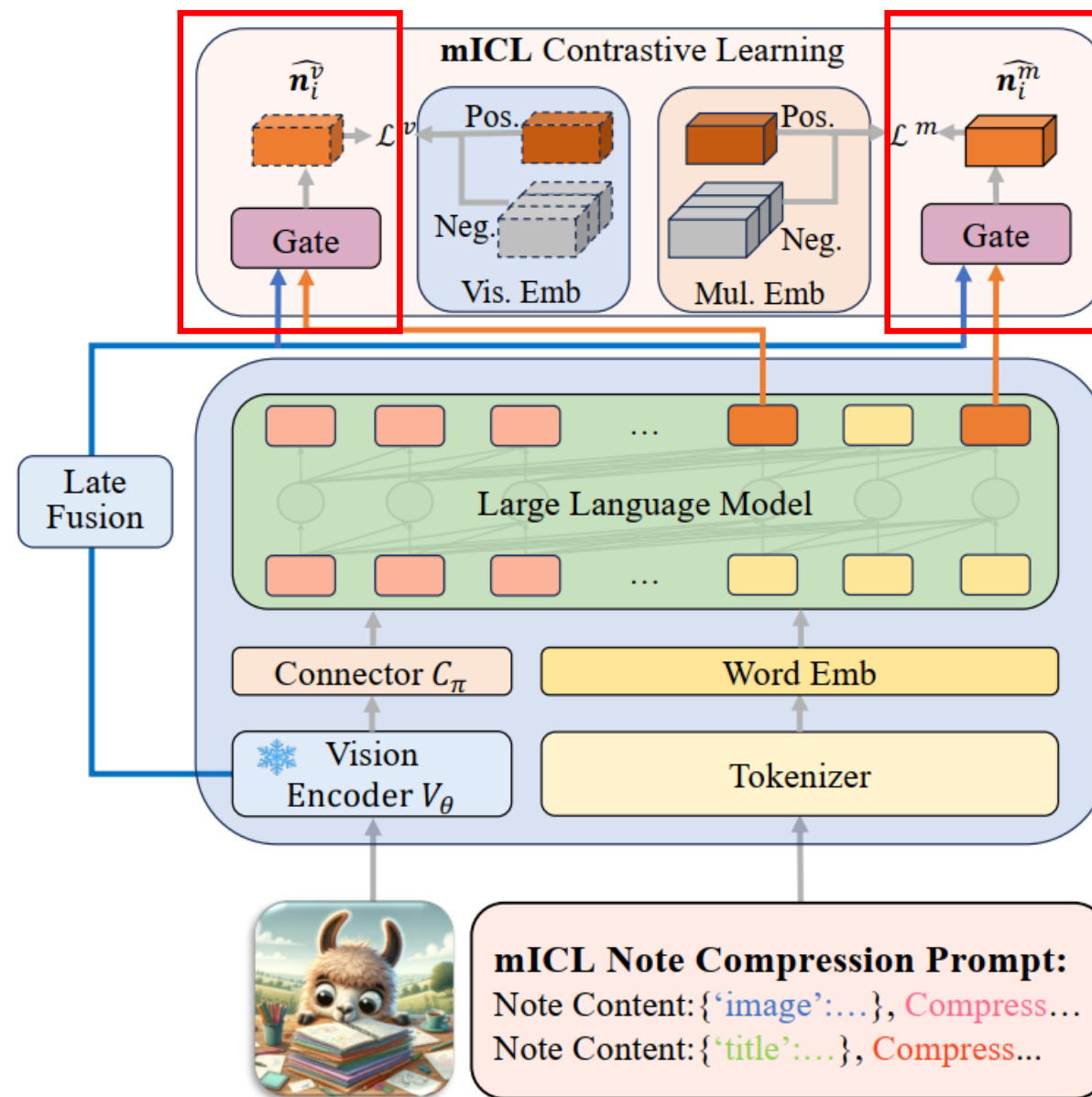
- **Collaborative Supervised Fine-tuning**
 - Generating hashtags/categories for each note

Multimodal In-Context Learning

- Separating multimodal content into visual & textual components
- Compressing into modality-compressed words

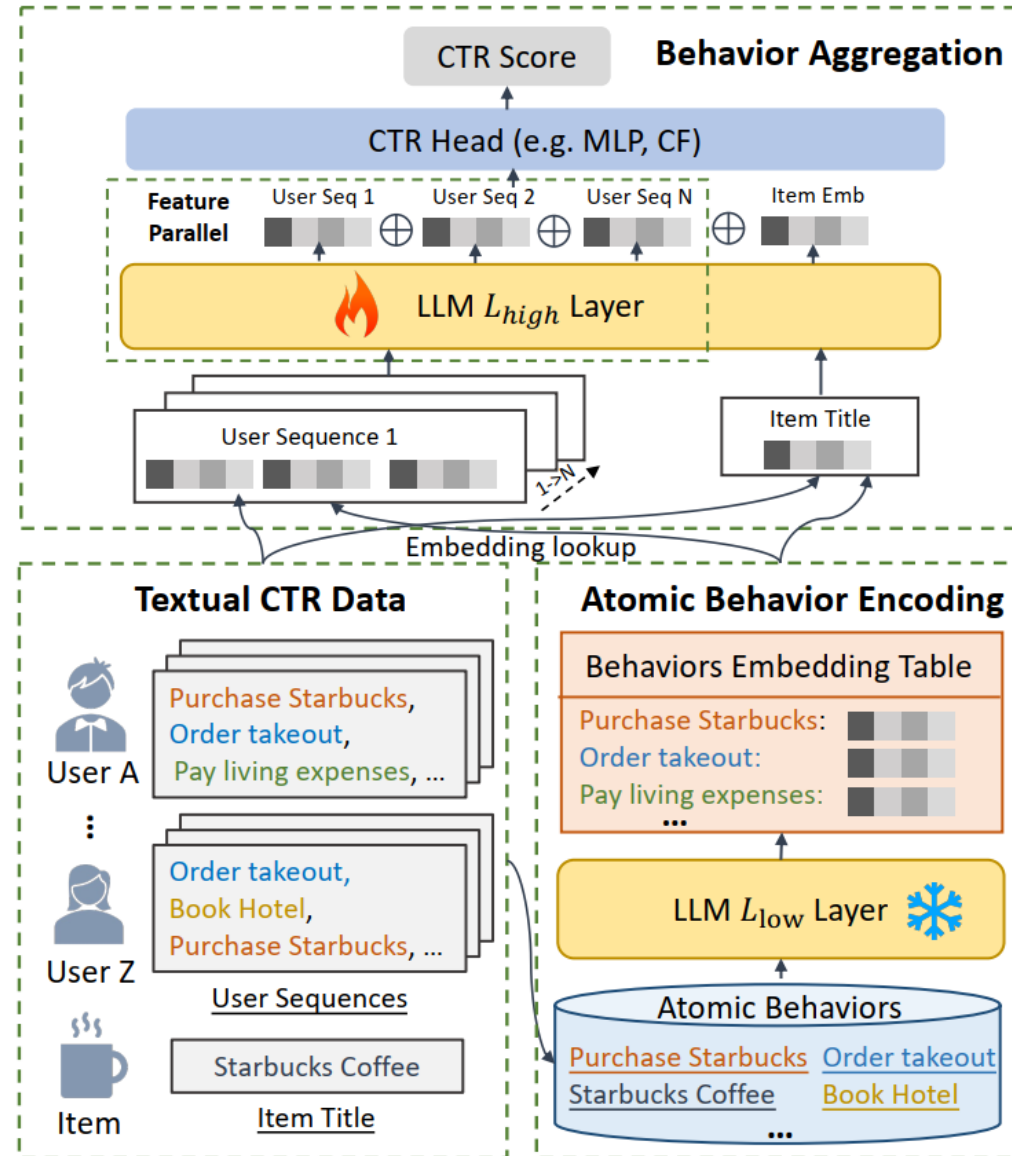


- **Late Fusion**
 - Multimodal Gating mechanism



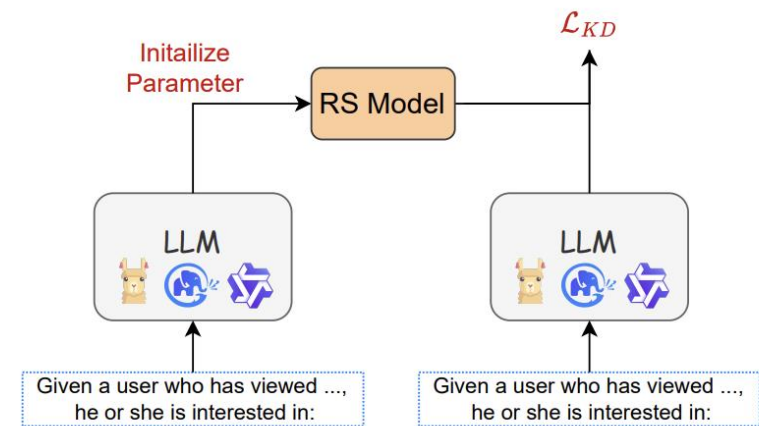
• Hierarchical Encoding

- Lower layers encoding item into atomic representation
- Higher layers generating high-level user representations



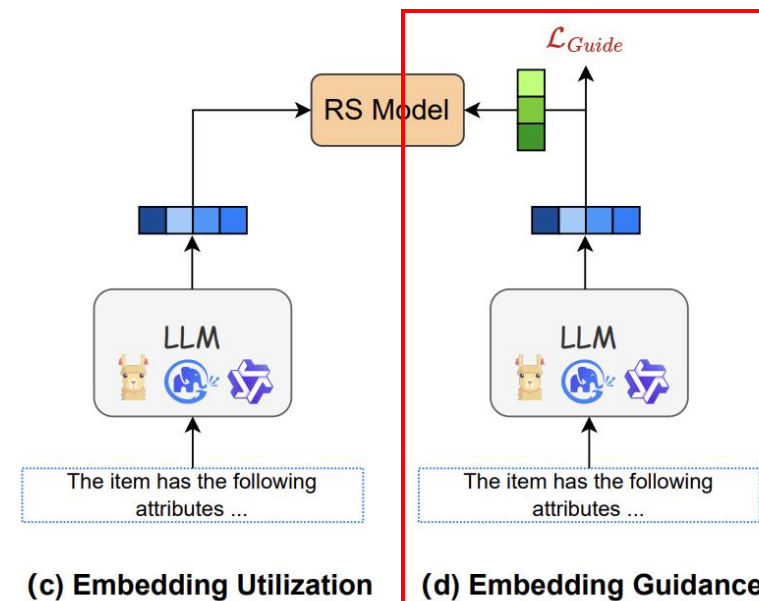
Embedding Guidance

- This subcategory refers to only using the LLM embeddings as the guidance for training or parameter synthesis.
- Categories
 - User Only
 - User & Item



(a) Model Initialization

(b) Model Distillation



(c) Embedding Utilization

(d) Embedding Guidance

• Multi-Scenario Modeling

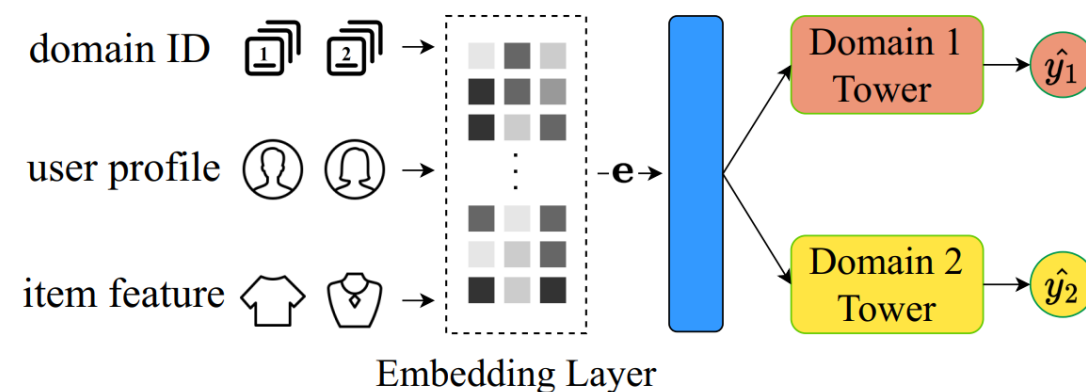
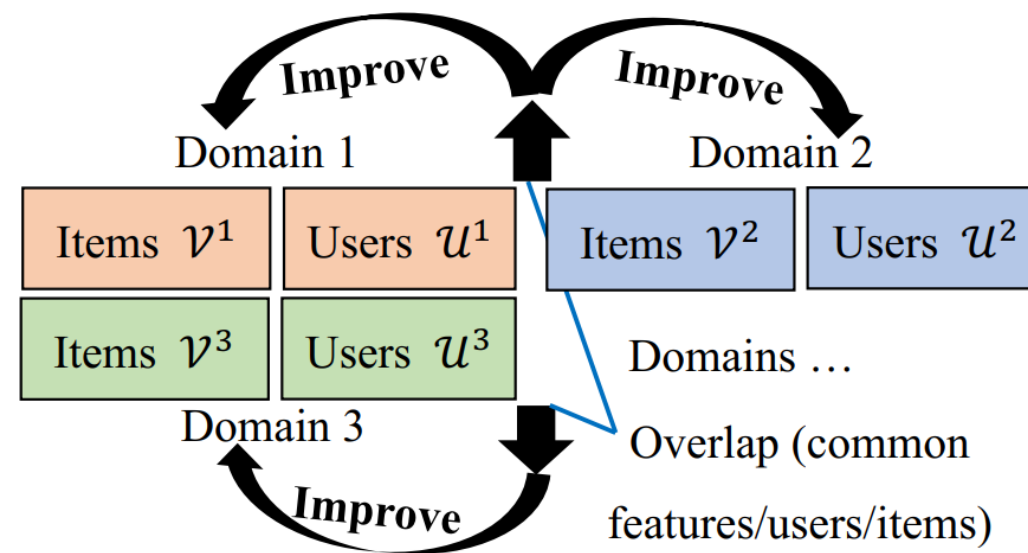
- Tackling data sparsity
- Reducing computation cost
- High efficiency

• Challenges

- Insufficient scenario knowledge
- Ignoring cross-scenario preferences

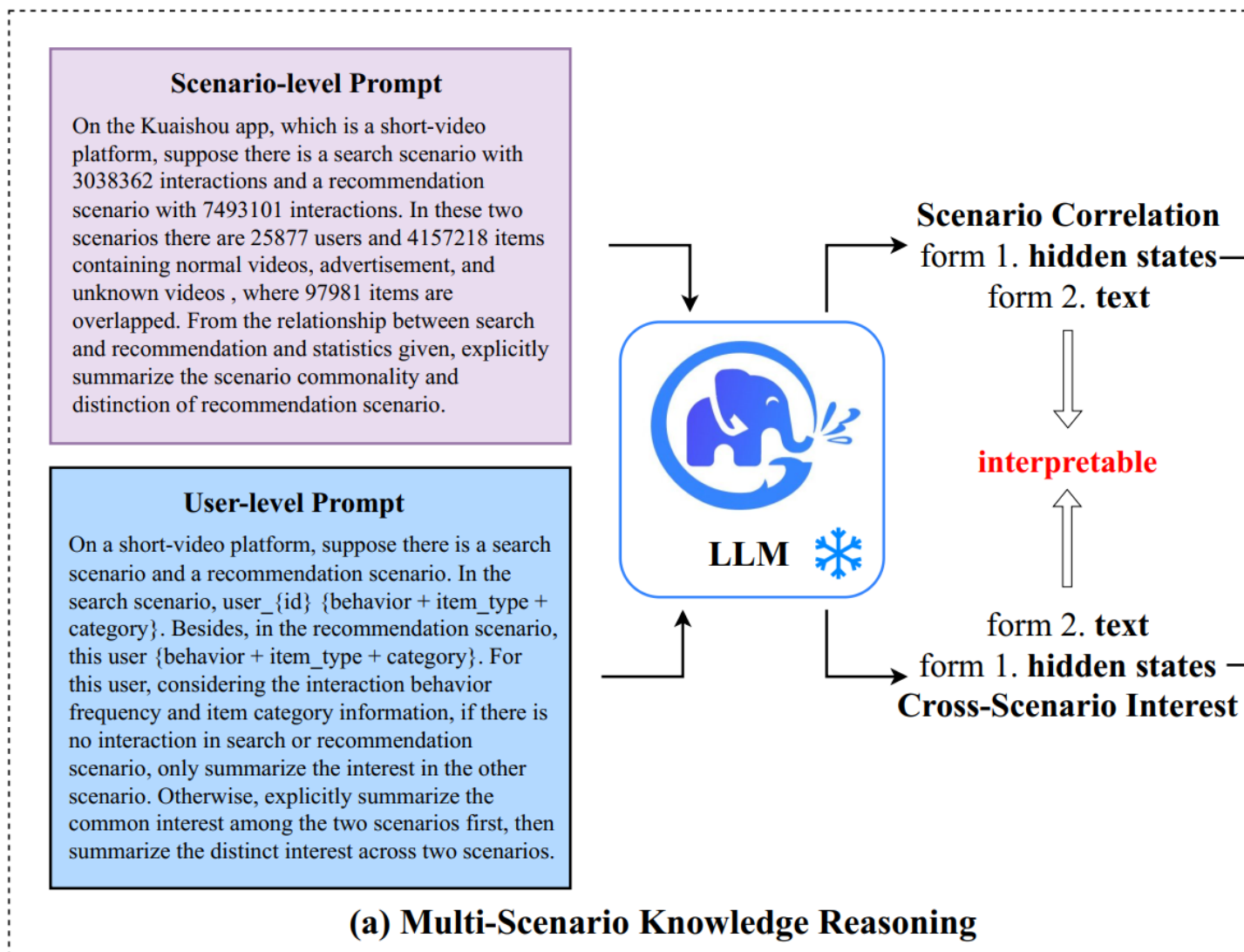
• Motivation

- Incorporating LLM to improve conventional recommender system
- LLM as reasoner + encoder



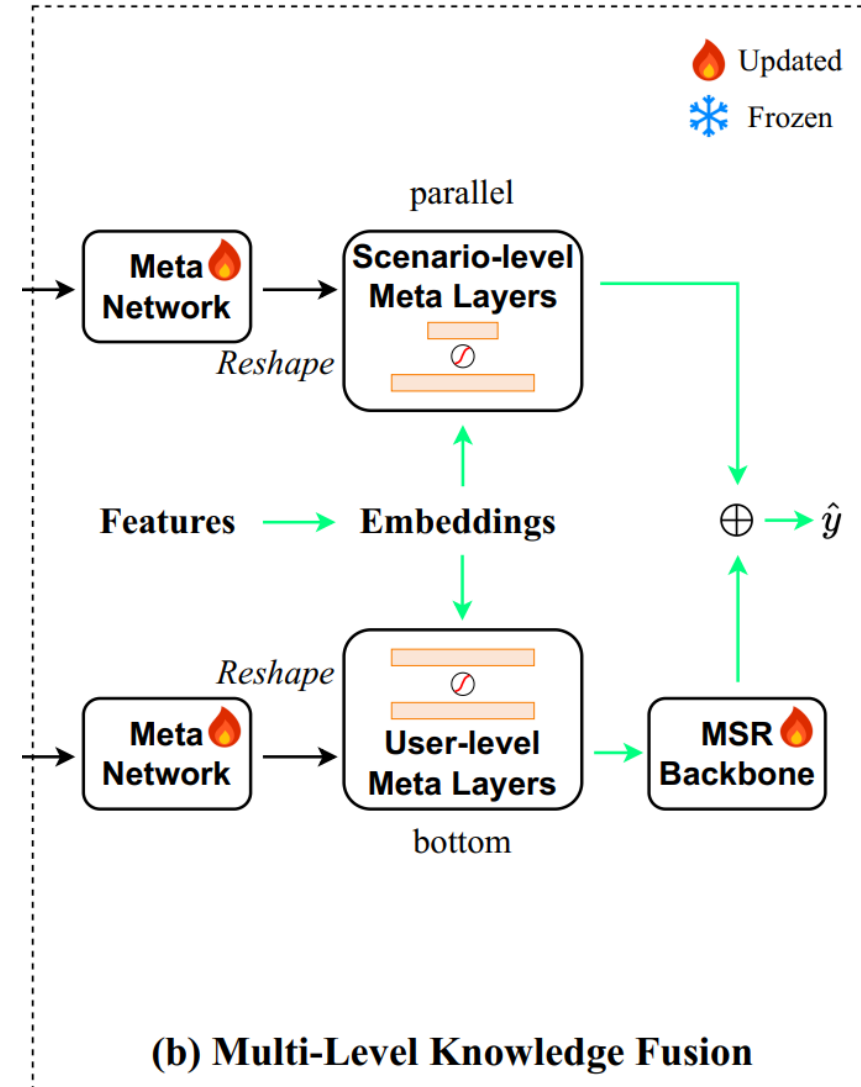
- **Multi-scenario reasoning**

- Scenario-level prompt
- User-level prompt



• Multi-level Fusion

- Meta-networks generating meta layers
- Hierarchical bottom + parallel structure



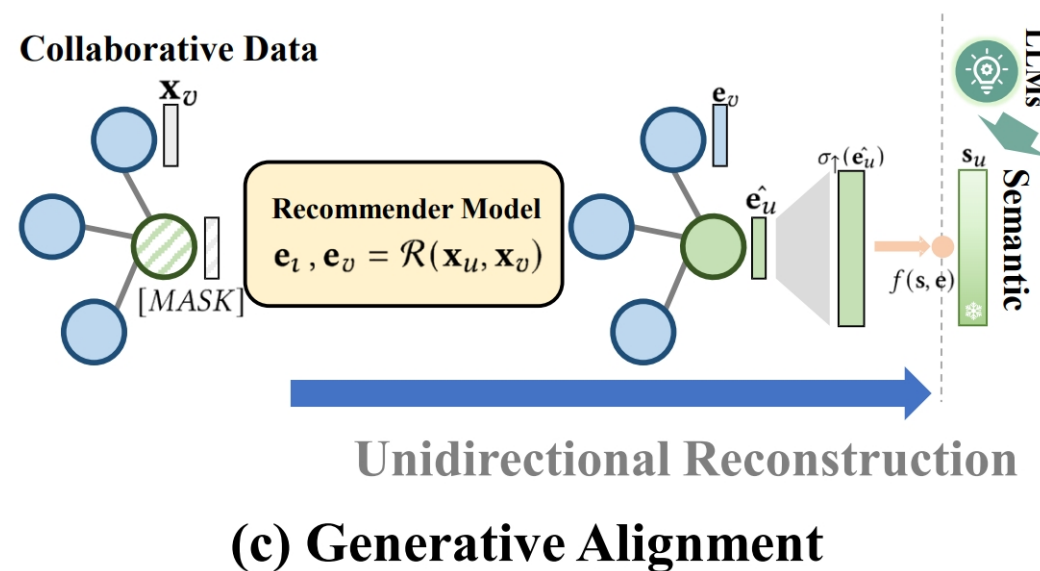
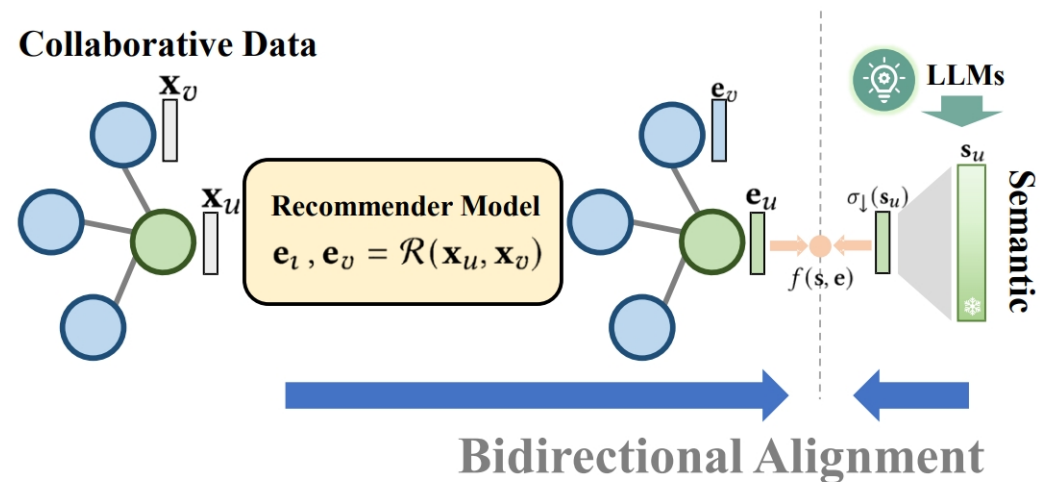
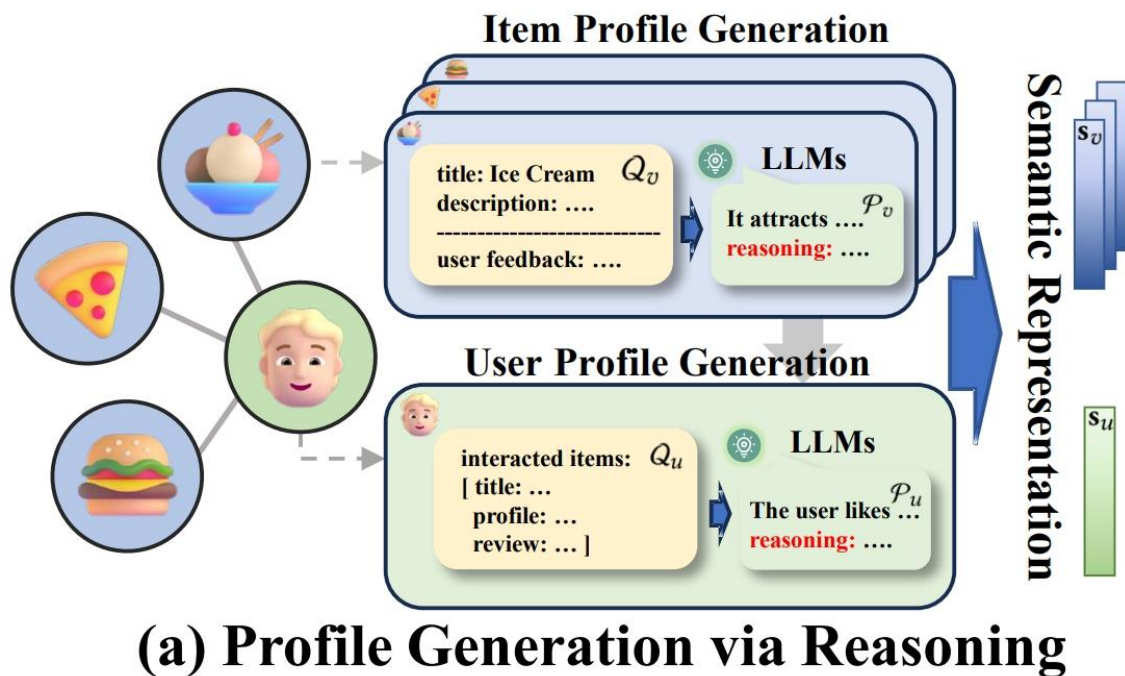
- LLM4MSR achieves an increase of 1.5%, 1%, and 40% in AUC on three datasets
 - Domain correlation & Personalized interest + Adaptive meta network

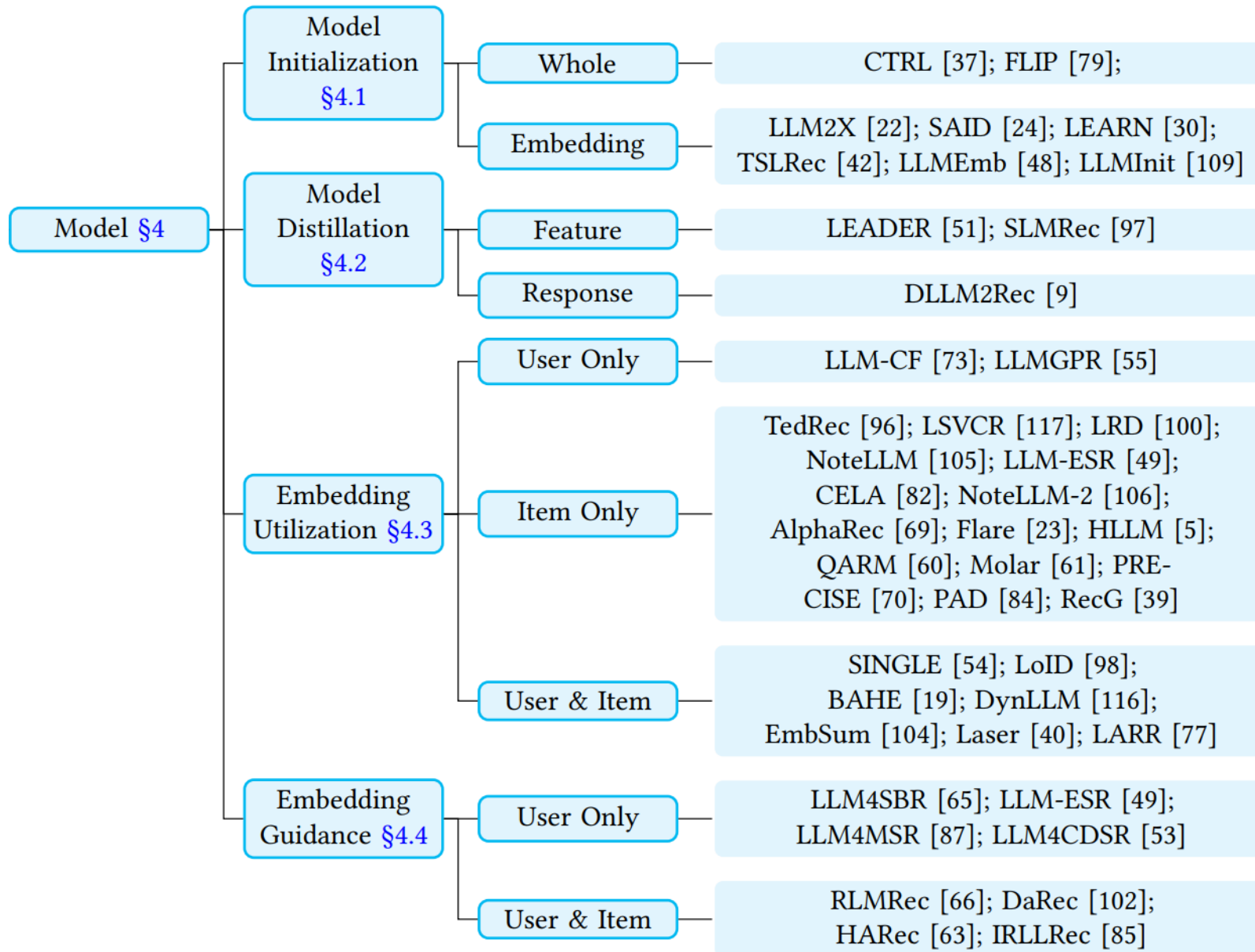
Backbones	KuaiSAR-small		KuaiSAR		Amazon		
	Rec	Search	Rec	Search	Rec#1	Rec#2	Rec#3
STAR	0.7225	0.6089	0.7387	0.6268	0.6156	0.6320	0.6225
STAR_DN	<u>0.7241</u>	<u>0.6116</u>	<u>0.7404</u>	<u>0.6270</u>	<u>0.6306</u>	<u>0.6350</u>	0.6355
STAR_EP	0.7230	0.6082	0.7388	0.6266	0.6211	0.6302	<u>0.6378</u>
STAR_ours	0.7276★	0.6181★	0.7408★	0.6332★	0.8720★	0.6364★	0.7543★
OMoE	0.7241	0.6163	0.7394	0.6310	0.5995	0.6043	0.6252
OMoE_DN	<u>0.7249</u>	<u>0.6183</u>	<u>0.7402</u>	0.6319	0.6027	0.6171	0.6289
OMoE_EP	0.7246	0.6179	0.7392	<u>0.6330</u>	<u>0.6143</u>	<u>0.6176</u>	<u>0.6312</u>
OMoE_ours	0.7265★	0.6186★	0.7413★	0.6332★	0.8182★	0.6180★	0.6823★
MMoE	0.7235	0.6150	0.7392	0.6313	0.5907	0.5999	0.6032
MMoE_DN	<u>0.7246</u>	0.6164	0.7394	<u>0.6330</u>	<u>0.6310</u>	0.6201	<u>0.6371</u>
MMoE_EP	0.7245	0.6178	<u>0.7397</u>	0.6326	0.6218	<u>0.6221</u>	0.6315
MMoE_ours	0.7264★	<u>0.6166★</u>	0.7410★	0.6341★	0.7860★	0.6222★	0.6854★
PLE	0.7249	0.6149	0.7396	0.6313	0.6059	0.6127	0.5995
PLE_DN	<u>0.7258</u>	<u>0.6165</u>	0.7400	0.6333	0.6066	0.6195	0.6162
PLE_EP	0.7252	0.6184	0.7402	0.6339	0.6173	0.6229	0.6263
PLE_ours	0.7269★	0.6184★	0.7408★	0.6343★	0.8265★	0.6250★	0.7074★
AITM	0.7236	0.6154	0.7398	0.6329	0.6039	0.6126	0.6046
AITM_DN	<u>0.7243</u>	0.6172	<u>0.7398</u>	0.6318	0.6057	0.6151	0.6166
AITM_EP	0.7237	<u>0.6177</u>	0.7389	<u>0.6337</u>	<u>0.6064</u>	<u>0.6163</u>	<u>0.6238</u>
AITM_ours	0.7273★	0.6178★	0.7407★	0.6349★	0.8196★	0.6178★	0.7089★
Shared Bottom	0.7228	0.6169	0.7389	0.6321	0.6073	0.6077	0.6268
Shared Bottom_DN	<u>0.7250</u>	<u>0.6176</u>	<u>0.7396</u>	0.6323	0.6081	<u>0.6253</u>	<u>0.6294</u>
Shared Bottom_EP	0.7243	0.6182	0.7392	<u>0.6331</u>	<u>0.6268</u>	0.6174	0.6233
Shared Bottom_ours	0.7269★	0.6182★	0.7400★	0.6341★	0.8323★	0.6262★	0.7182★

- LLM4MSR enhances various MSR backbone models, showing great **compatibility** and **deployability**

Backbones	KuaiSAR-small		KuaiSAR		Amazon		
	Rec	Search	Rec	Search	Rec#1	Rec#2	Rec#3
STAR	0.7225	0.6089	0.7387	0.6268	0.6156	0.6320	0.6225
STAR_DN	<u>0.7241</u>	<u>0.6116</u>	<u>0.7404</u>	<u>0.6270</u>	<u>0.6306</u>	<u>0.6350</u>	0.6355
STAR_EP	0.7230	0.6082	0.7388	0.6266	0.6211	0.6302	0.6378
STAR_ours	0.7276★	0.6181★	0.7408★	0.6332★	0.8720★	0.6364★	0.7543★
OMoE	0.7241	0.6163	0.7394	0.6310	0.5995	0.6043	0.6252
OMoE_DN	<u>0.7249</u>	<u>0.6183</u>	<u>0.7402</u>	0.6319	0.6027	0.6171	0.6289
OMoE_EP	0.7246	0.6179	0.7392	0.6330	0.6143	0.6176	0.6312
OMoE_ours	0.7265★	0.6186★	0.7413★	0.6332★	0.8182★	0.6180★	0.6823★
MMoE	0.7235	0.6150	0.7392	0.6313	0.5907	0.5999	0.6032
MMoE_DN	<u>0.7246</u>	0.6164	0.7394	<u>0.6330</u>	<u>0.6310</u>	0.6201	<u>0.6371</u>
MMoE_EP	0.7245	0.6178	0.7397	0.6326	0.6218	0.6221	0.6315
MMoE_ours	0.7264★	0.6166★	0.7410★	0.6341★	0.7860★	0.6222★	0.6854★
PLE	0.7249	0.6149	0.7396	0.6313	0.6059	0.6127	0.5995
PLE_DN	<u>0.7258</u>	<u>0.6165</u>	0.7400	0.6333	0.6066	0.6195	0.6162
PLE_EP	0.7252	0.6184	0.7402	0.6339	0.6173	0.6229	0.6263
PLE_ours	0.7269★	0.6184★	0.7408★	0.6343★	0.8265★	0.6250★	0.7074★
AITM	0.7236	0.6154	0.7398	0.6329	0.6039	0.6126	0.6046
AITM_DN	<u>0.7243</u>	0.6172	<u>0.7398</u>	0.6318	0.6057	0.6151	0.6166
AITM_EP	0.7237	0.6177	0.7389	0.6337	0.6064	0.6163	0.6238
AITM_ours	0.7273★	0.6178★	0.7407★	0.6349★	0.8196★	0.6178★	0.7089★
Shared Bottom	0.7228	0.6169	0.7389	0.6321	0.6073	0.6077	0.6268
Shared Bottom_DN	<u>0.7250</u>	<u>0.6176</u>	<u>0.7396</u>	0.6323	0.6081	<u>0.6253</u>	<u>0.6294</u>
Shared Bottom_EP	0.7243	0.6182	0.7392	0.6331	0.6268	0.6174	0.6233
Shared Bottom_ours	0.7269★	0.6182★	0.7400★	0.6341★	0.8323★	0.6262★	0.7182★

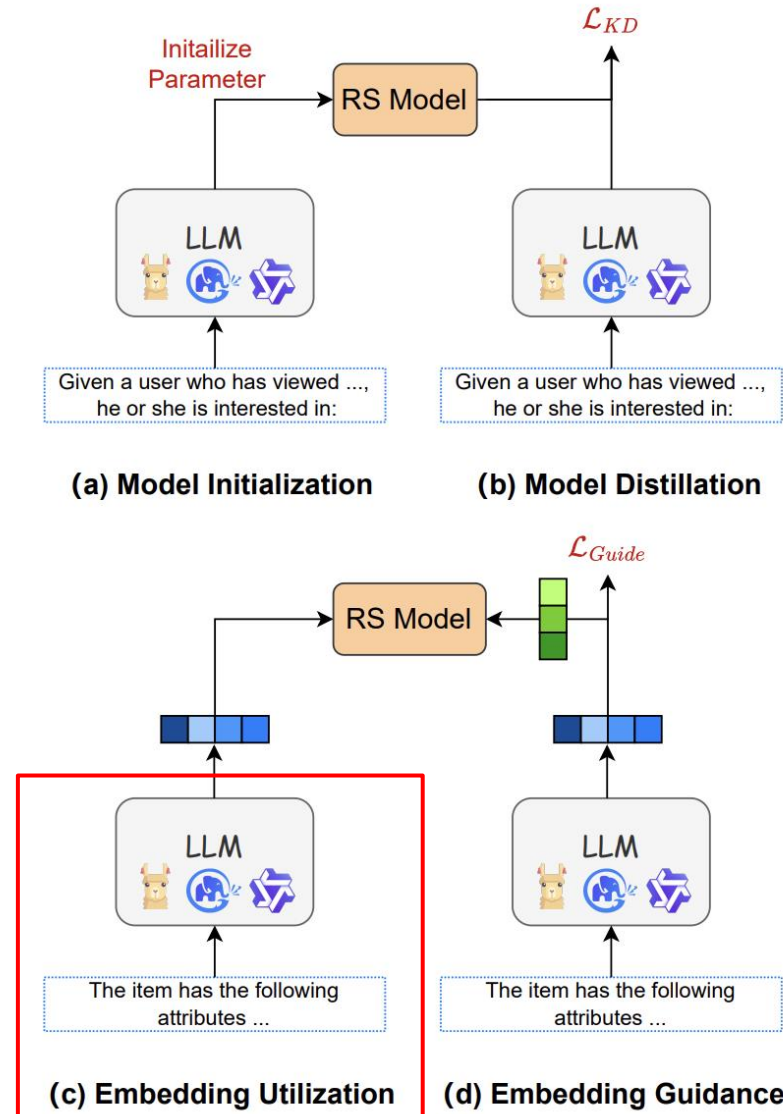
- **Generating profile**
- **Extra alignment loss**



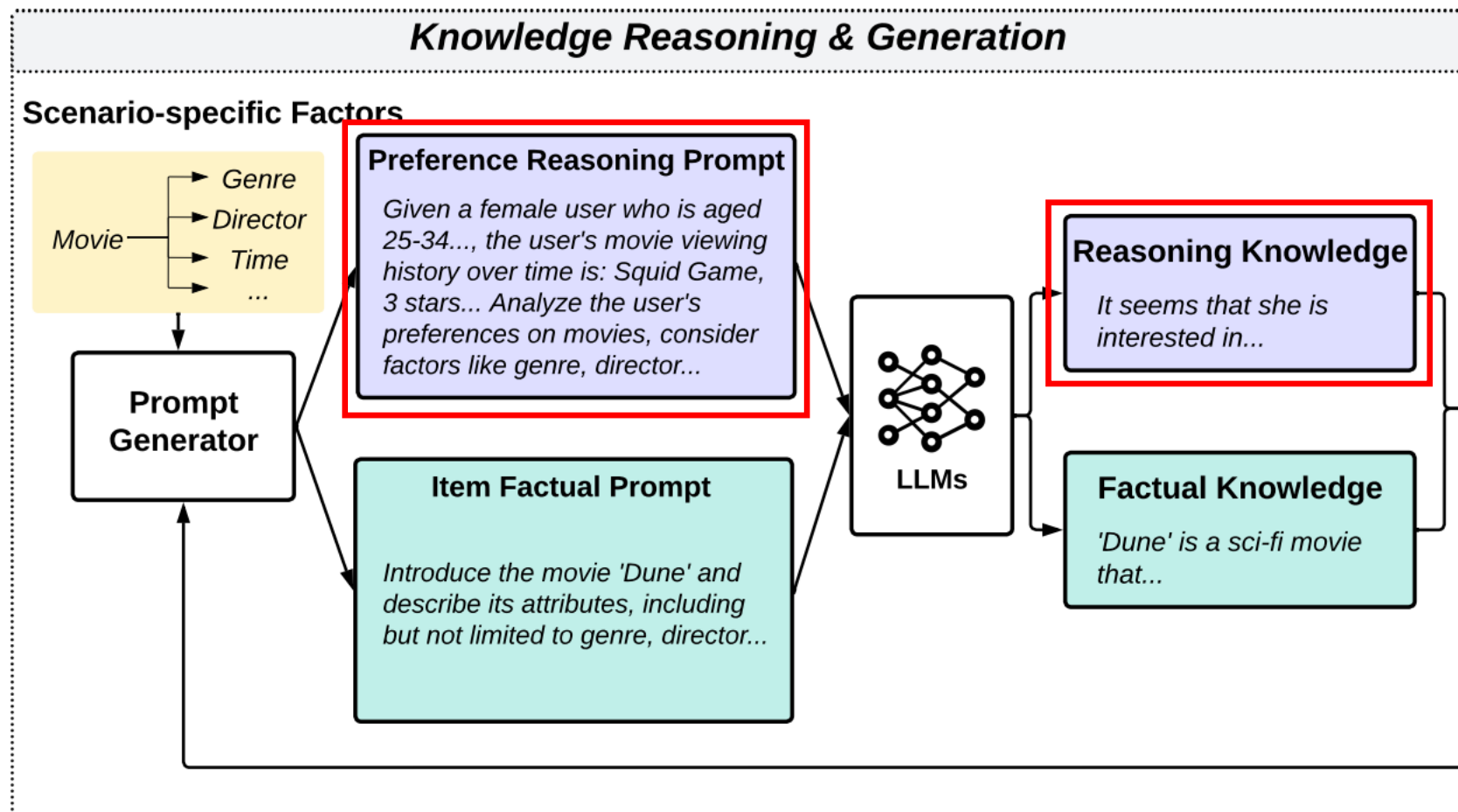


Embedding Utilization

- This subcategory refers to utilizing the embeddings derived from LLM to enhance traditional RS as a semantic supplement.
- This approach tackles the deficiency that textual outputs are often difficult and inefficient to be integrated into RS directly.
- Categories
 - User Only
 - Item Only
 - User & Item

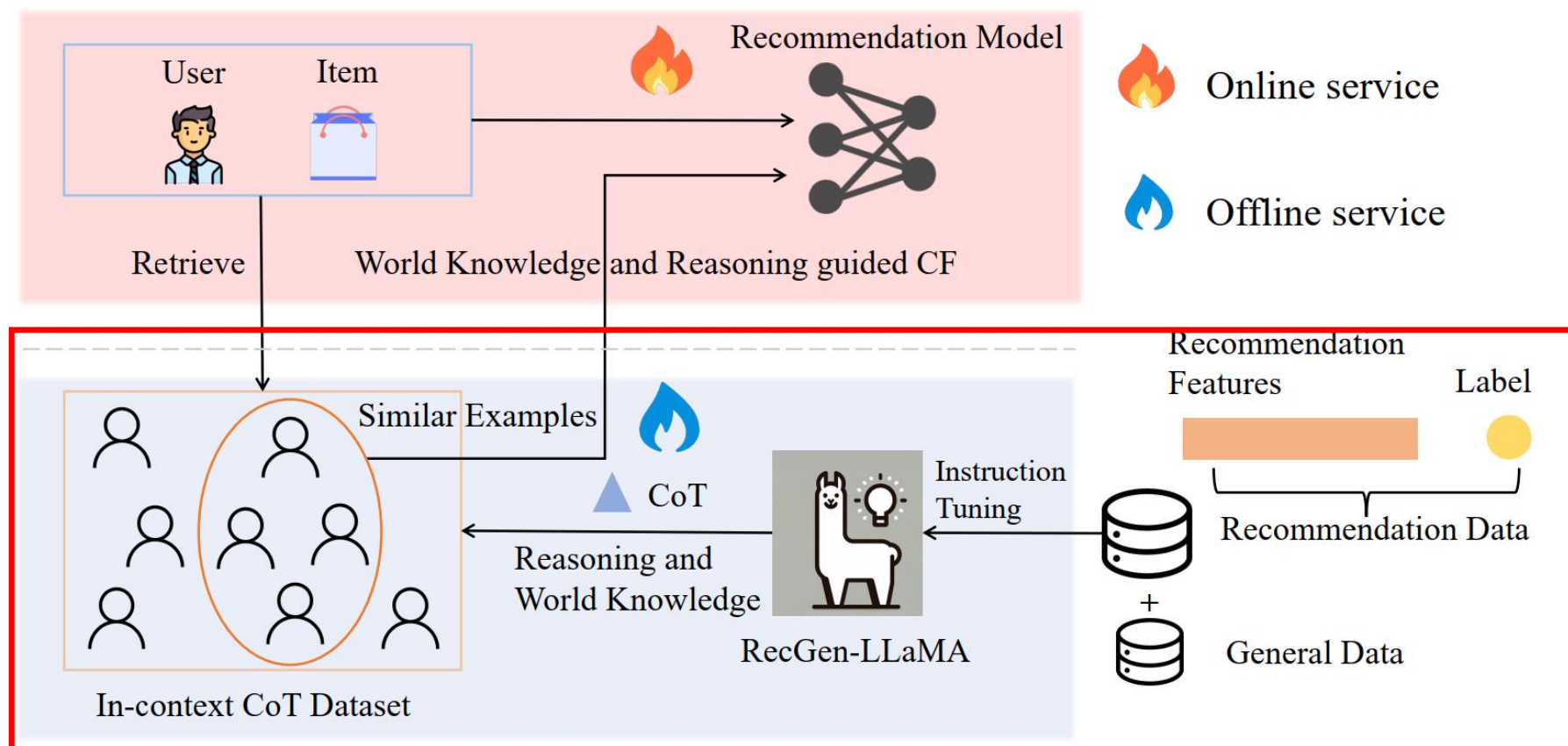


- Reasoning about user preferences given **user profile** and **interactions**



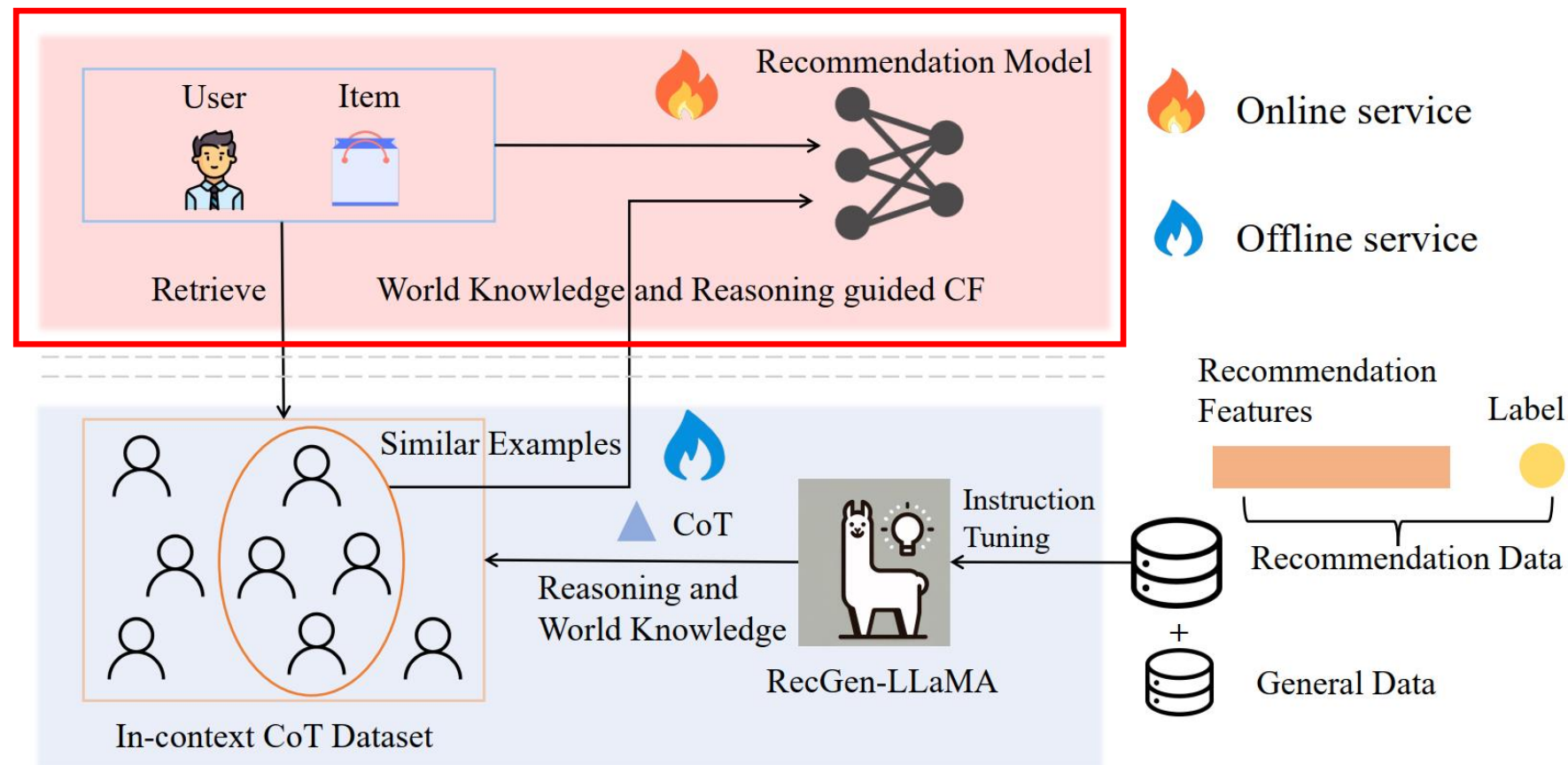
Offline service

- Chain of Thought (CoT) reasoning
- In-context CoT dataset construction



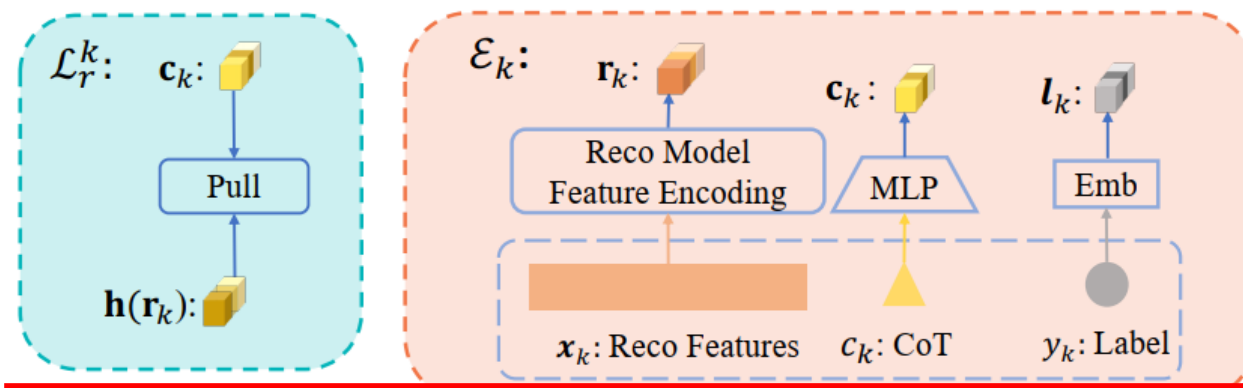
• Online service

- In-context CoT examples Retrieval
- World knowledge & reasoning guided CF
- Feature enhanced RS



• CoT Prompt

- Generating CoT of the user's decision making on the target item



Chain of Thought Prompt

<<SYS>> As an AI model developed for analyzing consumer behavior, your task is to generate a chain of thought that considers the following points:

1. Utilize the user's interaction history and review comments to summarize their profiles.
2. Introduce the target new item and detail its features precisely. In addition, integrate information about items related to the current target new item ...
3. Contemplate the alignment between the user's profile and the features of the target item.
4. Reflect on the user's potential desire for diversity in their purchases.

Your output should be a clear and logical chain of thought... Ensure your analysis is impartial... Focus should be on understanding factors that influence the user's decision-making regarding the target item. </SYS>

Please generate a chain of thought based on the user's ... considering how these might relate to their interest in the target new item.

{Recommendation Features}

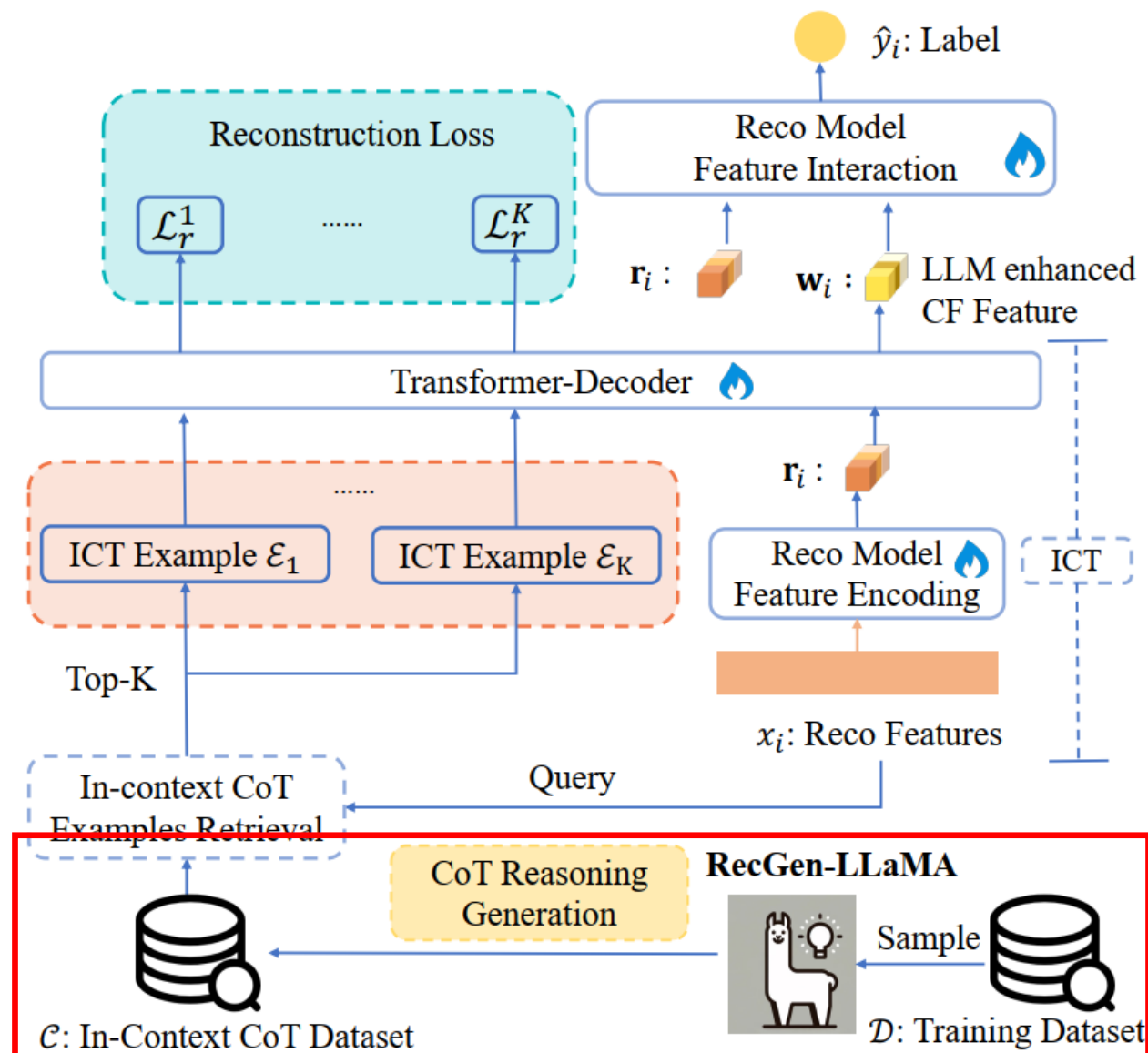
User's decision-making: The user {Ground-Truth Label} the target new item.

Let's think step by step and develop the chain of thought for the above considerations.

Commence with the chain of thought immediately:

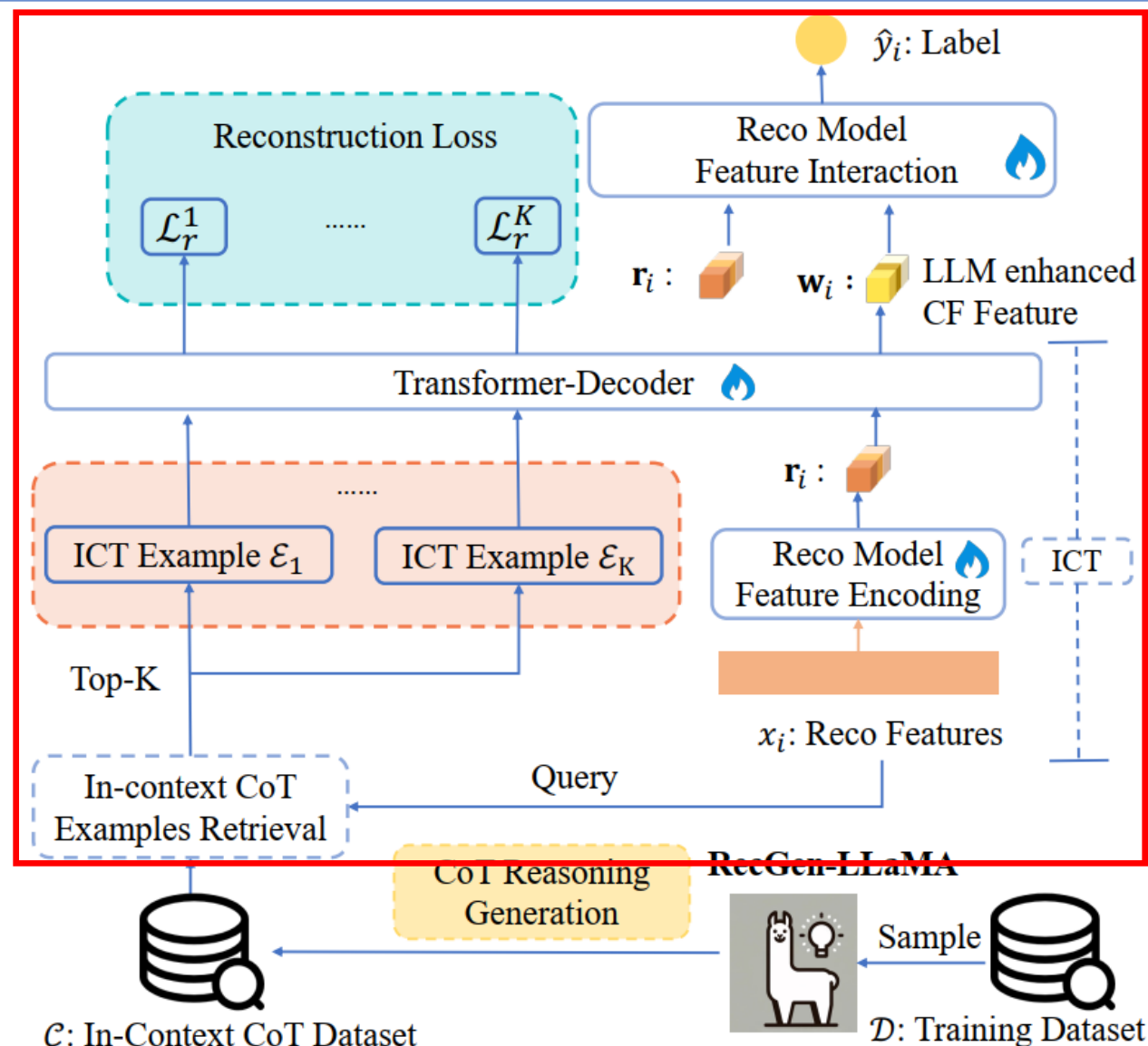
• Offline Service

- Instruction tuning on LLaMA2
- RecGen-LLaMA generating CoT forming ICT dataset



• Online Service

- Embedding-based retrieval forming ICT examples
- Learning world-knowledge and reasoning guided CF feature



Overall performance

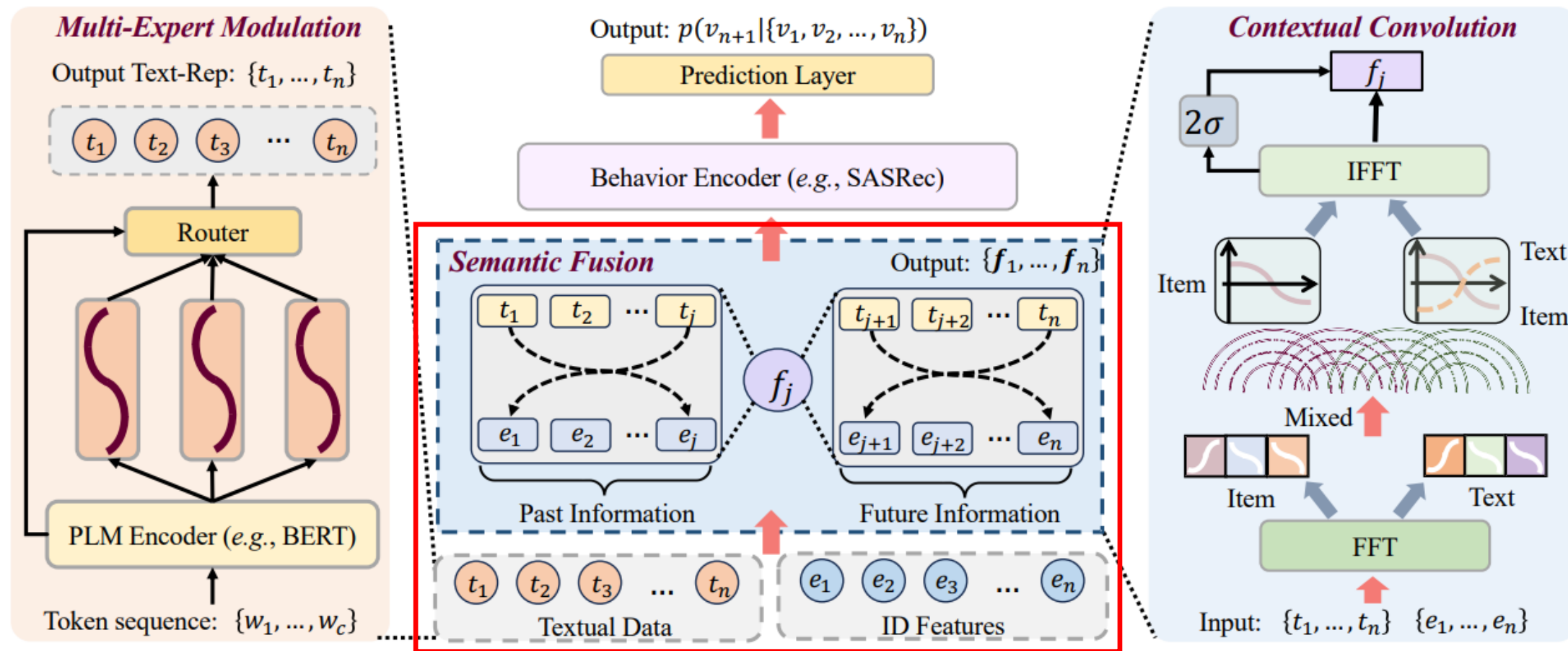


August 3-7, 2025



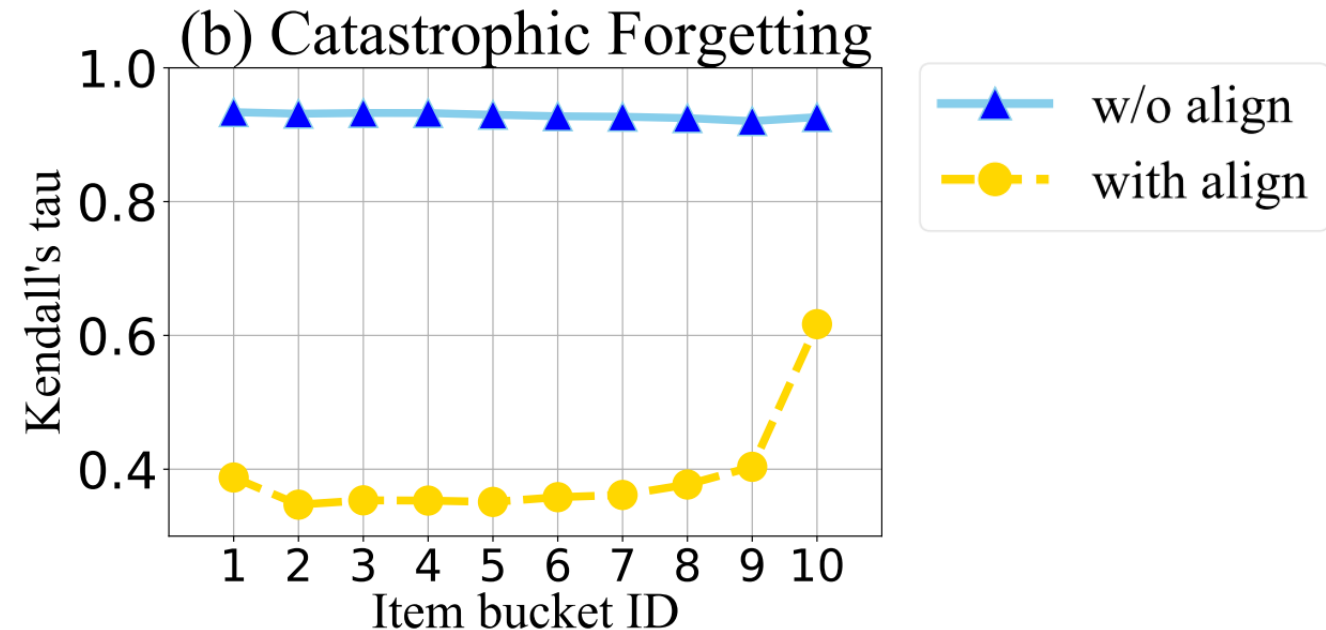
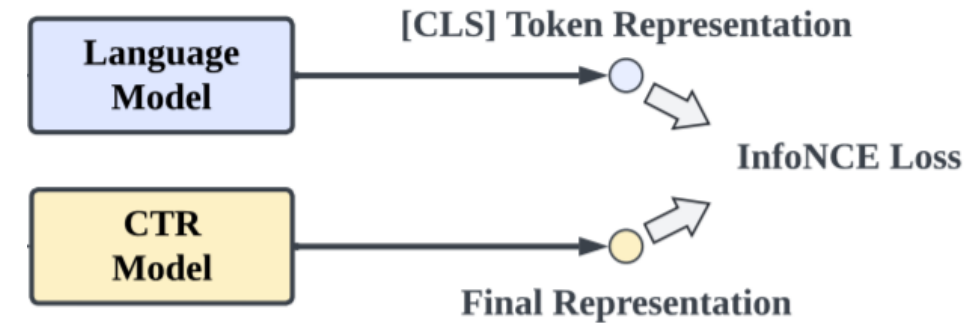
Backbone	Framework	Sports			Beauty			Toys		
		AUC↑	Logloss↓	RelaImpr↑	AUC↑	Logloss↓	RelaImpr↑	AUC↑	Logloss↓	RelaImpr↑
DeepFM	None	0.7990	0.5471	0.000%	0.7853	0.5545	0.000%	0.7681	0.5770	0.000%
	KD	<u>0.8043</u>	<u>0.5404</u>	1.773%	<u>0.7959</u>	<u>0.5442</u>	3.715%	<u>0.7713</u>	<u>0.5716</u>	1.194%
	KAR	0.7991	0.5469	0.033%	0.7870	0.5546	0.596%	0.7698	0.5718	0.634%
	LLM-CF	0.8137*	0.5306*	4.916%	0.8044*	0.5366*	6.695%	0.7881*	0.5581*	7.460%
xDeepFM	None	0.8158	0.5318	0.000%	0.8065	0.5359	0.000%	0.7836	0.5589	0.000%
	KD	<u>0.8169</u>	0.5298	0.348%	<u>0.8104</u>	0.5345	1.272%	0.7865	0.5553	1.023%
	KAR	0.8161	0.5279	0.094%	0.8101	0.5315	1.175%	0.7898	0.5529	2.186%
	LLM-CF	0.8196*	0.5248*	1.203%	0.8113	0.5311	1.566%	0.7947*	0.5473*	3.914%
AutoInt	None	0.8003	0.5444	0.000%	0.7949	0.5469	0.00%	0.7630	0.5770	0.000%
	KD	0.8012	0.5439	0.300%	<u>0.7961</u>	<u>0.5444</u>	0.407%	0.7635	0.5770	0.190%
	KAR	0.8039	0.5390	1.199%	0.7939	0.5476	-0.339%	0.7683	0.5741	2.015%
	LLM-CF	0.8088*	0.5391	2.831%	0.8090*	0.5321*	4.781%	0.7754*	0.5685*	4.714%
DCNv1	None	0.8023	0.5442	0.000%	0.8146	0.5255	0.000%	0.7621	0.5831	0.000%
	KD	<u>0.8040</u>	<u>0.5441</u>	0.562%	0.8147	0.5286	0.031%	<u>0.7652</u>	0.5847	1.183%
	KAR	0.8024	0.5469	0.033%	0.8165	0.5229	0.604%	0.7651	0.5821	1.144%
	LLM-CF	0.8092*	0.5368*	2.282%	0.8182*	0.5216*	1.144%	0.7702*	0.5745*	3.090%
DCNv2	None	0.8110	0.5331	0.000%	0.8028	0.5378	0.00%	0.7774	0.5650	0.000%
	KD	<u>0.8112</u>	<u>0.5320</u>	0.064%	0.8057	0.5343	0.958%	0.7827	0.5609	1.911%
	KAR	0.8087	0.5363	-0.739%	0.8003	0.5404	-0.825%	0.7759	0.5662	-0.541%
	LLM-CF	0.8131*	0.5307*	0.675%	0.8033	0.5372	0.165%	0.7812	0.5619	1.370%
DIN	None	0.7986	0.5519	0.000%	0.7861	0.5613	0.000%	0.7586	0.5885	0.000%
	KD	<u>0.8023</u>	<u>0.5422</u>	1.239%	<u>0.7934</u>	<u>0.5518</u>	2.551%	<u>0.7652</u>	<u>0.5847</u>	2.552%
	KAR	0.7971	0.5525	-0.502%	0.7861	0.5604	0.000%	0.7620	0.5874	1.315%
	LLM-CF	0.8089*	0.5374*	3.449%	0.7967*	0.5492*	3.705%	0.7783*	0.5699*	7.618%

- LLM-CF achieves improvement on different CTR backbones.



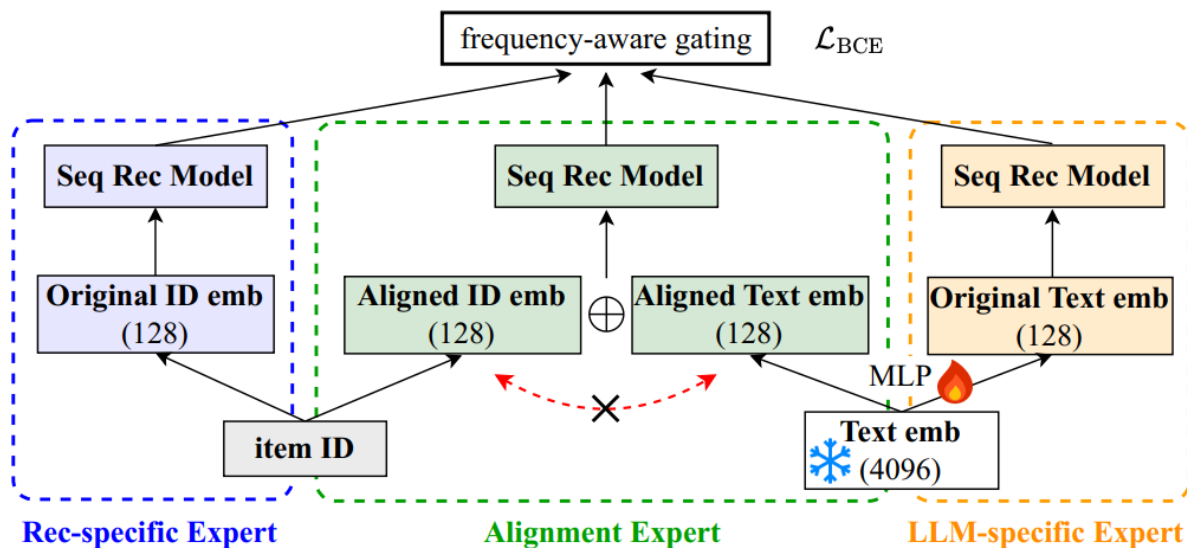
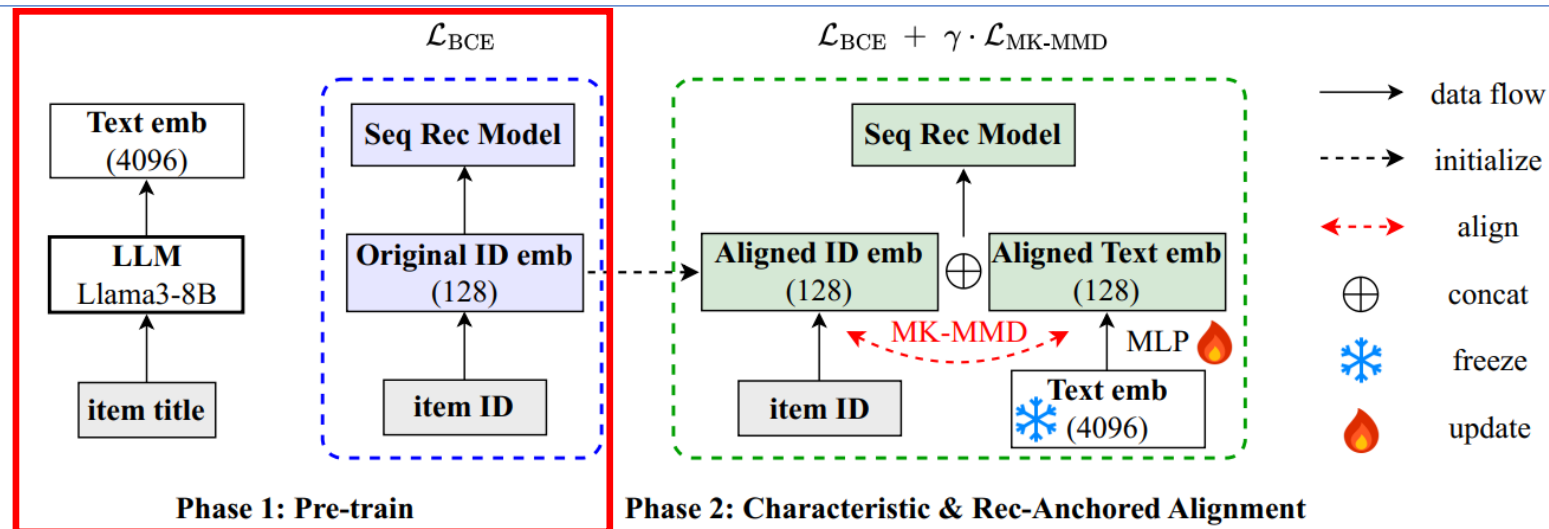
- TedRec applies Fourier Transform to conduct semantic fusion in the frequency domain.

- **Challenges for Alignment + SFT**
 - **Inability to Capture All Statistics of Data Distribution:** Contrastive learning with common cosine kernel is not optimal
 - **Catastrophic Forgetting in Alignment:** With only one set of collaborative embeddings, the alignment leads to catastrophic forgetting on the collaborative embeddings



- **Pre-train**

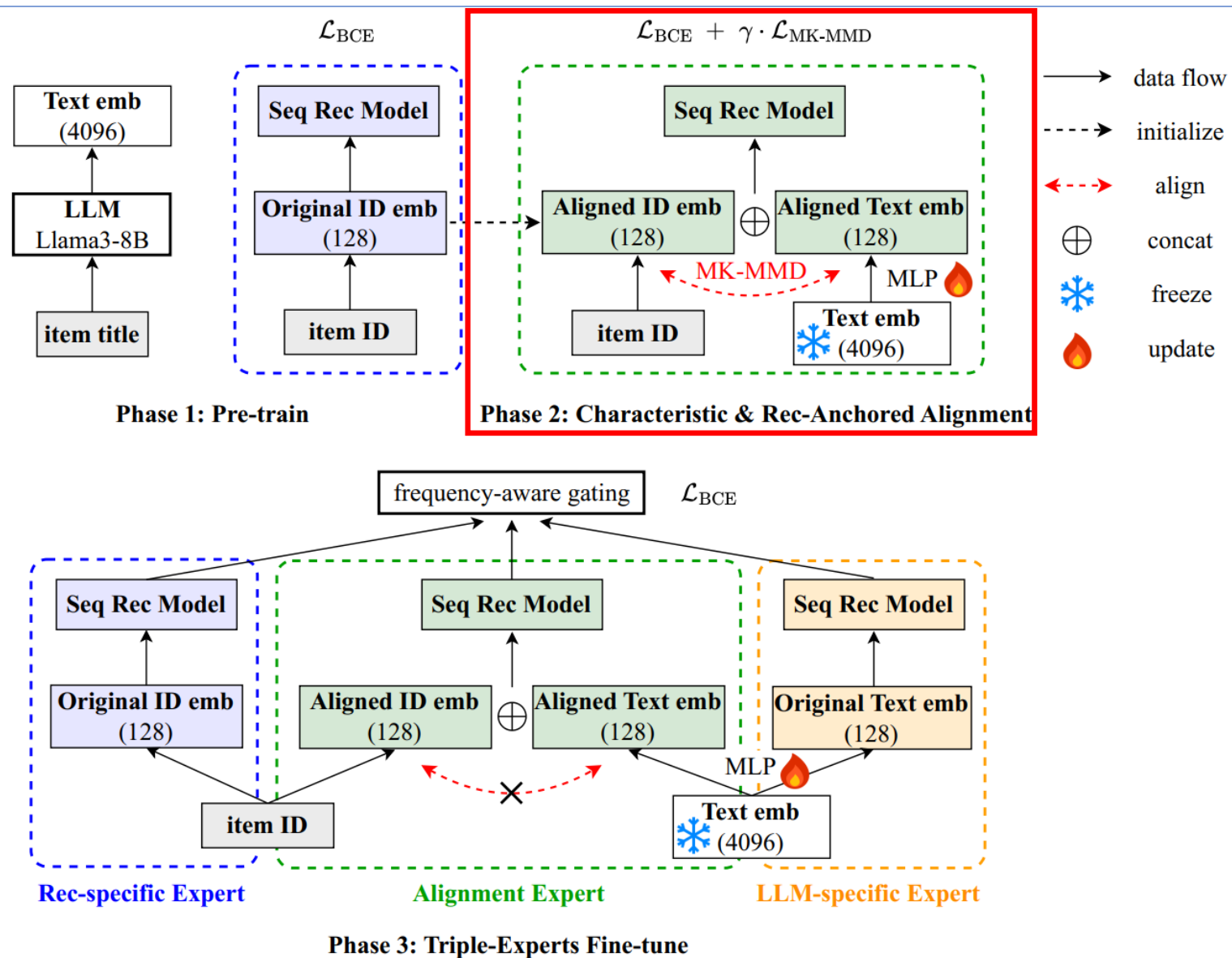
- Text & Collaborative emb



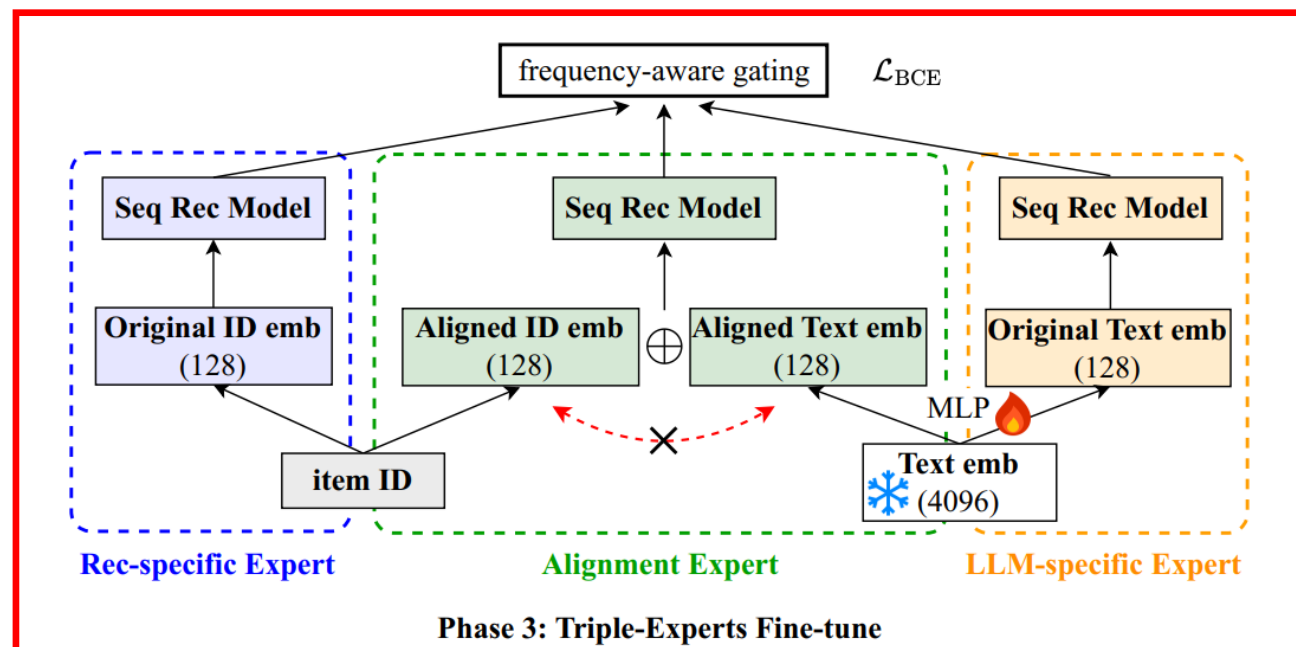
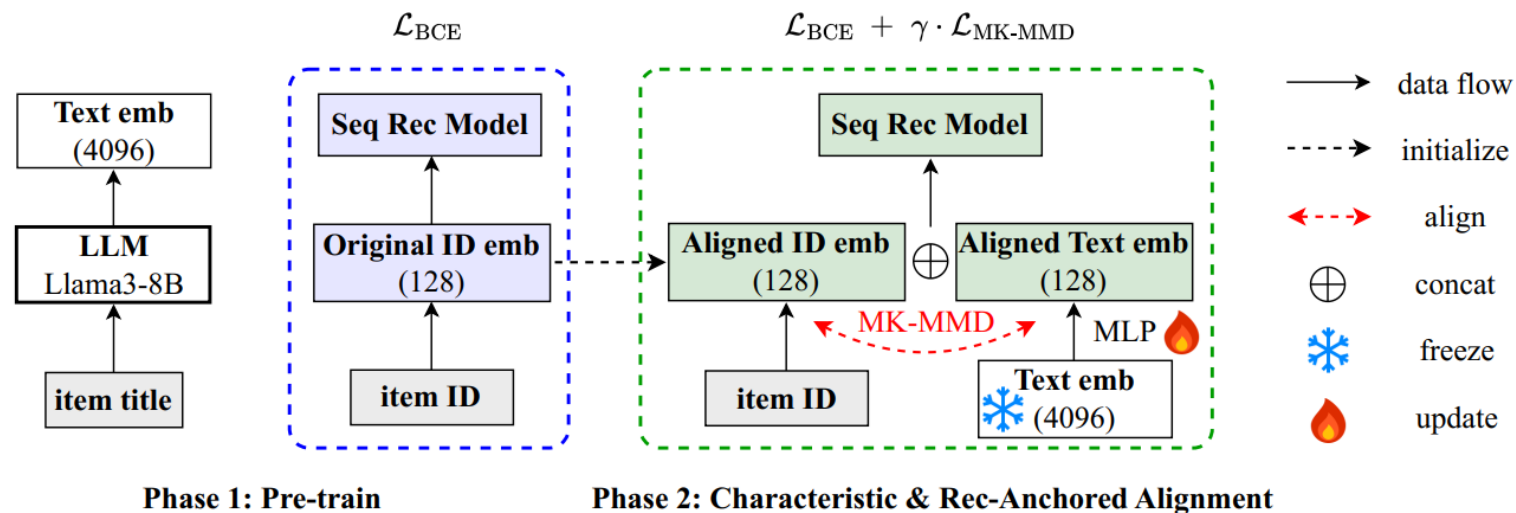
Phase 3: Triple-Experts Fine-tune

Align

- Recommendation as anchor
- Multi-kernel maximum mean discrepancy as alignment loss based on characteristic kernel



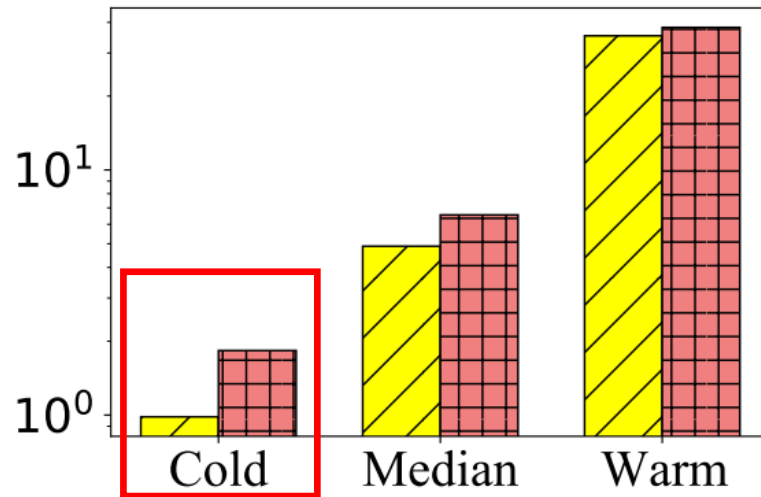
- **Disentangle**
 - Triple-Experts & Multi-Emb



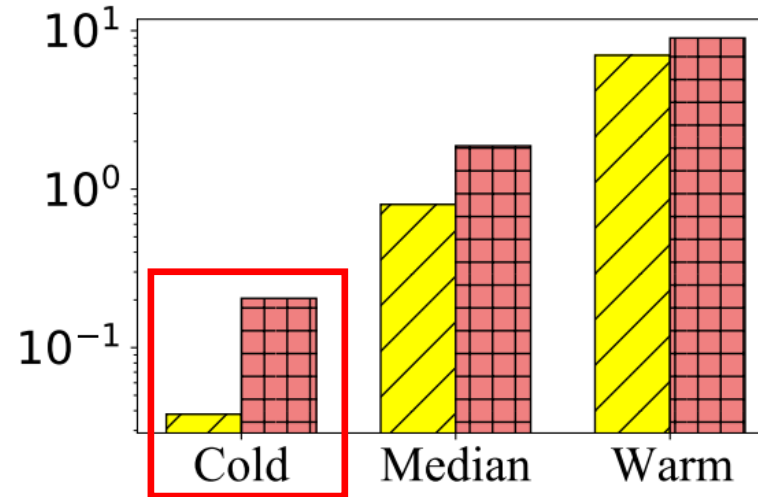
Datasets	Metric	SASRec	Hybrid	MoRec	CTRL	MAKE	DisCo	SMEM	Ours	Impr.
MIND	HR@10	16.8437	17.3385	14.1235	18.1318	16.4317	17.3367	<u>18.4319</u>	18.6703[★]	10.84%
	nDCG@10	9.0520	9.3321	7.7090	9.7996	8.9684	9.4131	<u>10.0003</u>	10.1515[★]	12.15%
Electronics	HR@10	1.6754	1.5673	0.9562	1.9468	0.9495	1.8148	<u>2.3492</u>	2.4804[★]	48.05%
	nDCG@10	0.7938	0.8385	0.4721	0.9370	0.4611	0.9025	<u>1.4287</u>	1.5373[★]	93.67%
Prime Pantry	HR@10	2.6338	3.1000	3.1699	2.8048	3.3408	2.8358	<u>3.4341</u>	3.8303[★]	45.43%
	nDCG@10	1.2926	1.5606	1.6013	1.3700	1.6379	1.3451	<u>1.7440</u>	1.9104[★]	47.80%

- PAD surpasses all baselines and achieves significant improvement.

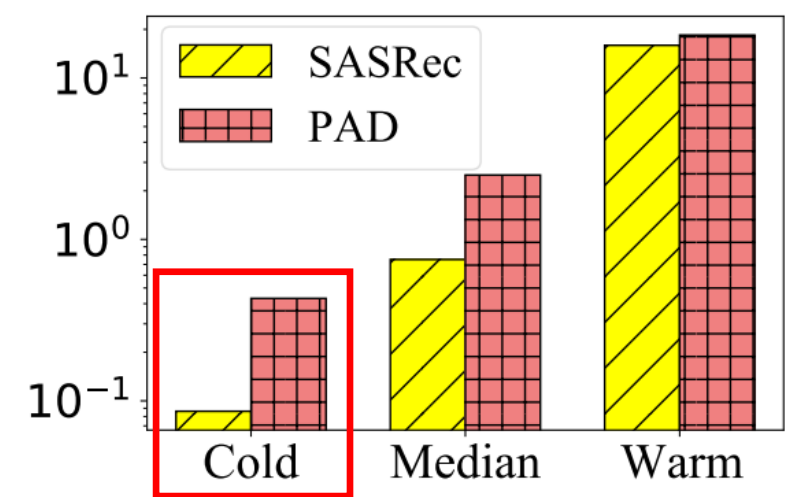
MIND



Electronics



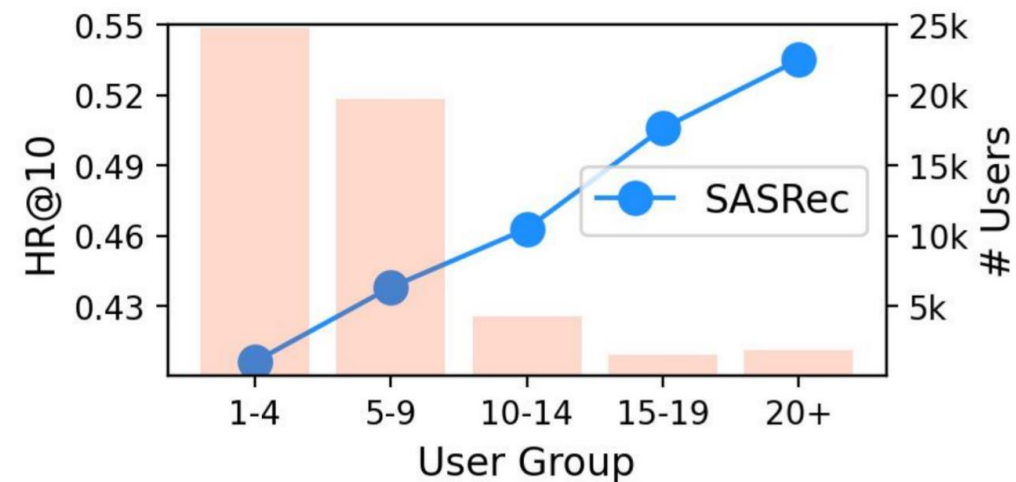
Prime Pantry



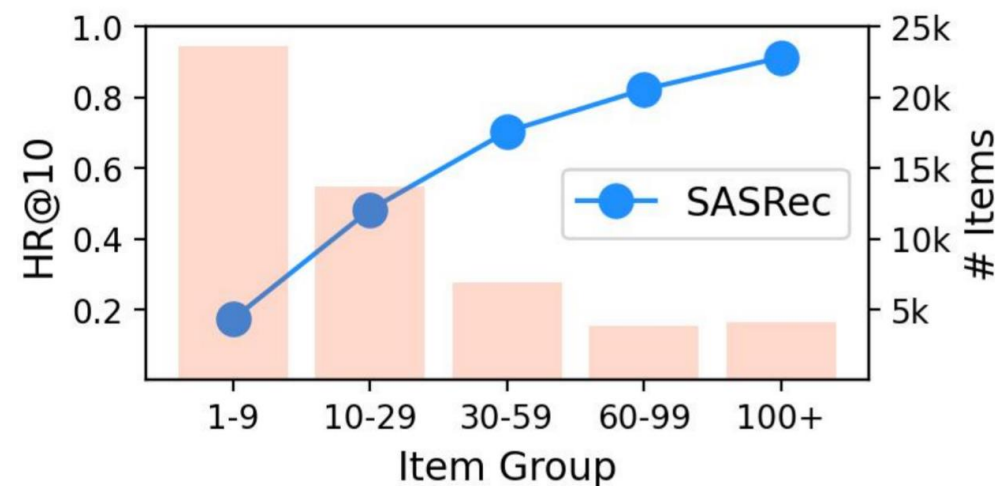
- PAD can mitigate cold-start problem with LLM knowledge.

• Long-tail Challenges for SR

- **Long-tail User Challenge:** The majority of users receive less than optimal recommendation services
→ Poor experience for **new users**
- **Long-tail Item Challenge:** Current SRS models often underperform those unpopular items
→ Less profits for **small sellers**
- **LLMs are promising to address the long-tail challenges from semantic view!**

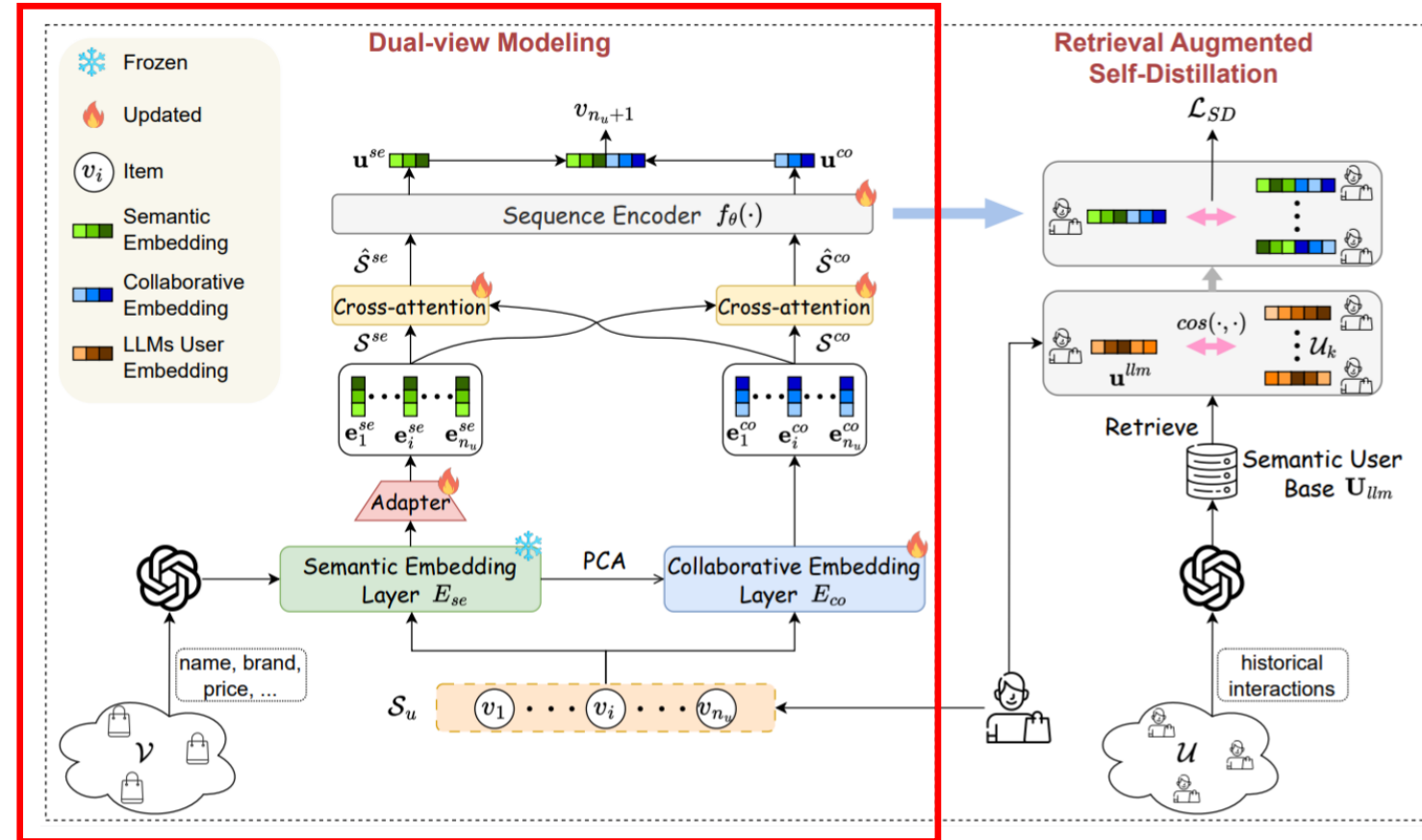


(a) Long-tail User Challenge

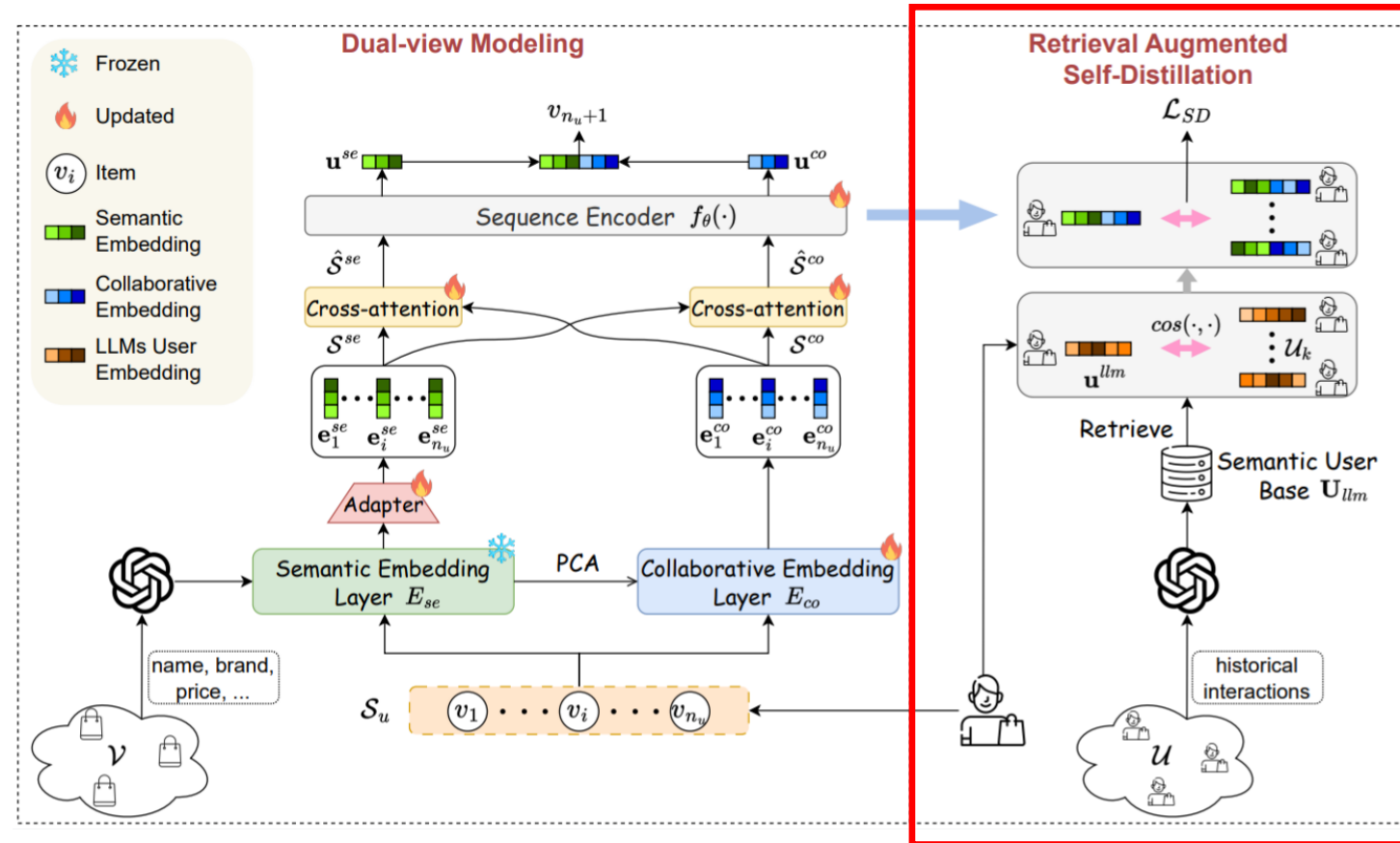


(b) Long-tail Item Challenge

- **Dual-view Modeling:** Consisting of semantic-view modeling and collaborative-view modeling to address the long-tail item issue



- Retrieval Augmented Self-Distillation:** Enhancing the SRS model to address the long-tail user problem

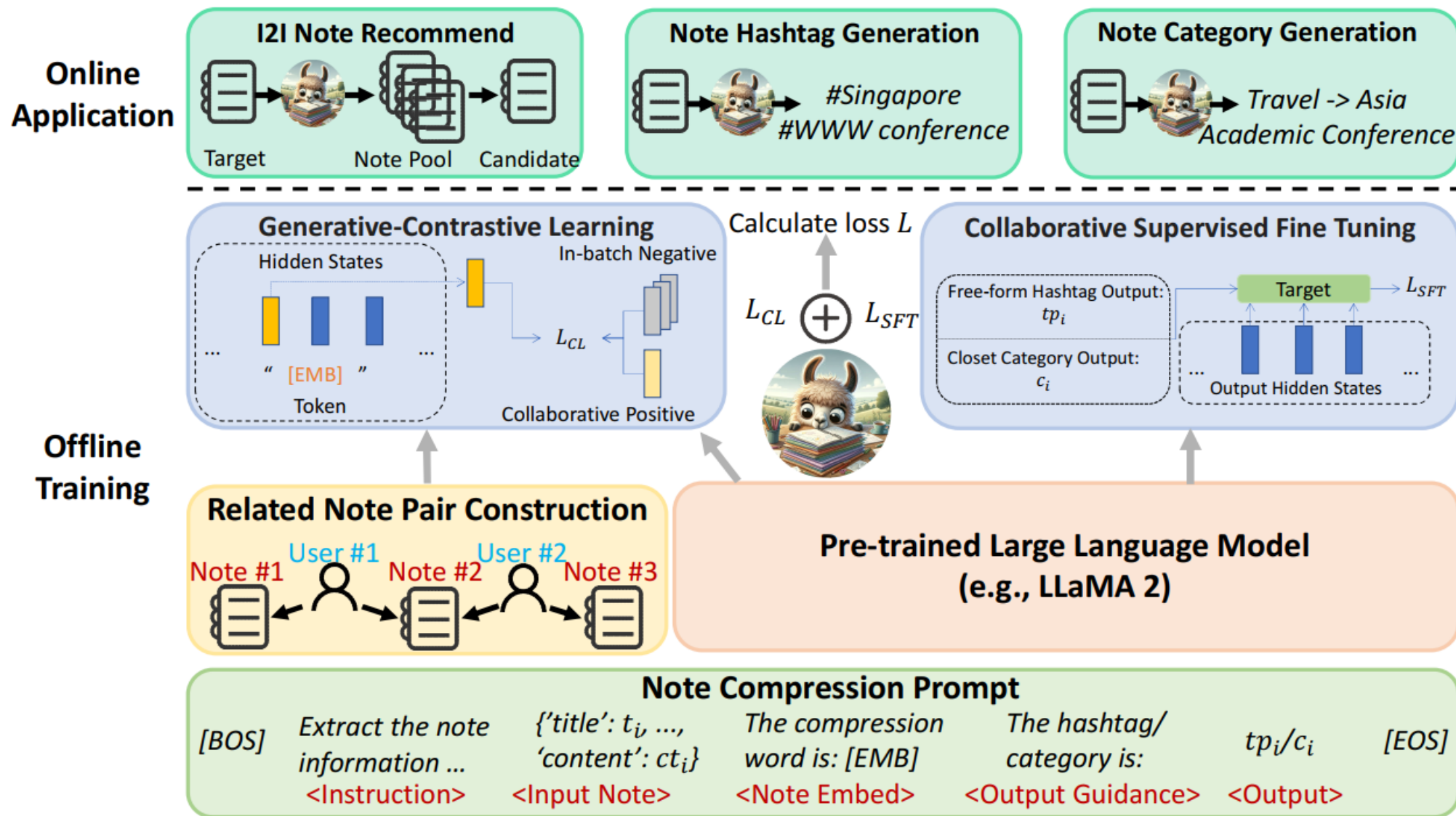


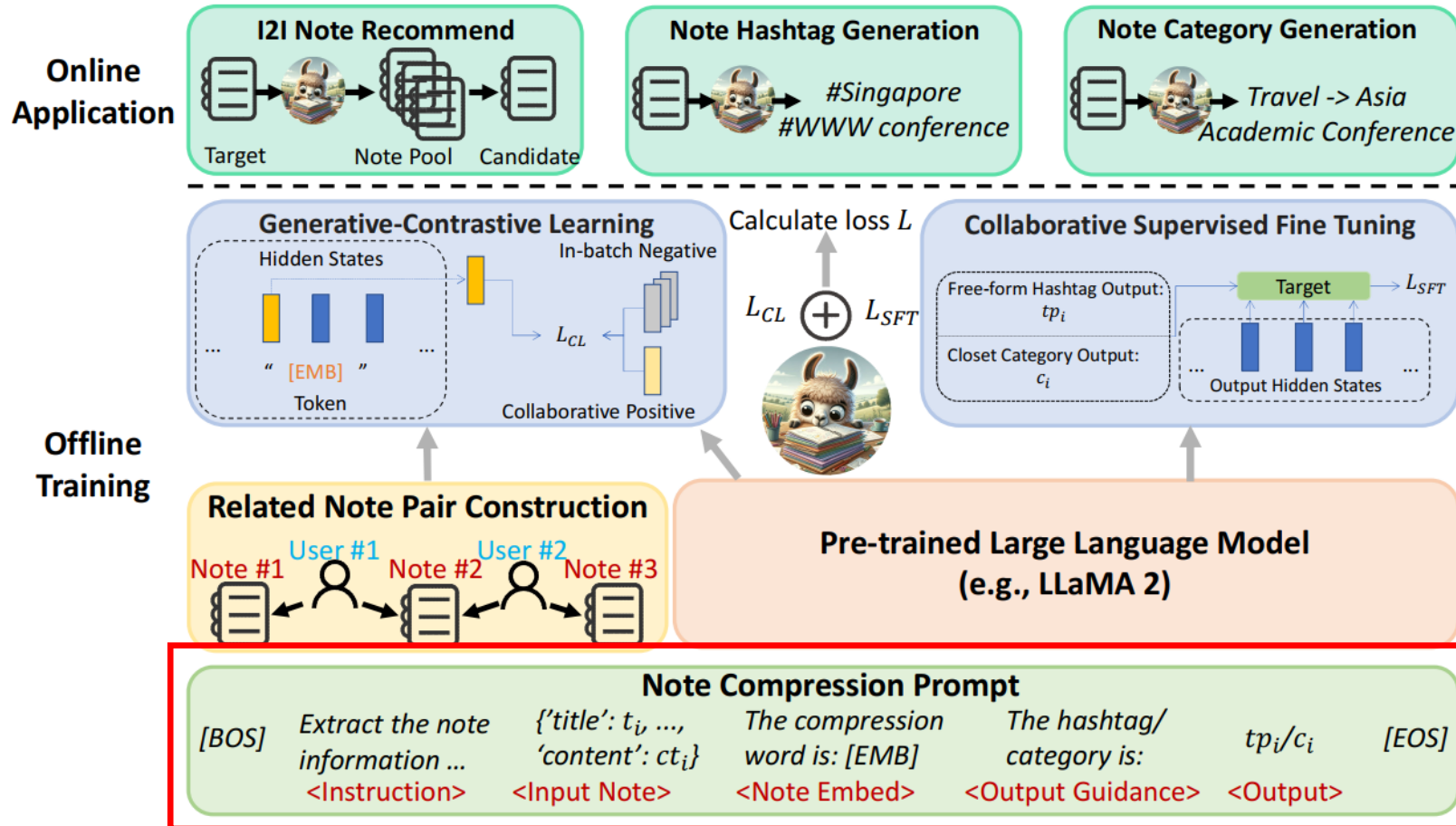
Dataset	Model	Overall		Tail Item		Head Item		Tail User		Head User	
		H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10
Yelp	GRU4Rec	0.4879	0.2751	0.0171	0.0059	0.6265	0.3544	0.4919	0.2777	0.4726	0.2653
	- CITIES	0.4898	0.2749	0.0134	0.0051	0.6301	0.3543	0.4936	0.2783	0.4756	0.2618
	- MELT	0.4985	0.2825	0.0201	0.0079	0.6393	0.3633	0.5046	0.2865	0.4750	0.2671
	- RLMRec	0.4886	0.2777	0.0188	0.0067	0.6269	0.3574	0.4920	0.2804	0.4756	0.2671
	- LLMInit	0.4872	0.2749	0.0201	0.0072	0.6246	0.3537	0.4908	0.2775	0.4732	0.2647
	- LLM-ESR	0.5724*	0.3413*	0.0763*	0.0318*	0.7184*	0.4324*	0.5782*	0.3456*	0.5501*	0.3247*
	Bert4Rec	0.5307	0.3035	0.0115	0.0044	0.6836	0.3916	0.5325	0.3047	0.5241	0.2988
	- CITIES	0.5249	0.3015	0.0041	0.0014	0.6783	0.3899	0.5274	0.3032	0.5155	0.2954
	- MELT	0.6206	0.3770	0.0429	0.0149	0.7907	0.4836	0.6210	0.3780	0.6191	0.3733
	- RLMRec	0.5306	0.3039	0.0104	0.0040	0.6938	0.3922	0.5351	0.3065	0.5137	0.2936
	- LLMInit	0.6199	0.3781	0.0874	0.0330	0.7766	0.4797	0.6204	0.3796	0.6178	0.3723
	- LLM-ESR	0.6623*	0.4222*	0.1227*	0.0500*	0.8212*	0.5318*	0.6637*	0.4247*	0.6571*	0.4127*
	SASRec	0.5940	0.3597	0.1142	0.0495	0.7353	0.4511	0.5893	0.3578	0.6122	0.3672
	- CITIES	0.5828	0.3540	0.1532	0.0700	0.7093	0.4376	0.5785	0.3511	0.5994	0.3649
	- MELT	0.6257	0.3791	0.1015	0.0371	0.7801	0.4799	0.6246	0.3804	0.6299	0.3744
	- RLMRec	0.5990	0.3623	0.0953	0.0412	0.7474	0.4568	0.5966	0.3613	0.6084	0.3658
	- LLMInit	0.6415	0.3997	0.1760	0.0789	0.7785	0.4941	0.6403	0.4010	0.6462	0.3948
	- LLM-ESR	0.6673*	0.4208*	0.1893*	0.0845*	0.8080*	0.5199*	0.6685*	0.4229*	0.6627*	0.4128*

- LLM-ESR leads the overall performance, which indicates better enhancing effects.

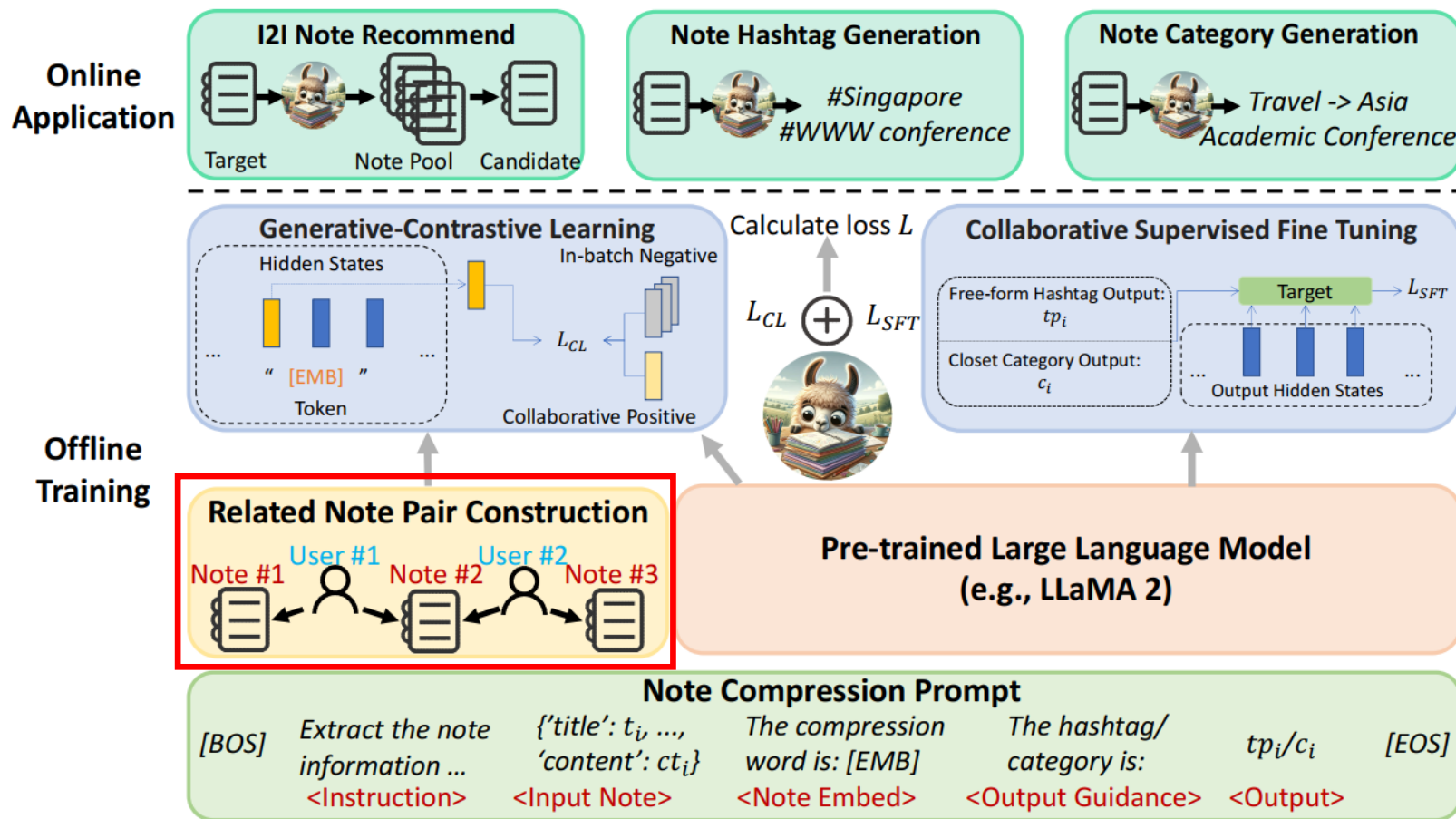
Dataset	Model	Overall		Tail Item		Head Item		Tail User		Head User	
		H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10	H@10	N@10
Yelp	GRU4Rec	0.4879	0.2751	0.0171	0.0059	0.6265	0.3544	0.4919	0.2777	0.4726	0.2653
	- CITIES	0.4898	0.2749	0.0134	0.0051	0.6301	0.3543	0.4936	0.2783	0.4756	0.2618
	- MELT	0.4985	0.2825	0.0201	0.0079	0.6393	0.3633	0.5046	0.2865	0.4750	0.2671
	- RLMRec	0.4886	0.2777	0.0188	0.0067	0.6269	0.3574	0.4920	0.2804	0.4756	0.2671
	- LLMInit	0.4872	0.2749	0.0201	0.0072	0.6246	0.3537	0.4908	0.2775	0.4732	0.2647
	- LLM-ESR	0.5724*	0.3413*	0.0763*	0.0318*	0.7184*	0.4324*	0.5782*	0.3456*	0.5501*	0.3247*
	Bert4Rec	0.5307	0.3035	0.0115	0.0044	0.6836	0.3916	0.5325	0.3047	0.5241	0.2988
	- CITIES	0.5249	0.3015	0.0041	0.0014	0.6783	0.3899	0.5274	0.3032	0.5155	0.2954
	- MELT	0.6206	0.3770	0.0429	0.0149	0.7907	0.4836	0.6210	0.3780	0.6191	0.3733
	- RLMRec	0.5306	0.3039	0.0104	0.0040	0.6938	0.3922	0.5351	0.3065	0.5137	0.2936
	- LLMInit	0.6199	0.3781	0.0874	0.0330	0.7766	0.4797	0.6204	0.3796	0.6178	0.3723
	- LLM-ESR	0.6623*	0.4222*	0.1227*	0.0500*	0.8212*	0.5318*	0.6637*	0.4247*	0.6571*	0.4127*
	SASRec	0.5940	0.3597	0.1142	0.0495	0.7353	0.4511	0.5893	0.3578	0.6122	0.3672
	- CITIES	0.5828	0.3540	0.1532	0.0700	0.7093	0.4376	0.5785	0.3511	0.5994	0.3649
	- MELT	0.6257	0.3791	0.1015	0.0371	0.7801	0.4799	0.6246	0.3804	0.6299	0.3744
	- RLMRec	0.5990	0.3623	0.0953	0.0412	0.7474	0.4568	0.5966	0.3613	0.6084	0.3658
	- LLMInit	0.6415	0.3997	0.1760	0.0789	0.7785	0.4941	0.6403	0.4010	0.6462	0.3948
	- LLM-ESR	0.6673*	0.4208*	0.1893*	0.0845*	0.8080*	0.5199*	0.6685*	0.4229*	0.6627*	0.4128*

- LLM-ESR achieves the best on tail and popular item group.
- LLM-ESR can augment the tail user group better.



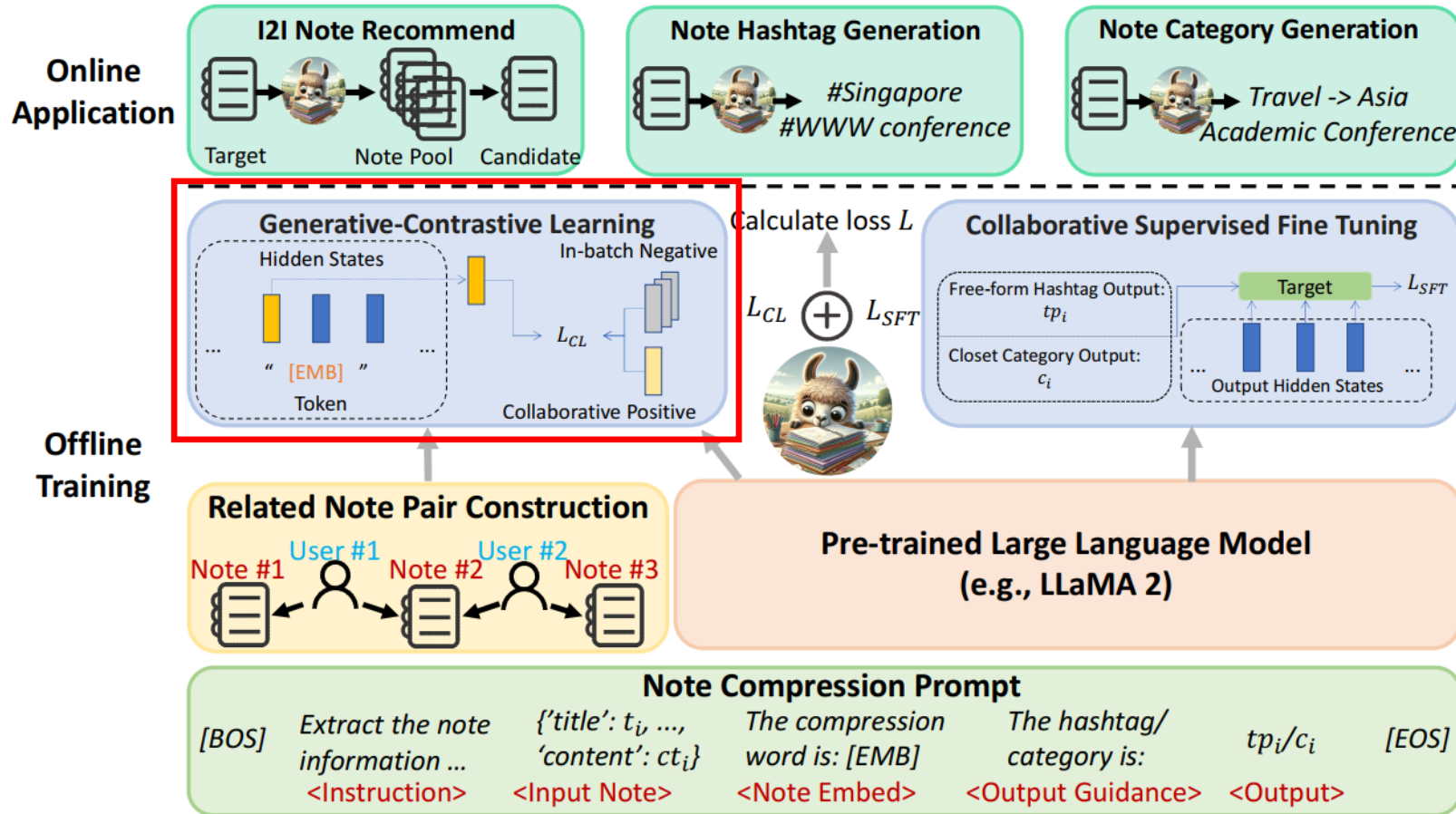


- **Note Compression Prompt**
 - Compressing the note content
 - Generating hashtags/categories

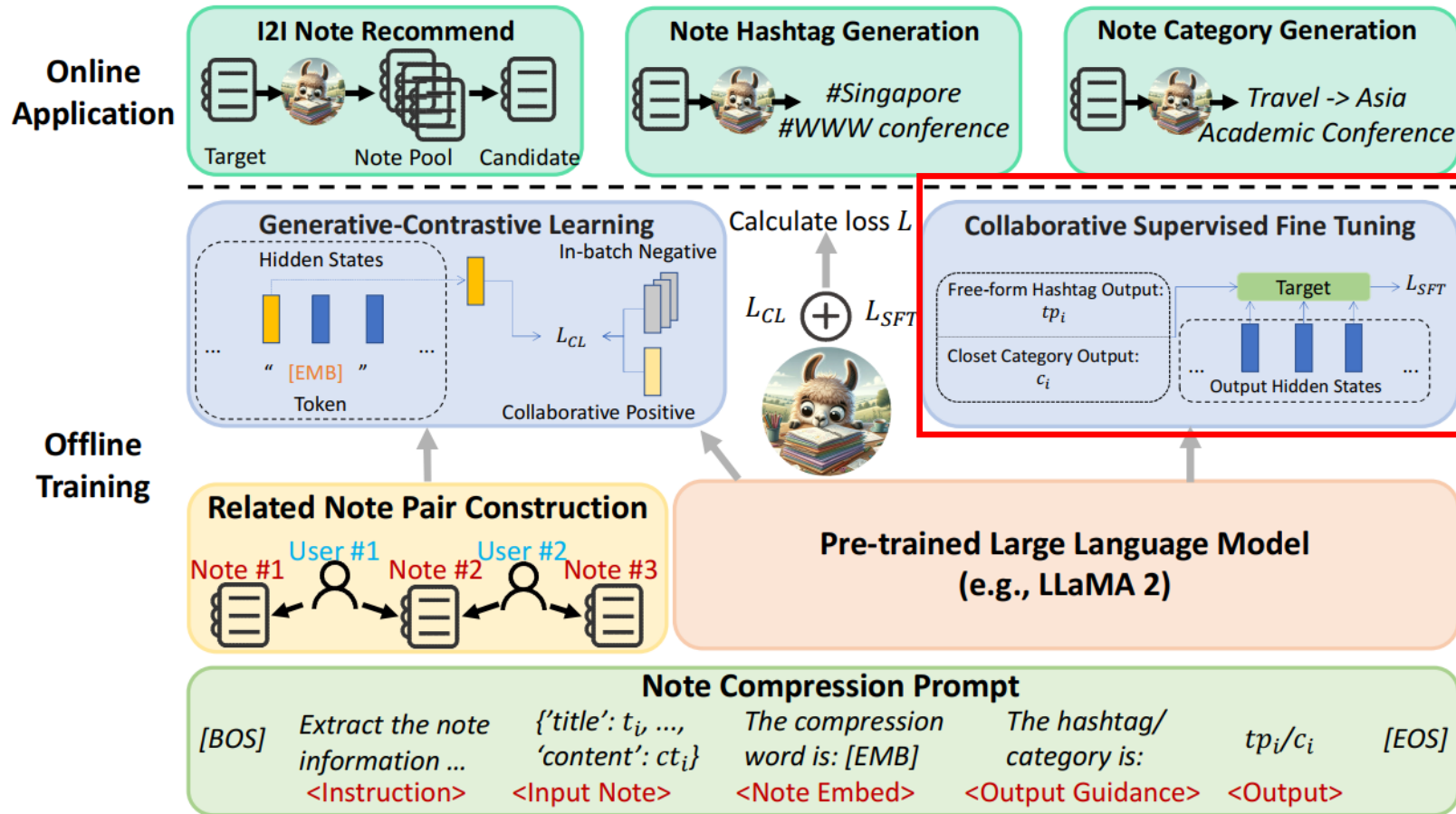


• Constructing related note pairs

- Counting co-occurrence scores
- Selecting notes with highest scores as related notes



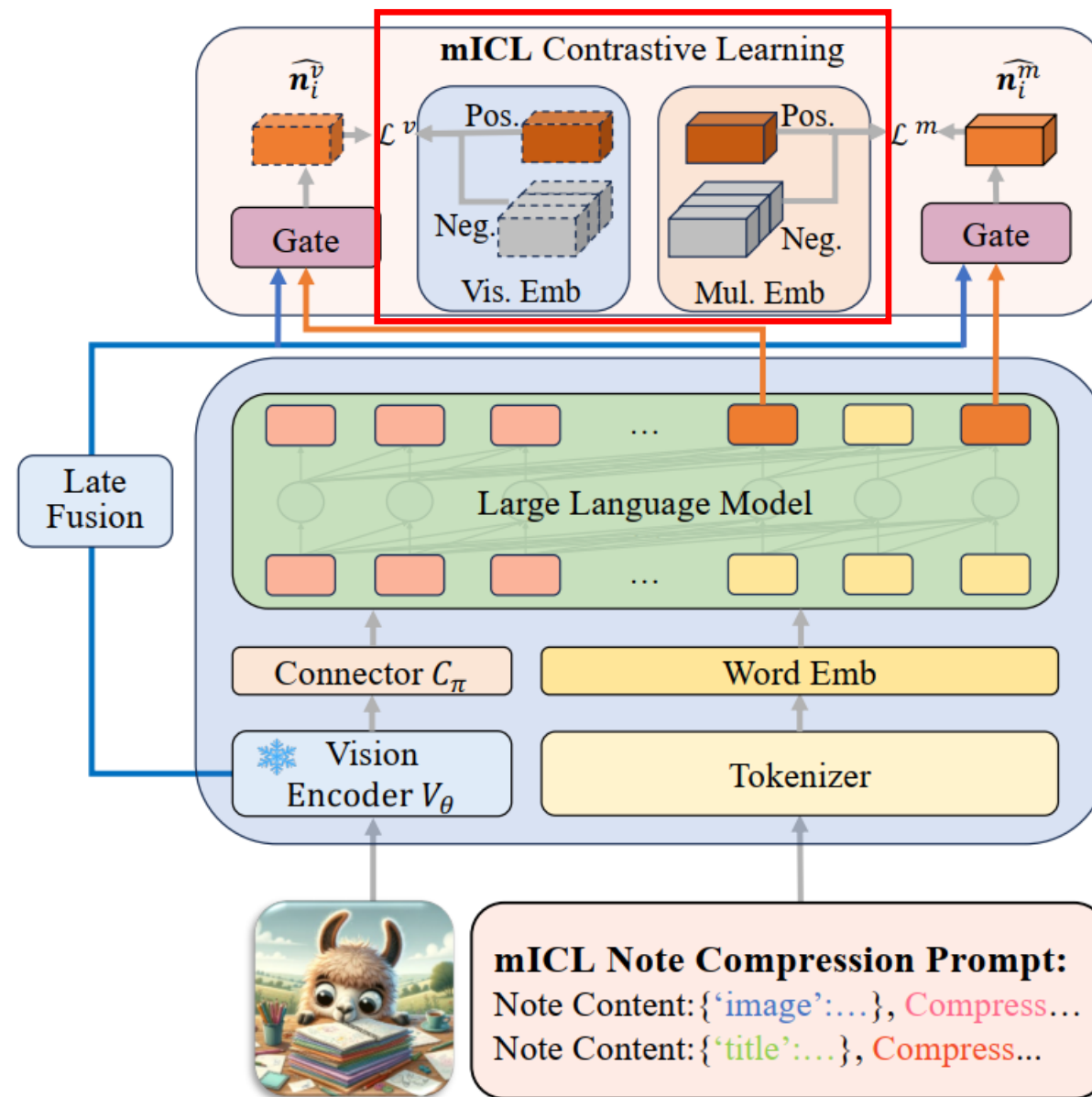
- **Generative-Contrastive Learning**
 - Identifying related notes from in-batch negatives



- **Collaborative Supervised Fine-tuning**
 - Generating hashtags/categories for each note

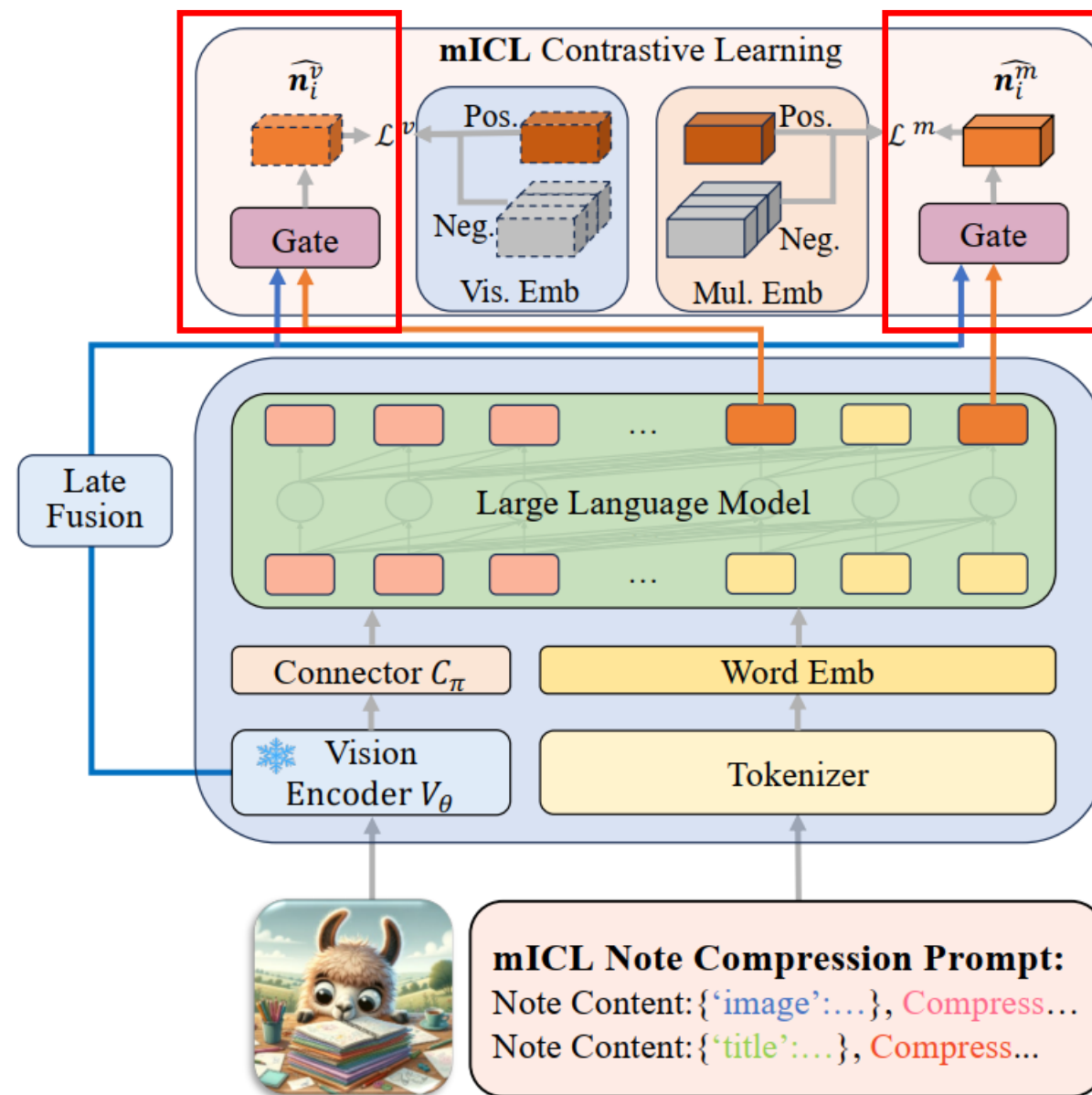
Multimodal In-Context Learning

- Separating multimodal content into visual & textual components
- Compressing into modality-compressed words



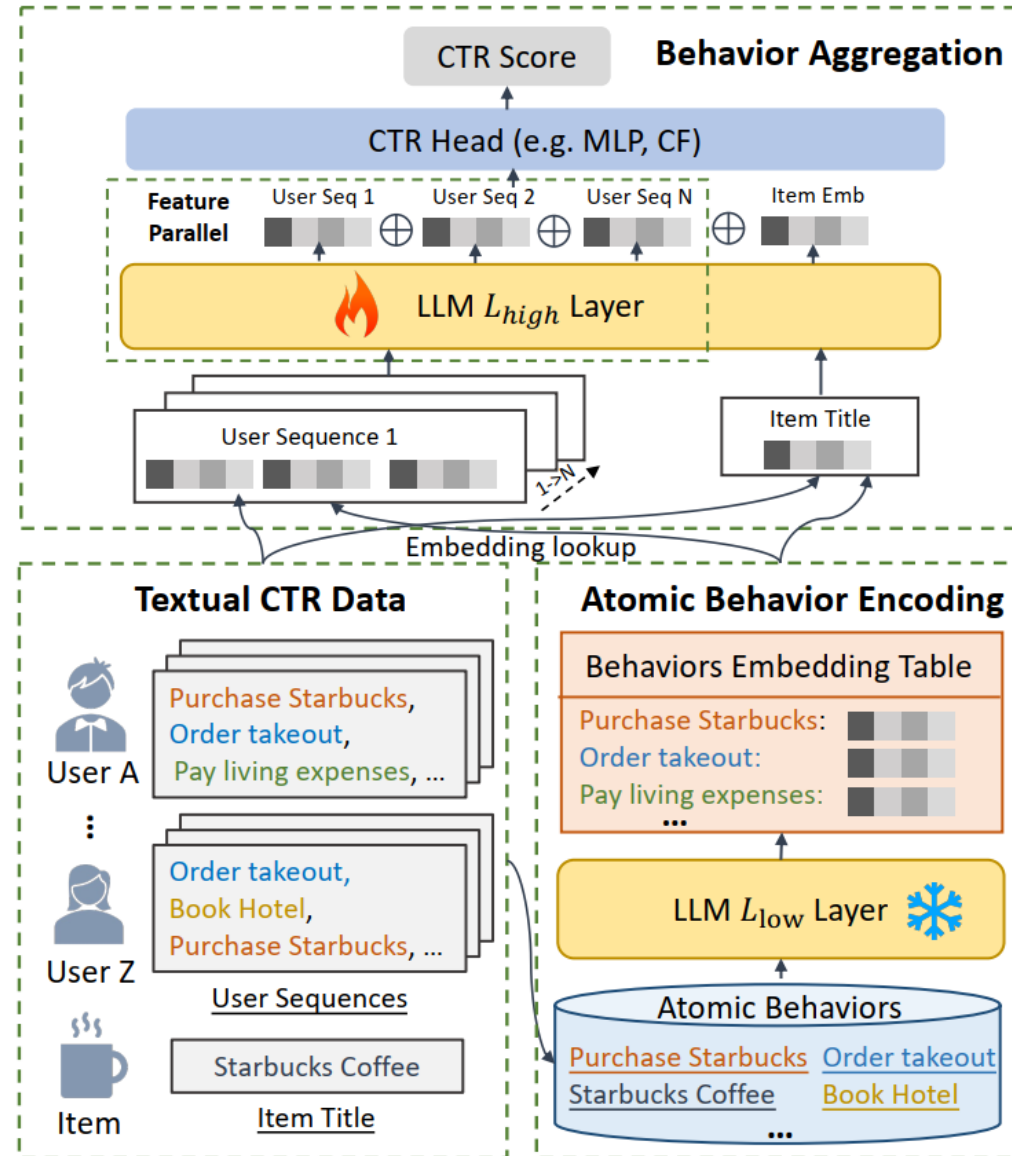
Late Fusion

- Multimodal Gating mechanism



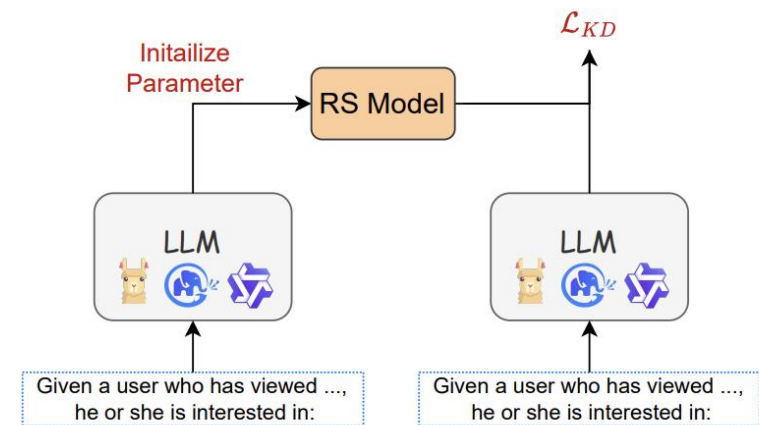
• Hierarchical Encoding

- Lower layers encoding item into atomic representation
- Higher layers generating high-level user representations



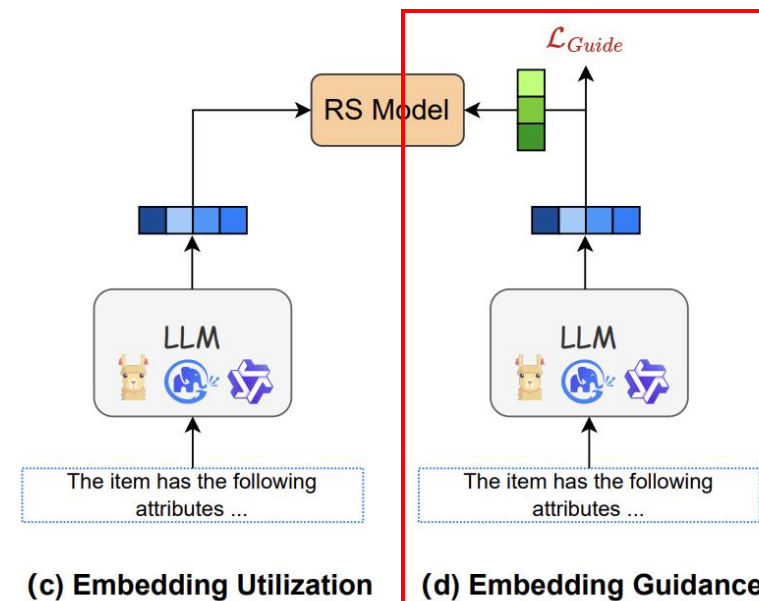
Embedding Guidance

- This subcategory refers to only using the LLM embeddings as the guidance for training or parameter synthesis.
- Categories
 - User Only
 - User & Item



(a) Model Initialization

(b) Model Distillation



(c) Embedding Utilization

(d) Embedding Guidance

• Multi-Scenario Modeling

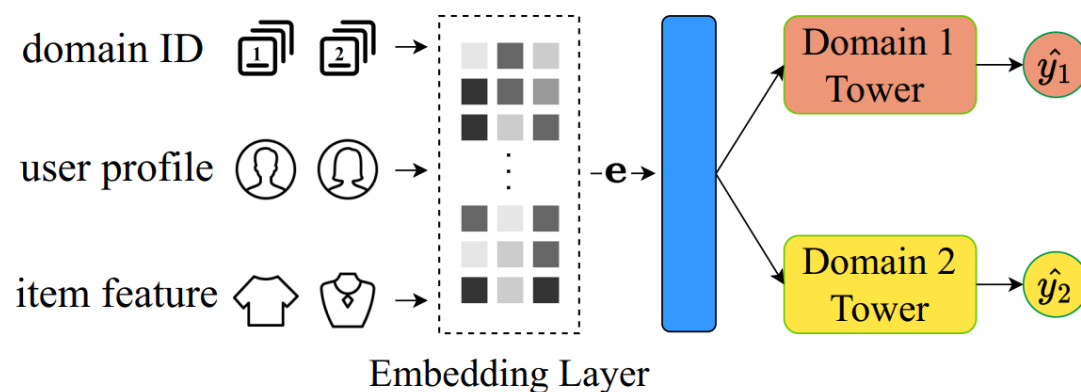
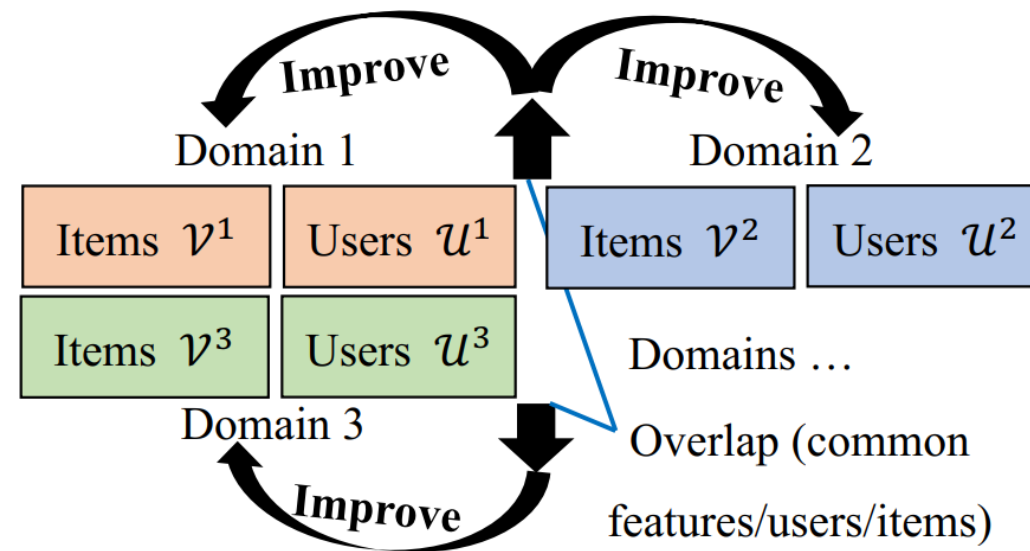
- Tackling data sparsity
- Reducing computation cost
- High efficiency

• Challenges

- Insufficient scenario knowledge
- Ignoring cross-scenario preferences

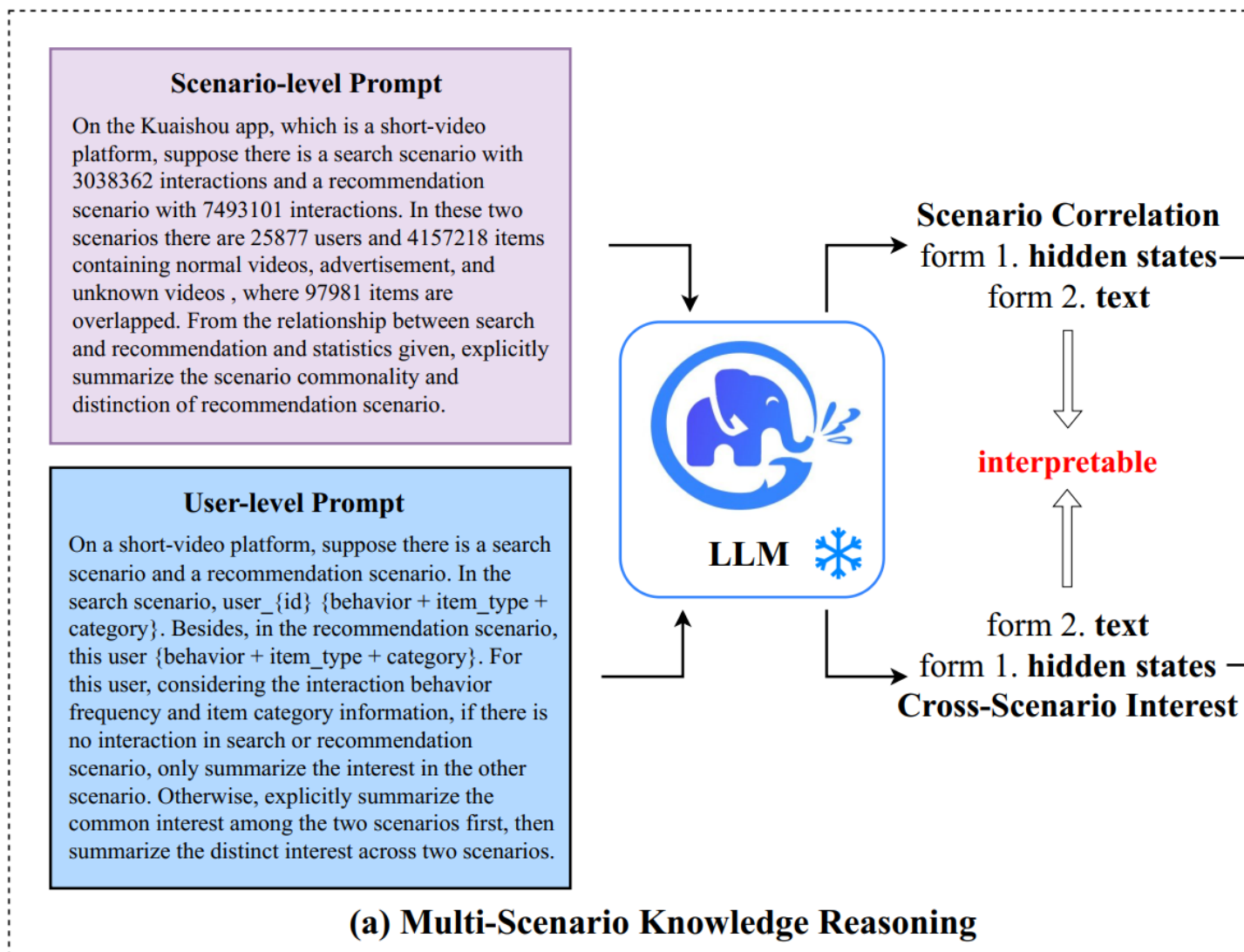
• Motivation

- Incorporating LLM to improve conventional recommender system
- LLM as reasoner + encoder



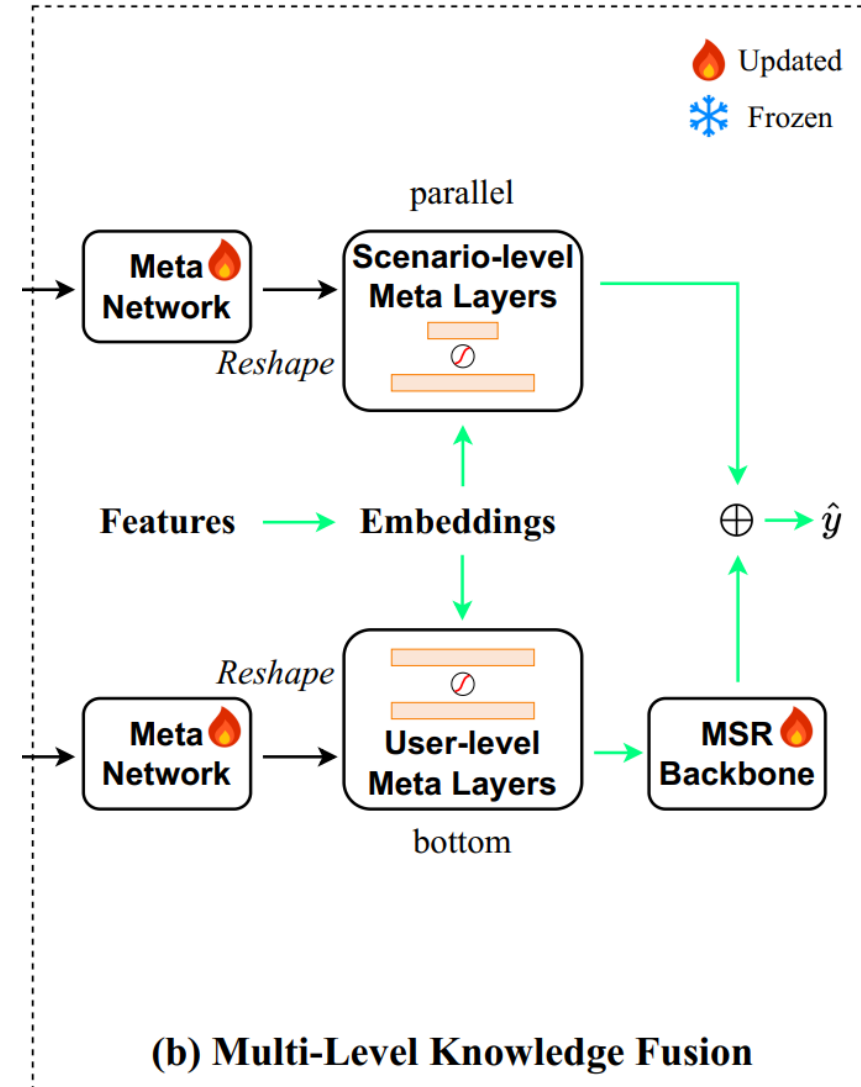
- **Multi-scenario reasoning**

- Scenario-level prompt
- User-level prompt



• Multi-level Fusion

- Meta-networks generating meta layers
- Hierarchical bottom + parallel structure



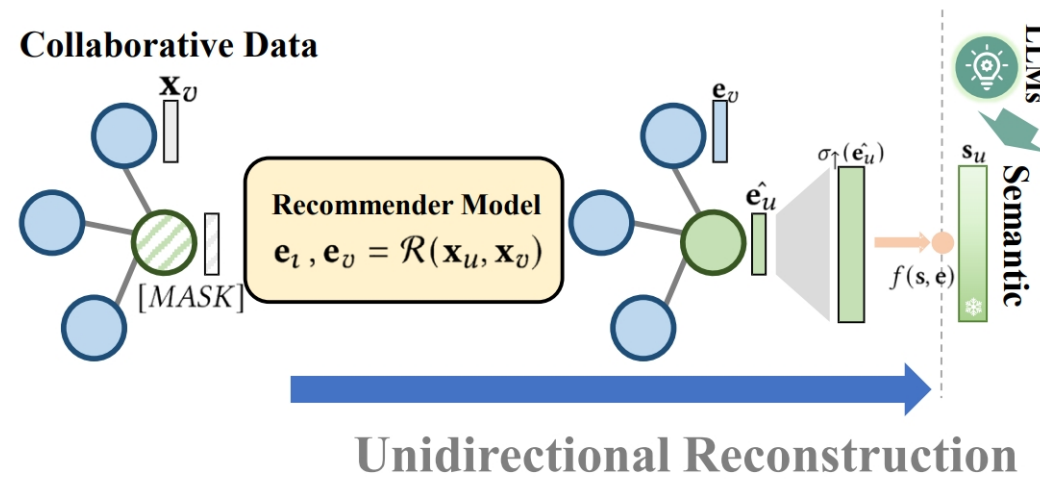
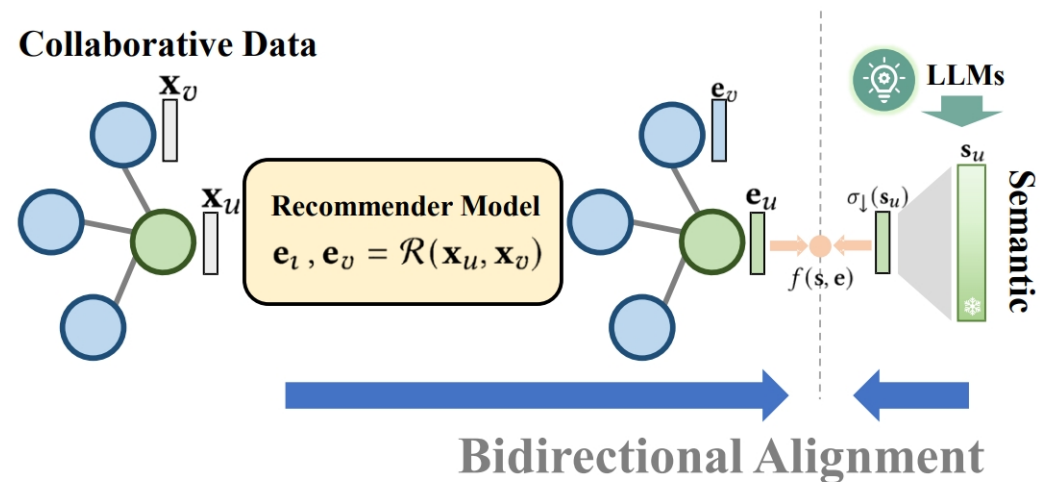
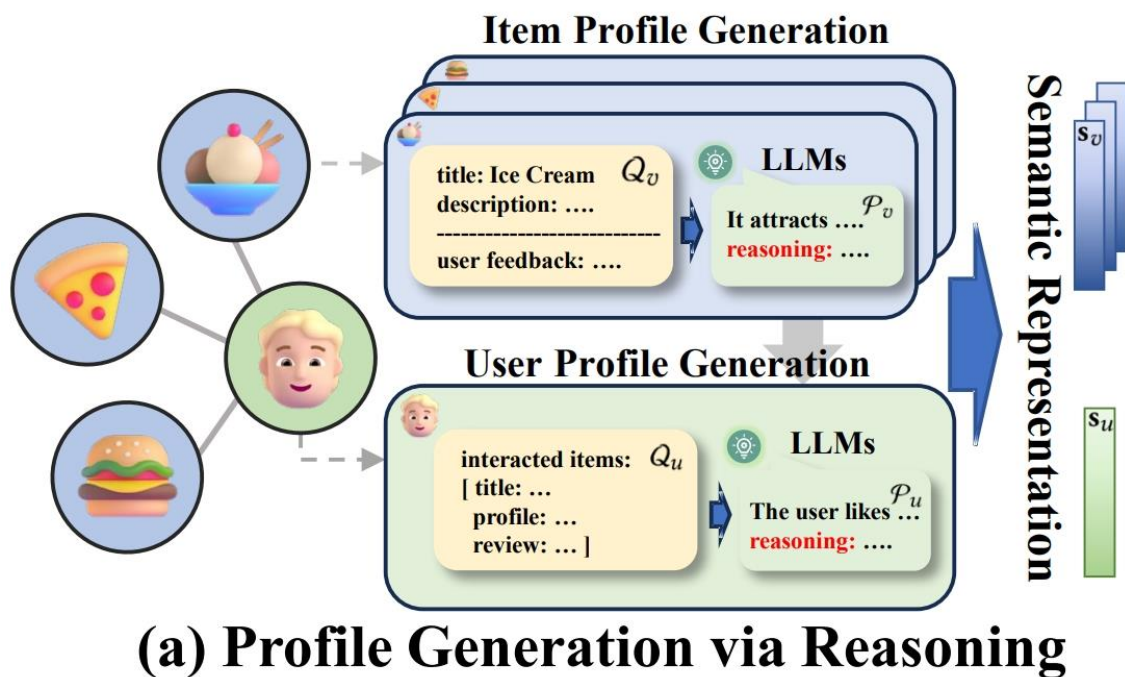
- LLM4MSR achieves an increase of 1.5%, 1%, and 40% in AUC on three datasets
 - Domain correlation & Personalized interest + Adaptive meta network

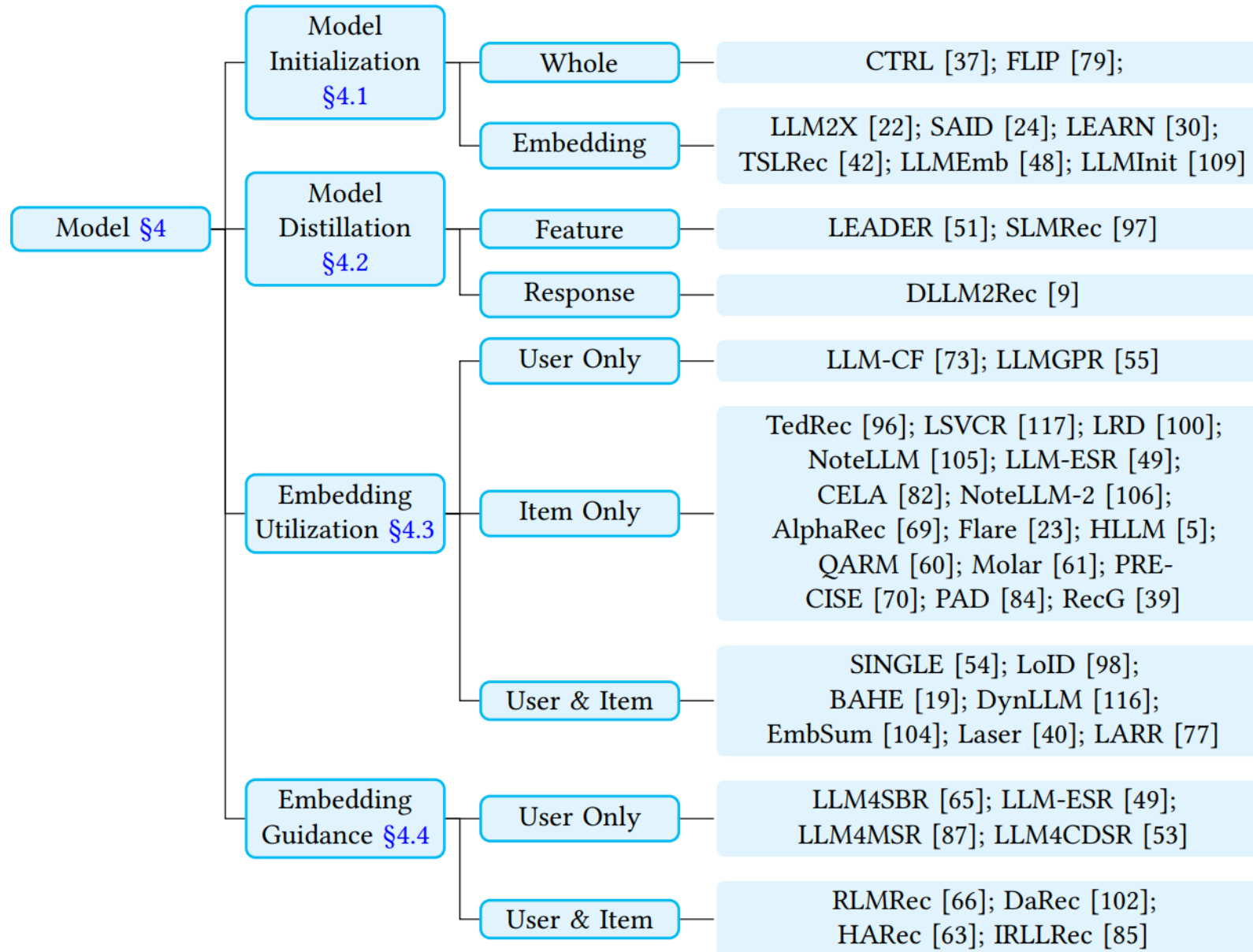
Backbones	KuaiSAR-small		KuaiSAR		Amazon		
	Rec	Search	Rec	Search	Rec#1	Rec#2	Rec#3
STAR	0.7225	0.6089	0.7387	0.6268	0.6156	0.6320	0.6225
STAR_DN	<u>0.7241</u>	<u>0.6116</u>	<u>0.7404</u>	<u>0.6270</u>	<u>0.6306</u>	<u>0.6350</u>	0.6355
STAR_EP	0.7230	0.6082	0.7388	0.6266	0.6211	0.6302	<u>0.6378</u>
STAR_ours	0.7276★	0.6181★	0.7408★	0.6332★	0.8720★	0.6364★	0.7543★
OMoE	0.7241	0.6163	0.7394	0.6310	0.5995	0.6043	0.6252
OMoE_DN	<u>0.7249</u>	<u>0.6183</u>	<u>0.7402</u>	0.6319	0.6027	0.6171	0.6289
OMoE_EP	0.7246	0.6179	0.7392	<u>0.6330</u>	<u>0.6143</u>	<u>0.6176</u>	<u>0.6312</u>
OMoE_ours	0.7265★	0.6186★	0.7413★	0.6332★	0.8182★	0.6180★	0.6823★
MMoE	0.7235	0.6150	0.7392	0.6313	0.5907	0.5999	0.6032
MMoE_DN	<u>0.7246</u>	0.6164	0.7394	<u>0.6330</u>	<u>0.6310</u>	0.6201	<u>0.6371</u>
MMoE_EP	0.7245	0.6178	<u>0.7397</u>	0.6326	0.6218	<u>0.6221</u>	0.6315
MMoE_ours	0.7264★	<u>0.6166★</u>	0.7410★	0.6341★	0.7860★	0.6222★	0.6854★
PLE	0.7249	0.6149	0.7396	0.6313	0.6059	0.6127	0.5995
PLE_DN	<u>0.7258</u>	<u>0.6165</u>	0.7400	0.6333	0.6066	0.6195	0.6162
PLE_EP	0.7252	0.6184	0.7402	0.6339	0.6173	0.6229	0.6263
PLE_ours	0.7269★	0.6184★	0.7408★	0.6343★	0.8265★	0.6250★	0.7074★
AITM	0.7236	0.6154	0.7398	0.6329	0.6039	0.6126	0.6046
AITM_DN	<u>0.7243</u>	0.6172	<u>0.7398</u>	0.6318	0.6057	0.6151	0.6166
AITM_EP	0.7237	<u>0.6177</u>	0.7389	<u>0.6337</u>	<u>0.6064</u>	<u>0.6163</u>	<u>0.6238</u>
AITM_ours	0.7273★	0.6178★	0.7407★	0.6349★	0.8196★	0.6178★	0.7089★
Shared Bottom	0.7228	0.6169	0.7389	0.6321	0.6073	0.6077	0.6268
Shared Bottom_DN	<u>0.7250</u>	<u>0.6176</u>	<u>0.7396</u>	0.6323	0.6081	<u>0.6253</u>	<u>0.6294</u>
Shared Bottom_EP	0.7243	0.6182	0.7392	<u>0.6331</u>	<u>0.6268</u>	0.6174	0.6233
Shared Bottom_ours	0.7269★	0.6182★	0.7400★	0.6341★	0.8323★	0.6262★	0.7182★

- LLM4MSR enhances various MSR backbone models, showing great **compatibility** and **deployability**

Backbones	KuaiSAR-small		KuaiSAR		Amazon		
	Rec	Search	Rec	Search	Rec#1	Rec#2	Rec#3
STAR	0.7225	0.6089	0.7387	0.6268	0.6156	0.6320	0.6225
STAR_DN	<u>0.7241</u>	<u>0.6116</u>	<u>0.7404</u>	<u>0.6270</u>	<u>0.6306</u>	<u>0.6350</u>	0.6355
STAR_EP	0.7230	0.6082	0.7388	0.6266	0.6211	0.6302	0.6378
STAR_ours	0.7276★	0.6181★	0.7408★	0.6332★	0.8720★	0.6364★	0.7543★
OMoE	0.7241	0.6163	0.7394	0.6310	0.5995	0.6043	0.6252
OMoE_DN	<u>0.7249</u>	<u>0.6183</u>	<u>0.7402</u>	0.6319	0.6027	0.6171	0.6289
OMoE_EP	0.7246	0.6179	0.7392	0.6330	0.6143	0.6176	0.6312
OMoE_ours	0.7265★	0.6186★	0.7413★	0.6332★	0.8182★	0.6180★	0.6823★
MMoE	0.7235	0.6150	0.7392	0.6313	0.5907	0.5999	0.6032
MMoE_DN	<u>0.7246</u>	0.6164	0.7394	<u>0.6330</u>	<u>0.6310</u>	0.6201	<u>0.6371</u>
MMoE_EP	0.7245	0.6178	0.7397	0.6326	0.6218	0.6221	0.6315
MMoE_ours	0.7264★	0.6166★	0.7410★	0.6341★	0.7860★	0.6222★	0.6854★
PLE	0.7249	0.6149	0.7396	0.6313	0.6059	0.6127	0.5995
PLE_DN	<u>0.7258</u>	<u>0.6165</u>	0.7400	0.6333	0.6066	0.6195	0.6162
PLE_EP	0.7252	0.6184	0.7402	0.6339	0.6173	0.6229	0.6263
PLE_ours	0.7269★	0.6184★	0.7408★	0.6343★	0.8265★	0.6250★	0.7074★
AITM	0.7236	0.6154	0.7398	0.6329	0.6039	0.6126	0.6046
AITM_DN	<u>0.7243</u>	0.6172	<u>0.7398</u>	0.6318	0.6057	0.6151	0.6166
AITM_EP	0.7237	0.6177	0.7389	0.6337	0.6064	0.6163	0.6238
AITM_ours	0.7273★	0.6178★	0.7407★	0.6349★	0.8196★	0.6178★	0.7089★
Shared Bottom	0.7228	0.6169	0.7389	0.6321	0.6073	0.6077	0.6268
Shared Bottom_DN	<u>0.7250</u>	<u>0.6176</u>	<u>0.7396</u>	0.6323	0.6081	<u>0.6253</u>	<u>0.6294</u>
Shared Bottom_EP	0.7243	0.6182	0.7392	0.6331	0.6268	0.6174	0.6233
Shared Bottom_ours	0.7269★	0.6182★	0.7400★	0.6341★	0.8323★	0.6262★	0.7182★

- **Generating profile**
- **Extra alignment loss**





Agenda

1 Introduction



Zijian Zhang



2 Knowledge Enhancement



Pengyue Jia



3 Interaction Enhancement



Ziwei Liu



4.1 Model Enhancement 1



Maolin Wang



4.2 Model Enhancement 2



Yuhao Wang



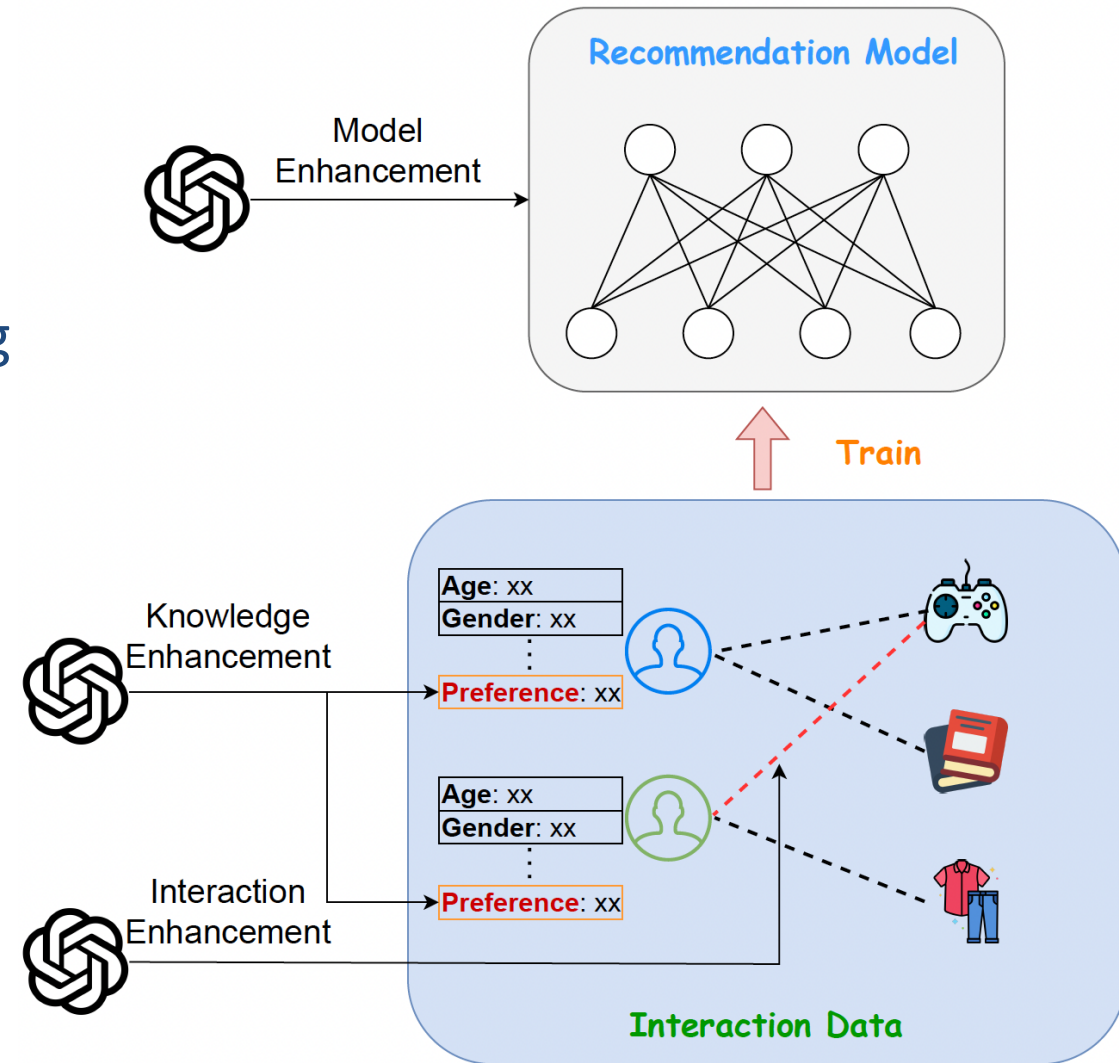
5 Conclusion

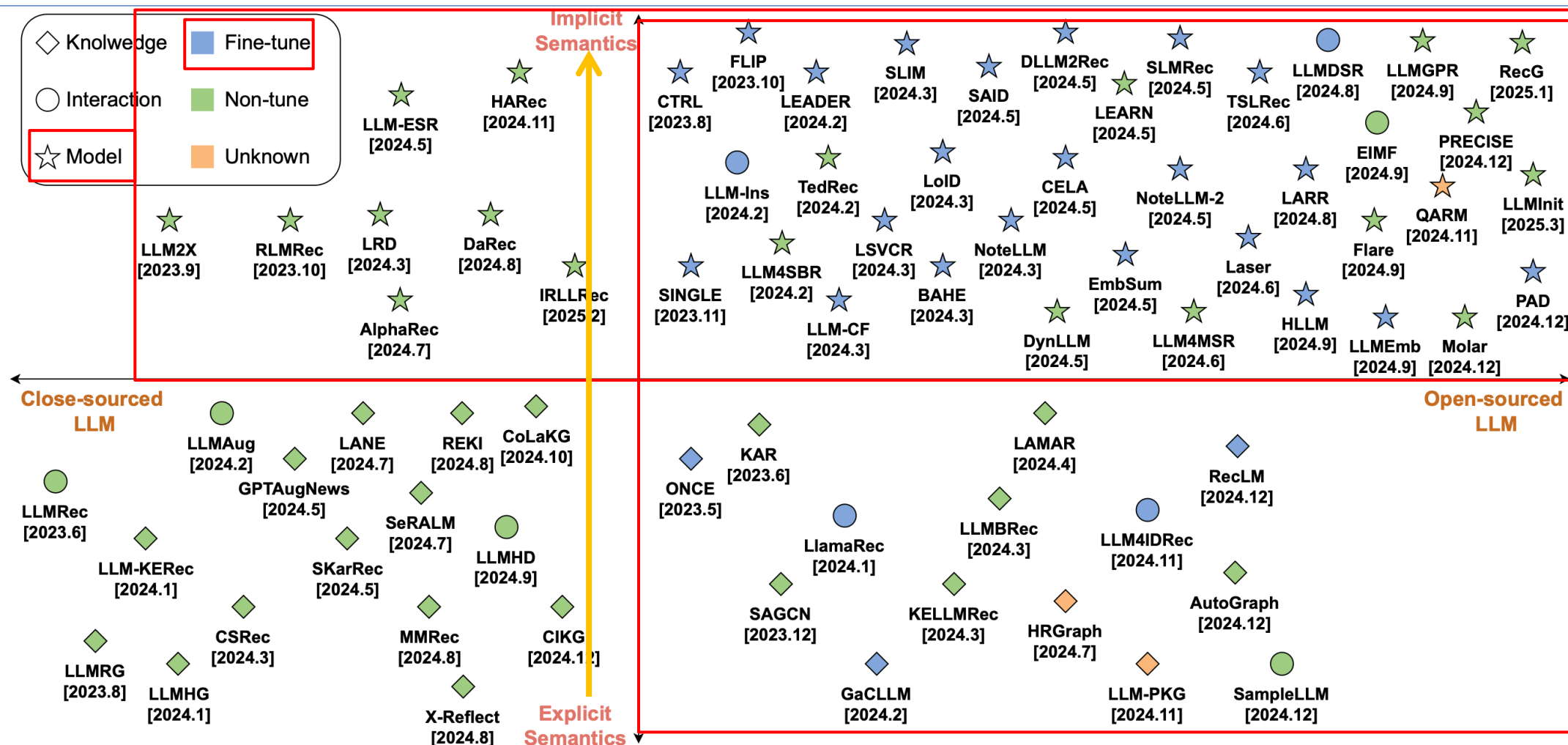


Yejing Wang

LLM-enhanced RS (LLMERS)

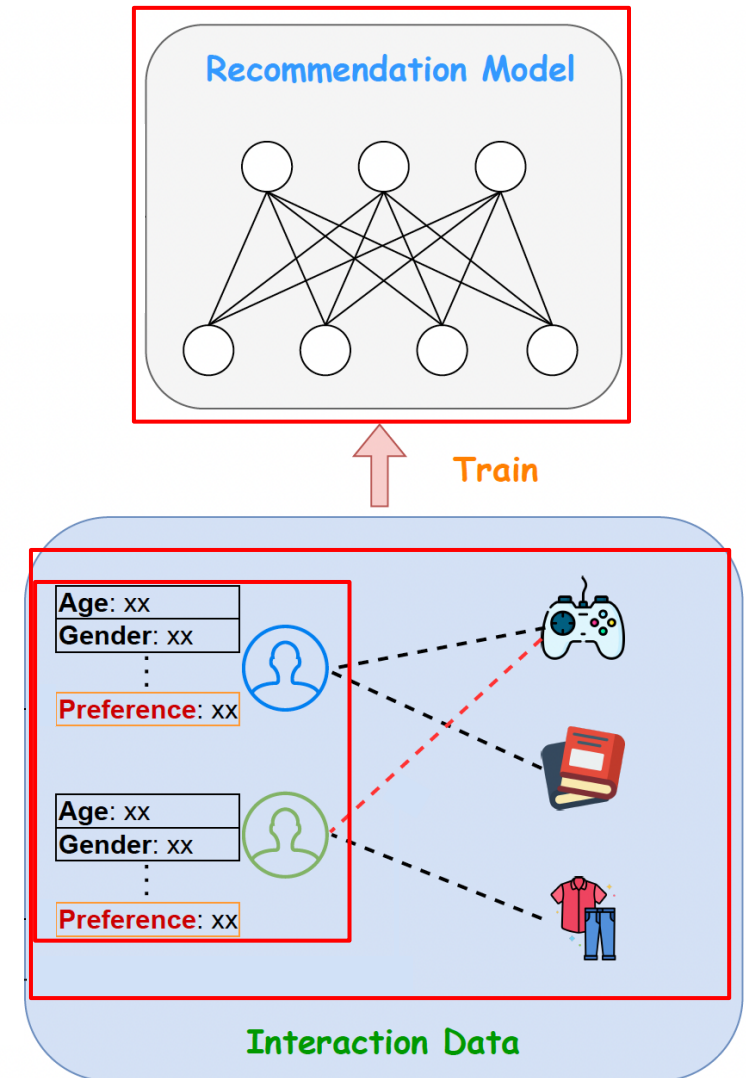
- Knowledge enhancement
 - World Knowledge, Reasoning
 - Explicit semantic -> requiring further encoding
- Interaction enhancement
 - Generation
 - Highly scalable
- Model enhancement
 - Representation, implicit semantic
 - (Small) pre-encoding & storage issues





- Semantics: explicit -> implicit, model enhancement solutions are dominant
- Fine-tuned open-source LLMs are more popular for better adaptability

- Extended to more recommendation tasks
 - Focused on collaborative filtering and sequential recommendation
 - Potential in other tasks like multi-task, cross-domain...
- Support for multimodal/text-free RS
 - Multimodal input: MLLM -> adaptive representation
 - E.g., phone case vs. dresses on E-commerce platforms
 - Text-free RS: tabular data comprehension
- User-centric enhancement
 - Relying on item description/representations -> lengthy prompt/behavior-unaware integration
 - Item-free user enhancement, behavior-aware integration



- Scalability
 - More efficient than “LLM as RS” solutions
 - Huge computational & storage burden for extremely numerous items, especially **knowledge & model enhancement** -> Limited ROI
- Explainability
 - Not addressed by LLM
 - Generating explanations while enhancing RS
- Evaluations
 - Only evaluated combined with traditional RS
 - Lack of benchmarks and metrics for LLM **enhancement**

